

Recurrent Neural Networks

Wei Xu

(many slides from Greg Durrett)

This Lecture

- ▶ Recurrent neural networks
- ▶ Vanishing gradient problem
- ▶ LSTMs / GRUs
- ▶ Applications — RNN language models

Readings

- ▶ Reading: RNNs
 - ▶ Eisenstein 7.6
 - ▶ Jurafsky and Martin, Chapter 14
 - ▶ Goldberg 10, 11

A Primer on Neural Network Models for Natural Language Processing

Yoav Goldberg
Draft as of October 5, 2015.

The most up-to-date version of this manuscript is available at <http://www.cs.biu.ac.il/~yogo/nnlp.pdf>. Major updates will be published on arxiv periodically. I welcome any comments you may have regarding the content and presentation. If you spot a missing reference or have relevant work you'd like to see mentioned, do let me know. first.last@gmail

Abstract

Over the past few years, neural networks have re-emerged as powerful machine-learning models, yielding state-of-the-art results in fields such as image recognition and speech processing. More recently, neural network models started to be applied also to textual natural language signals, again with very promising results. This tutorial surveys neural network models from the perspective of natural language processing research, in an attempt to bring natural-language researchers up to speed with the neural techniques. The tutorial covers input encoding for natural language tasks, feed-forward networks, convolutional networks, recurrent networks and recursive networks, as well as the computation graph abstraction for automatic gradient computation.

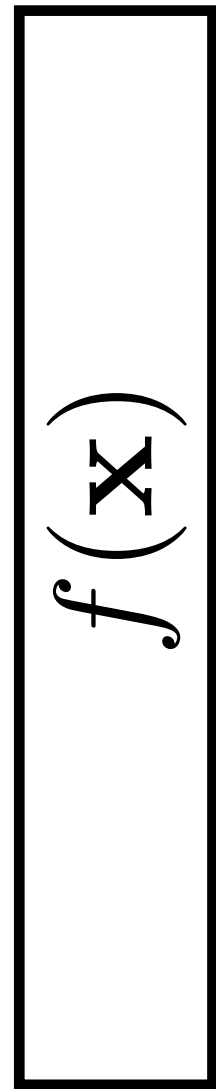
Recap: Feedforward Neural Networks

$$P(\mathbf{y}|\mathbf{x}) = \text{softmax}(Wg(Vf(\mathbf{x})))$$

We can think of a neural network classifier with one hidden layer as building a vector $\mathbf{z}$ which is a hidden layer representation (i.e. latent features) of the input, and then running standard logistic regression on the features that the network develops in $\mathbf{z}$.

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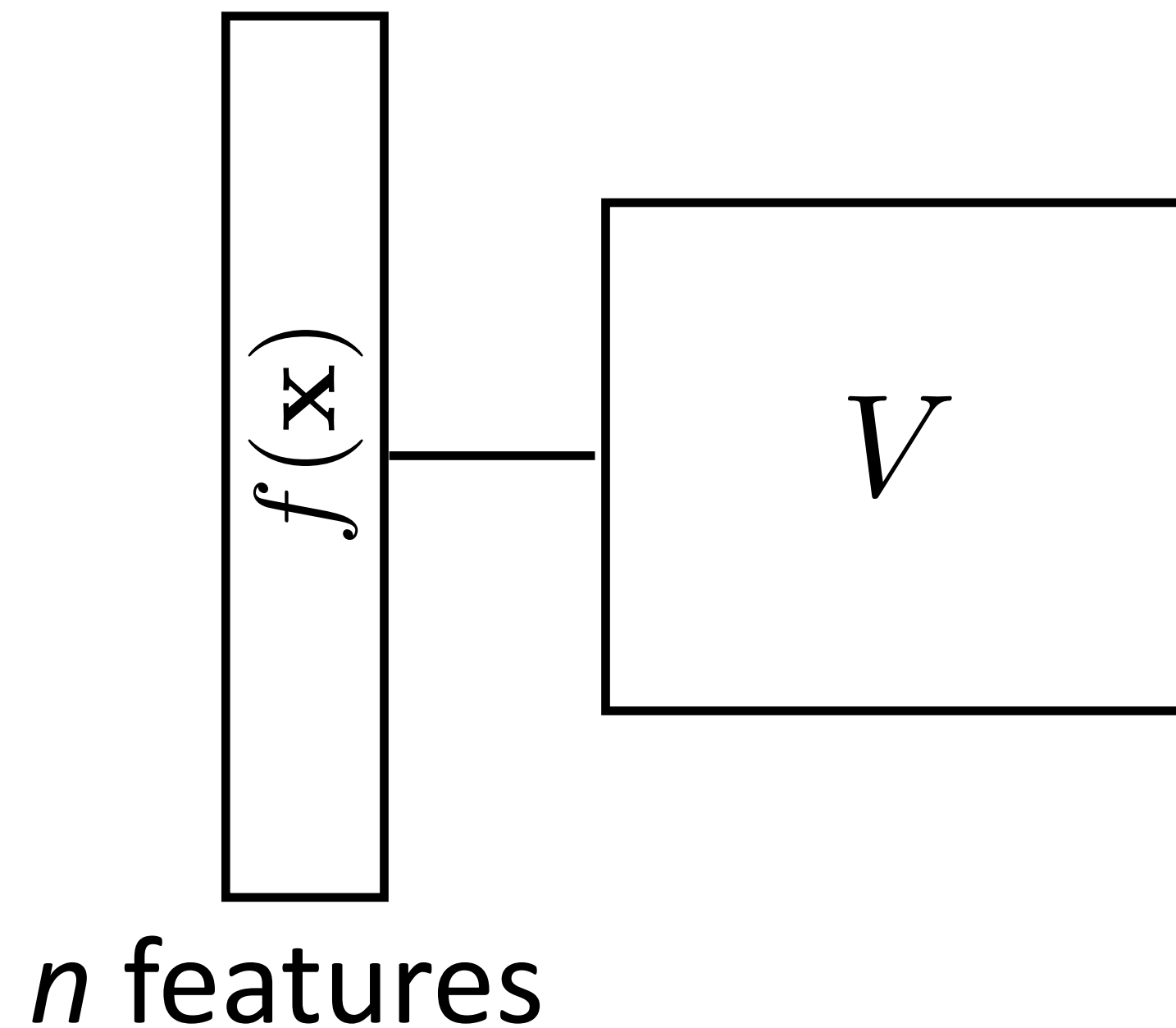


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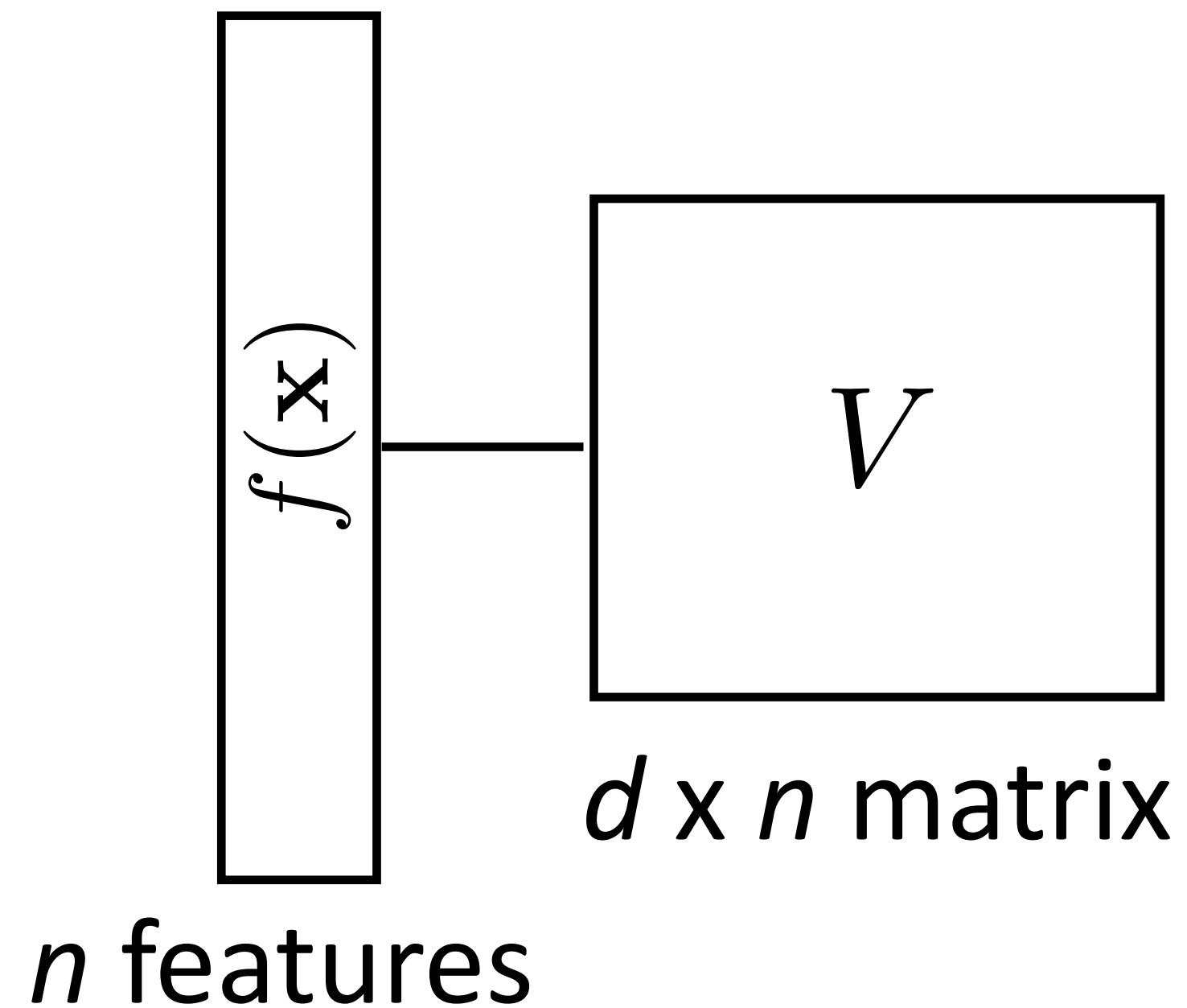
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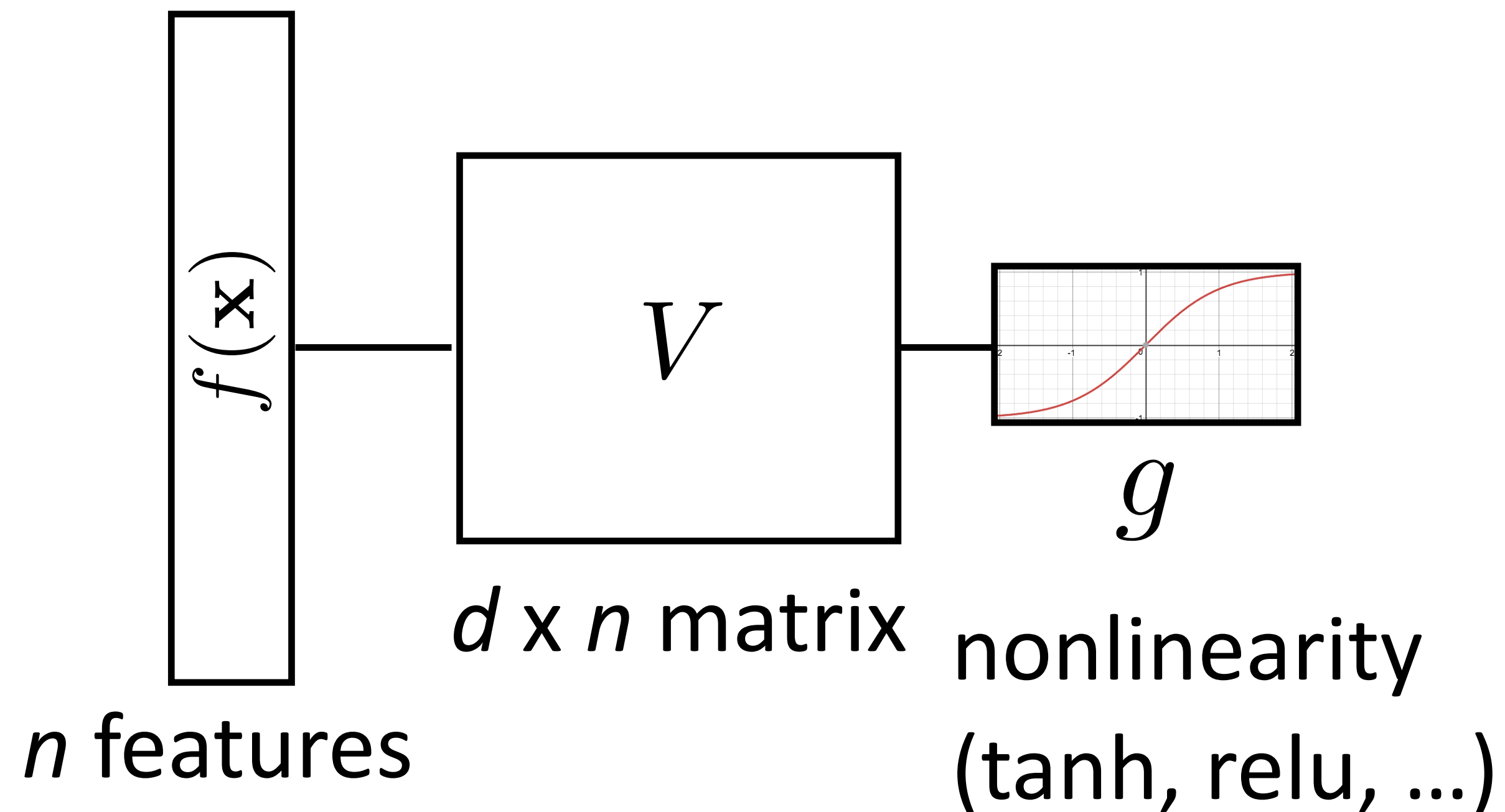
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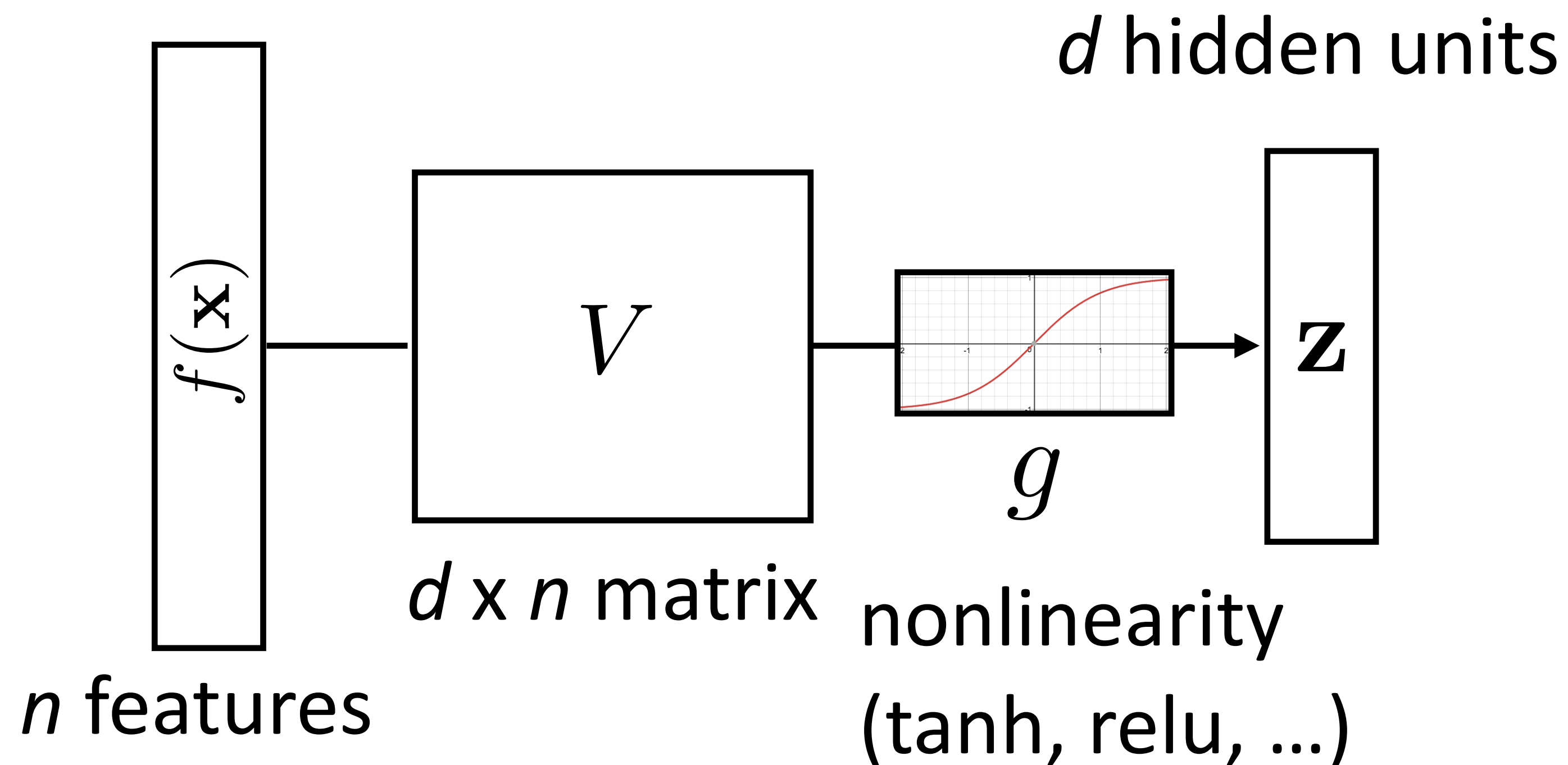
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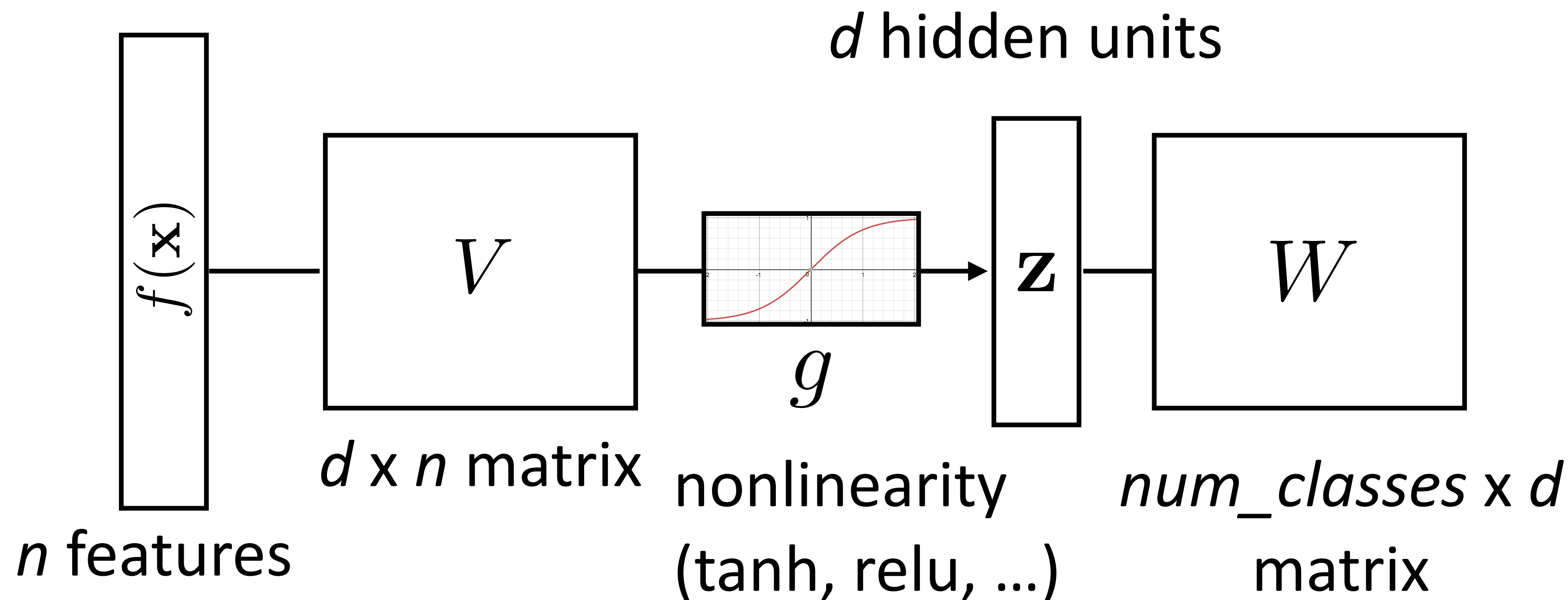
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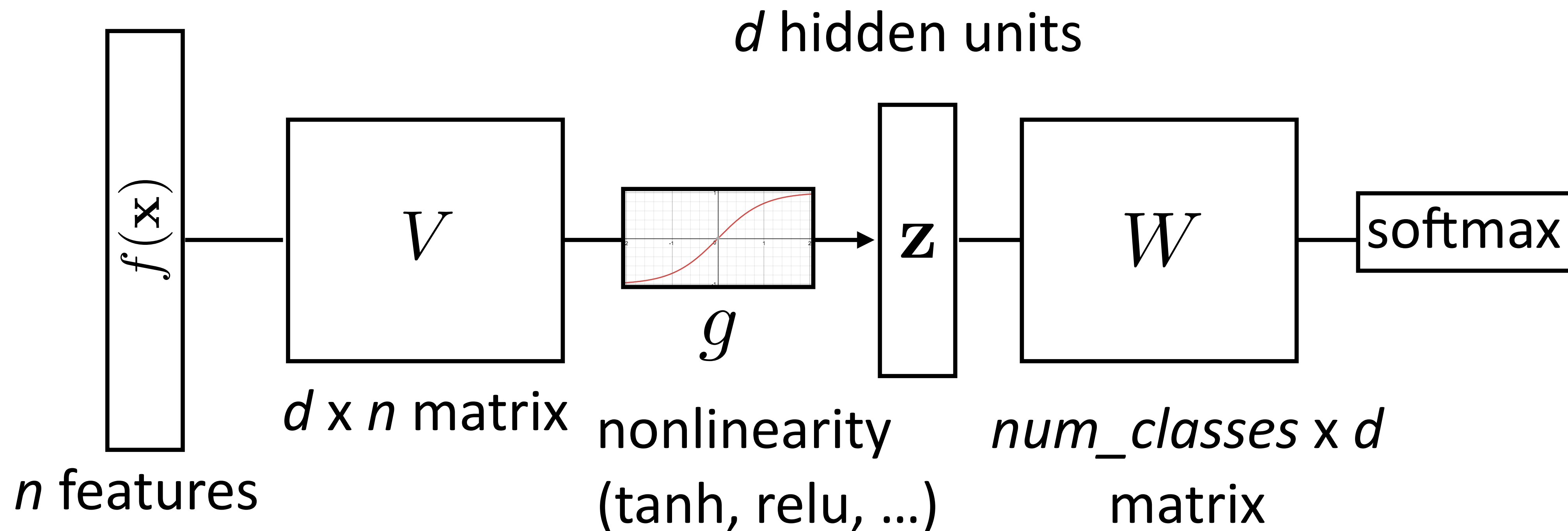
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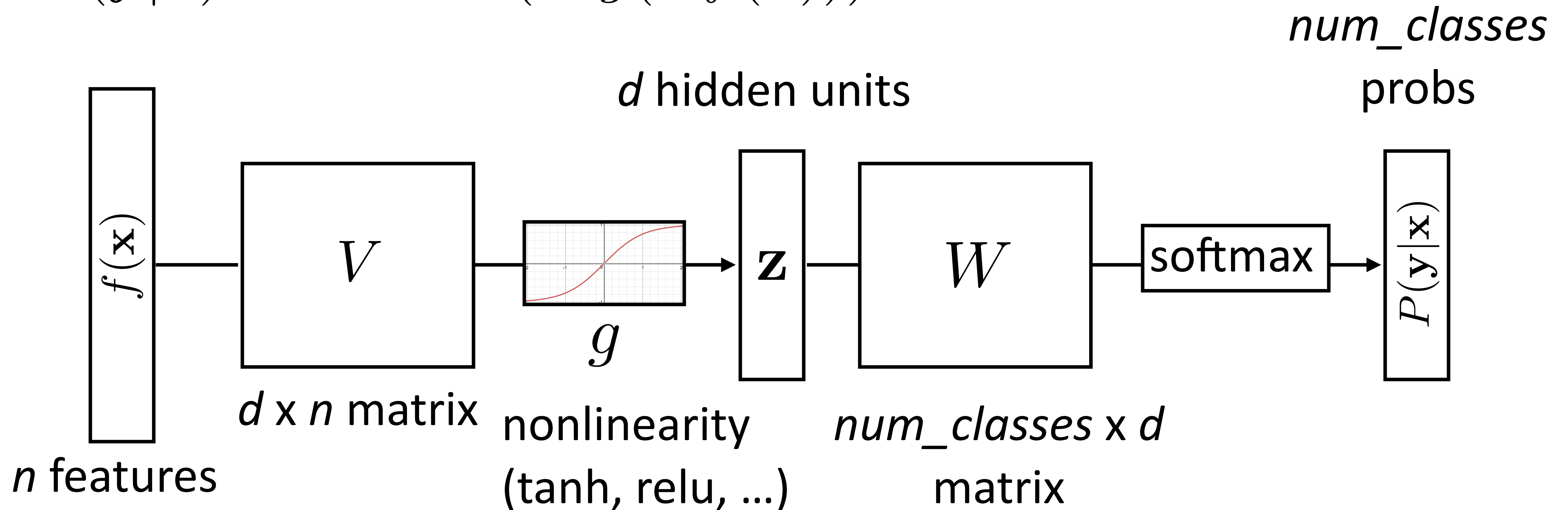
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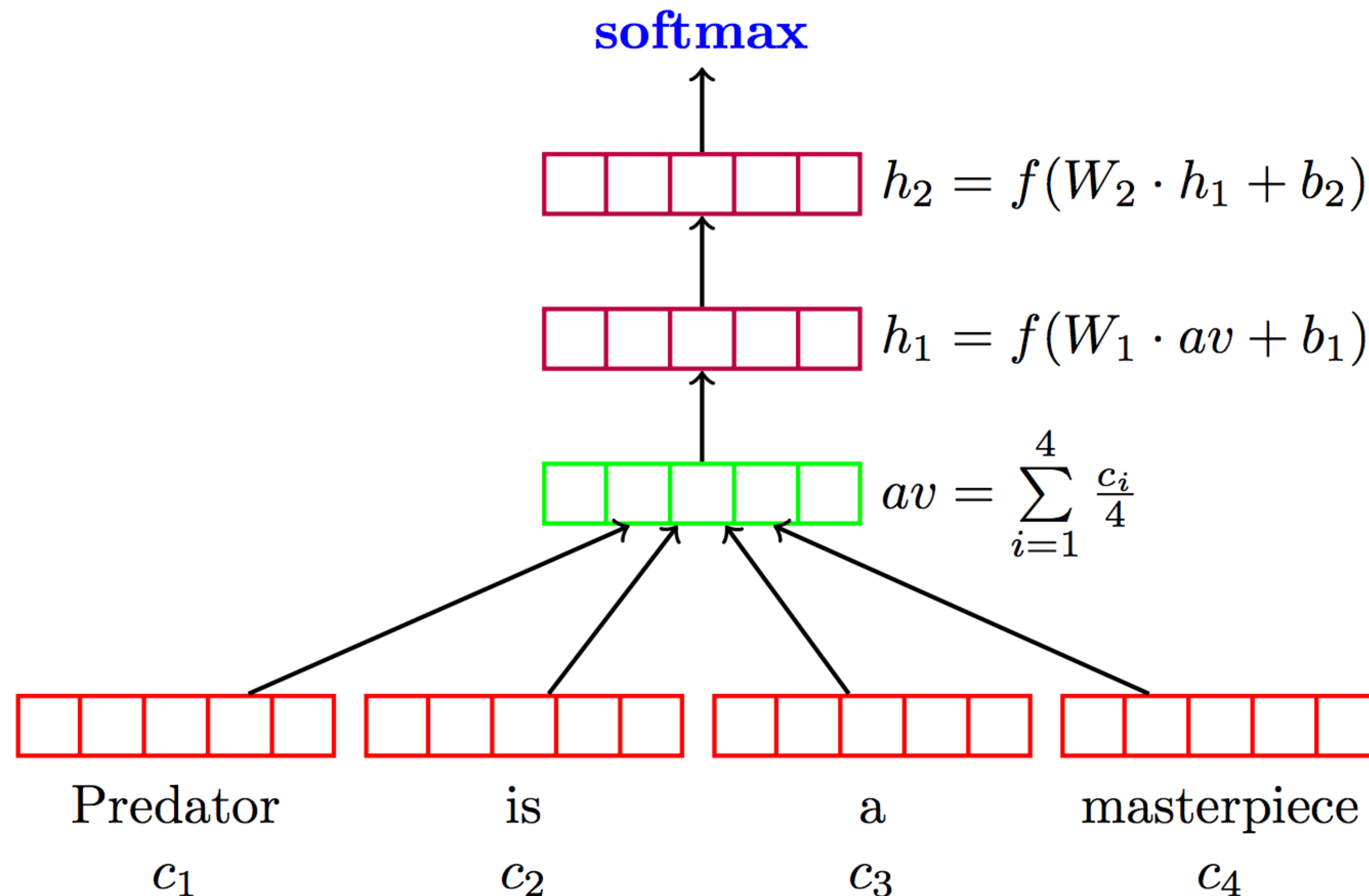
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Recap: Neural Bag-of-Words

- ▶ feedforward neural network on average of word embeddings from input



RNN Basics

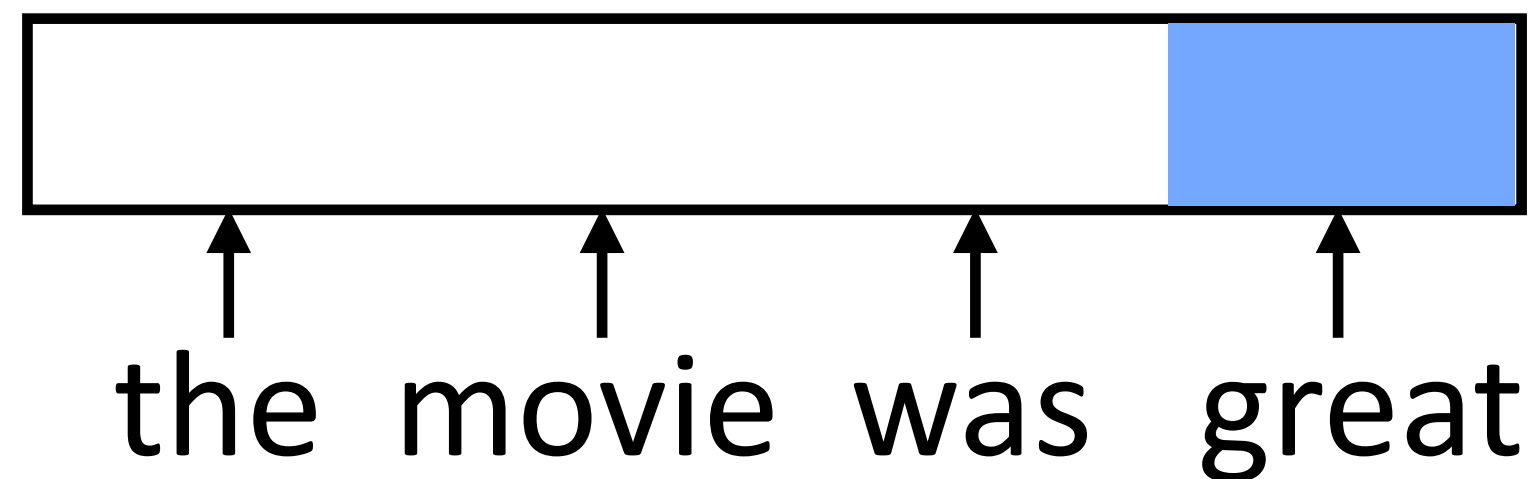
RNN Motivation

- ▶ Feedforward NNs isn't the best to handle sentences with variable length, words with multiple senses, or same words appear at different positions

the movie was great

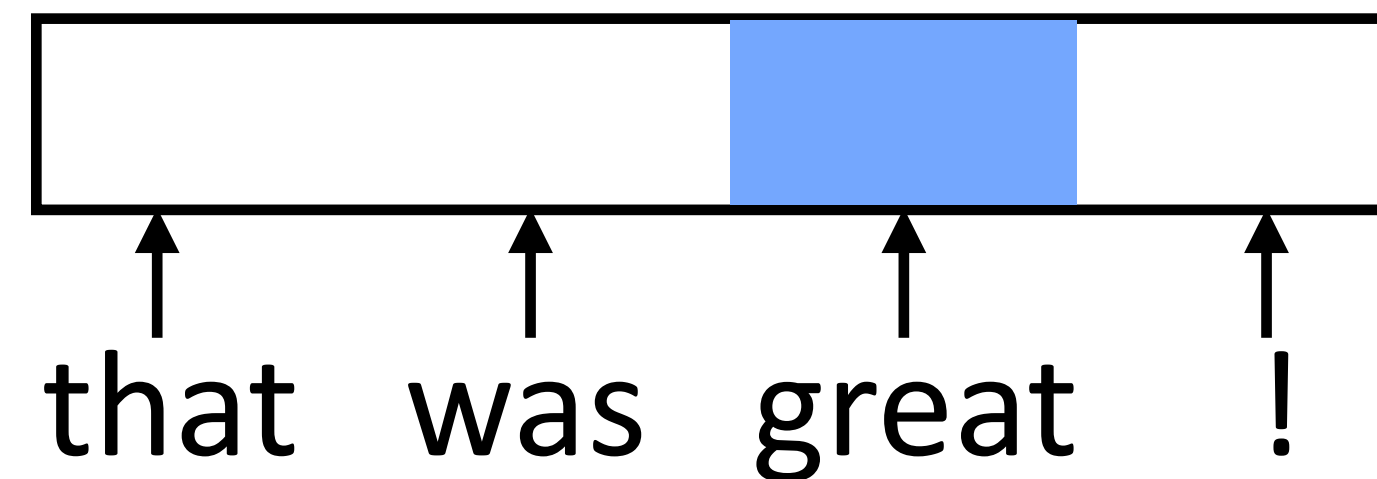
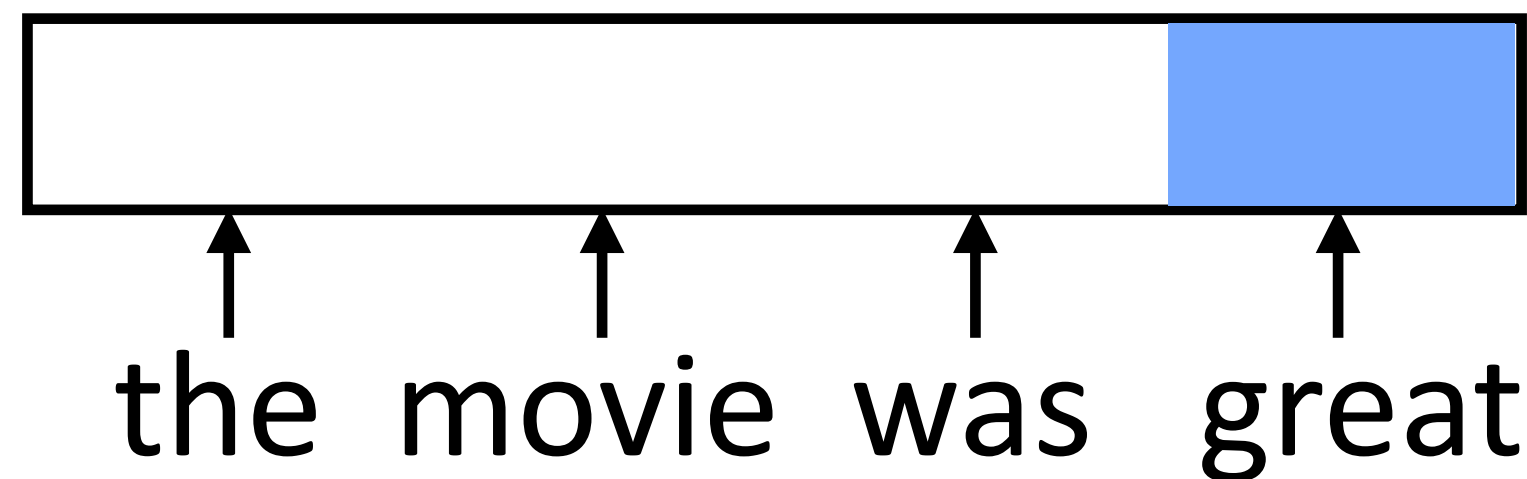
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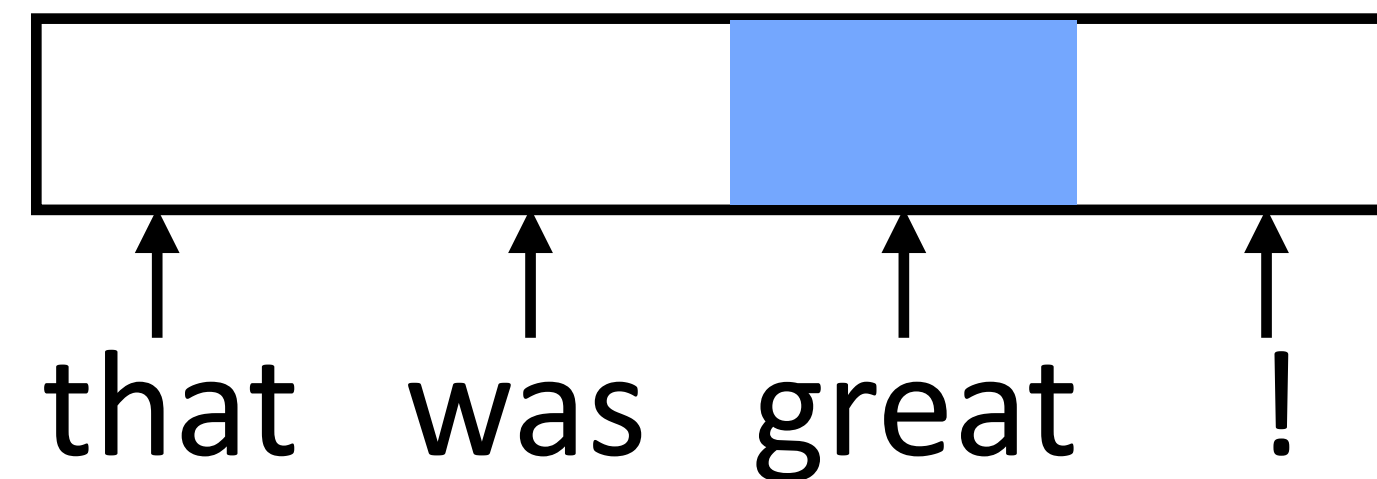
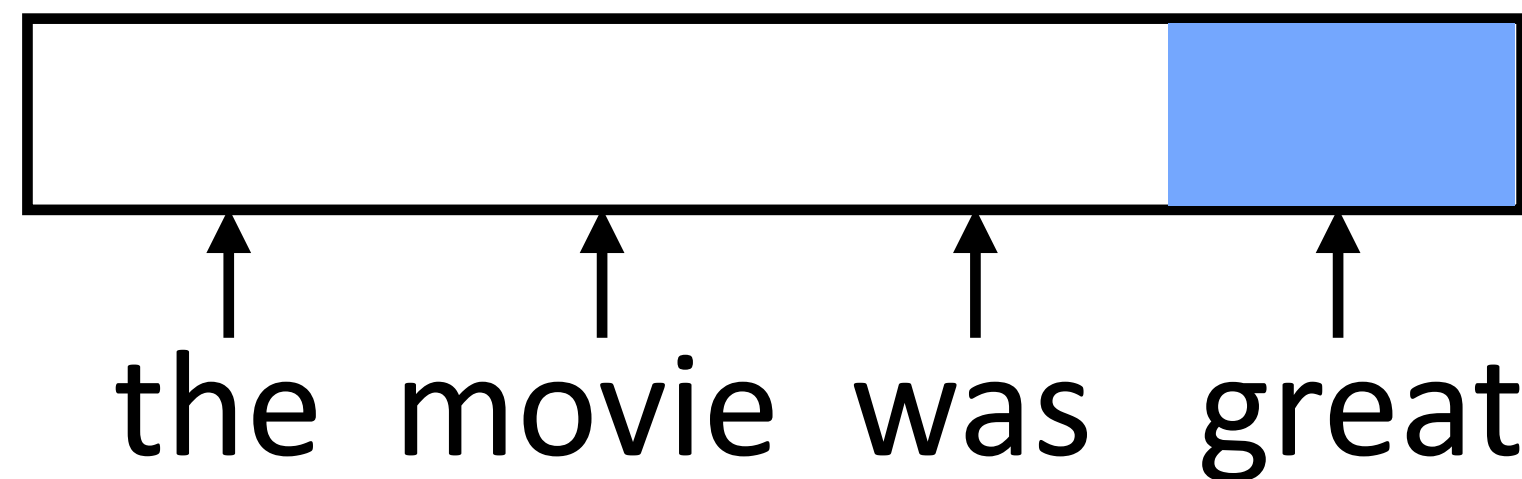
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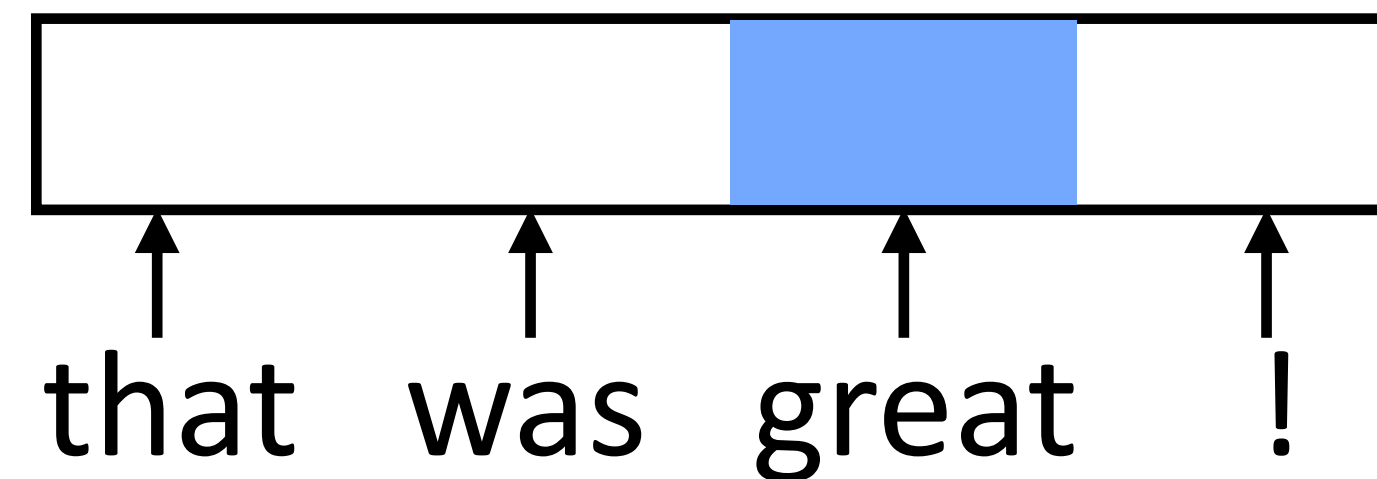
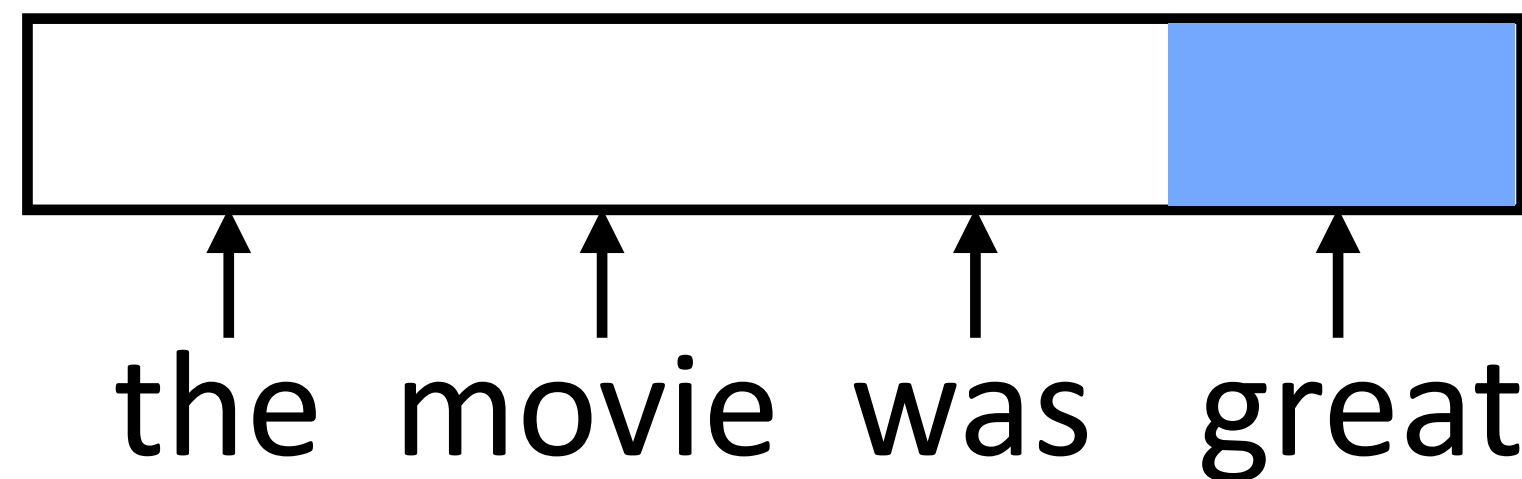
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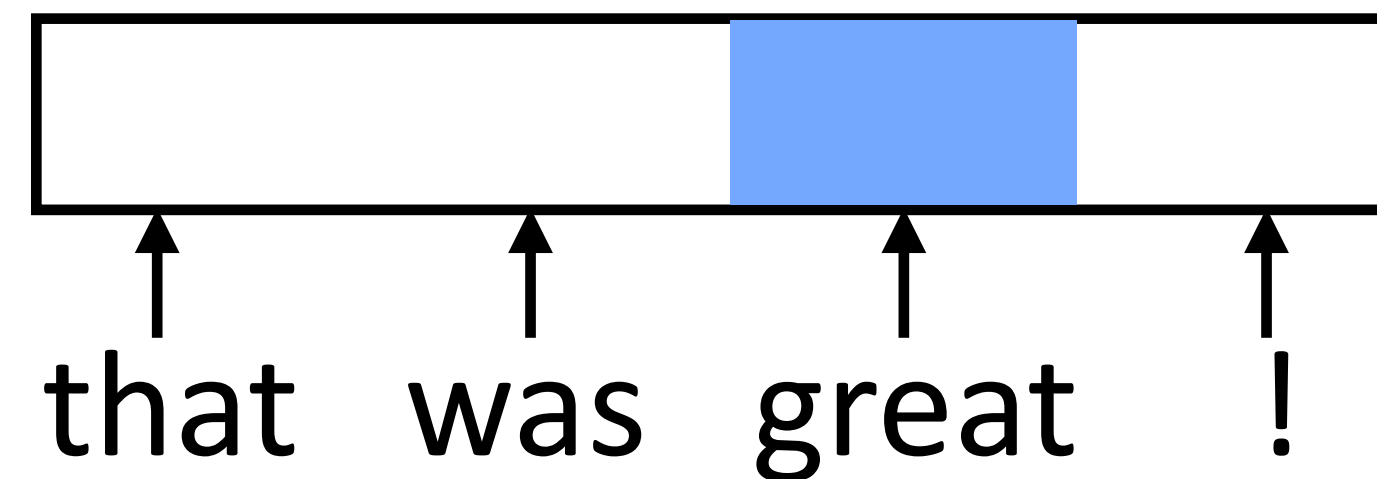
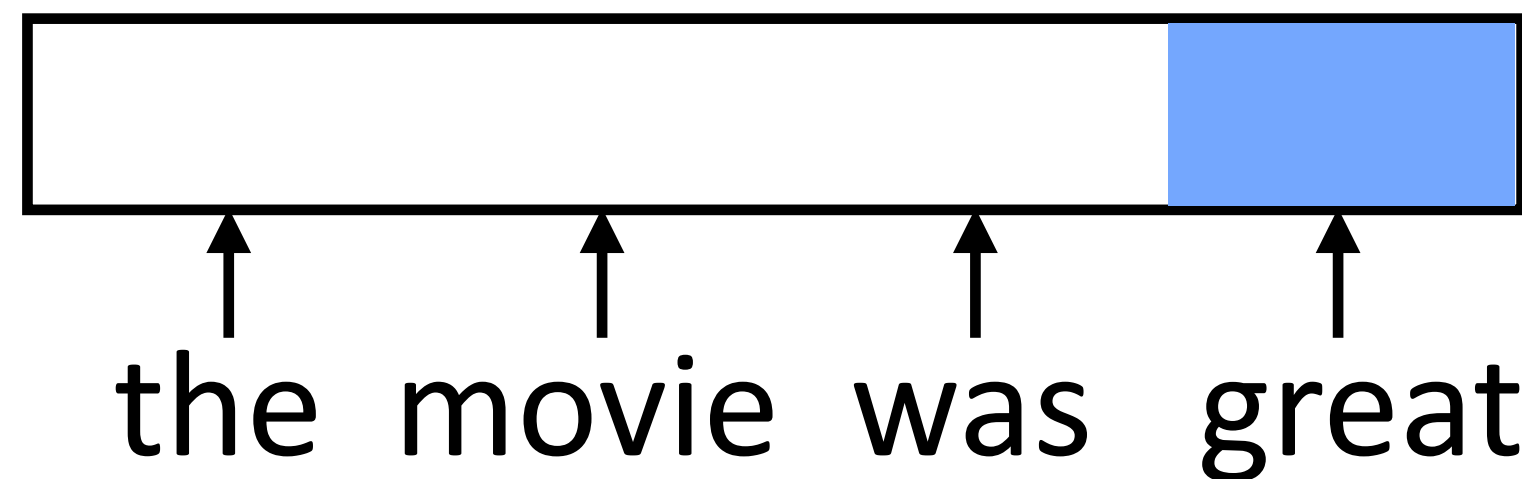
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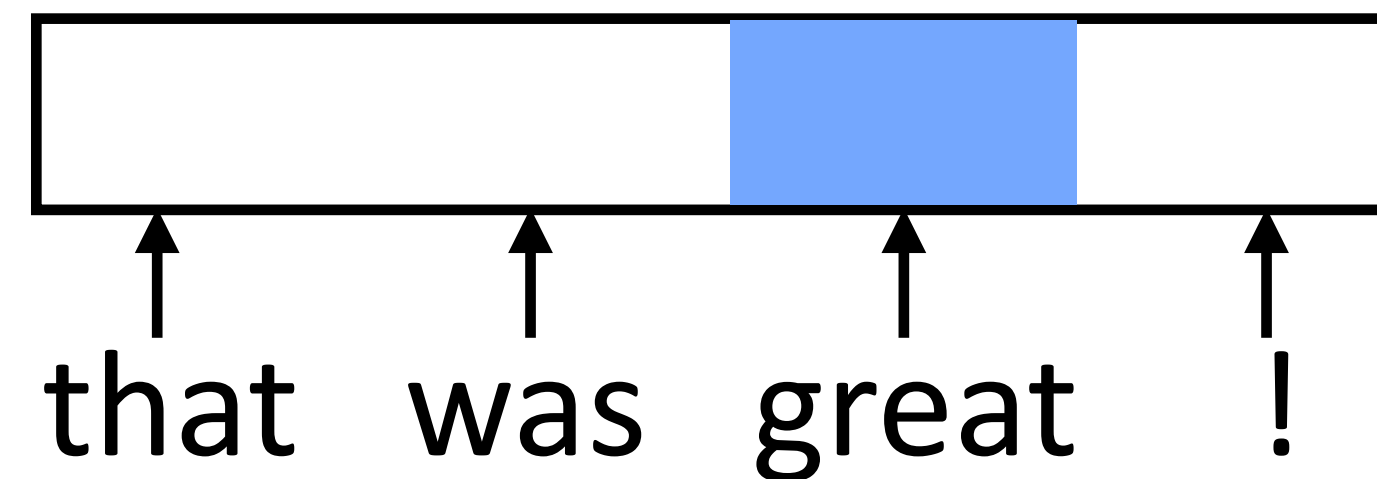
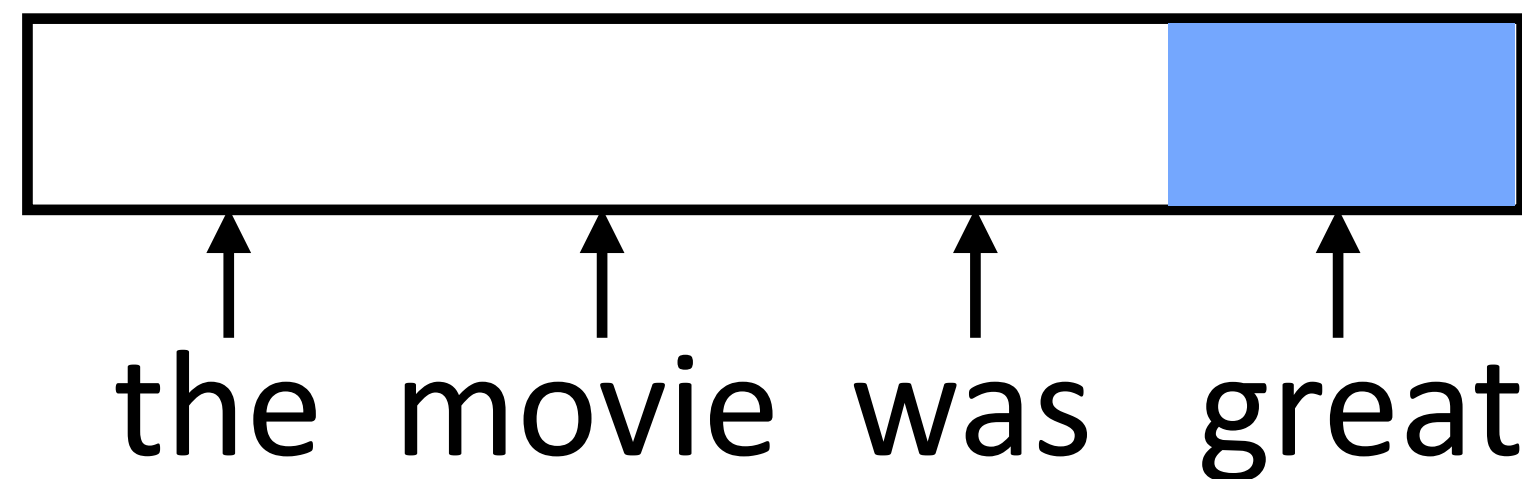
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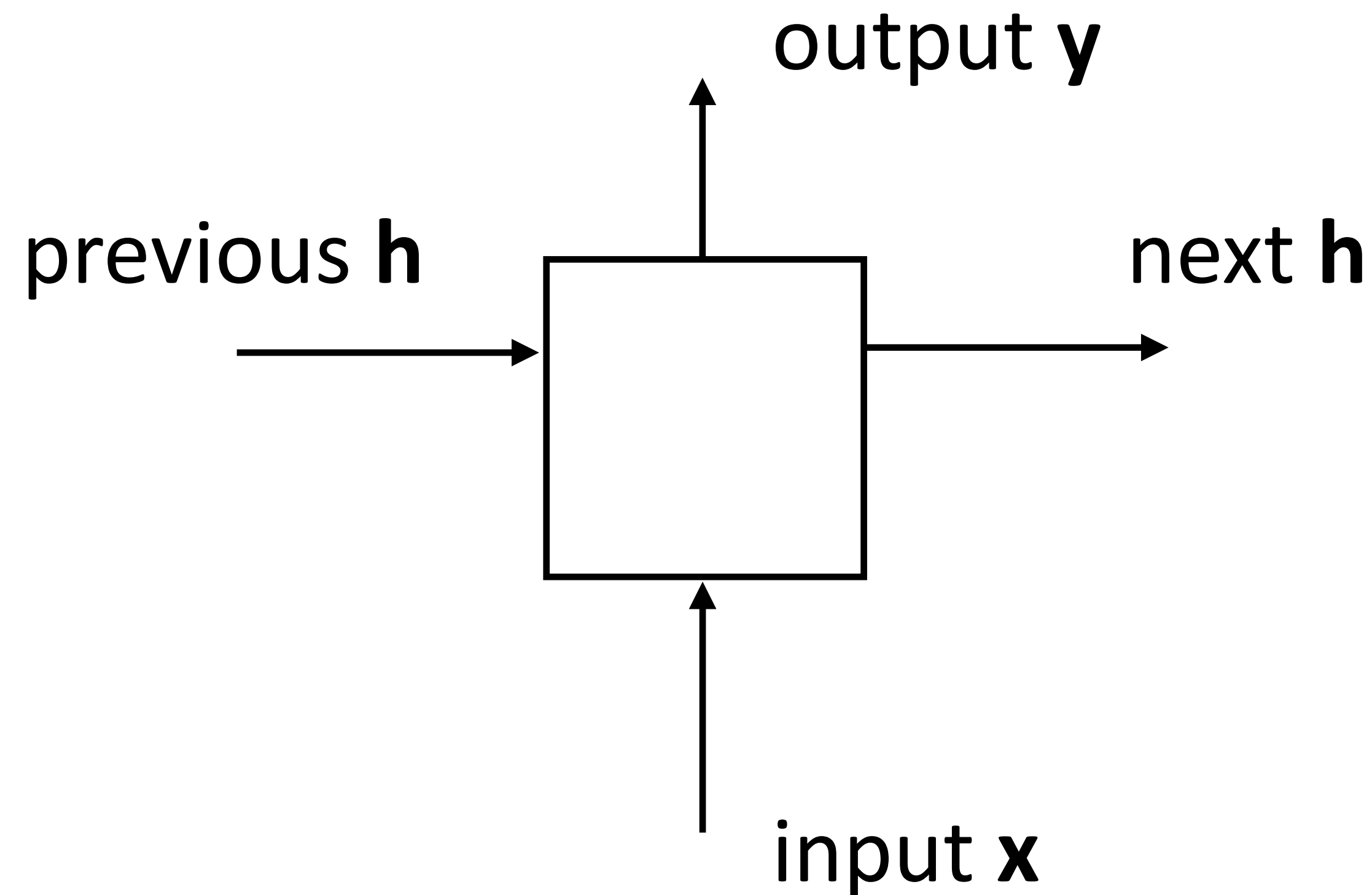
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- ▶ These don't look related (*great* is in two different orthogonal subspaces)
- ▶ Instead, we need to:
 - 1) Process each word in a uniform way
 - 2) ...while still exploiting the context that that token occurs in

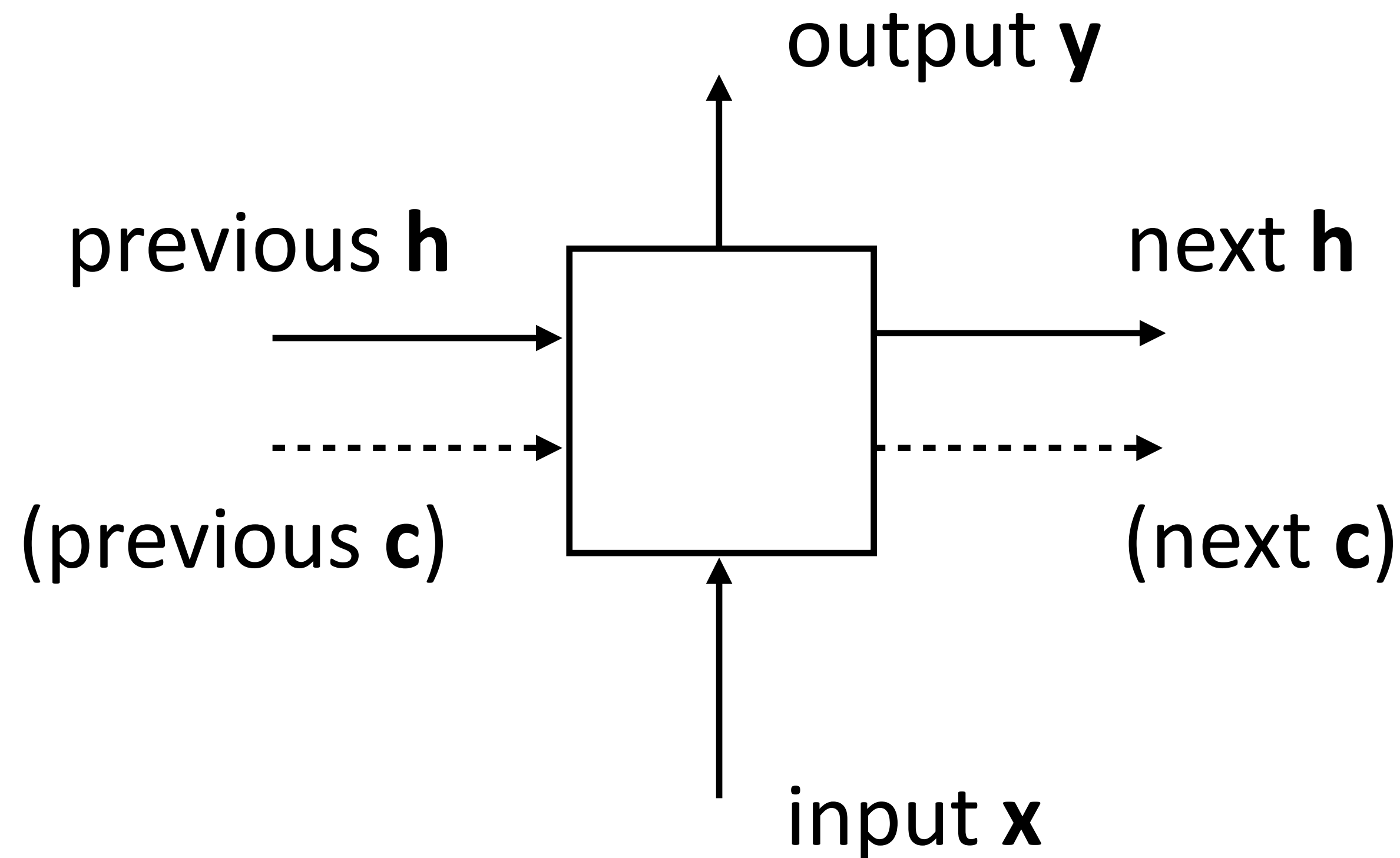
RNN Abstraction

- ▶ Cell that takes some input $\mathbf{x}$, has some hidden state $\mathbf{h}$, and updates that hidden state and produces output $\mathbf{y}$ (all vector-valued)

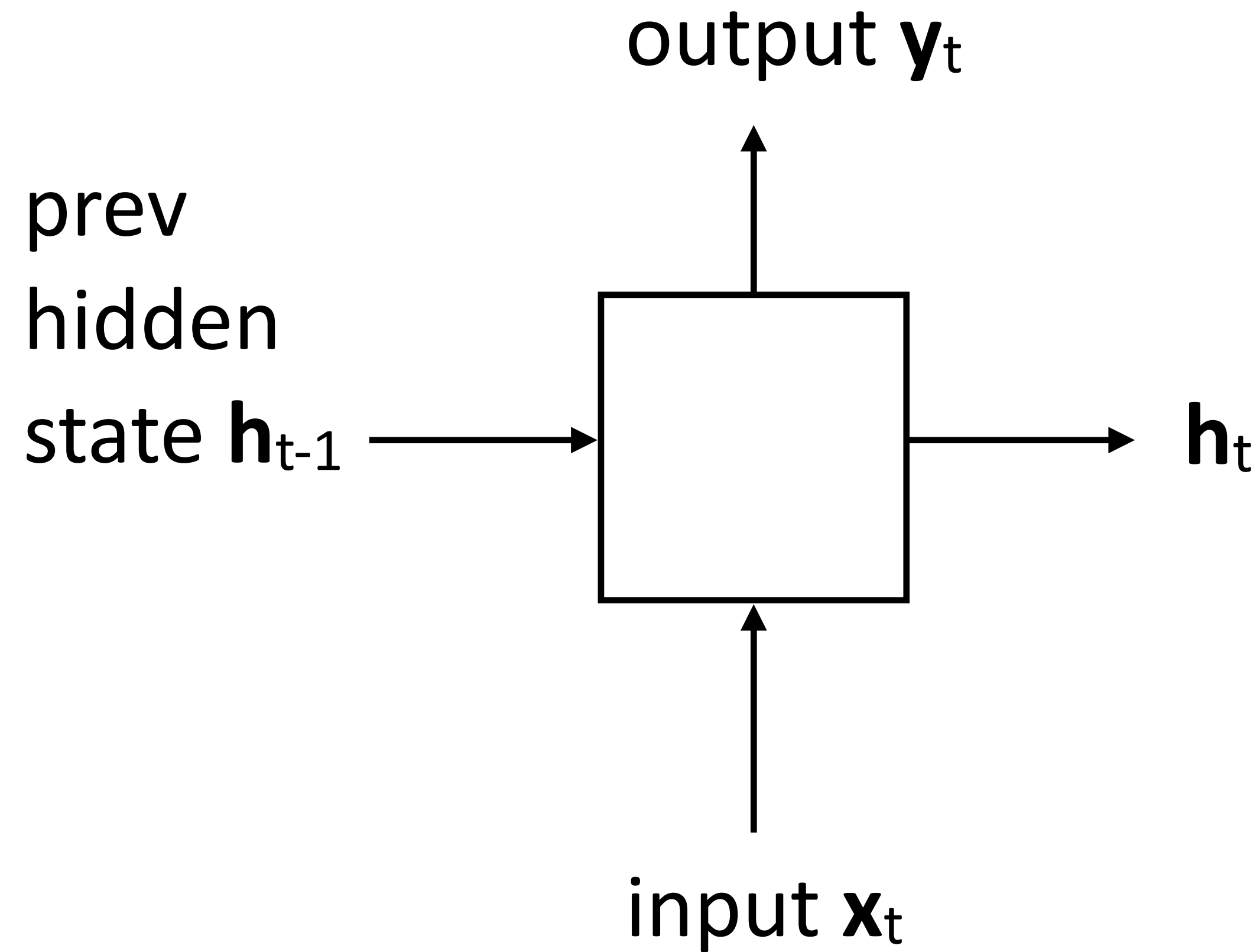


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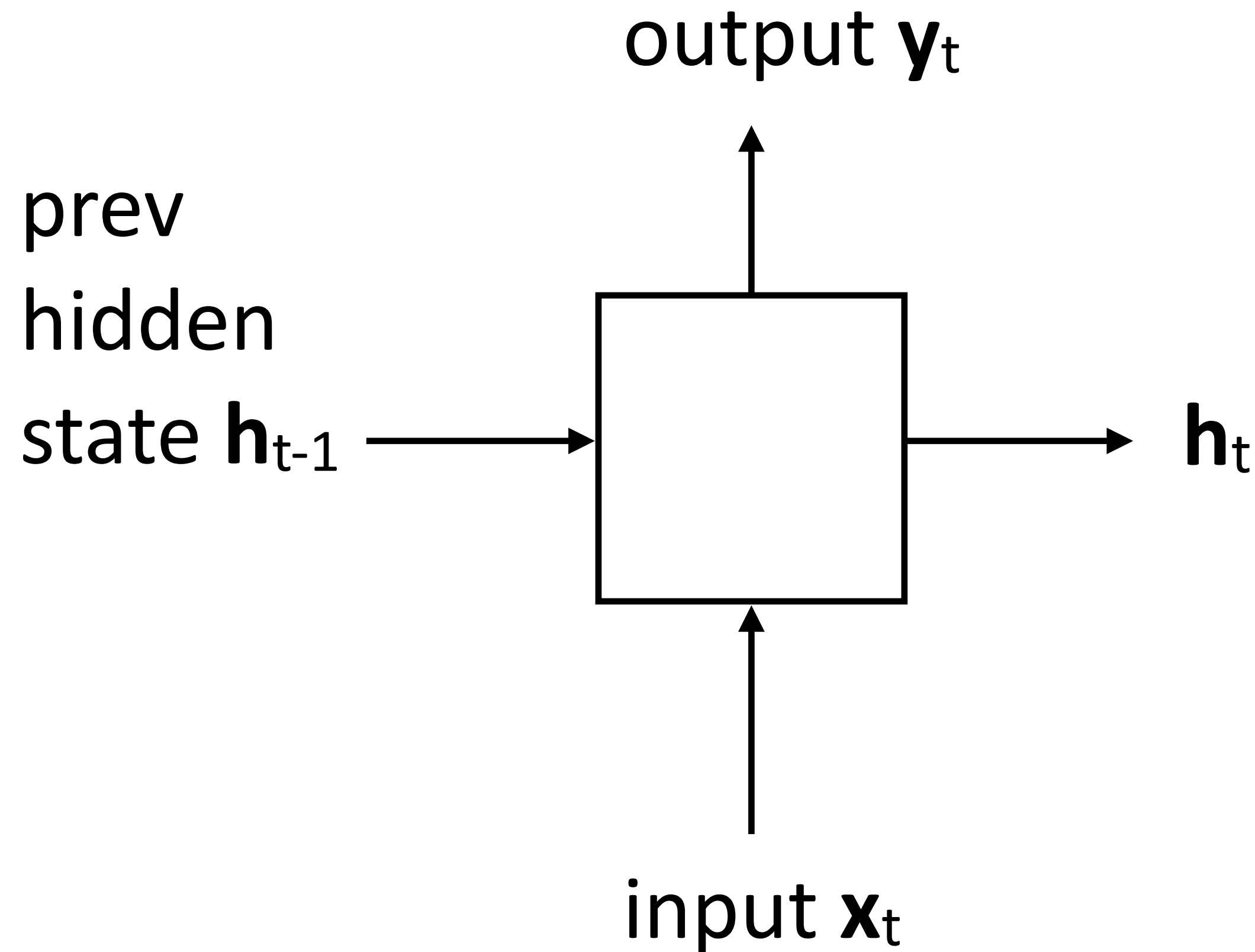
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Elman Networks



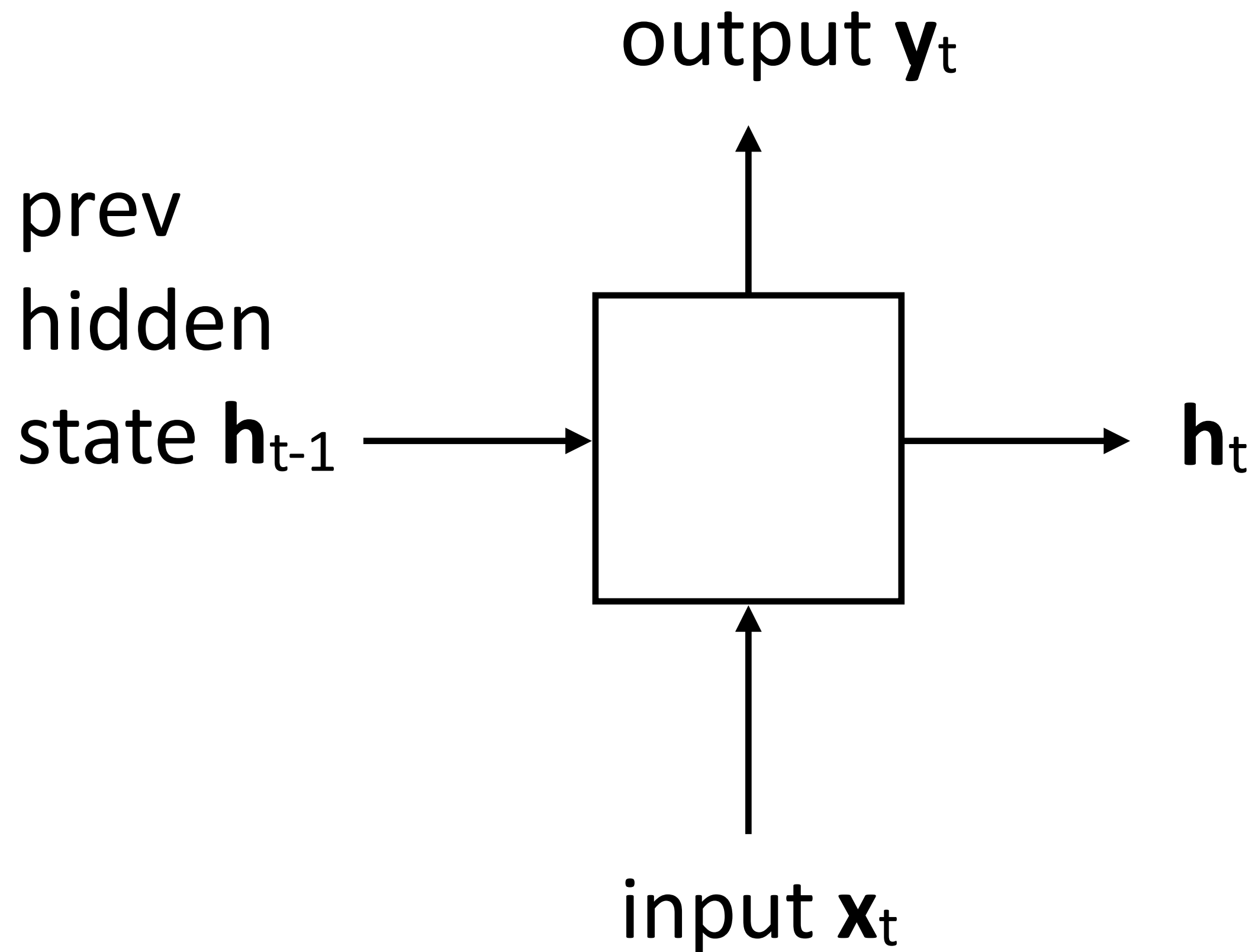
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- Updates hidden state based on input and current hidden state

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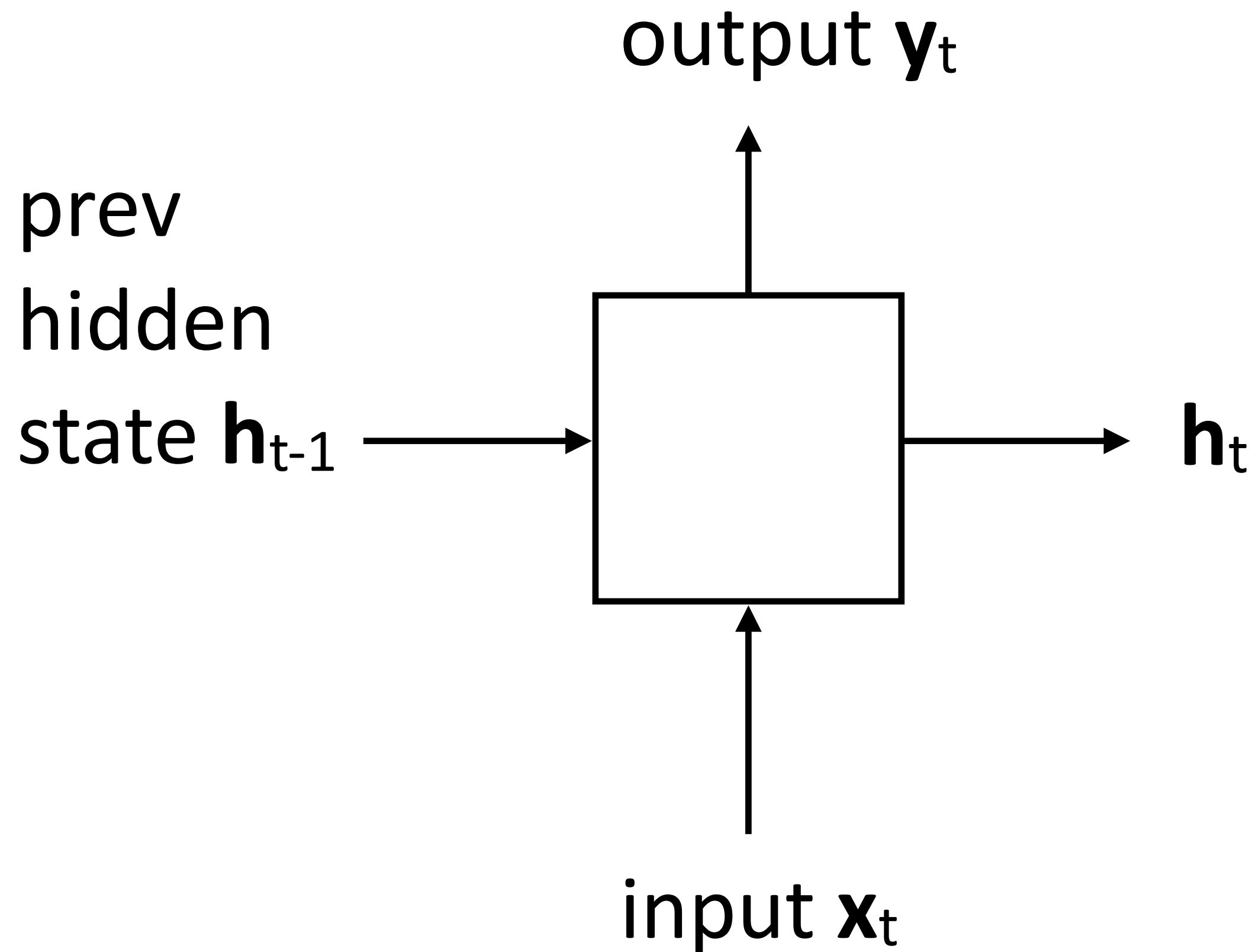
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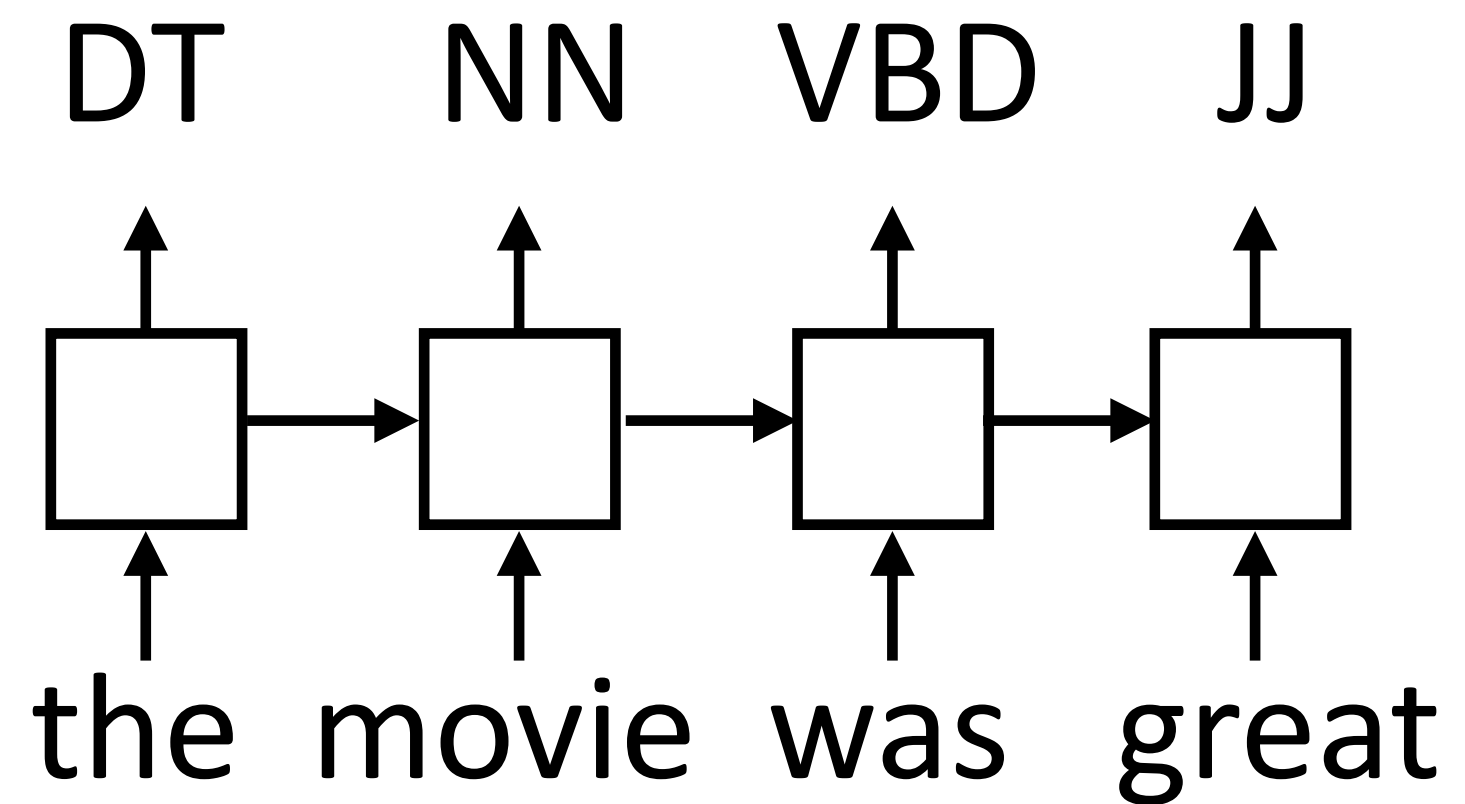
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- Long history! (invented in the late 1980s)

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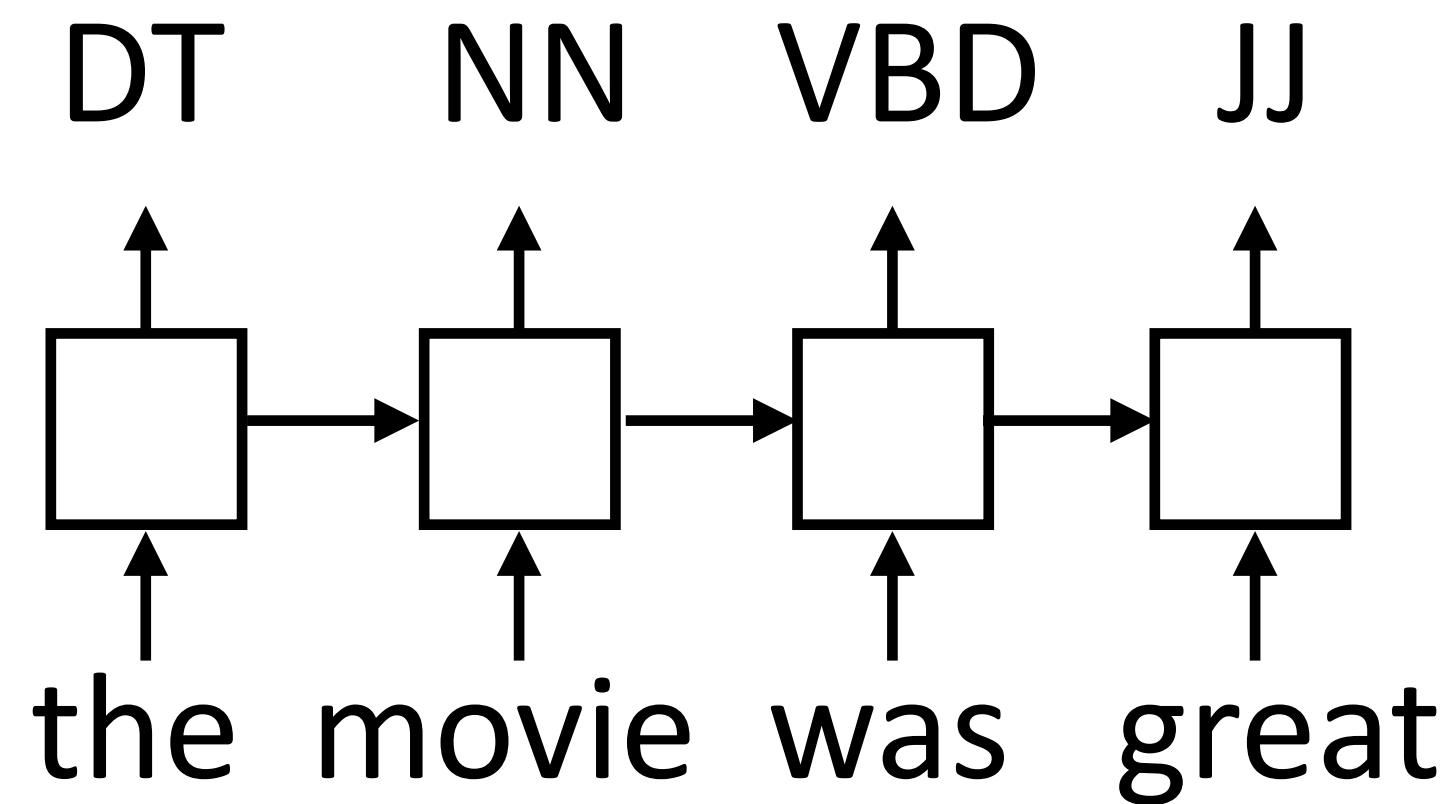
- ▶ Transducer: make some prediction for each element in a sequence



output $\mathbf{y}$ = score for each tag, then softmax

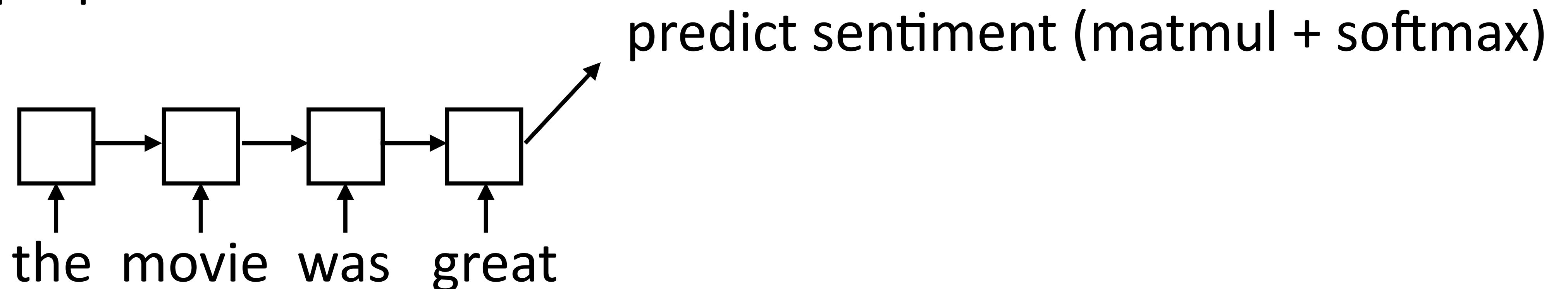
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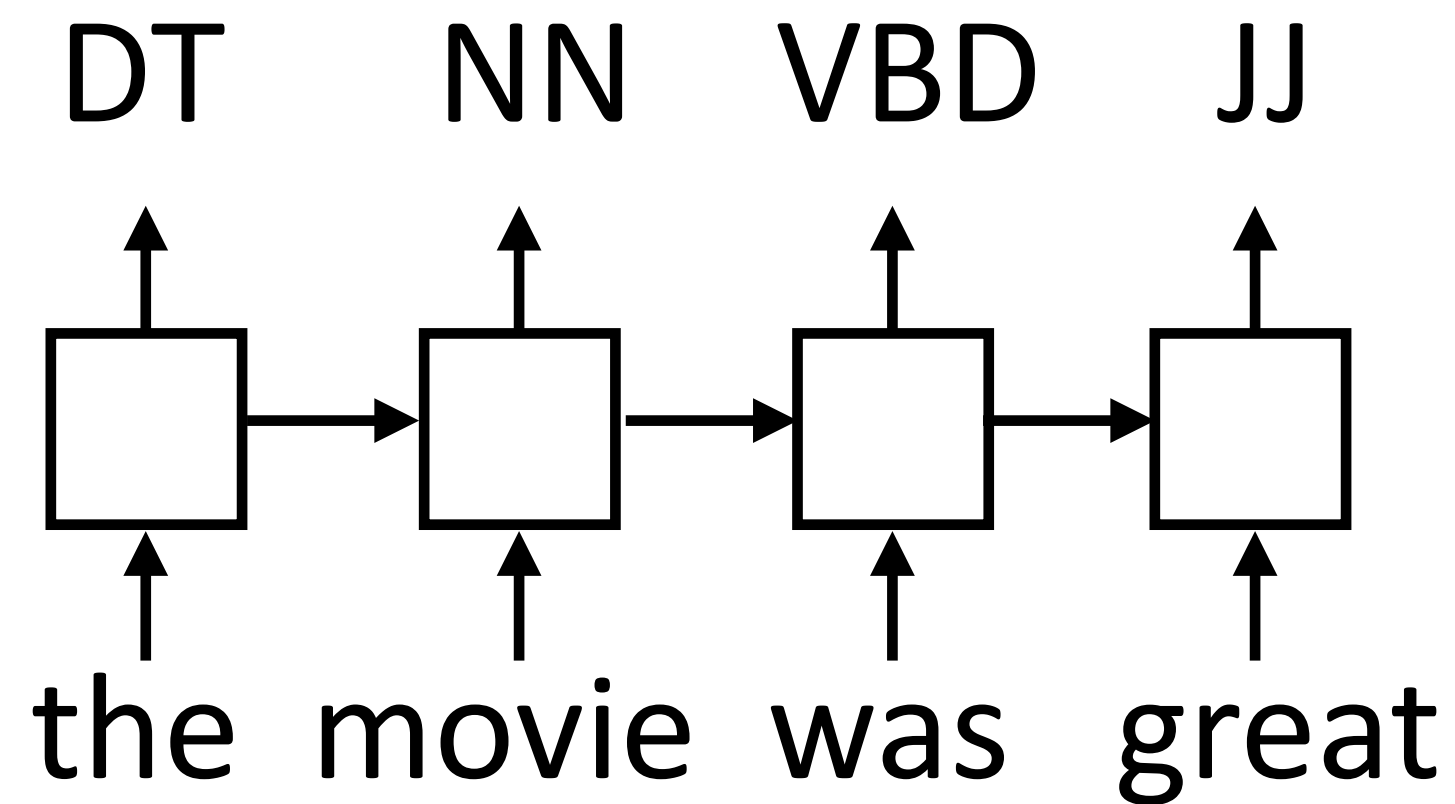
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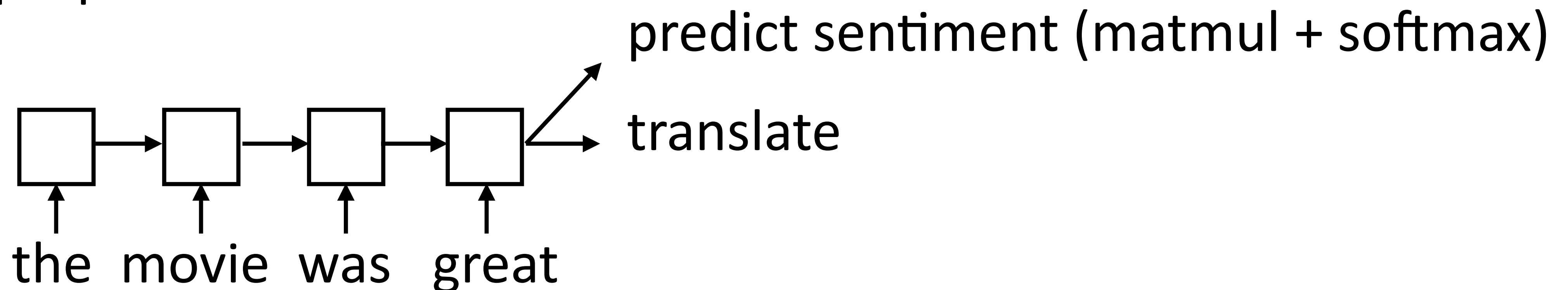
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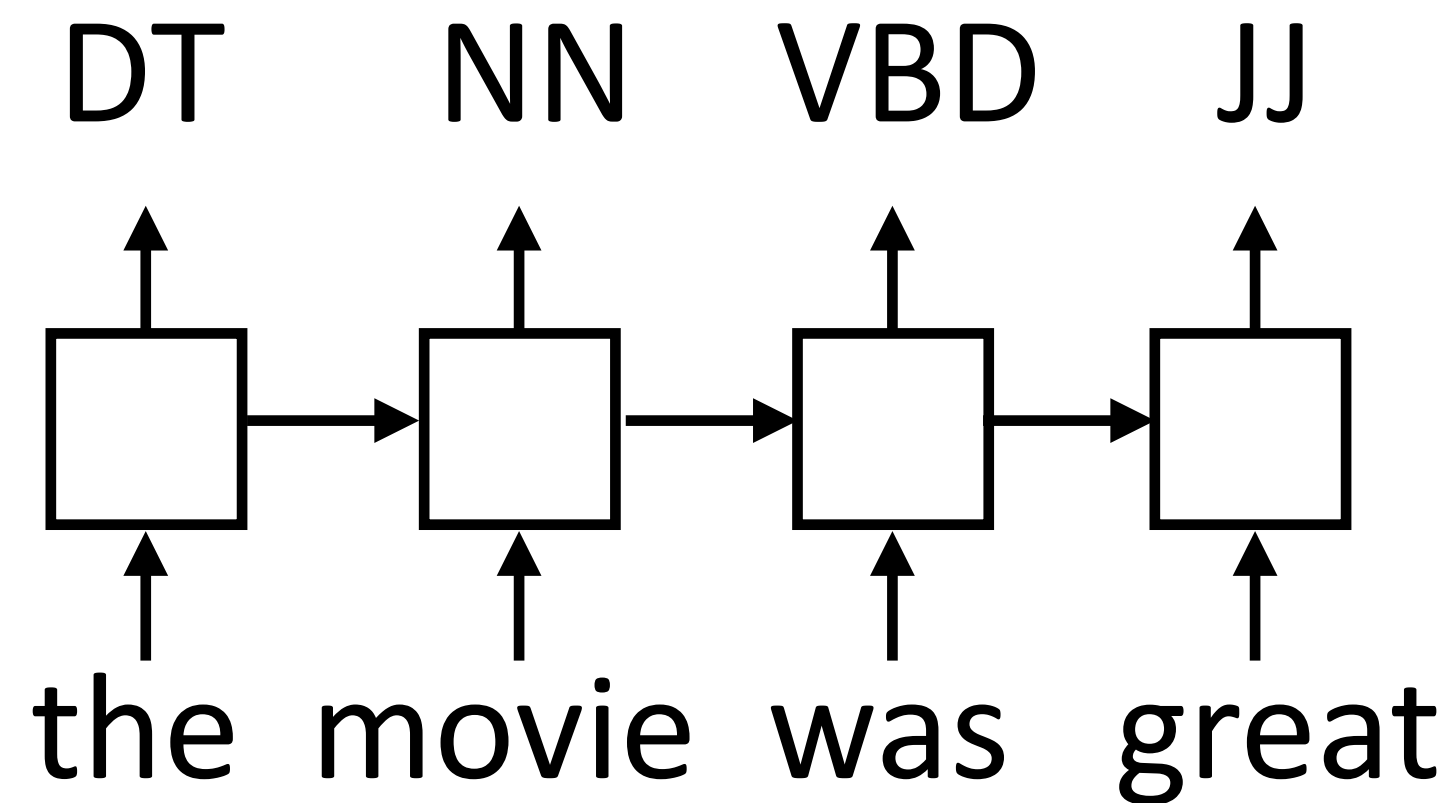
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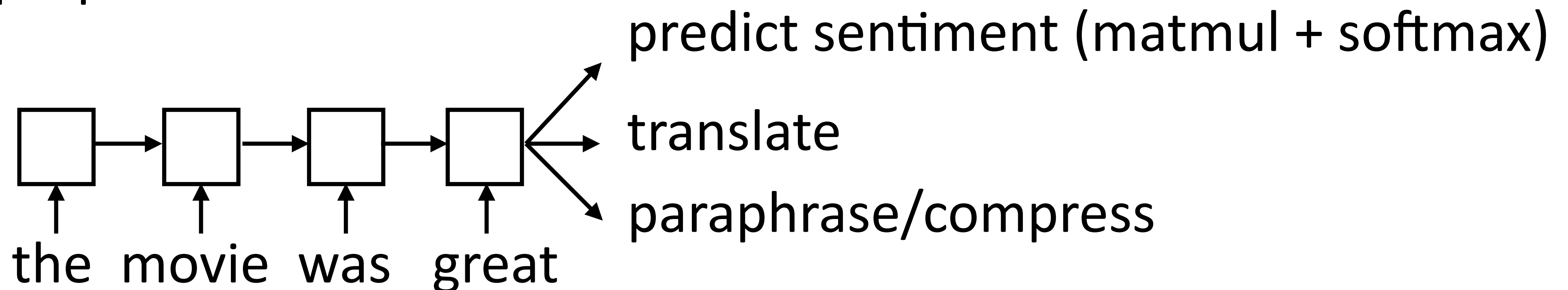
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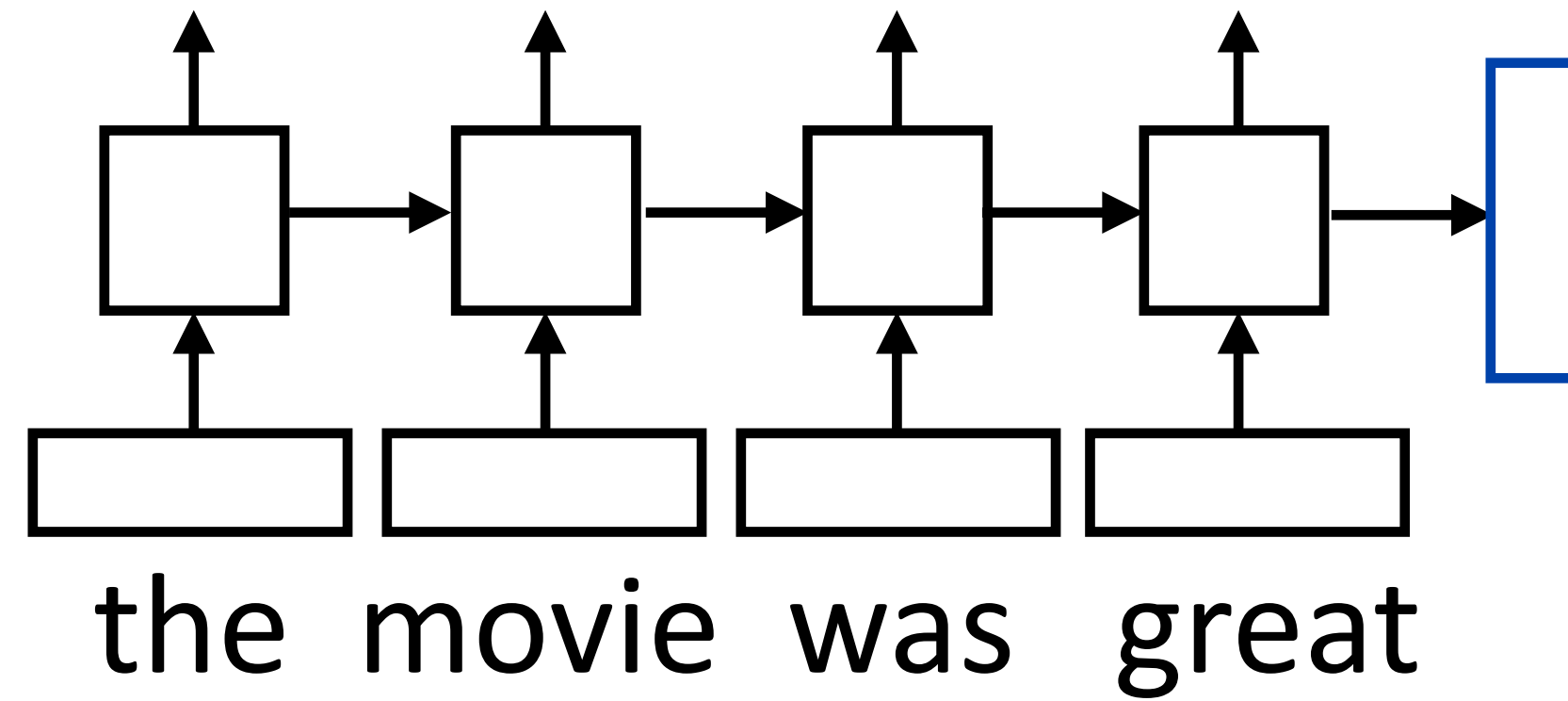


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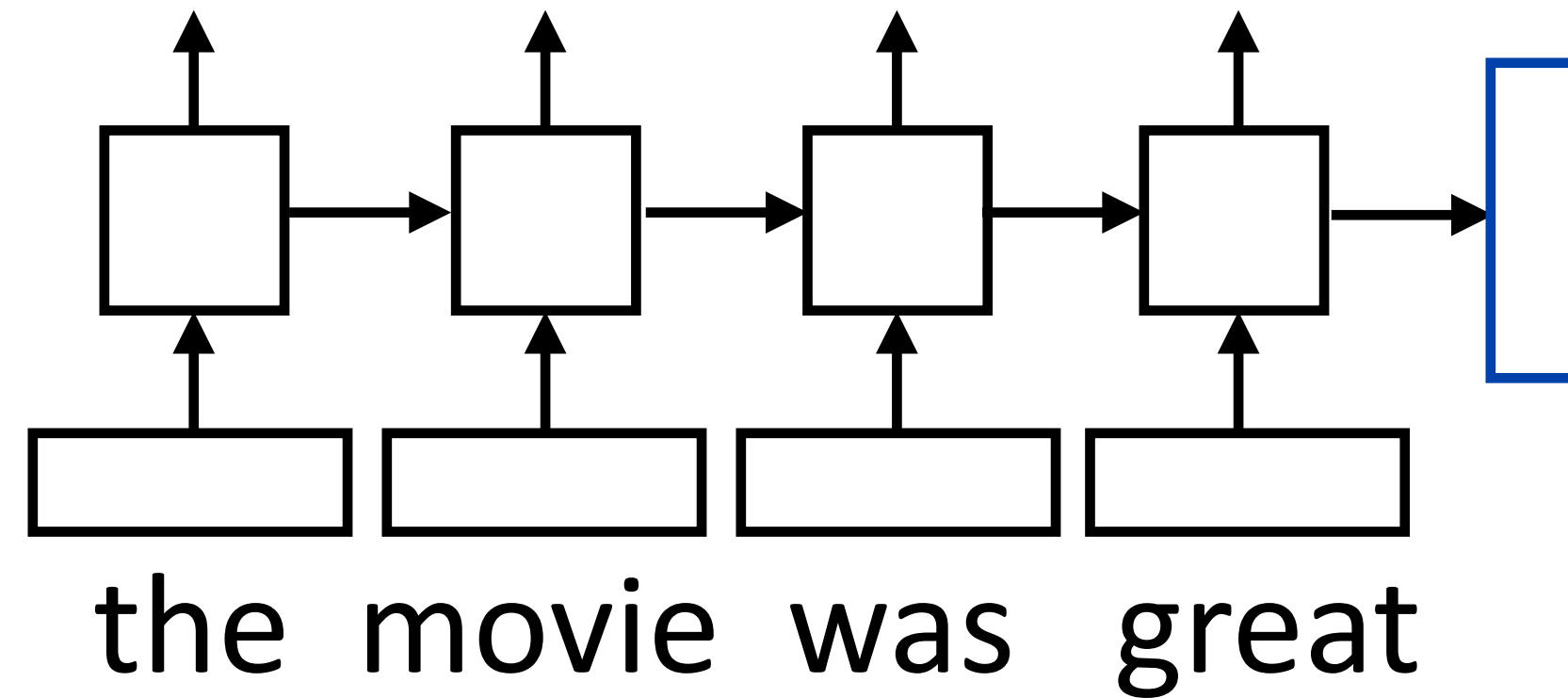
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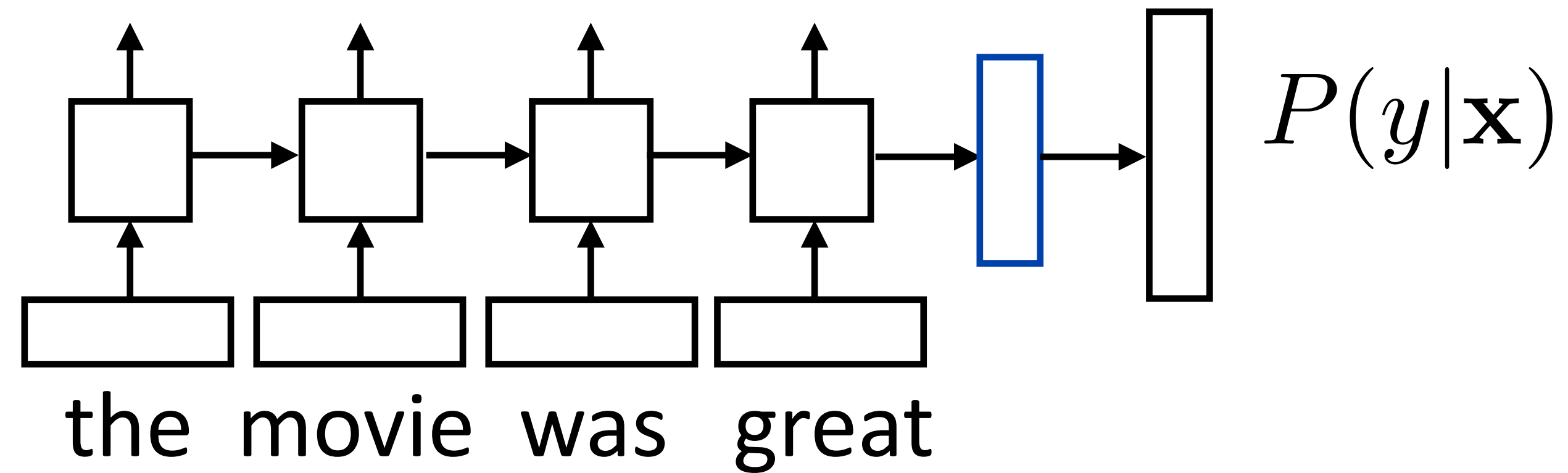


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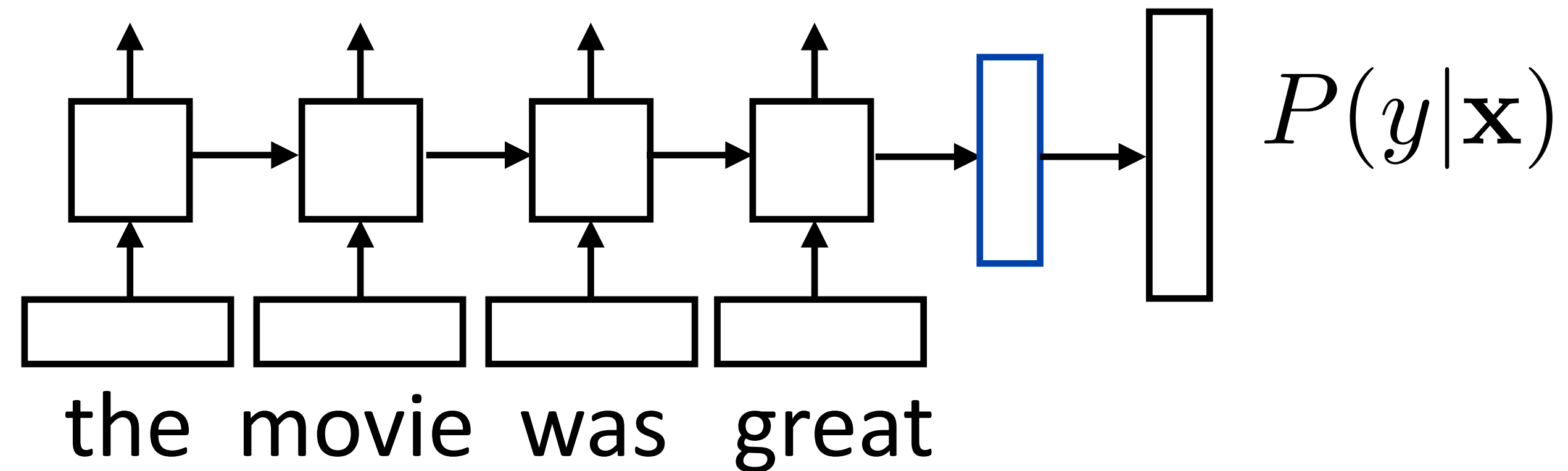
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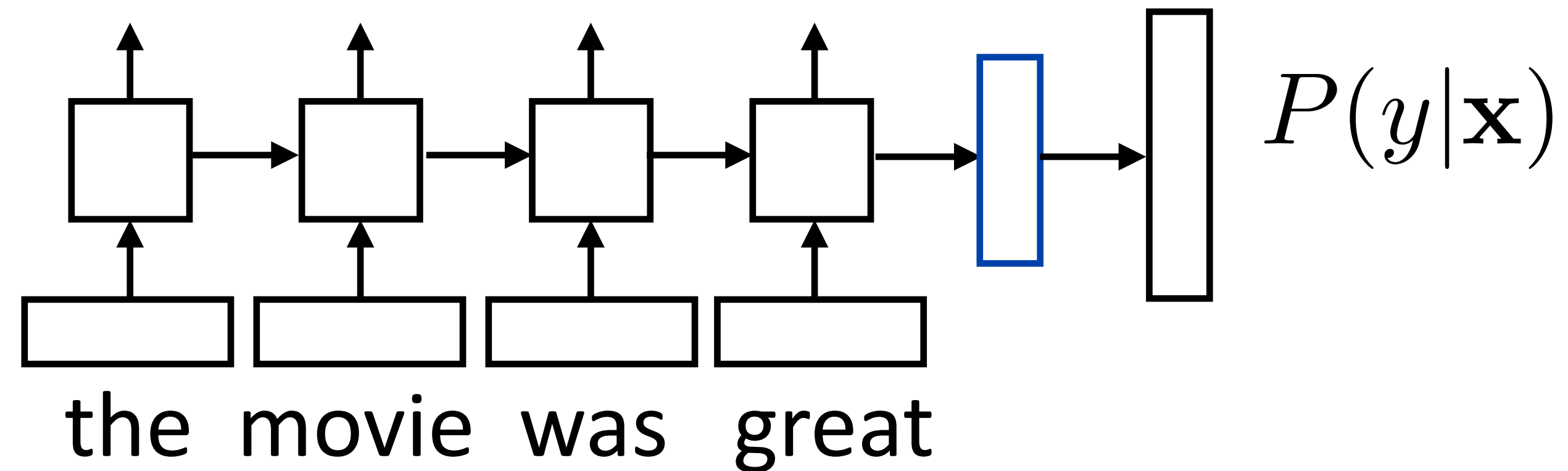
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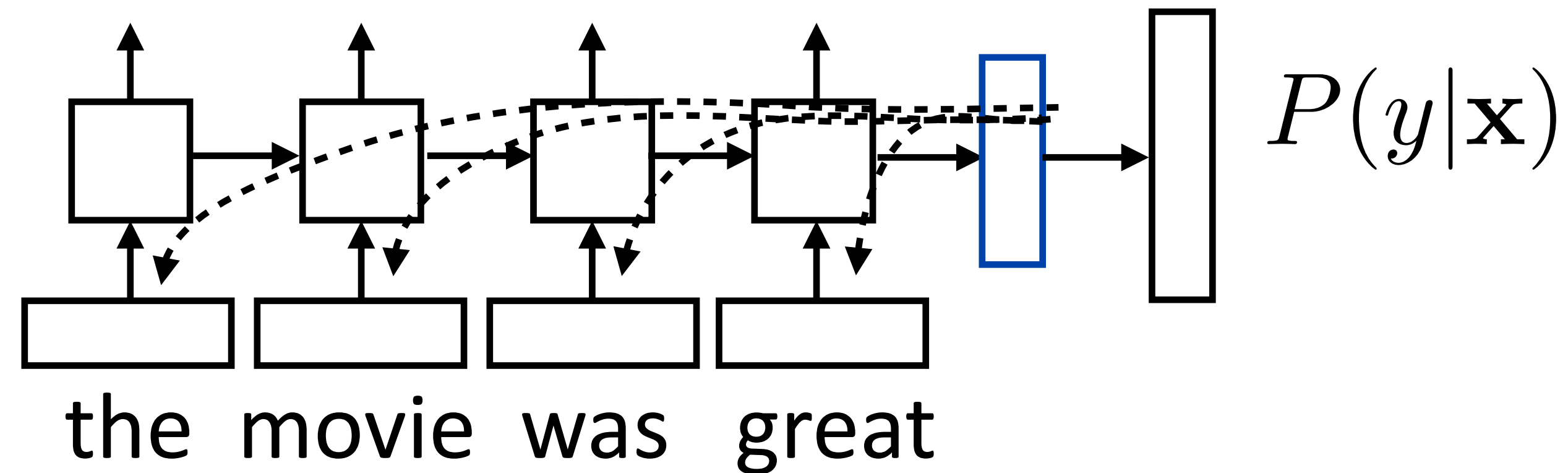
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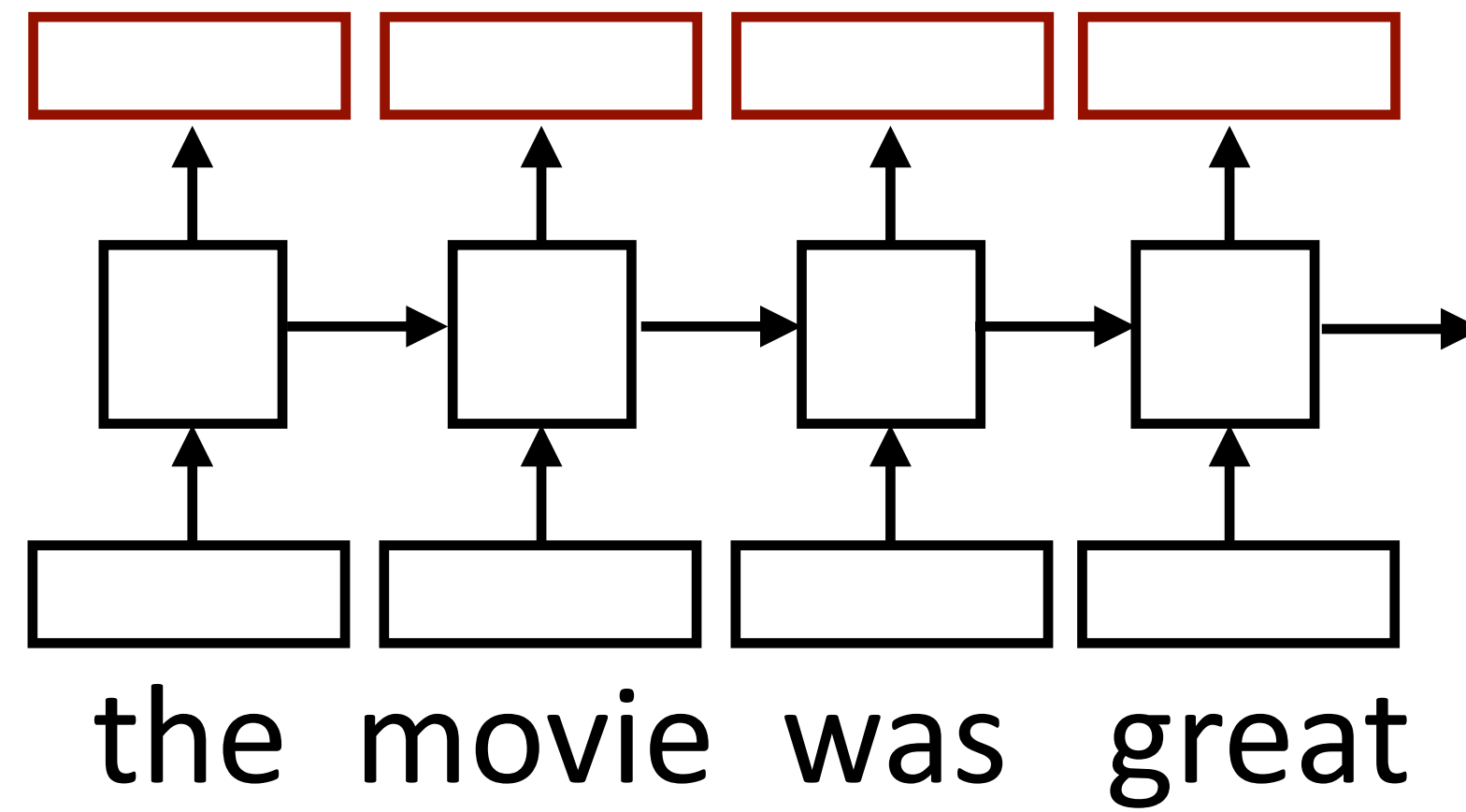
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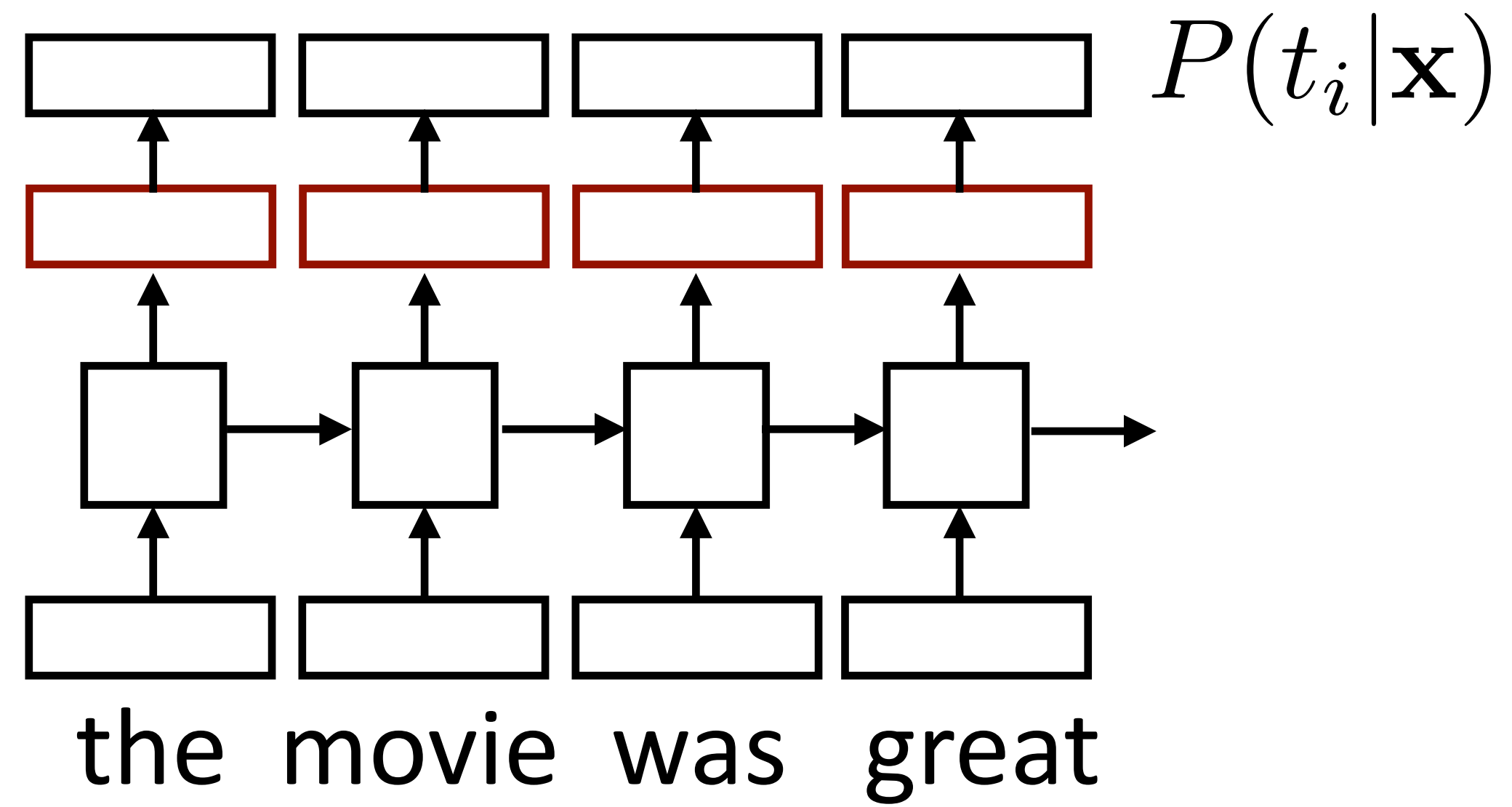


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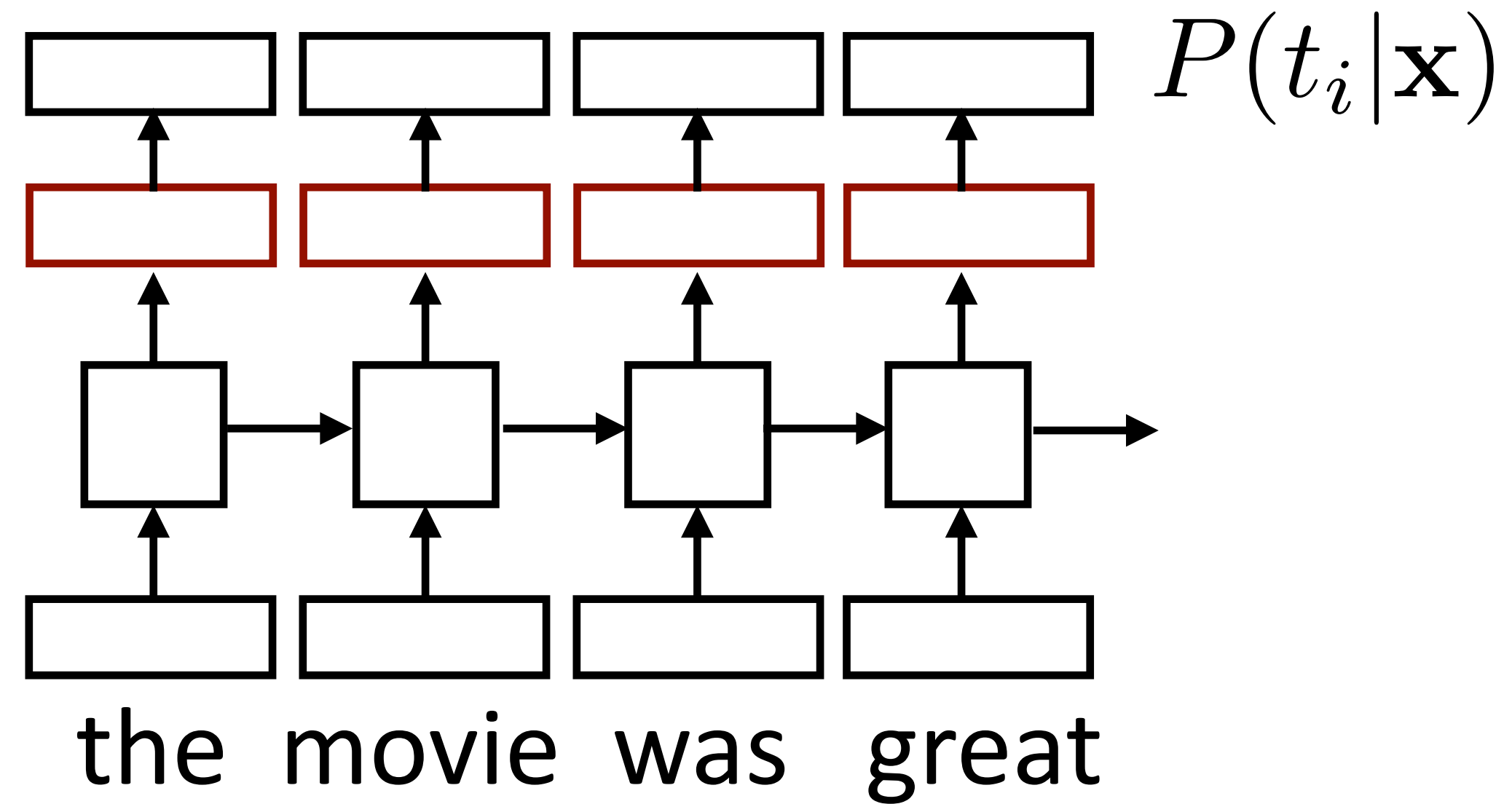
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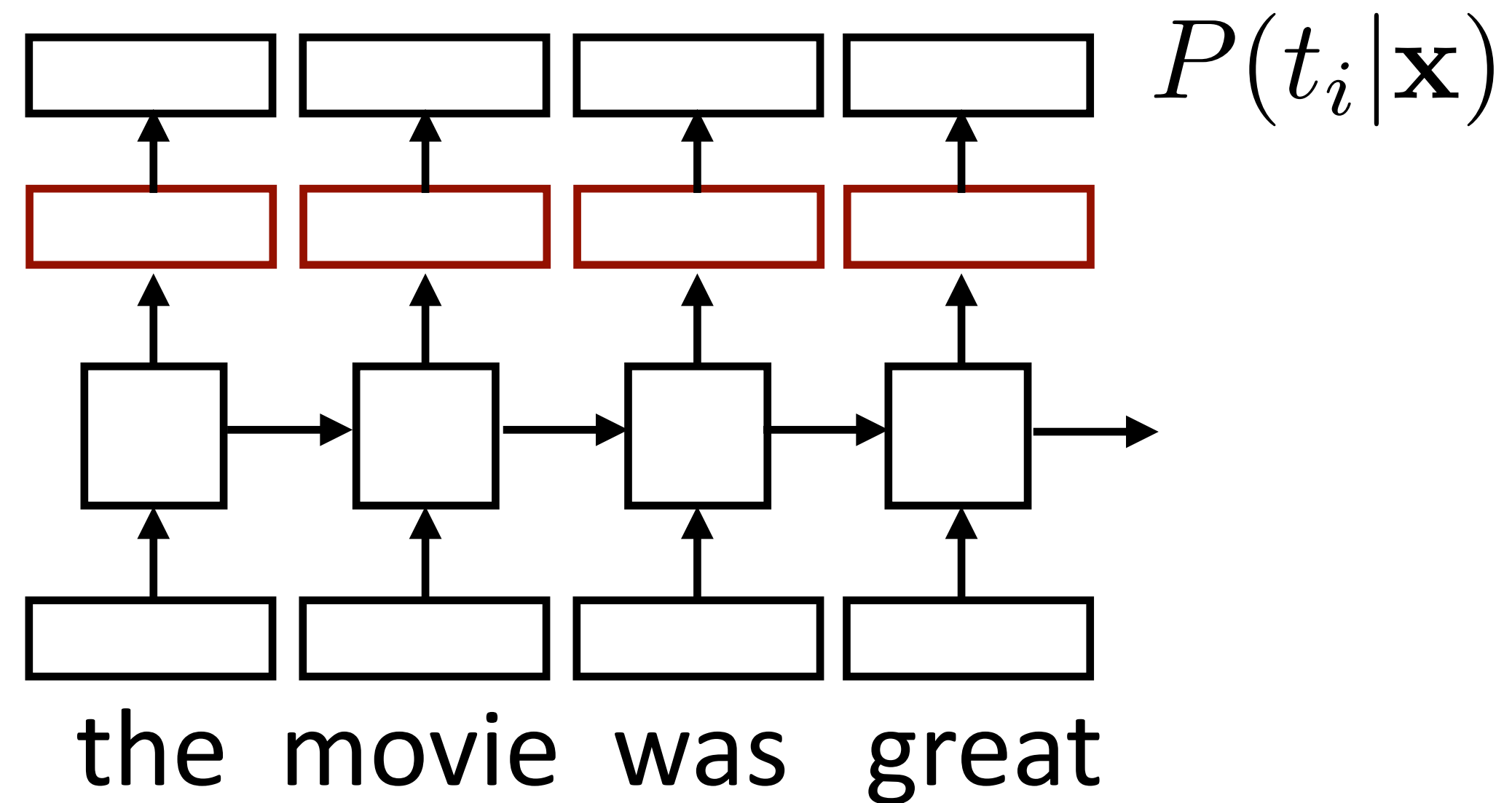


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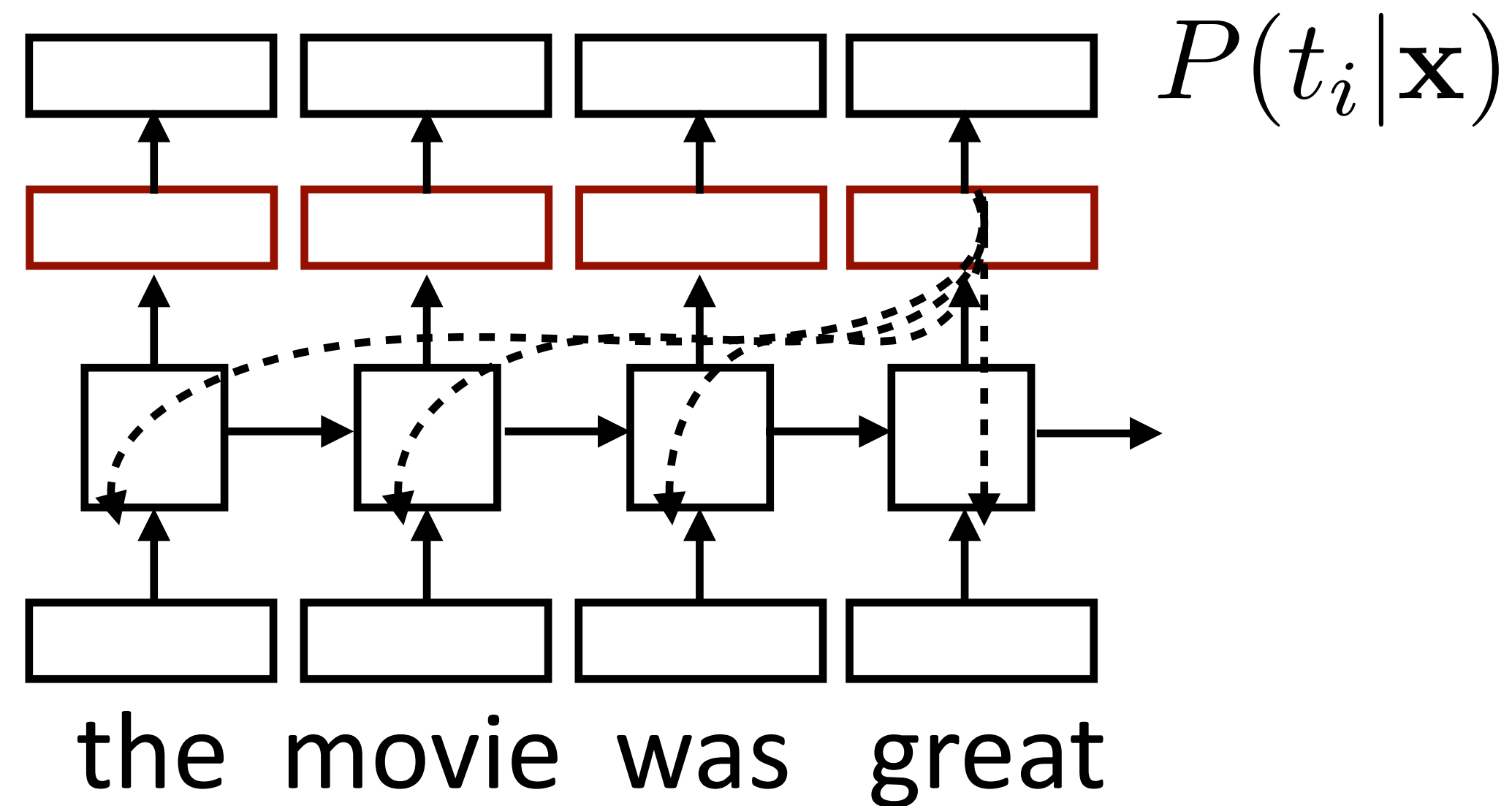
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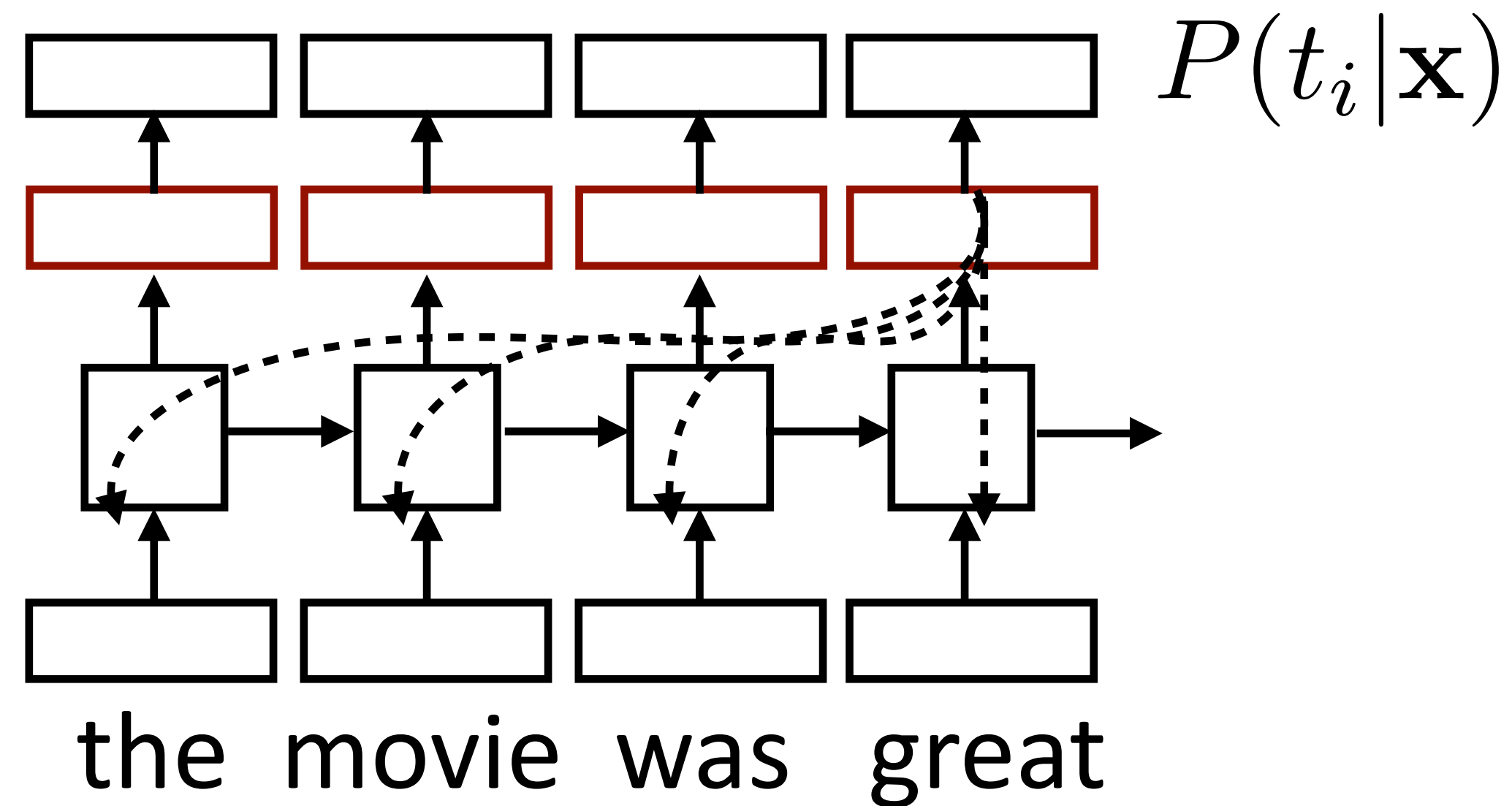
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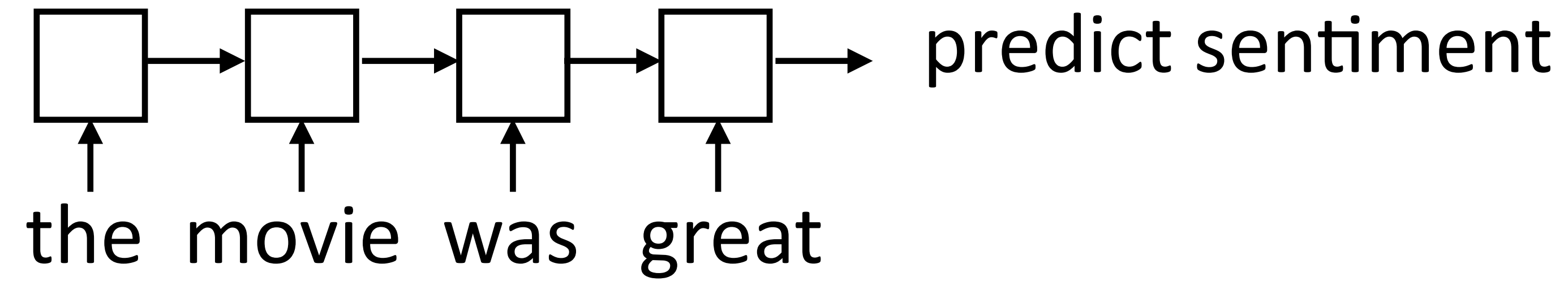
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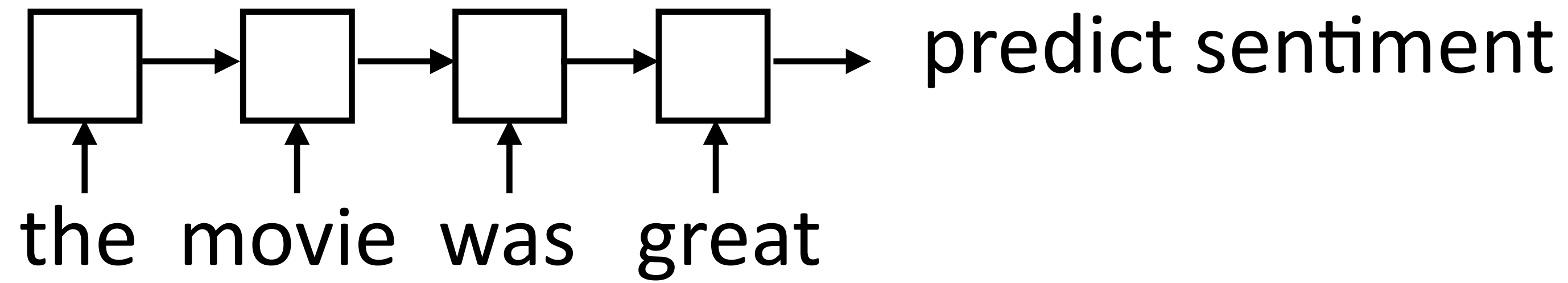


- ▶ Example: POS tagging, language modeling (predict next word given context)
- ▶ Loss = negative log likelihood of probability of gold predictions, summed over the tags
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RNN Intuition



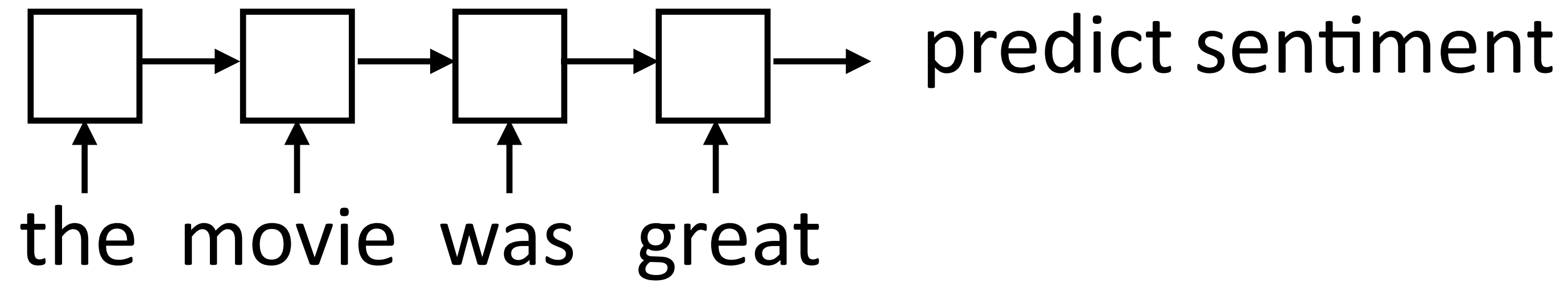
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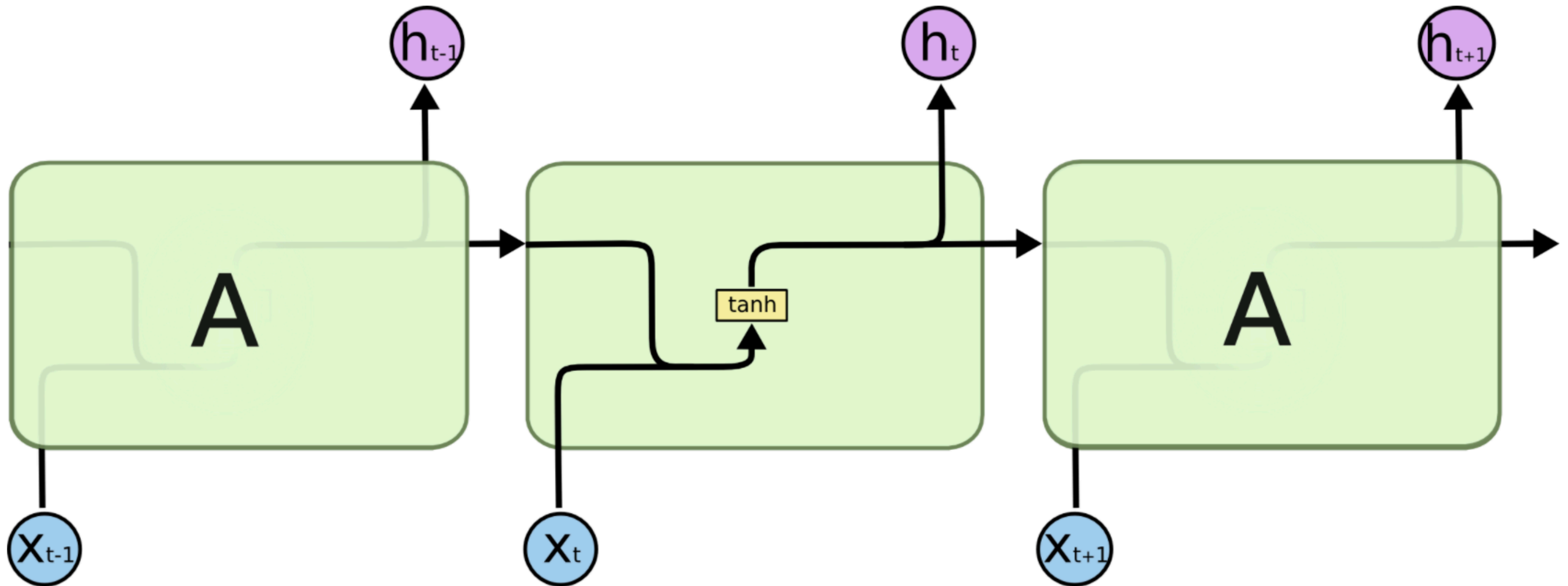


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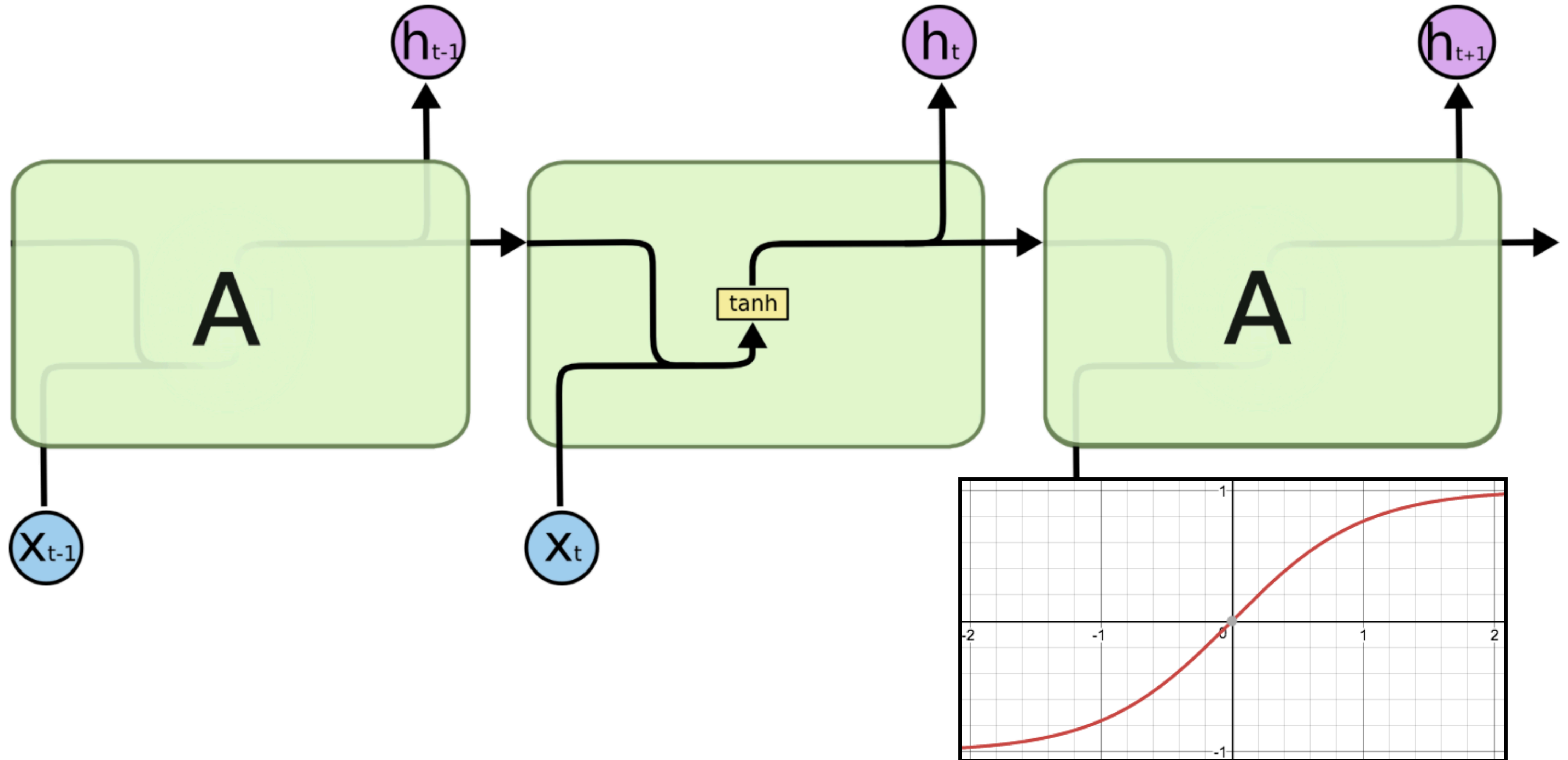
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- ▶ “Correct” parameter update is to do a better job of remembering the sentiment of *favorite*

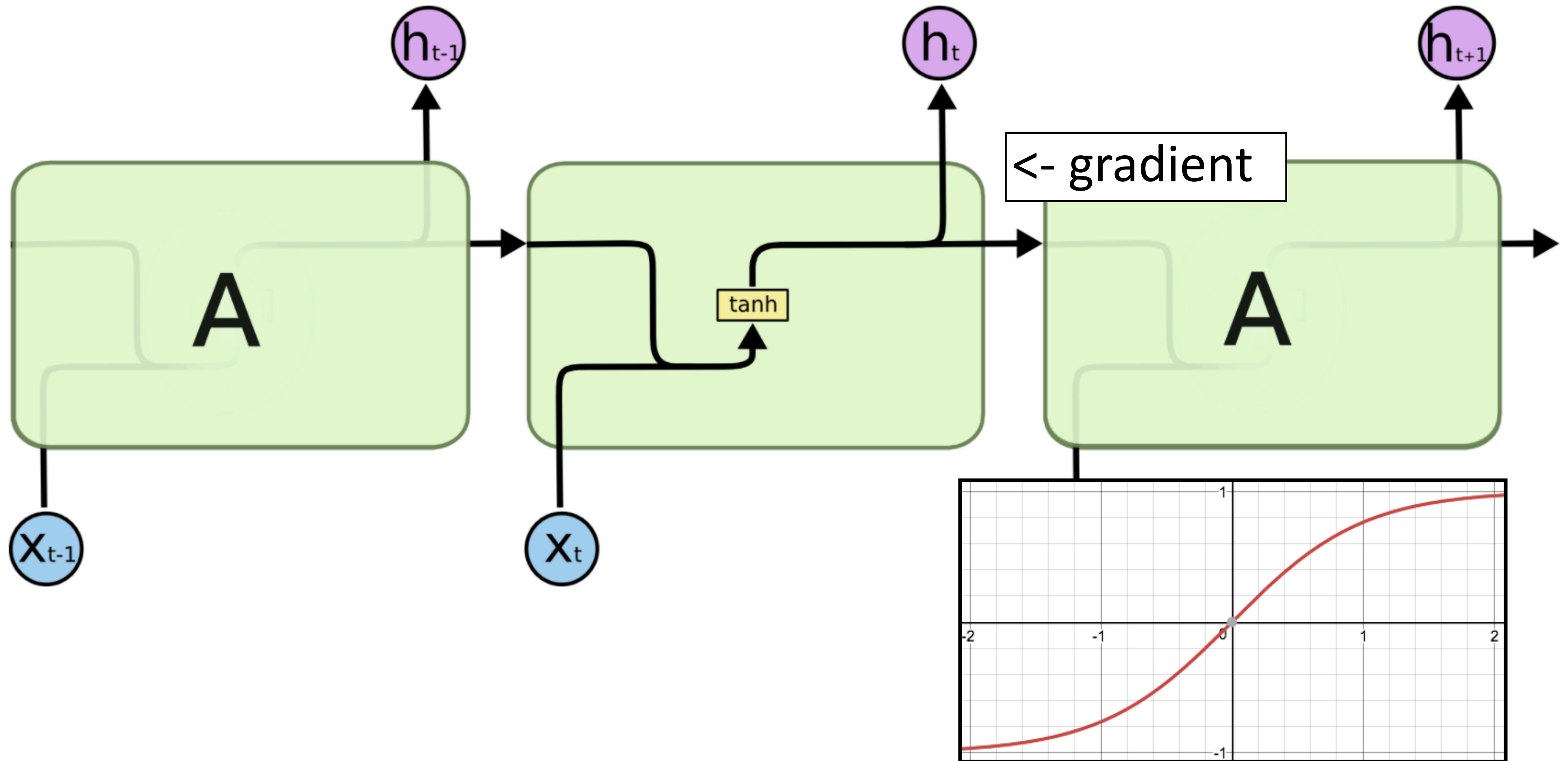
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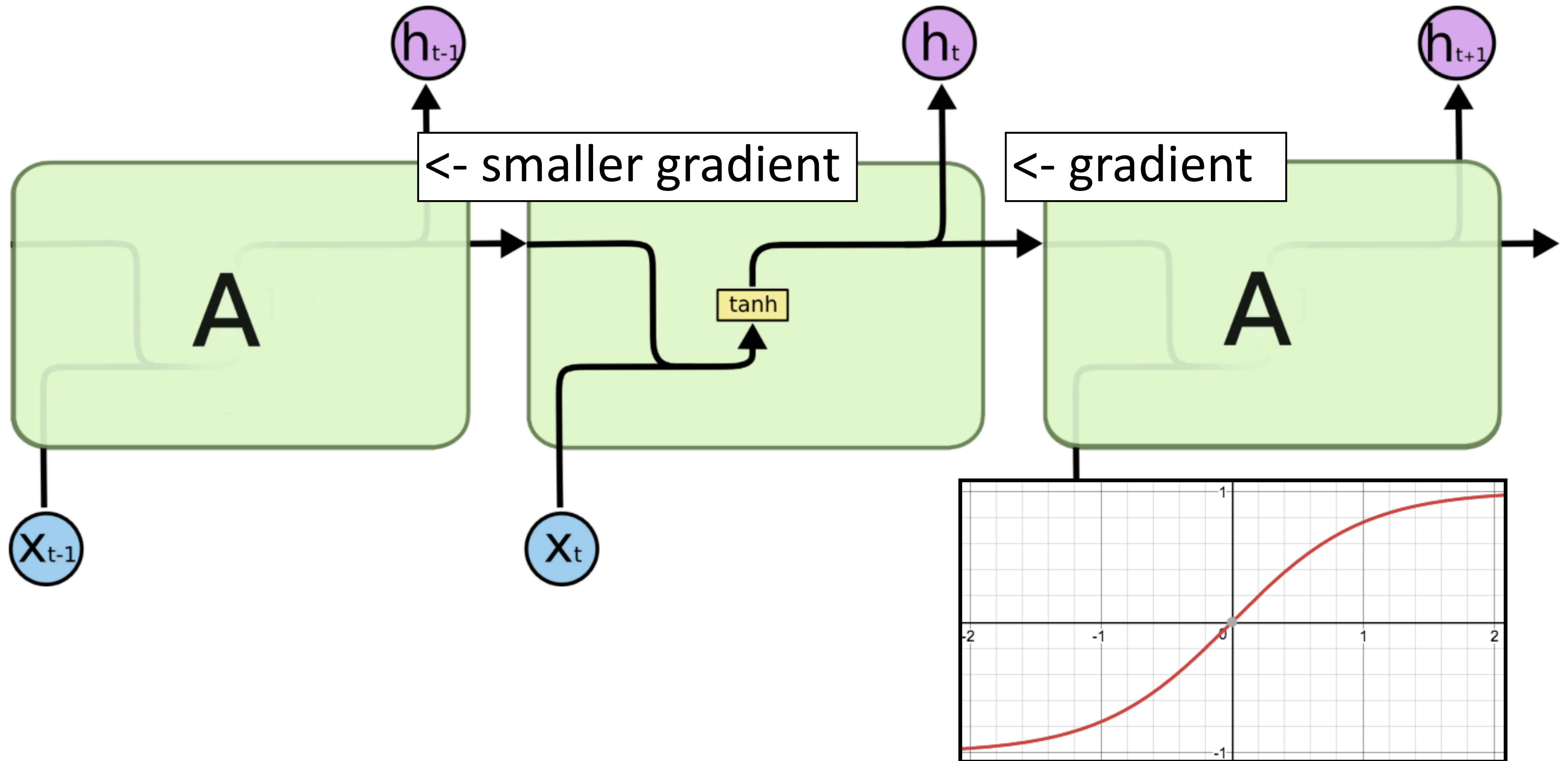
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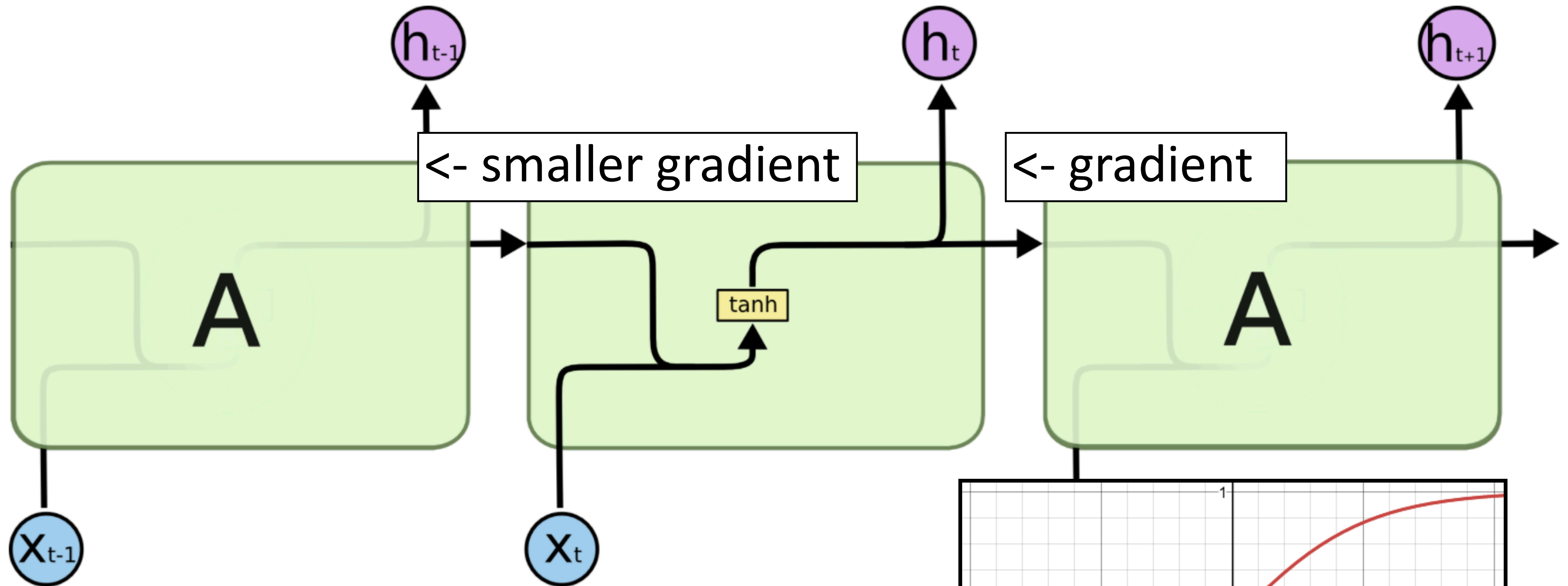
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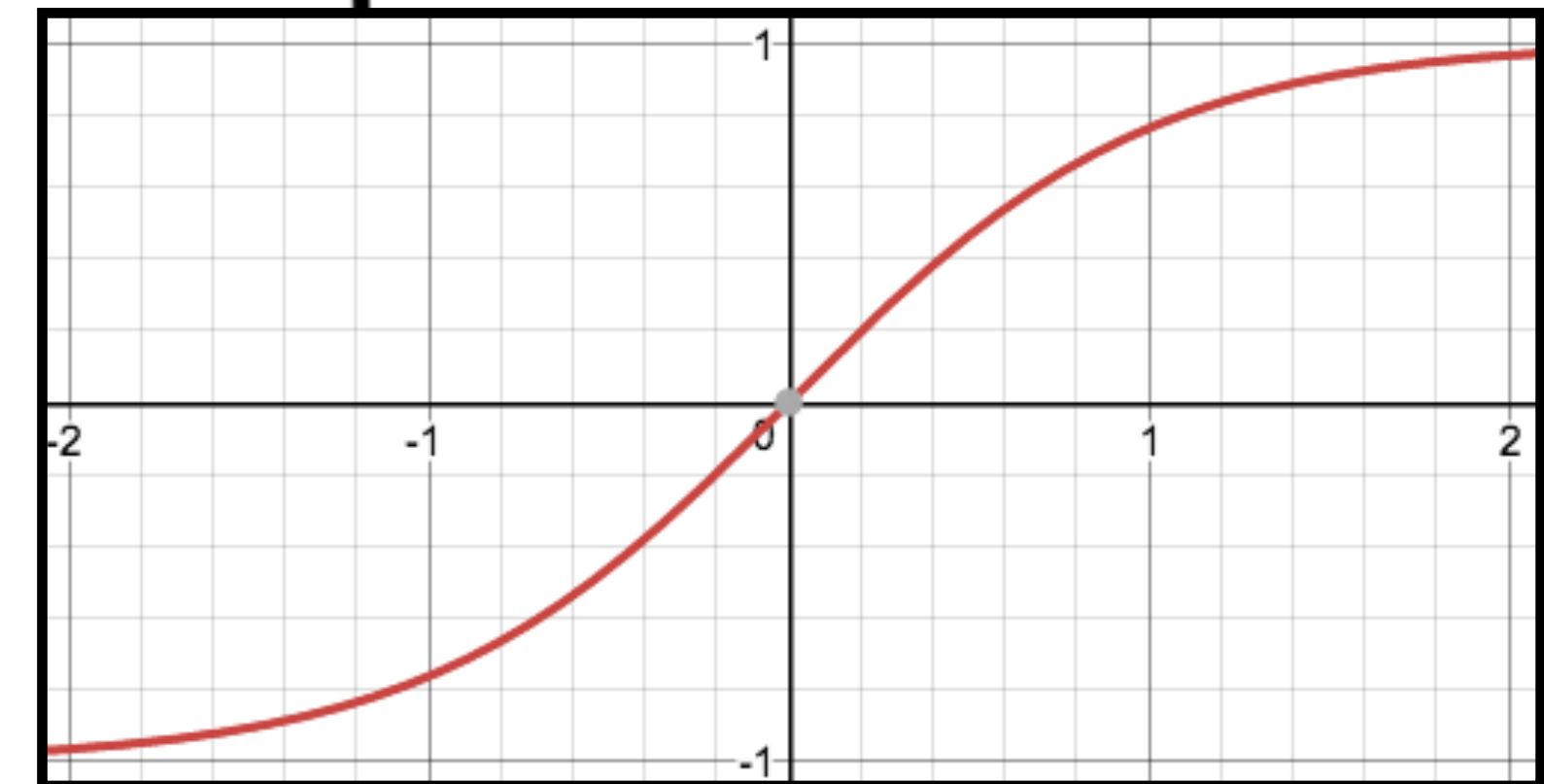
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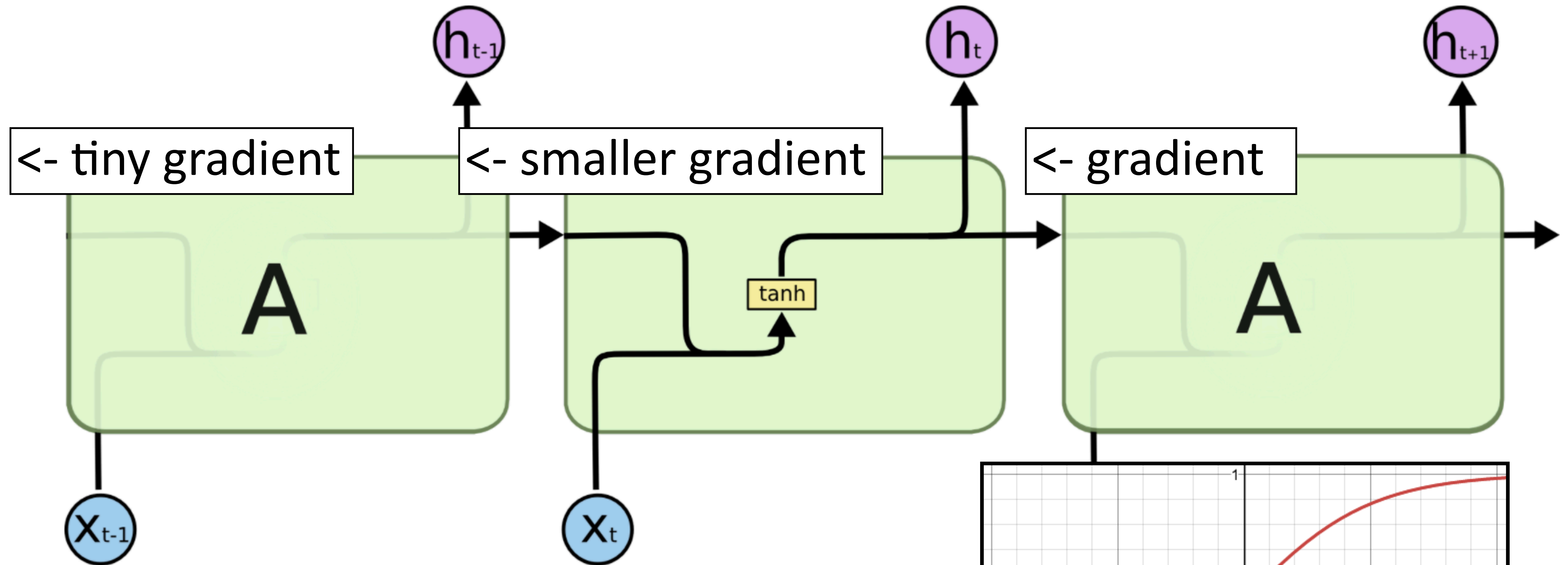
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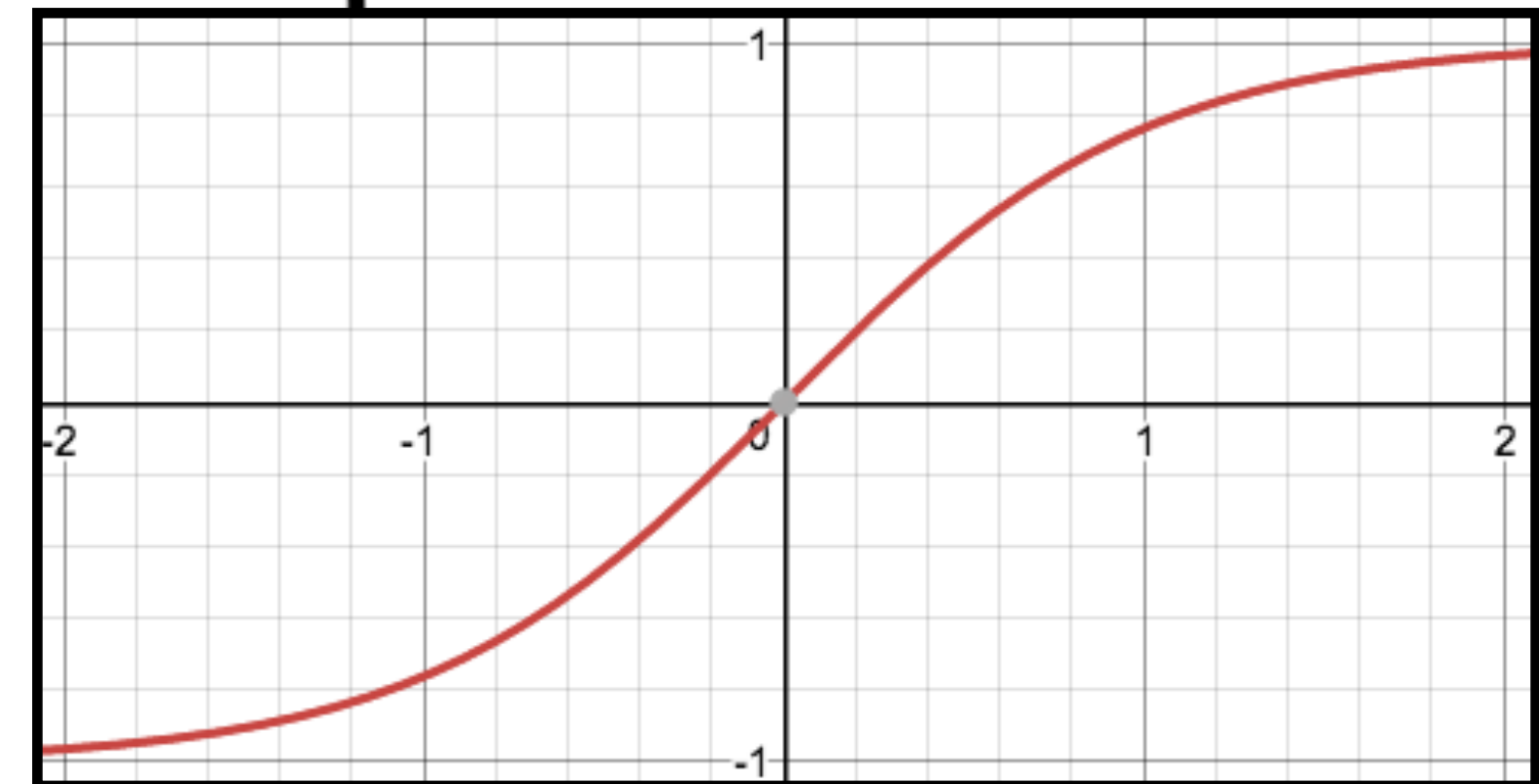
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Vanishing Gradient



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LSTMs/GRUs

A Bit of History

- ▶ Long Short-term Memory (Hochreiter & Schmidhuber, 1997)

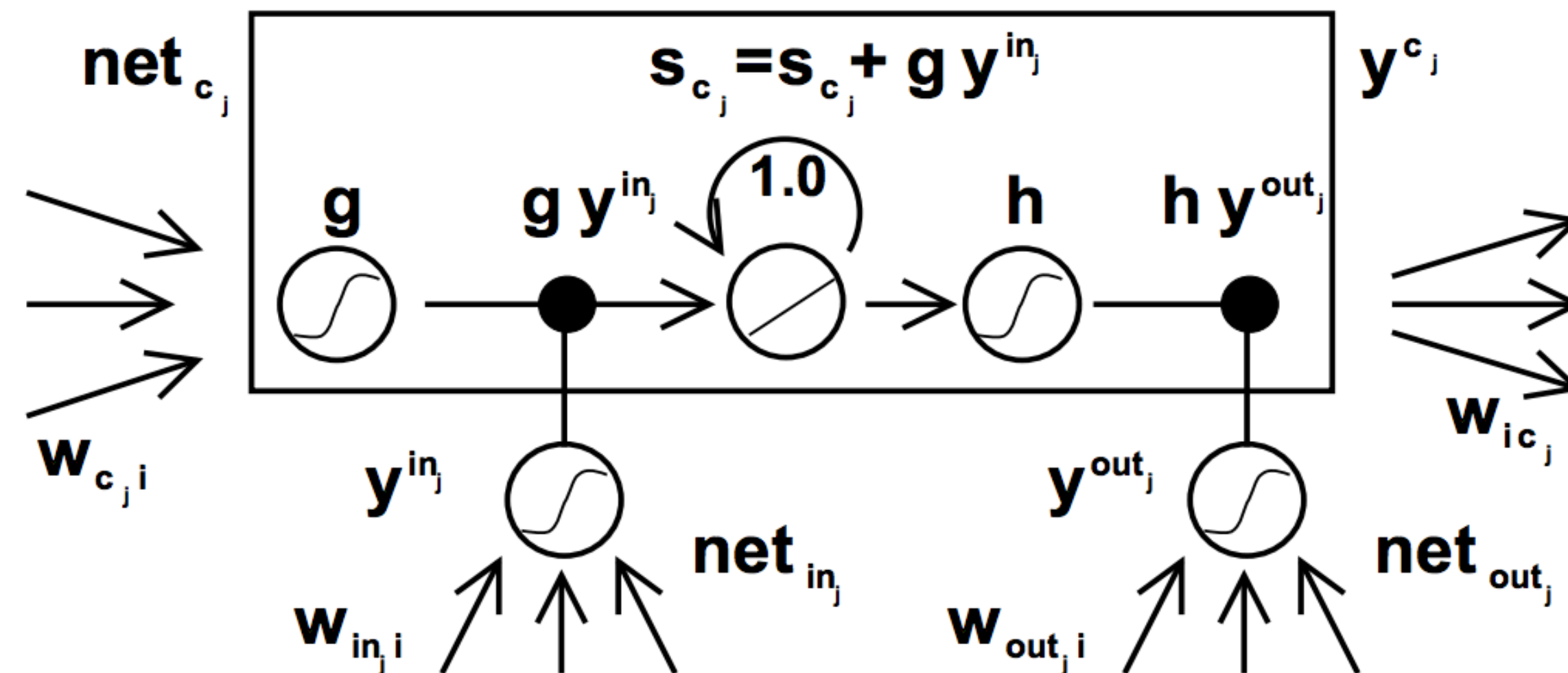


Figure 1: Architecture of memory cell c_j (the box) and its gate units in_j, out_j . The self-recurrent connection (with weight 1.0) indicates feedback with a delay of 1 time step. It builds the basis of the “constant error carrousel” CEC. The gate units open and close access to CEC. See text and appendix A.1 for details.

Gated Connections

- ▶ Designed to fix “vanishing gradient” problem using *gates*

$$\mathbf{h}_t = \mathbf{h}_{t-1} \odot \mathbf{f} + \text{func}(\mathbf{x}_t)$$

gated

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Elman

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gated Elman

- ▶ Vector-valued “forget gate” $\mathbf{f}$ computed based on input and previous hidden state

$$\mathbf{f} = \sigma(W^{xf}\mathbf{x}_t + W^{hf}\mathbf{h}_{t-1})$$

- ▶ Sigmoid: elements of $\mathbf{f}$ are in $(0, 1)$

Gated Connections

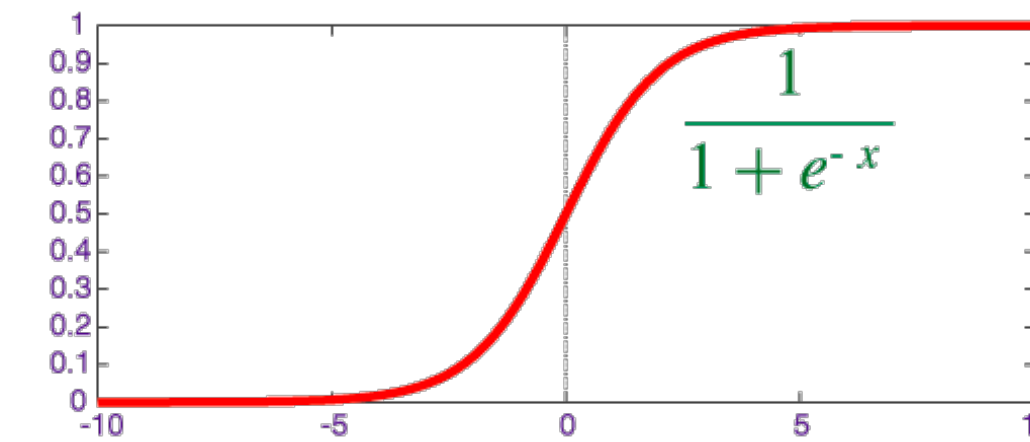
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$$\mathbf{h}_t = \tanh(W\mathbf{x}_t + V\mathbf{h}_{t-1} + \mathbf{b}_h)$$

Elman



- ▶ Vector-valued “forget gate” $\mathbf{f}$ computed based on input and previous hidden state

$$\mathbf{f} = \sigma(W^{xf}\mathbf{x}_t + W^{hf}\mathbf{h}_{t-1})$$

- ▶ Sigmoid: elements of $\mathbf{f}$ are in (0, 1)

Gated Connections

- Designed to fix “vanishing gradient” problem using *gates*

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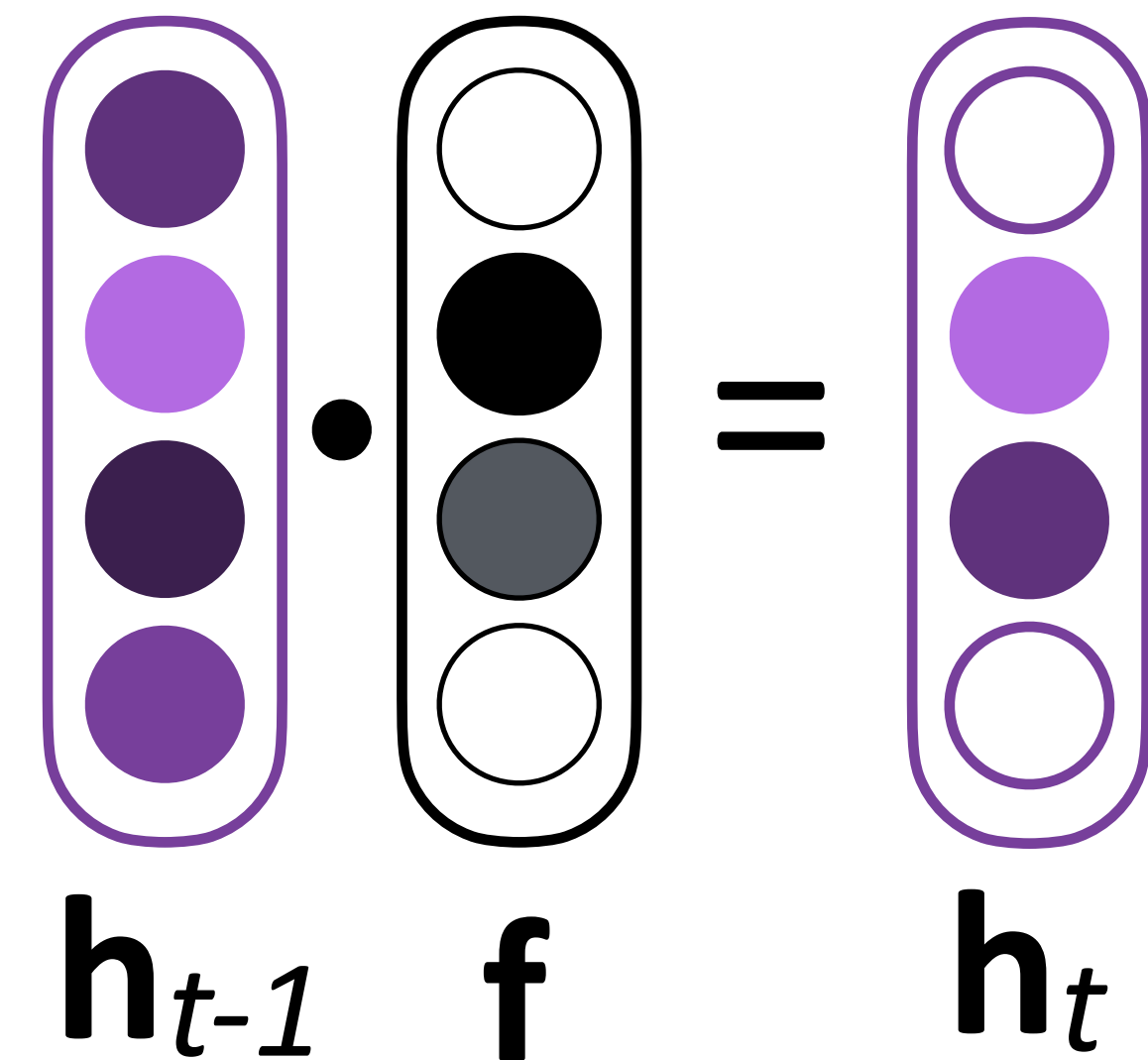
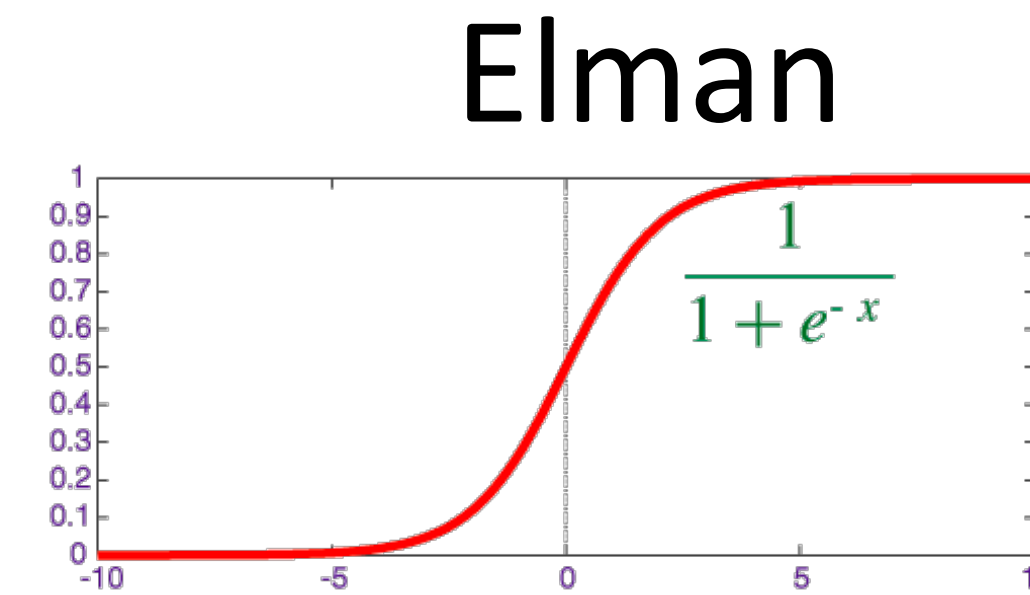
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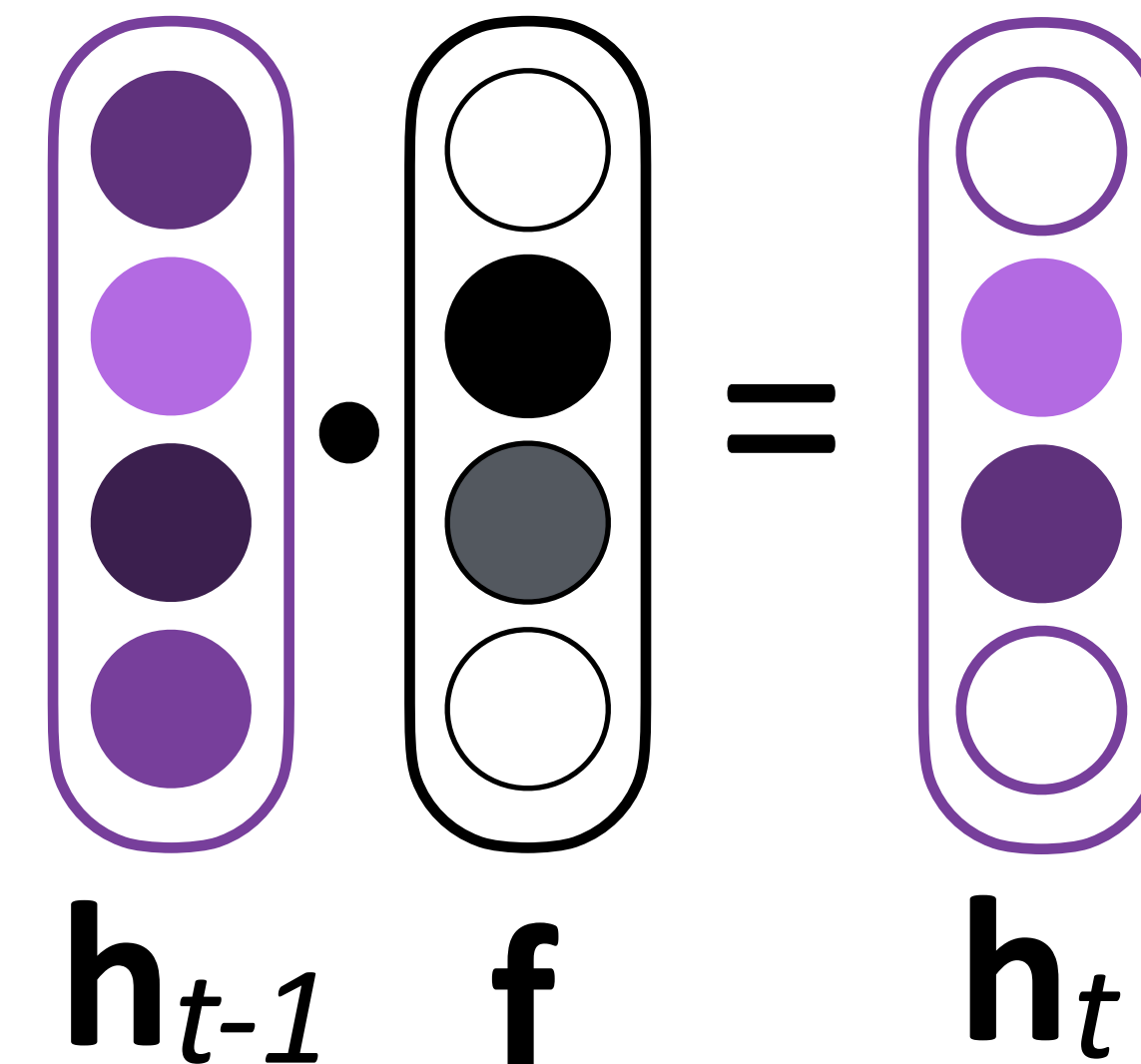
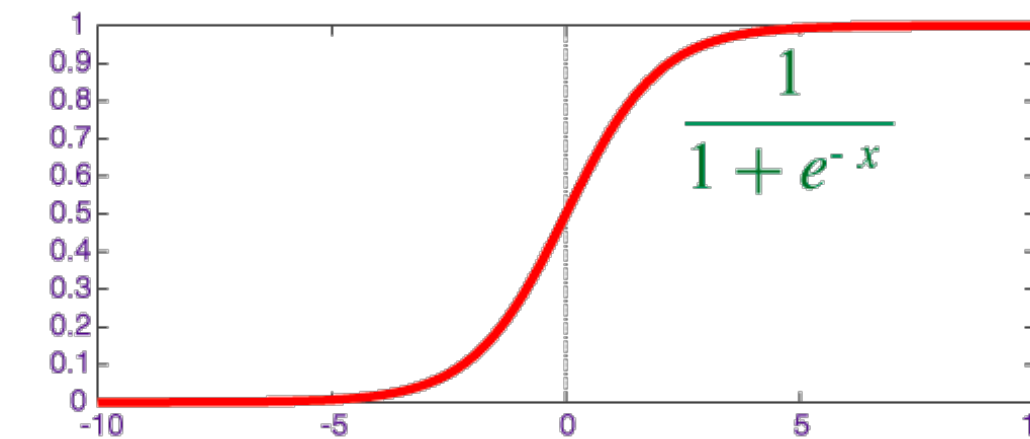
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- Sigmoid: elements of $\mathbf{f}$ are in $(0, 1)$
- If $\mathbf{f} \approx \mathbf{1}$, we simply sum up a function of all inputs — gradient doesn’t vanish!

Elman



LSTMs

- ▶ Long short-term memory network: hidden state as a “short-term” memory
- ▶ “Cell” $\mathbf{c}$ in addition to hidden state $\mathbf{h}$

$$\mathbf{c}_t = \mathbf{c}_{t-1} \odot \mathbf{f} + \text{func}(\mathbf{x}_t, \mathbf{h}_{t-1})$$

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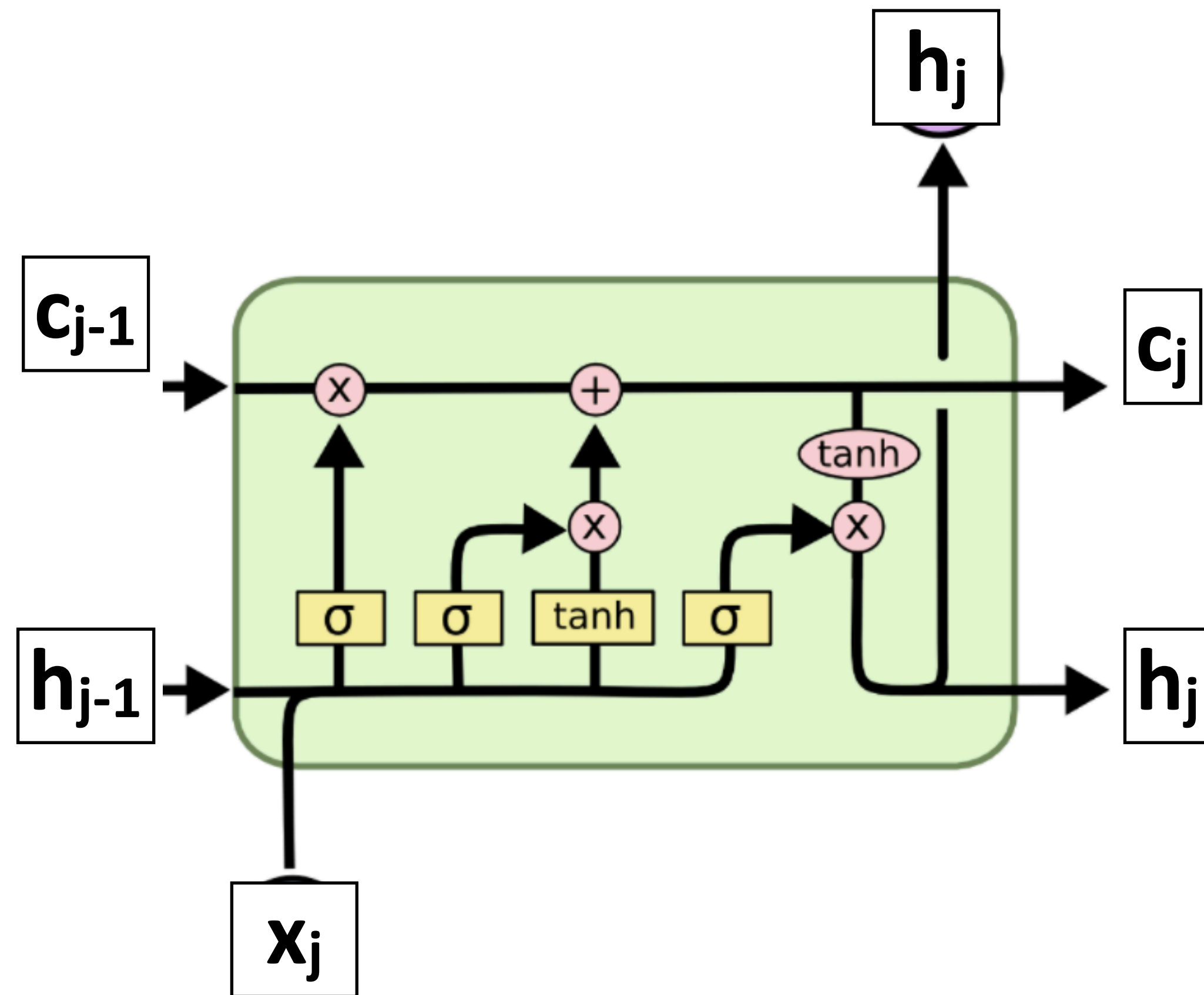
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- ▶ Basic communication flow: $\mathbf{x} \rightarrow \mathbf{c} \rightarrow \mathbf{h} \rightarrow \text{output}$, each step of this process is gated in addition to gates from previous timesteps

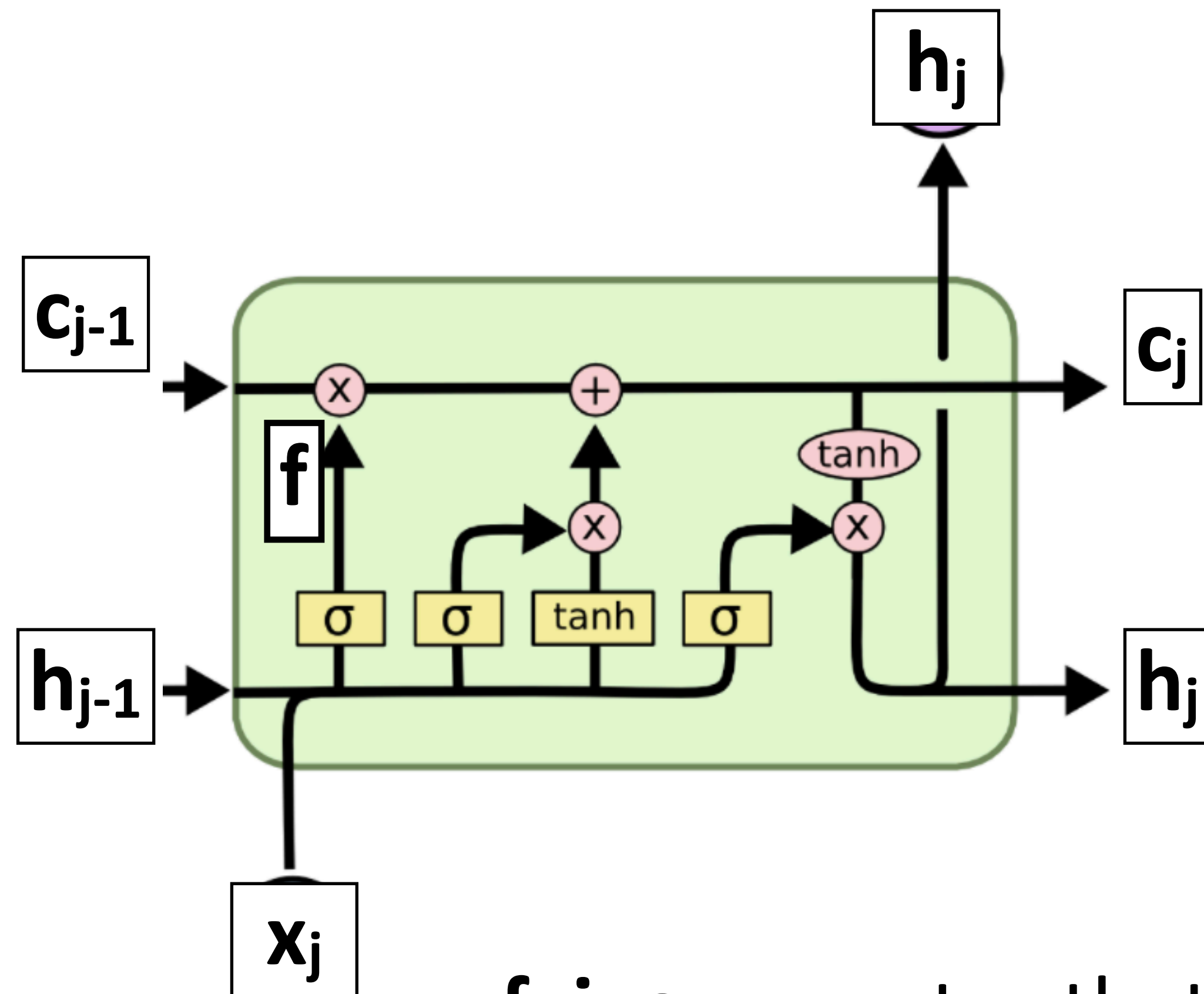
LSTMs



Hochreiter & Schmidhuber (1997)

<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

LSTMs



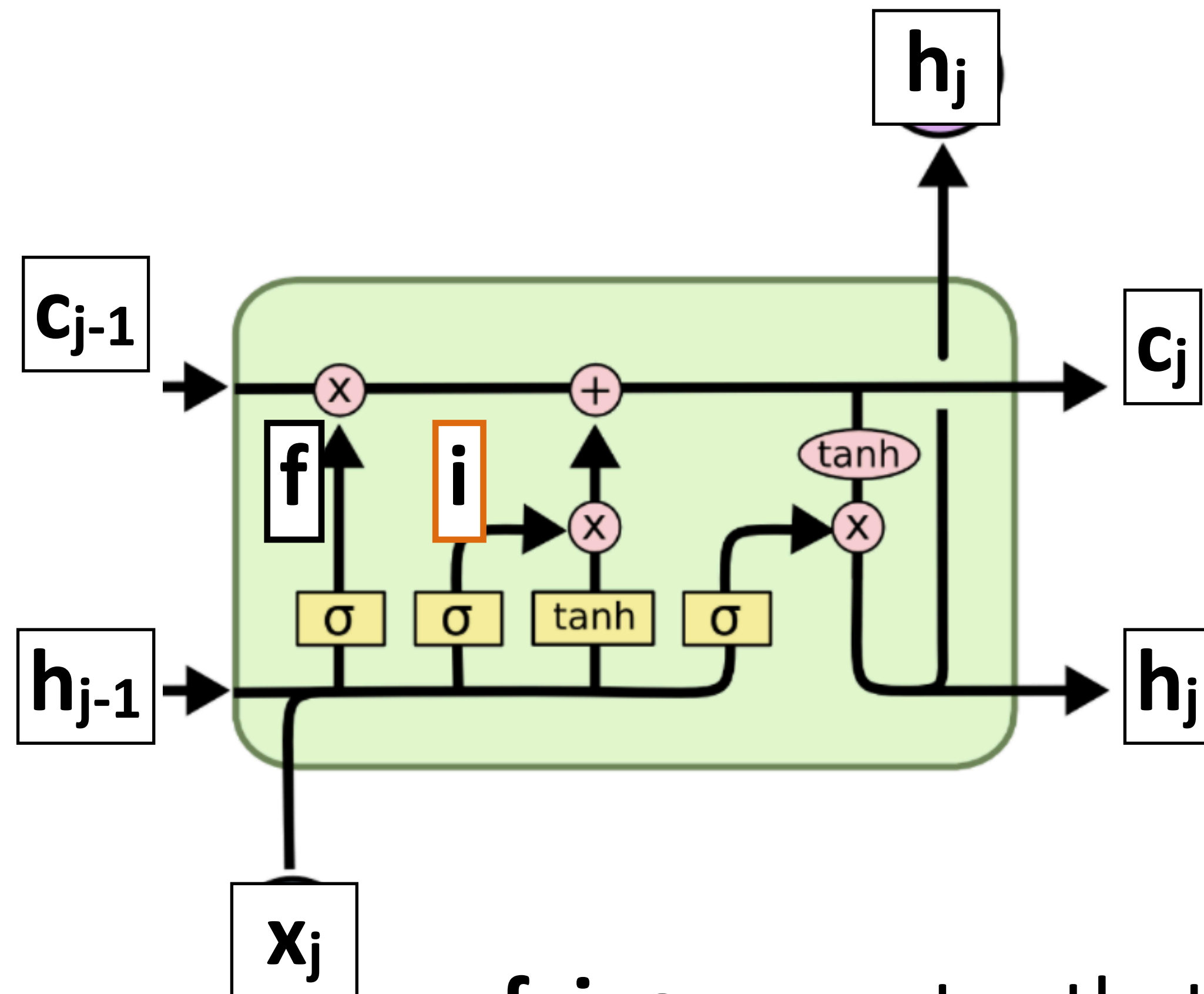
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- ▶ **f, i, o** are gates that control information flow

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LSTMs



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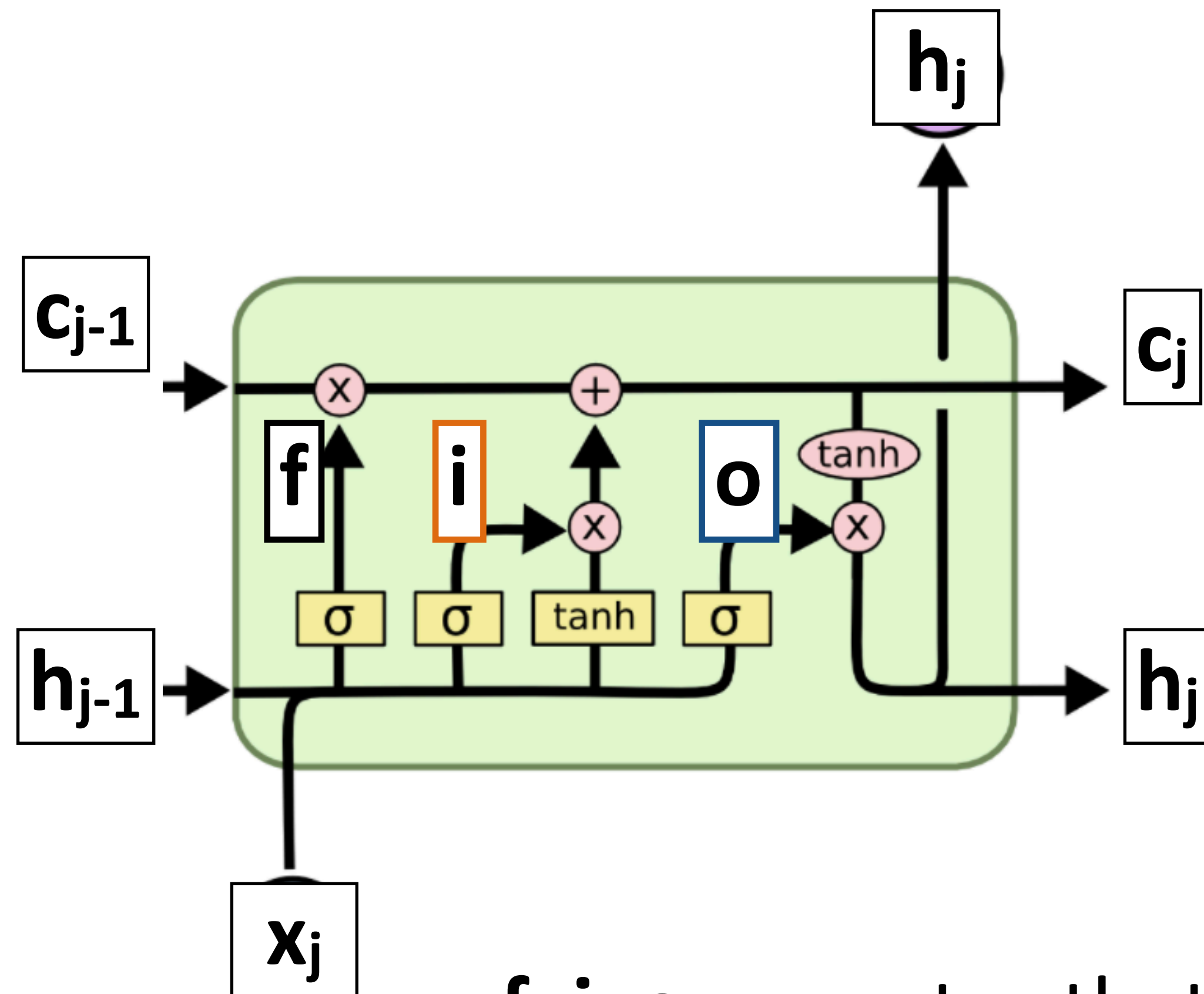
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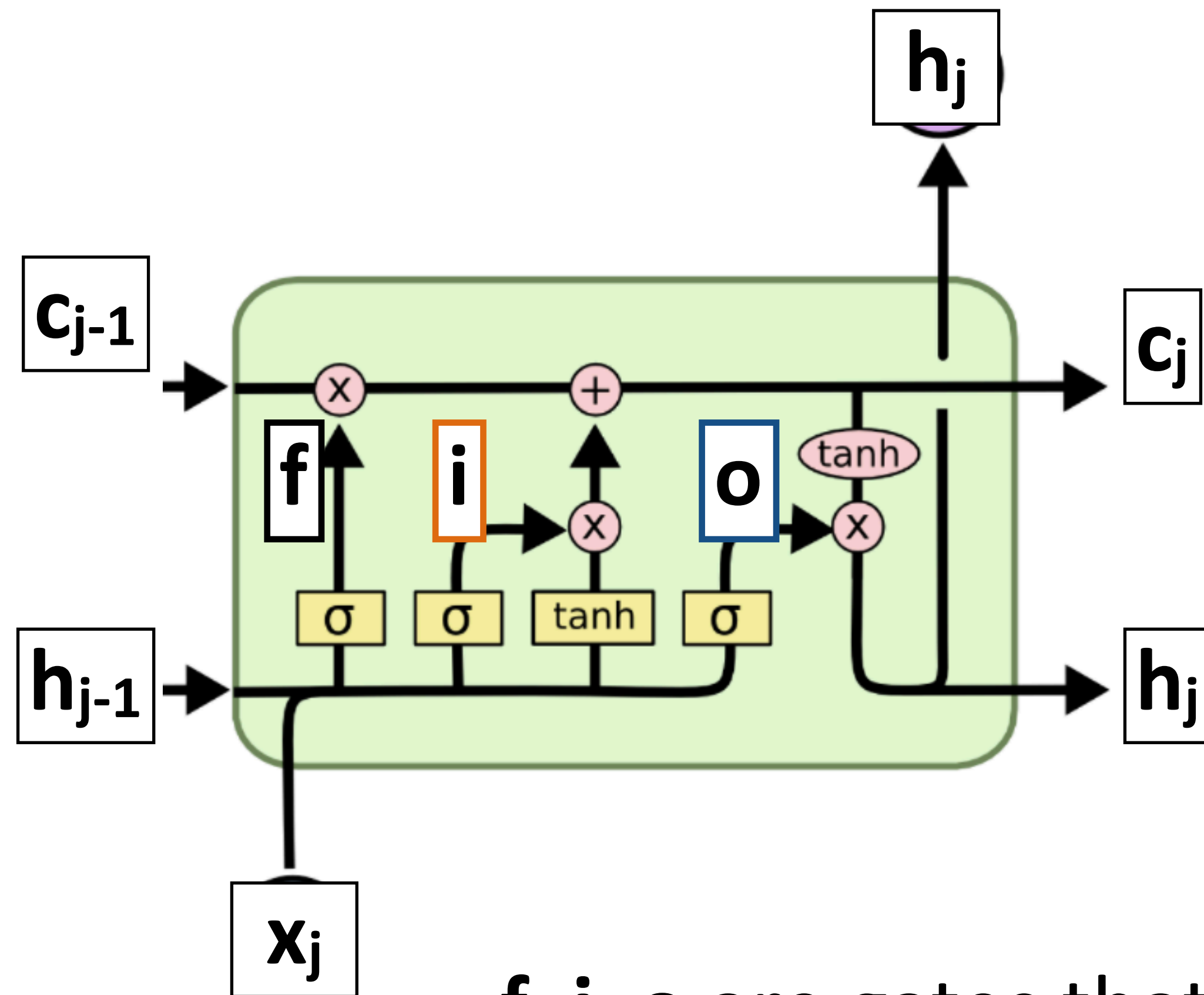
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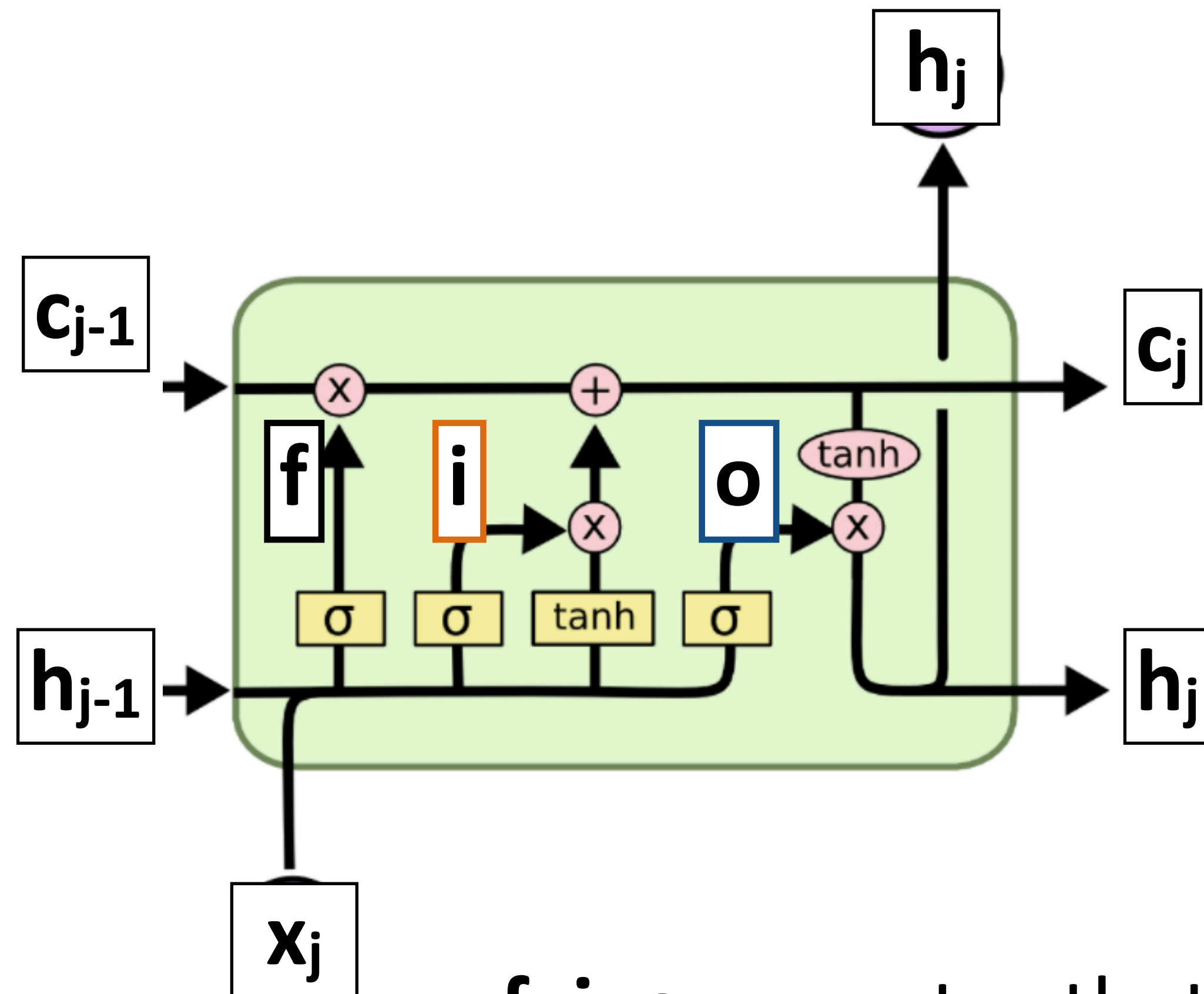
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LSTMs



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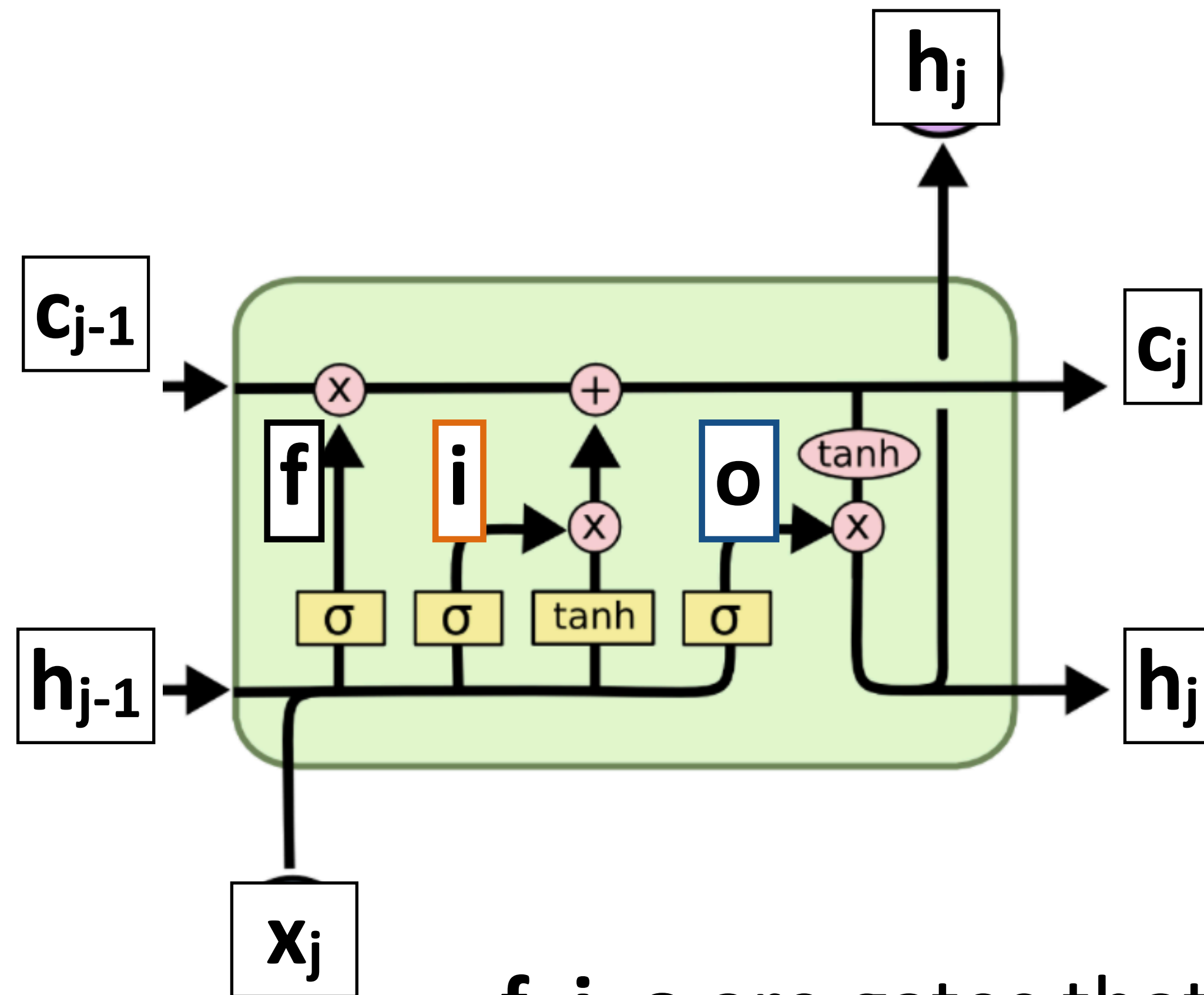
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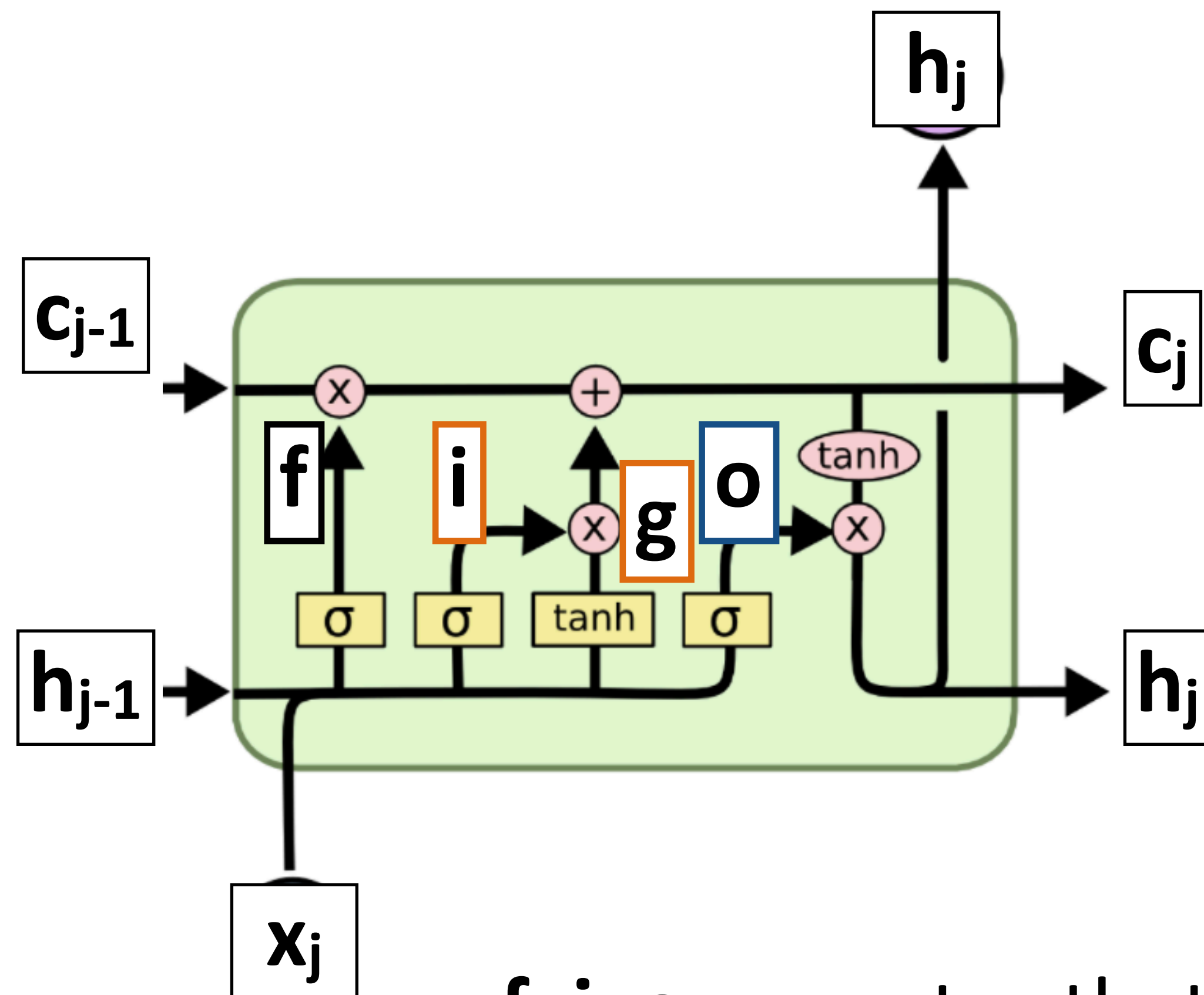
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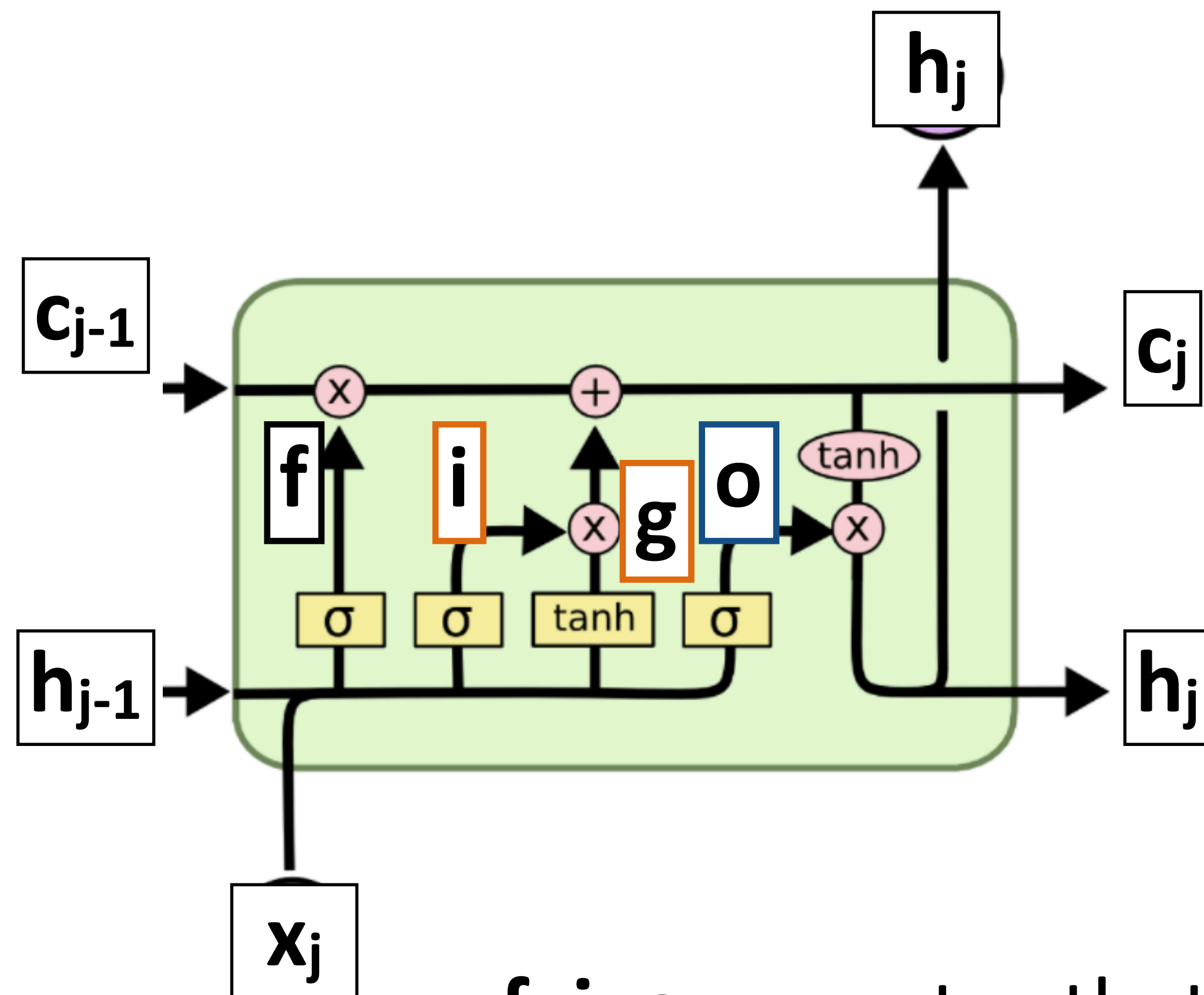
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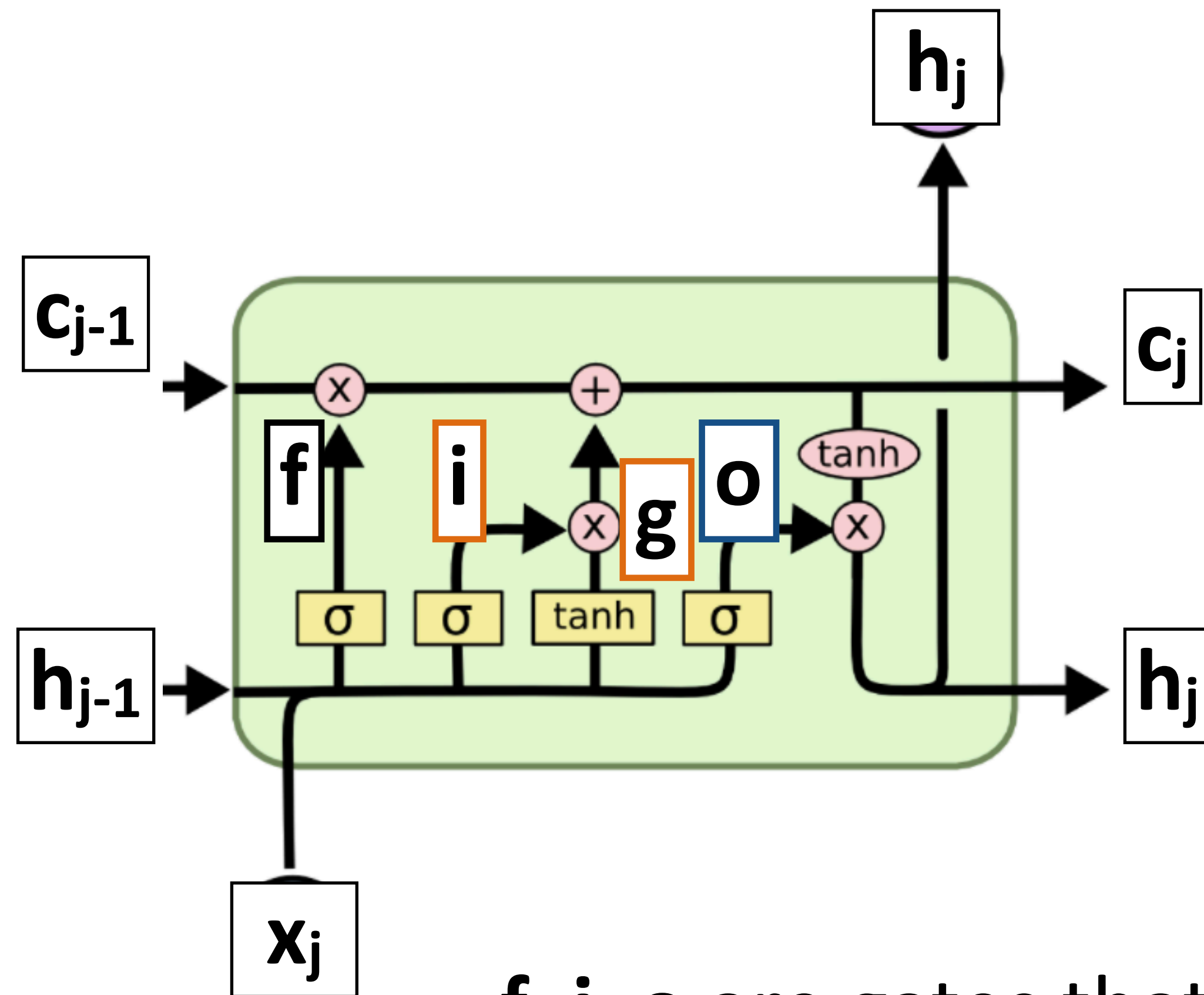
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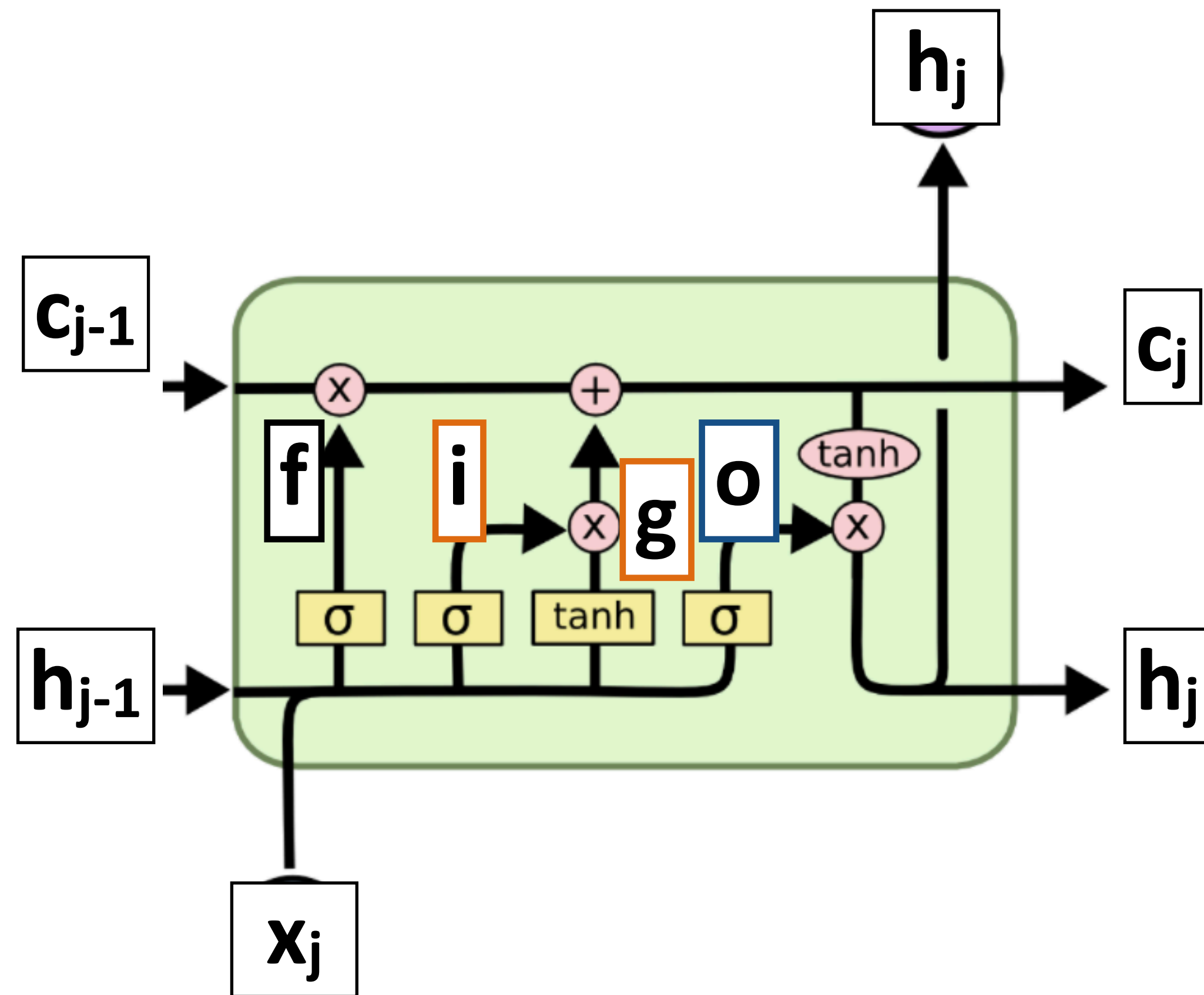
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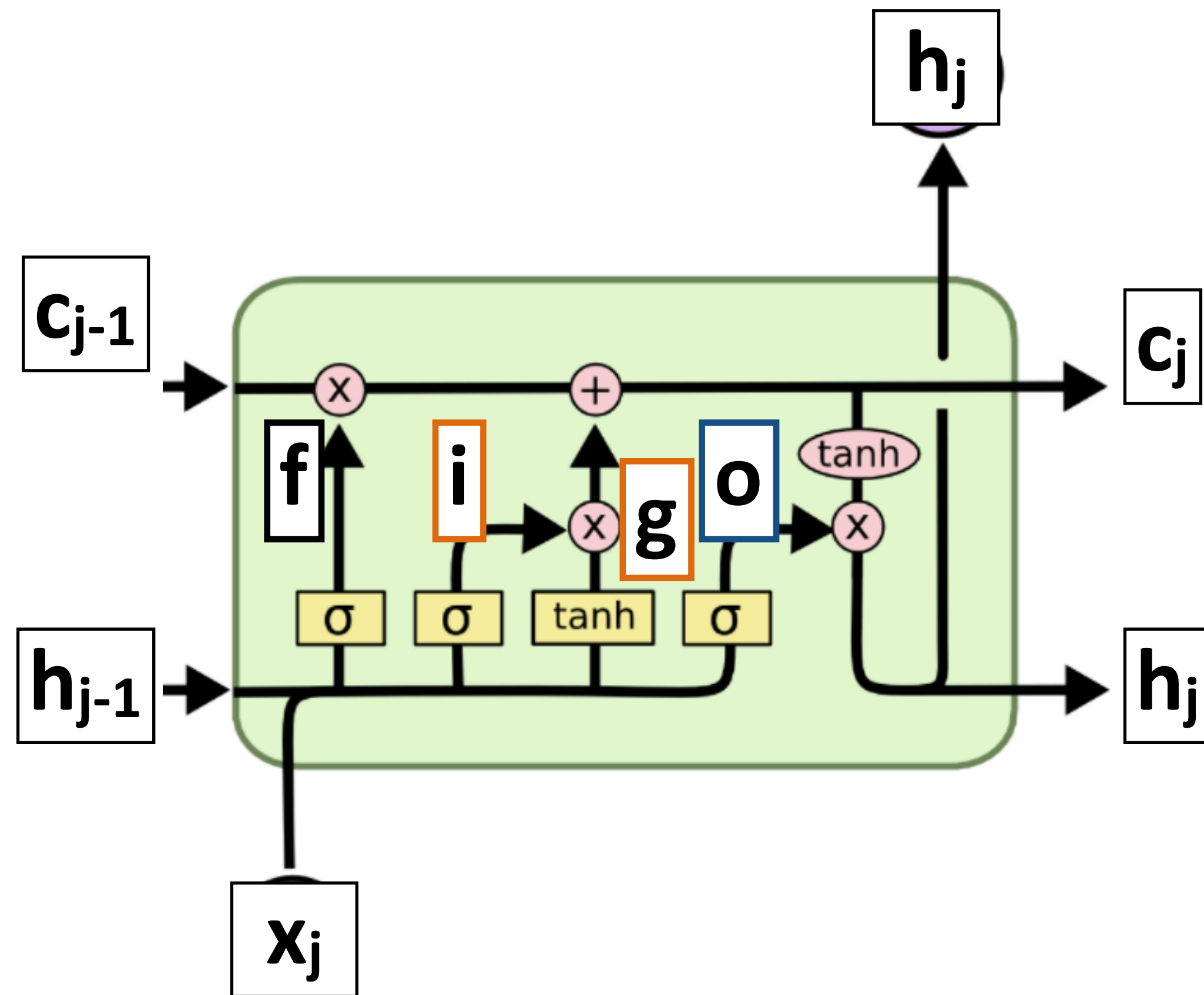
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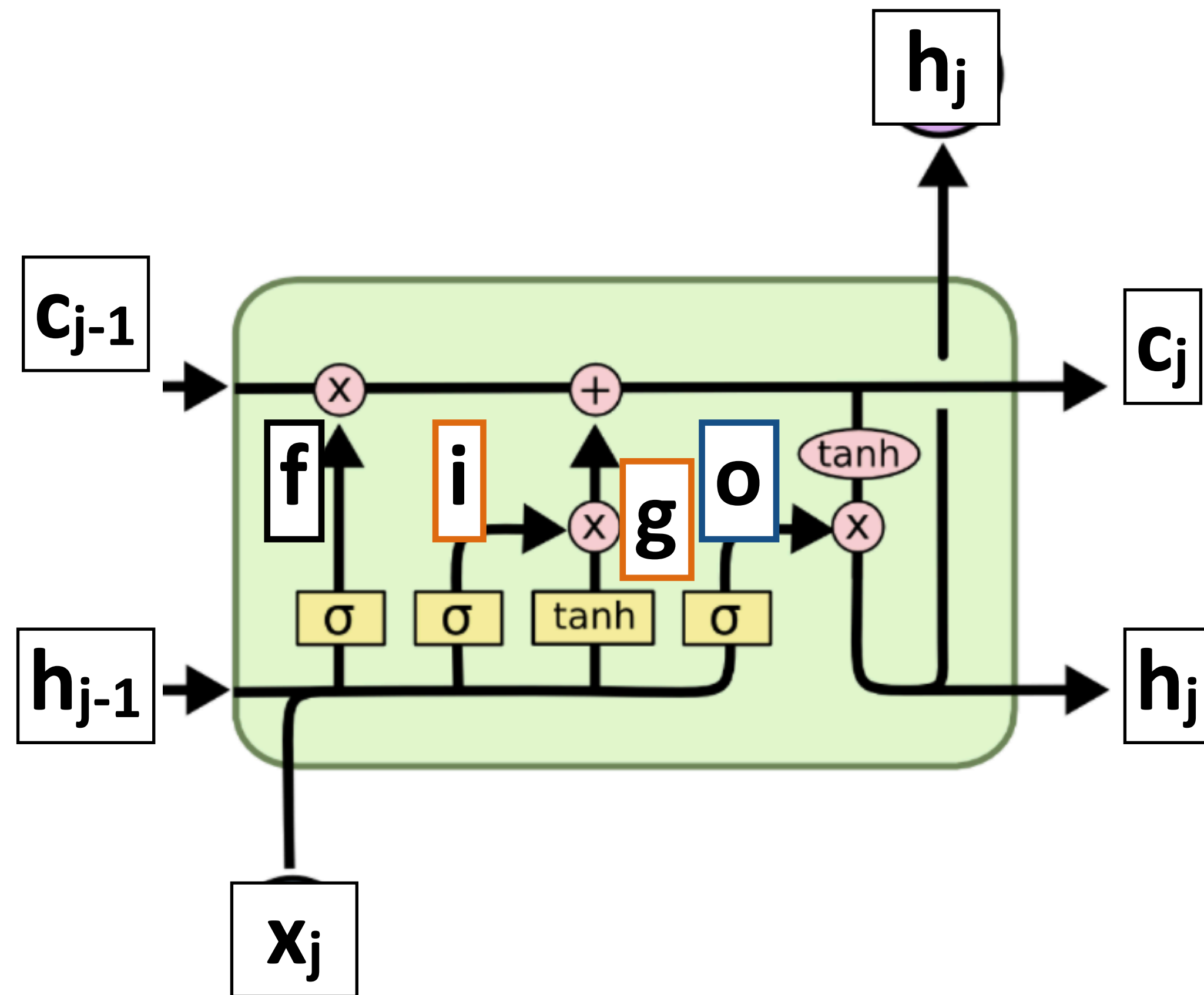
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LSTMs



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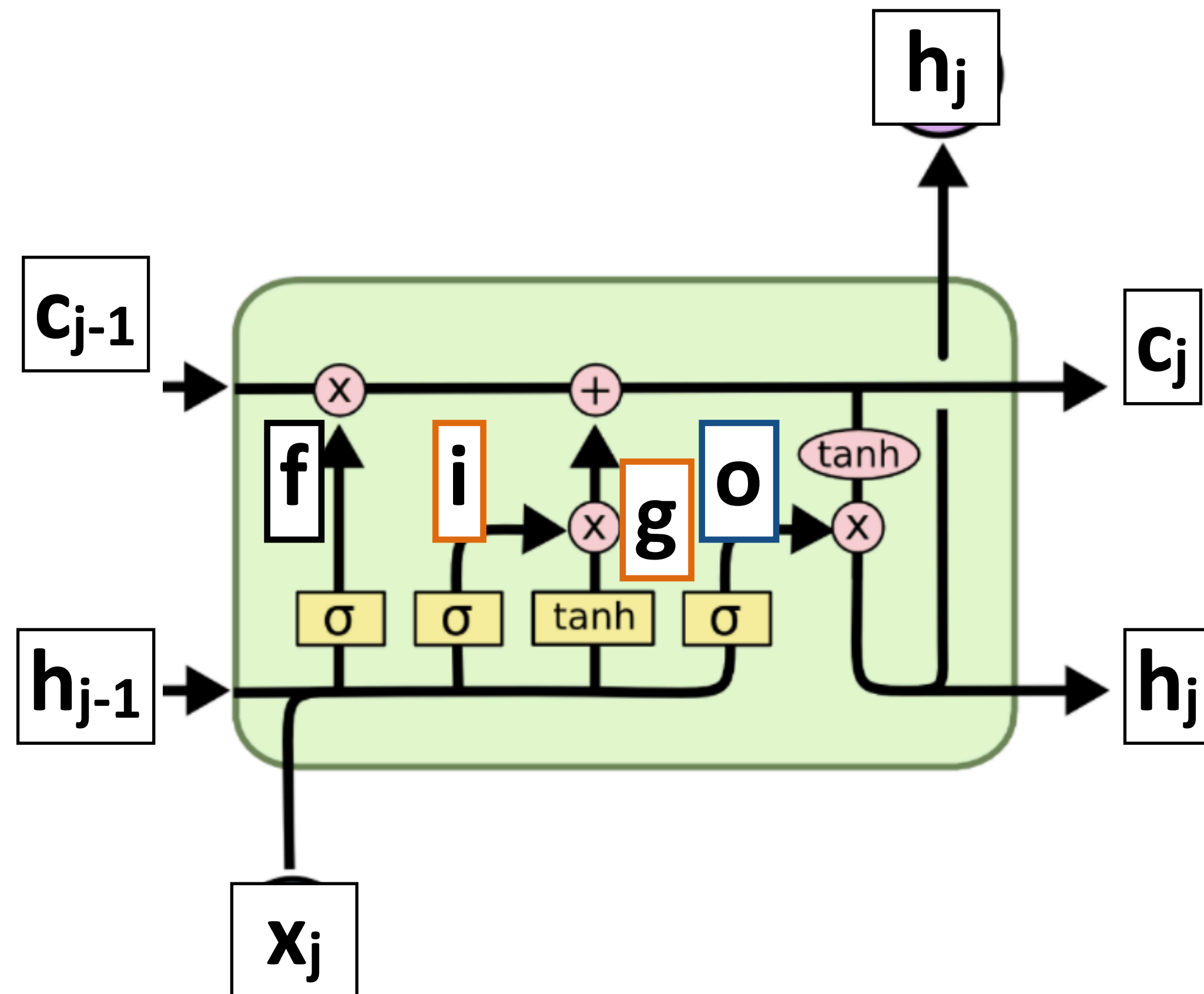
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LSTMs



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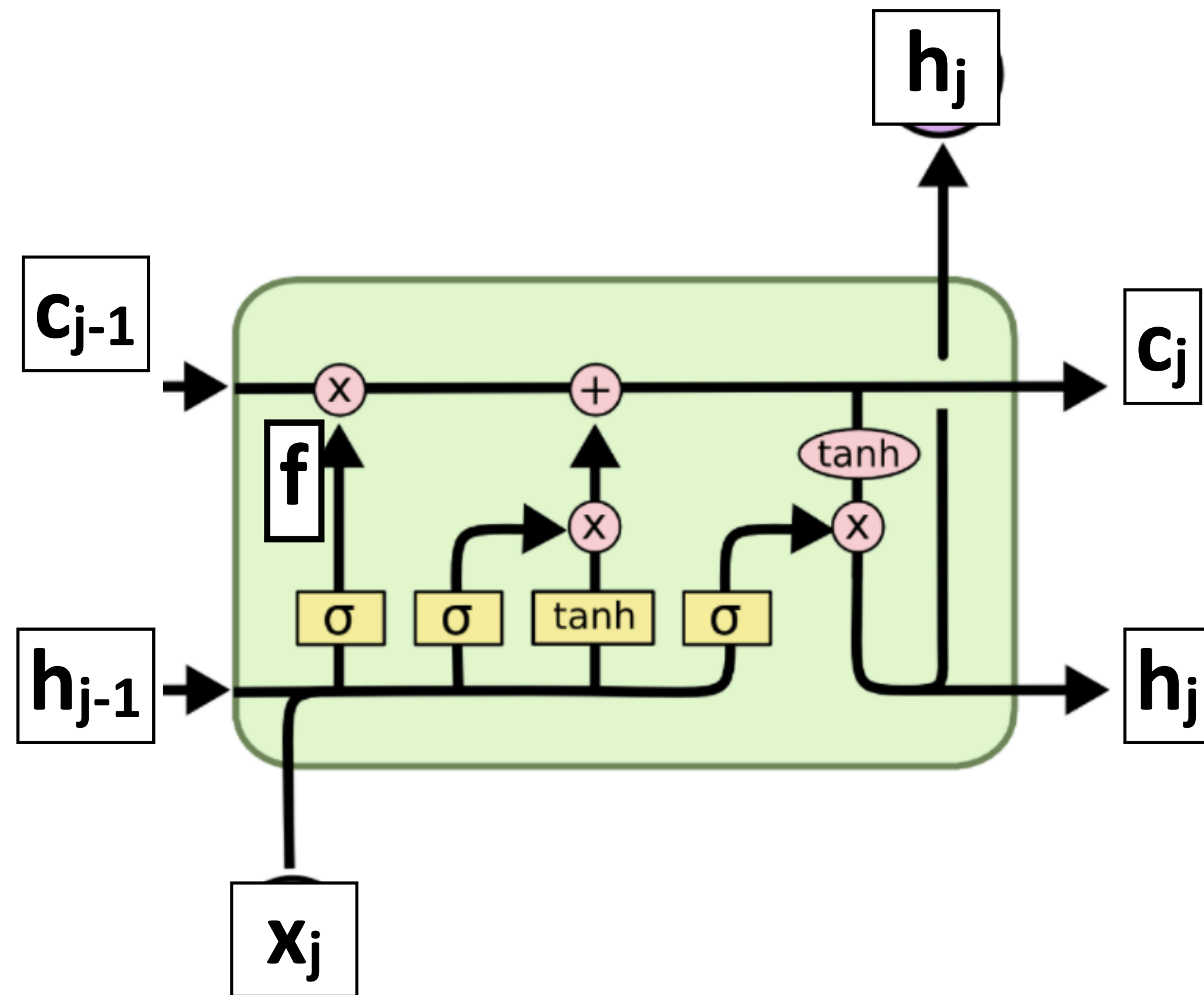
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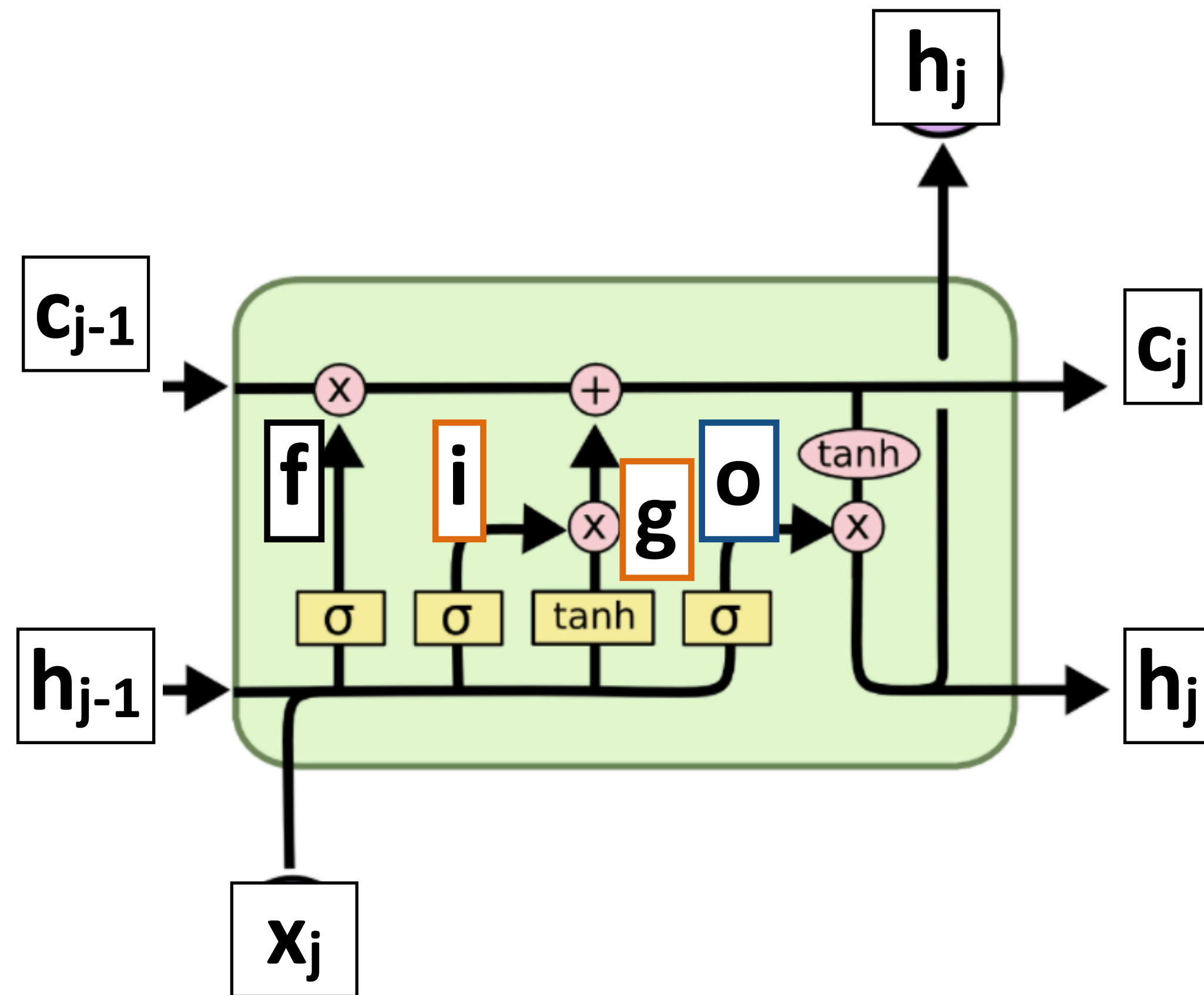
- ▶ Can we ignore the old value of c for this timestep?
- ▶ Can an LSTM sum up its inputs x ?
- ▶ Can we ignore a particular input x ?

LSTMs



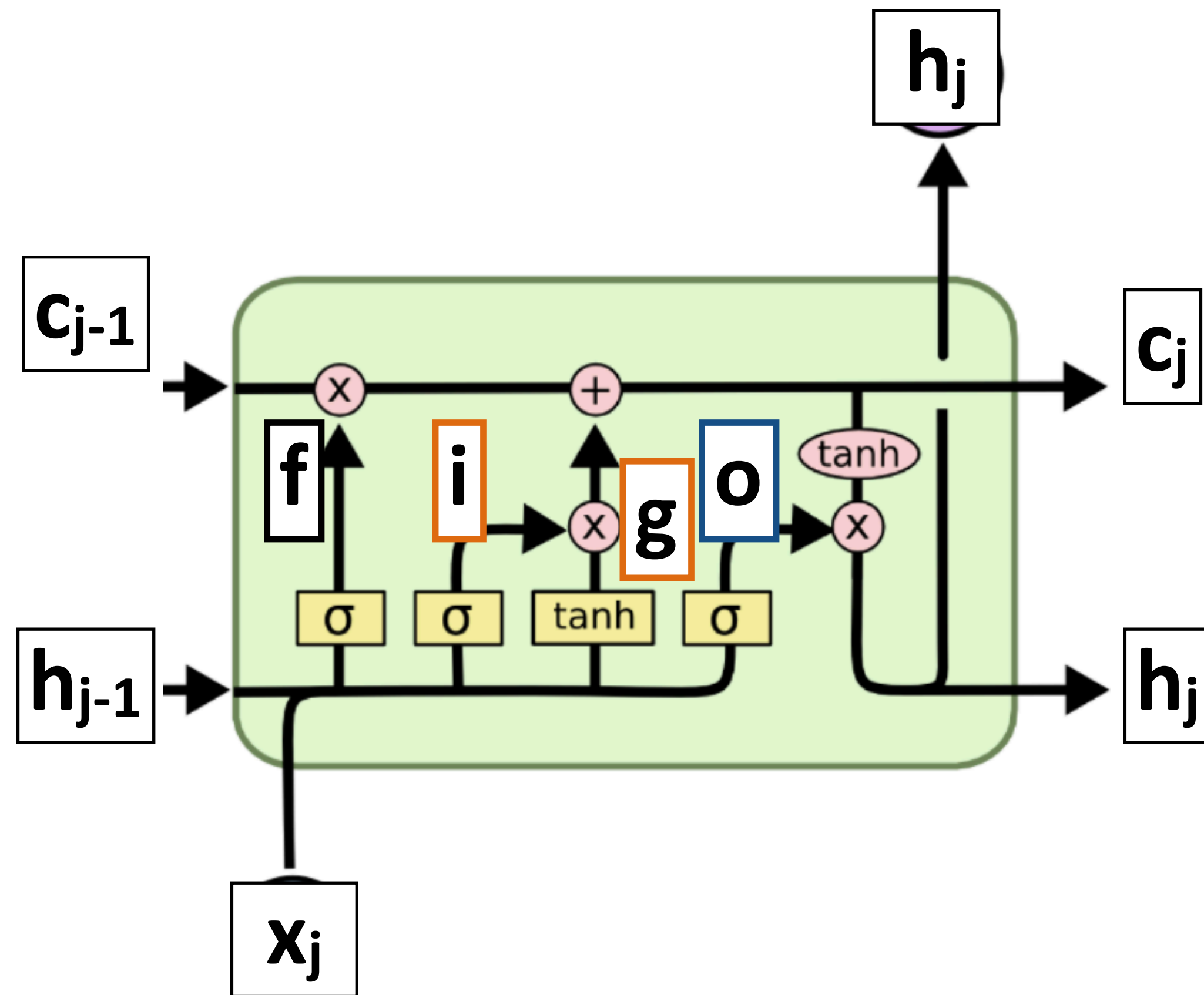
- ▶ Ignoring recurrent state entirely:
 - ▶ Lets us get feedforward layer over token

LSTMs



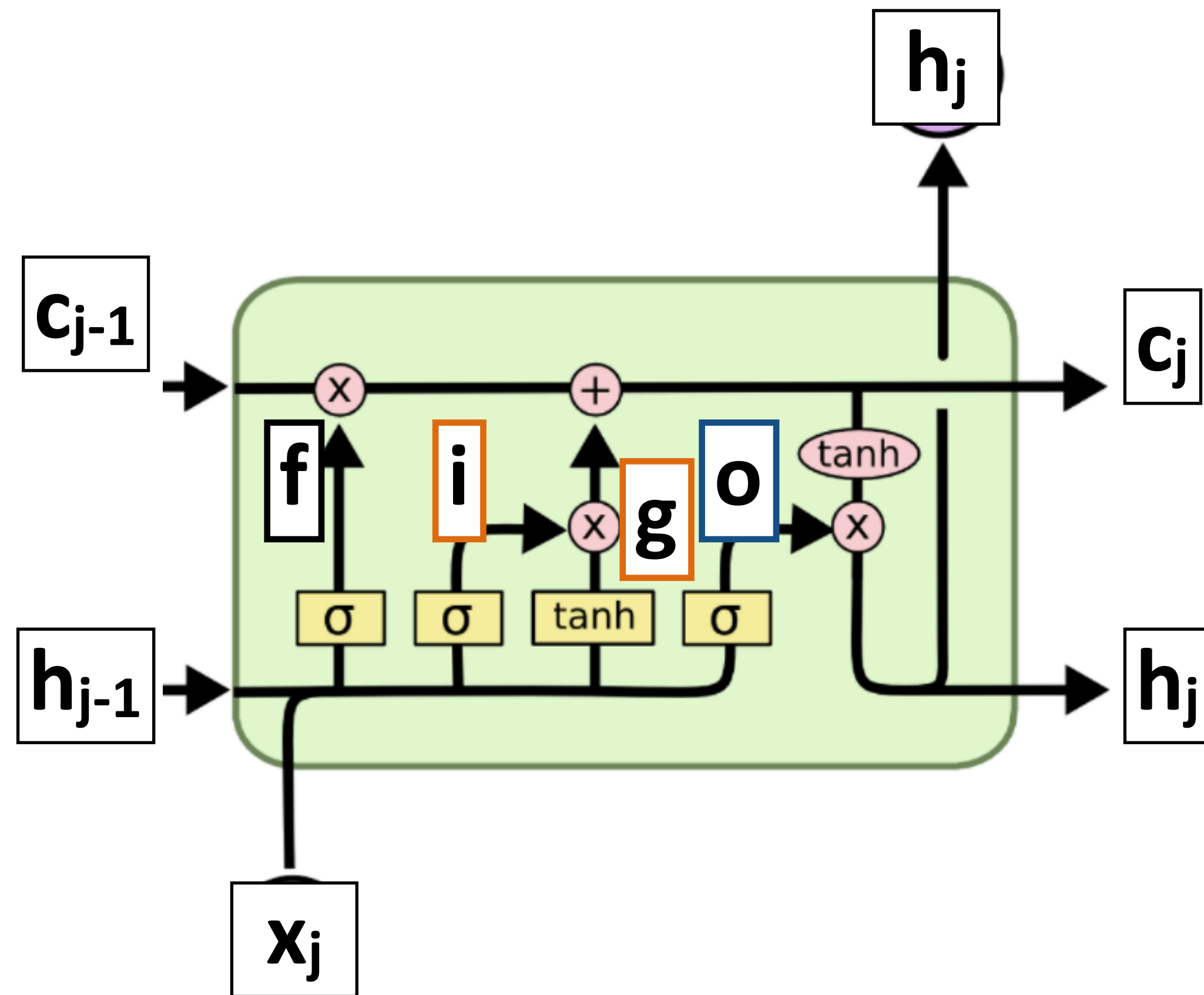
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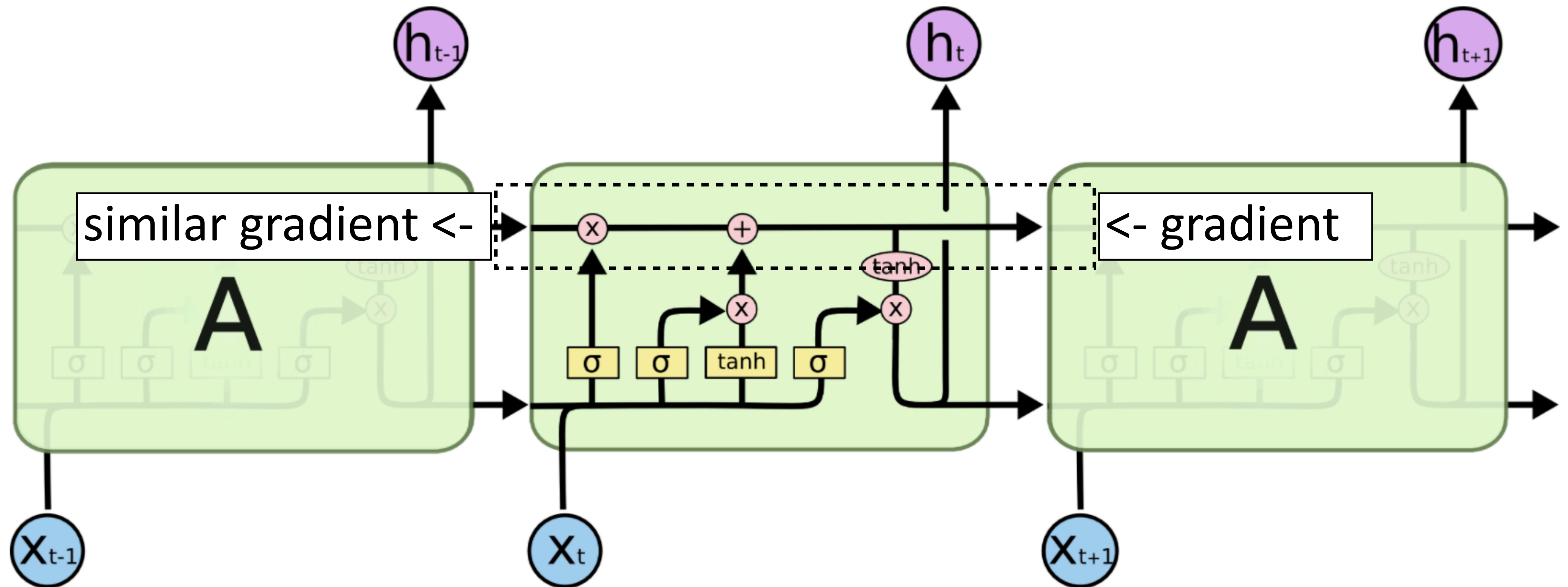
- ▶ Ignoring recurrent state entirely:
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- ▶ Ignoring input:
 - ▶ Lets us discard stopwords

LSTMs



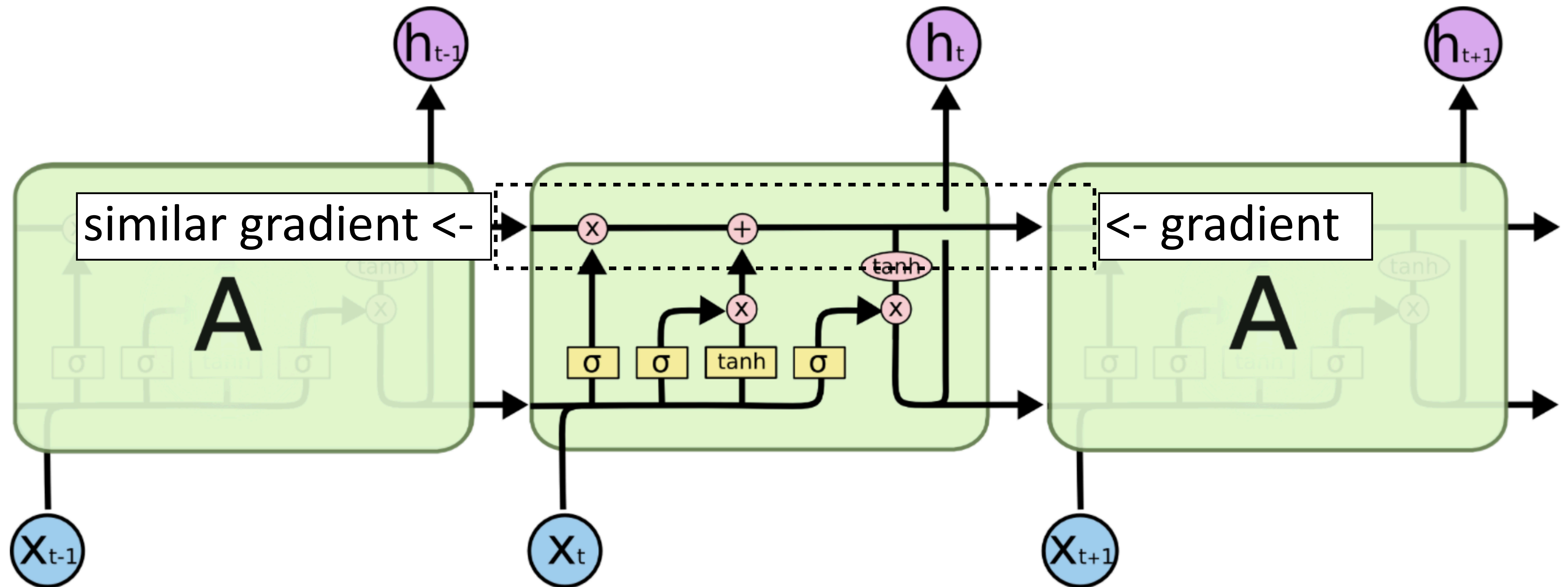
- ▶ Ignoring recurrent state entirely:
 - ▶ Lets us get feedforward layer over token
- ▶ Ignoring input:
 - ▶ Lets us discard stopwords
- ▶ Summing inputs:
 - ▶ Lets us compute a bag-of-words representation

LSTMs



"A Gentle Tutorial of Recurrent Neural Network with Error Backpropagation" Gang Chen (2018)
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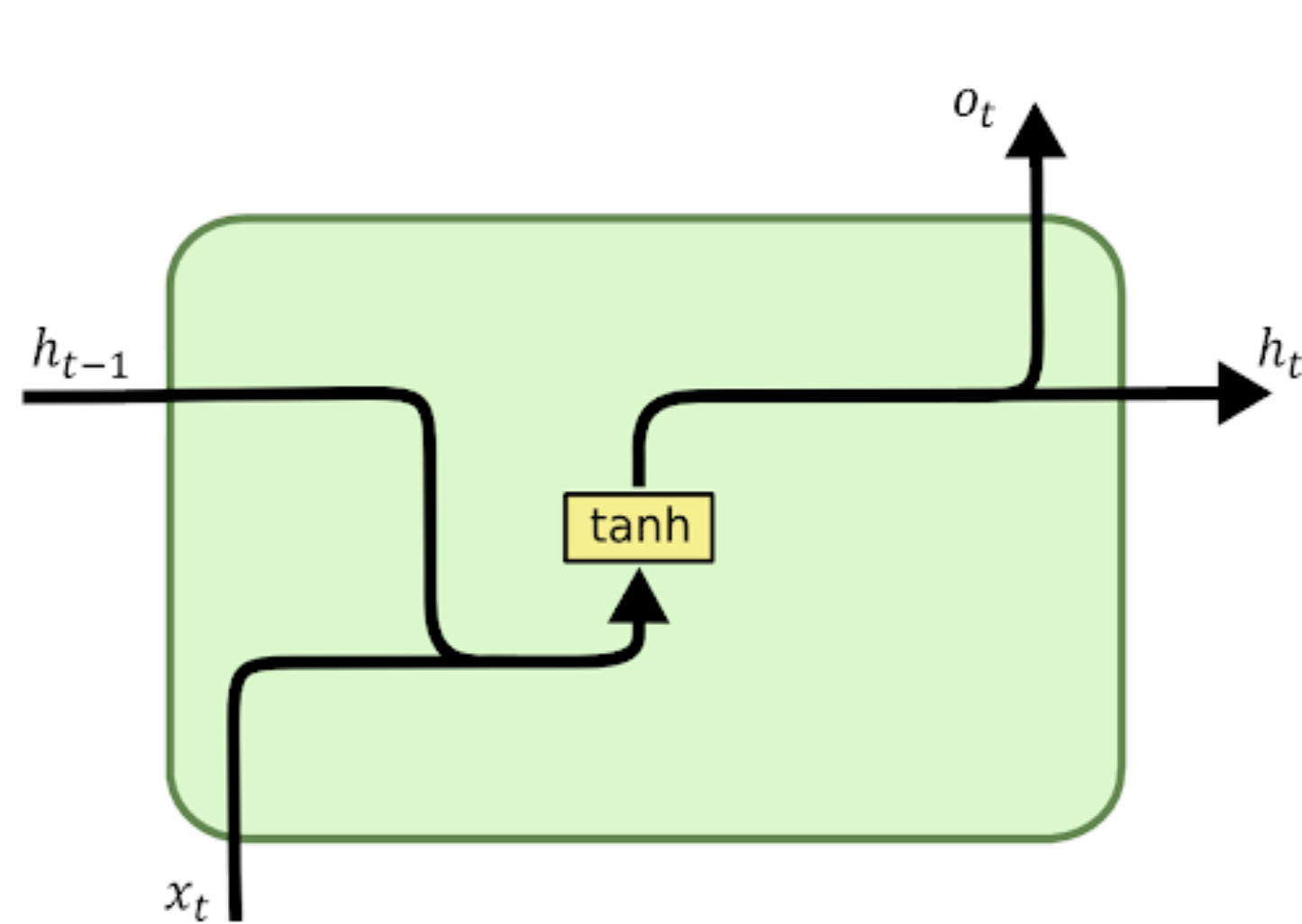
LSTMs



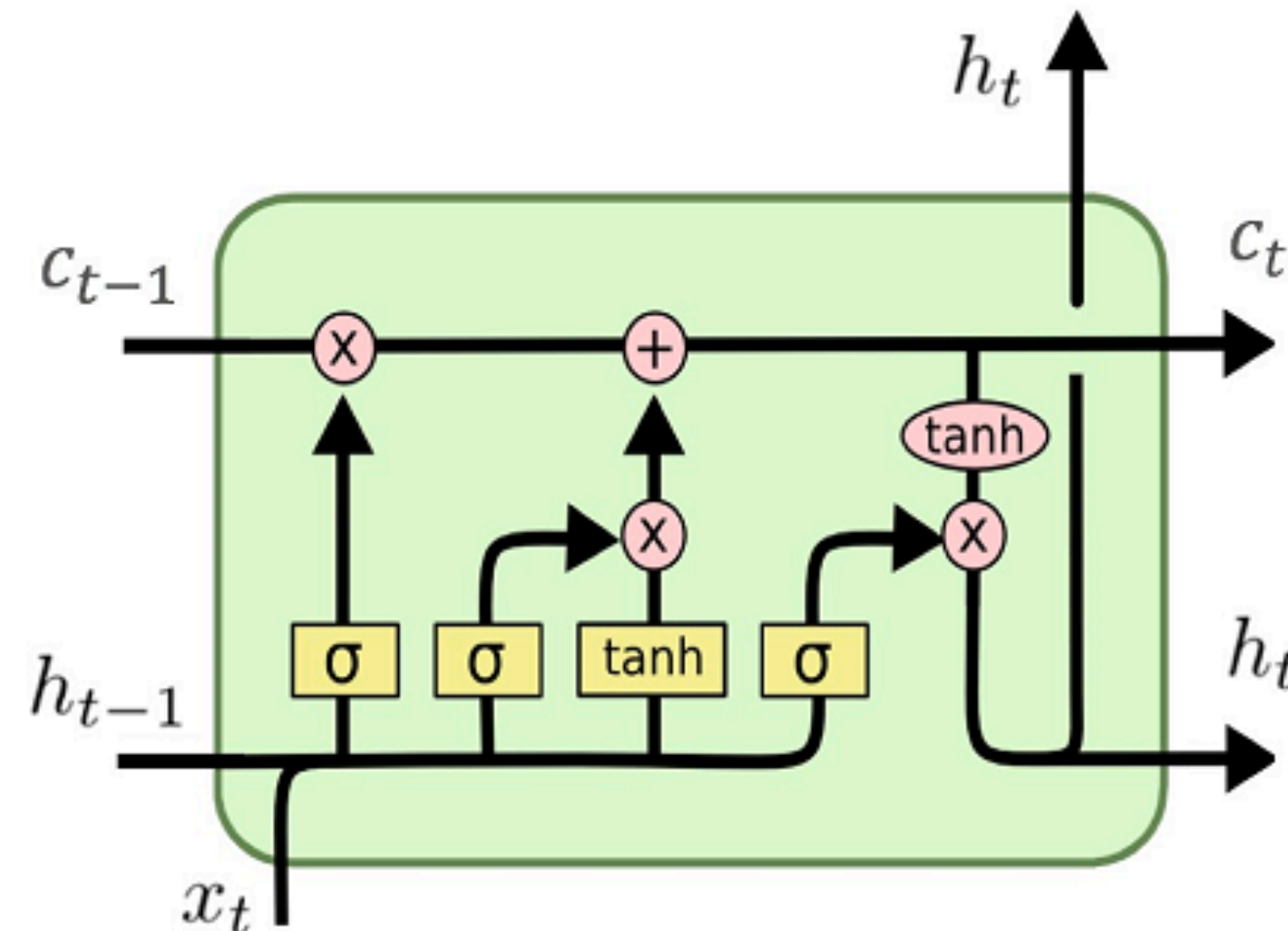
- ▶ Gradient still diminishes, but in a controlled way and generally by less — usually initialize forget gate = 1 to remember everything to start

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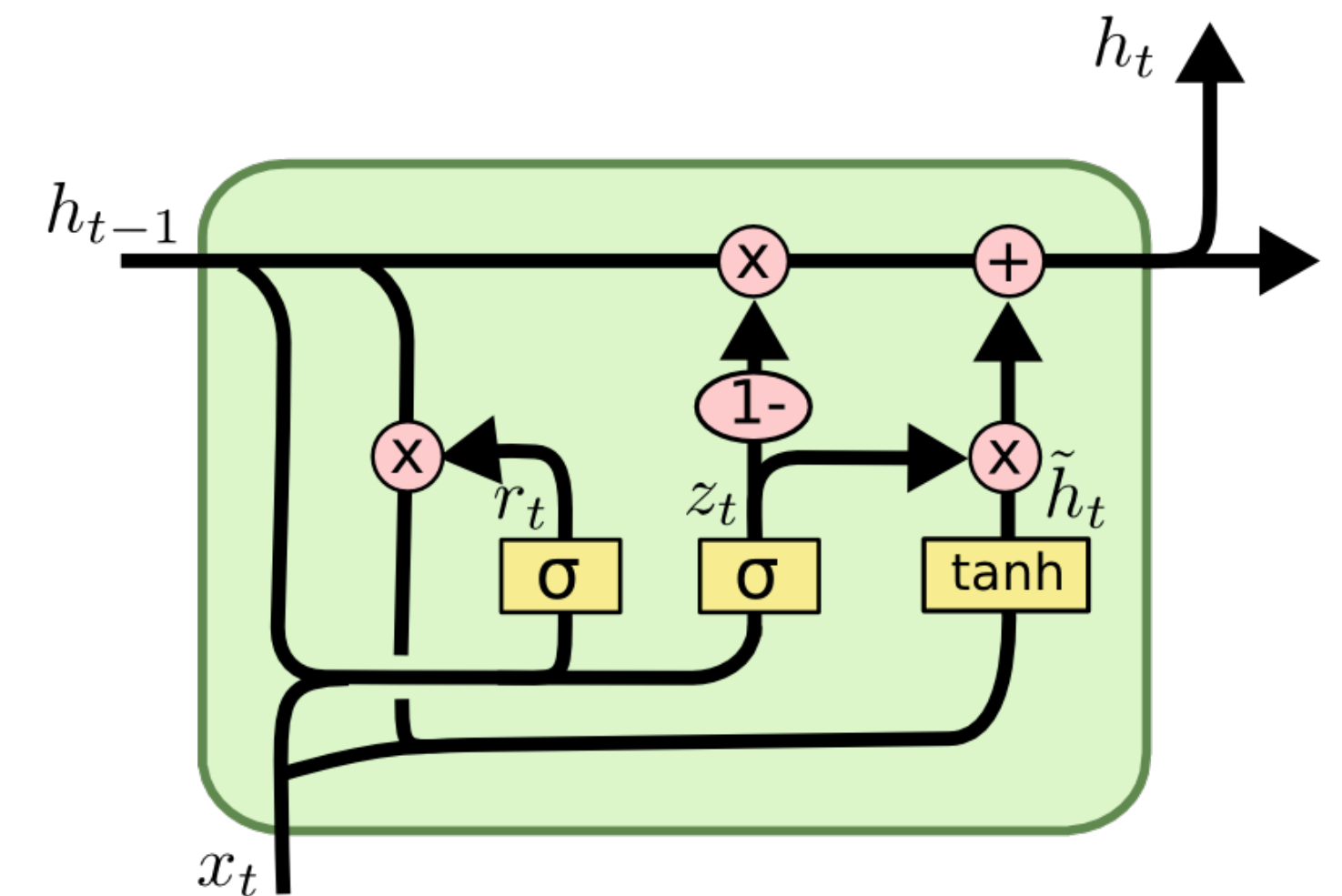
Standard RNN vs. LSTM vs. GRU



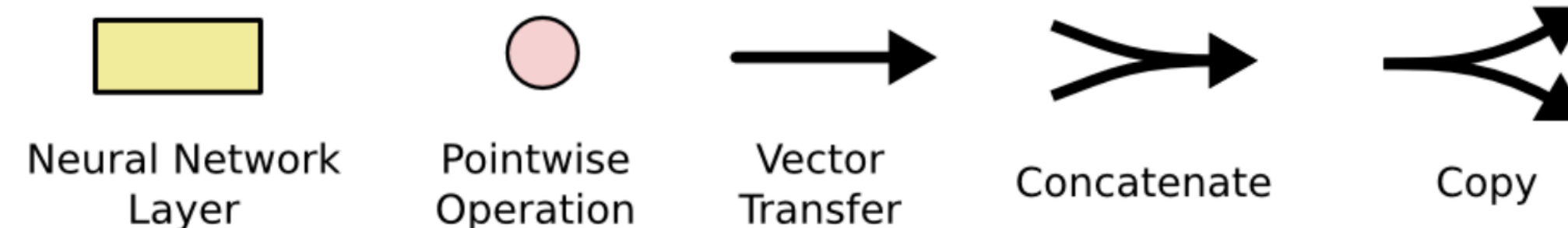
Standard RNN
(with a single layer)



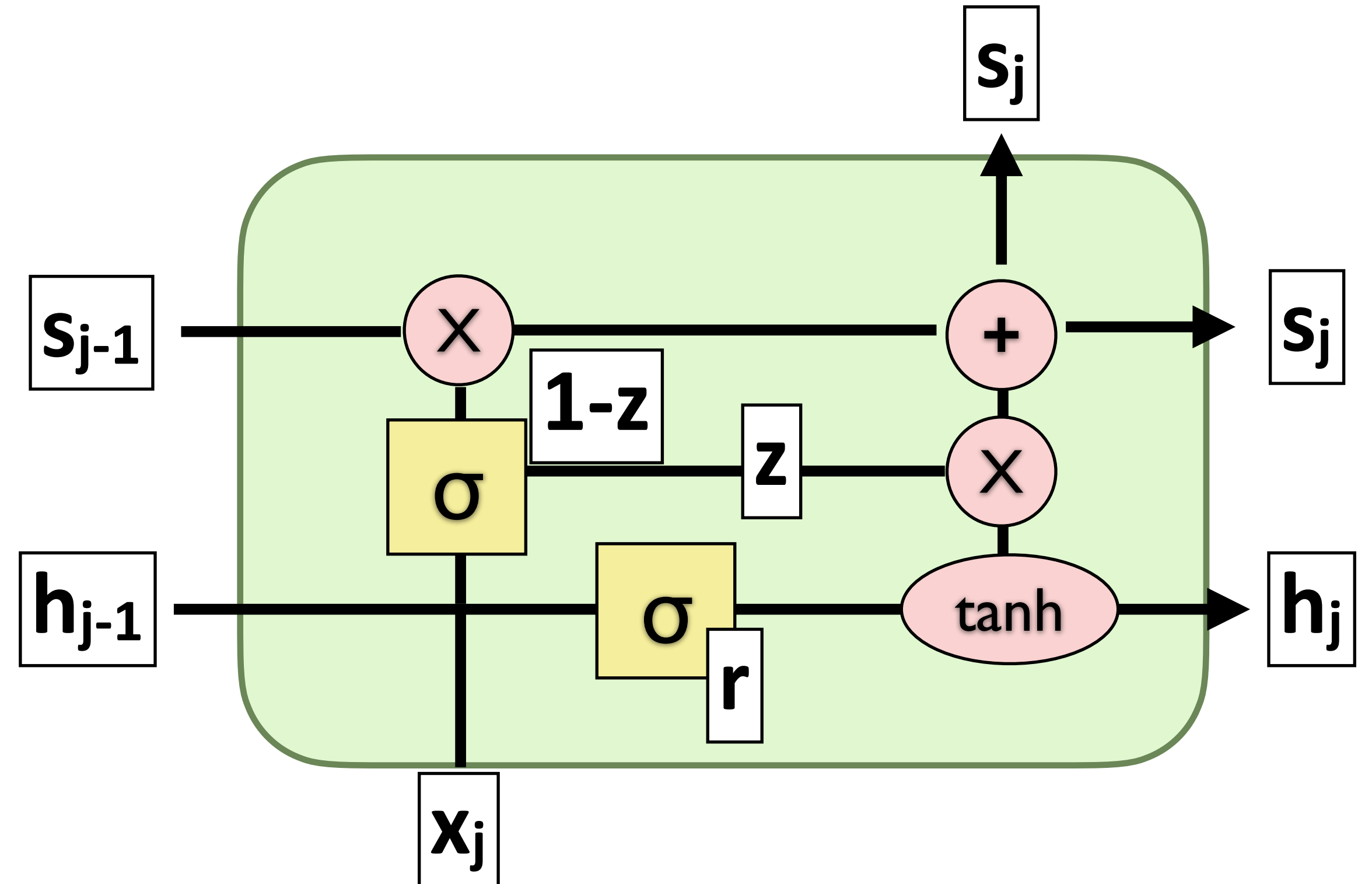
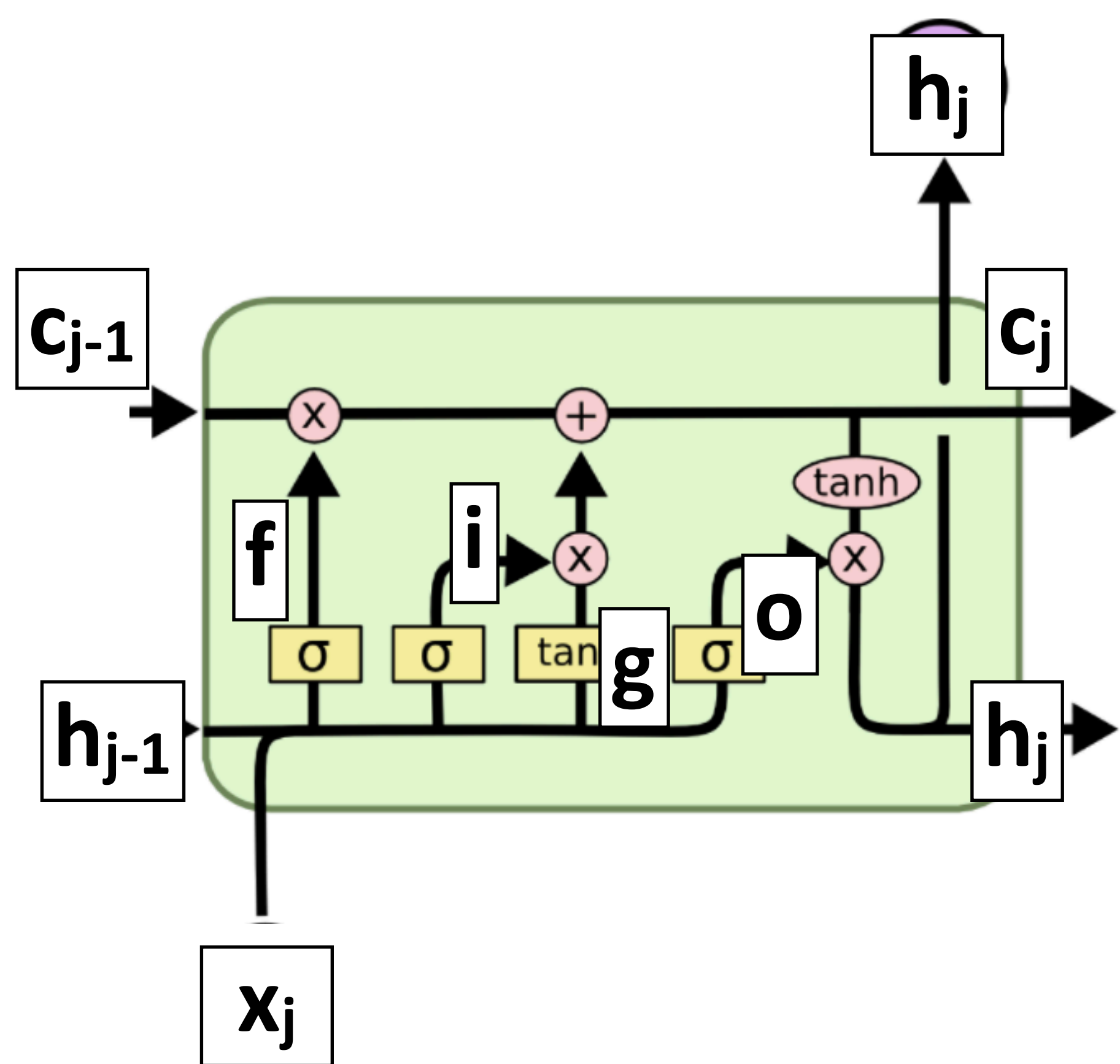
LSTM
(Long Short-term Memory)



GRU
(Gated Recurrent Unit)

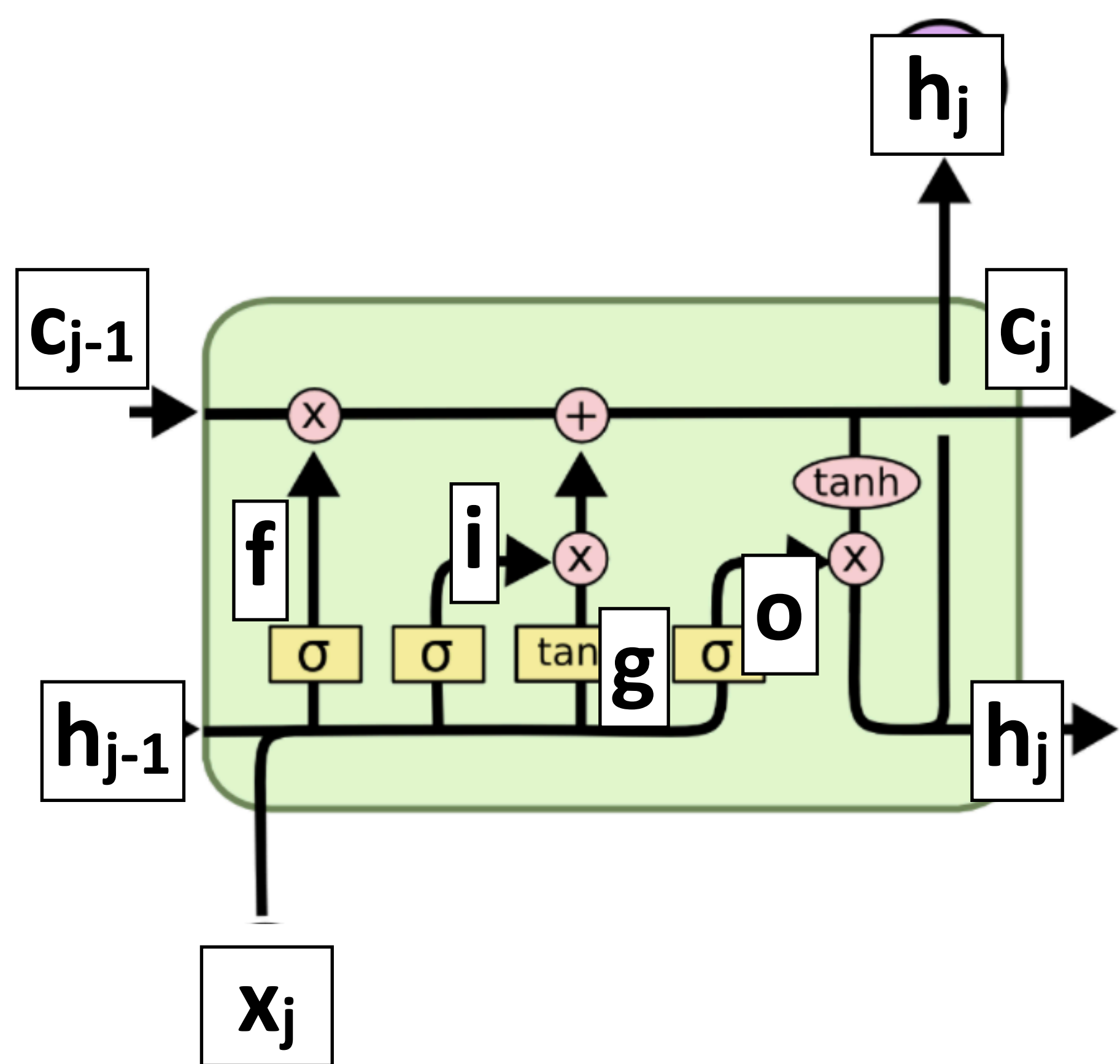


Gated Recurrent Units (GRUs)

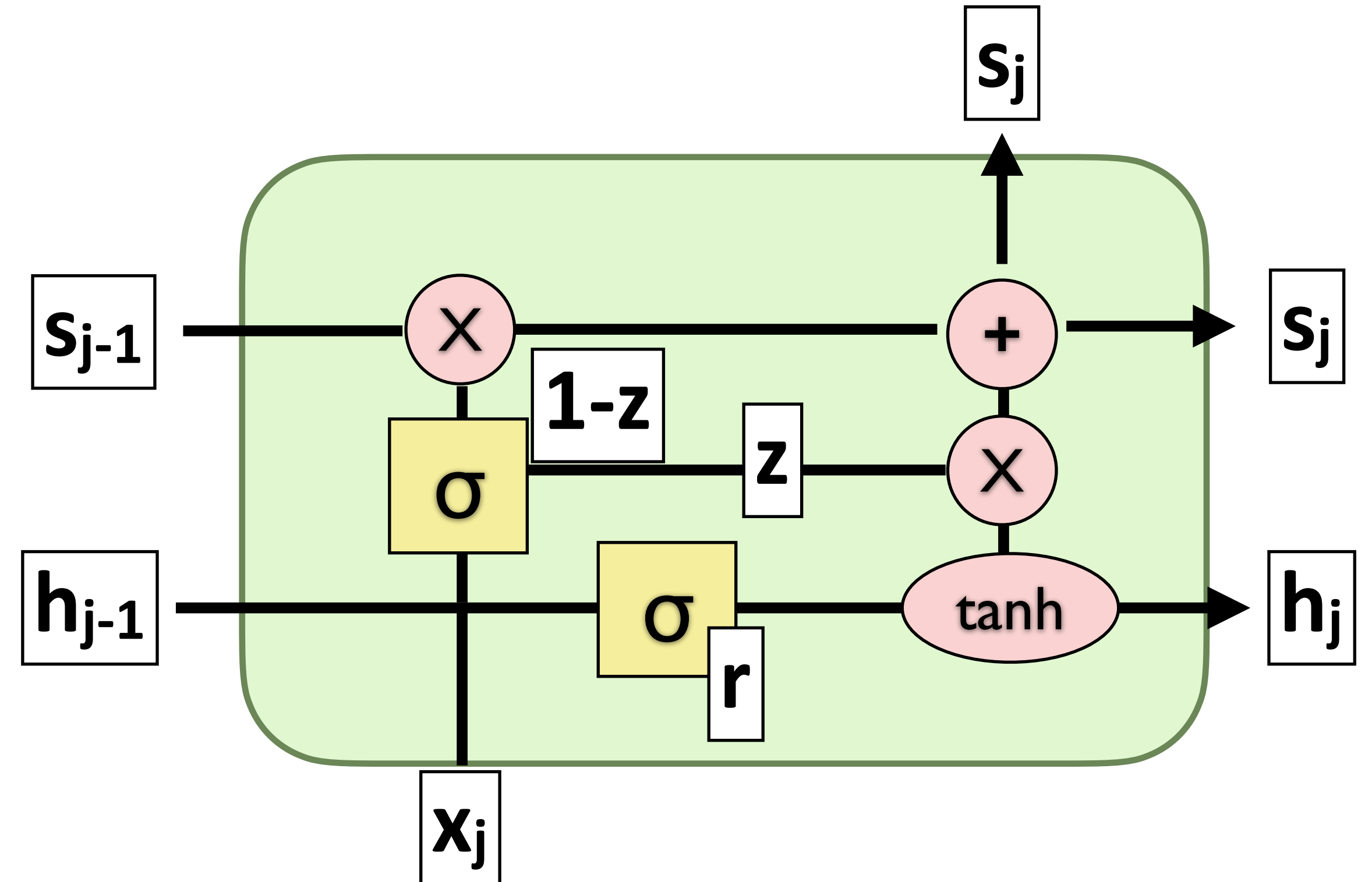


- ▶ LSTM: more complex and slower, may work a bit better

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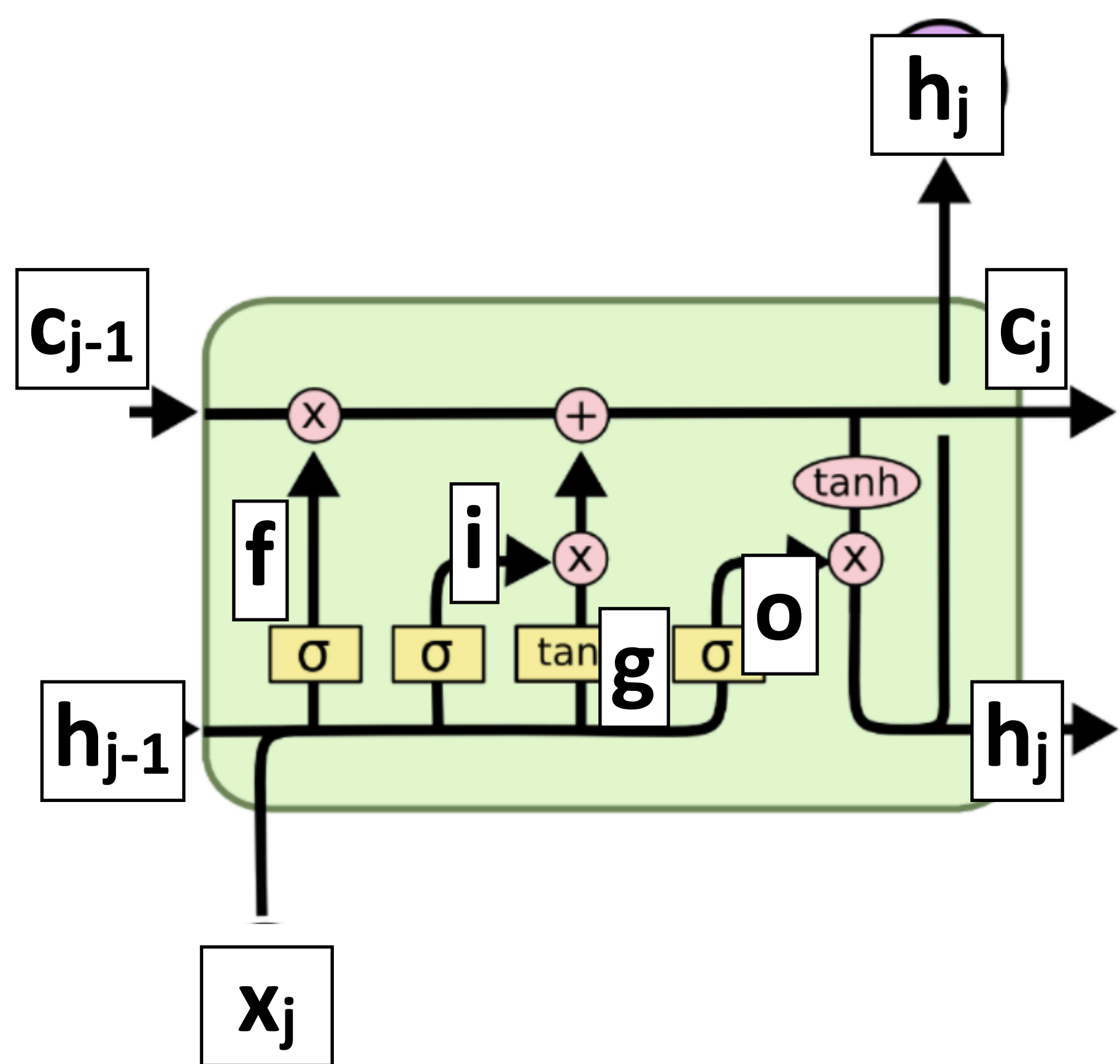


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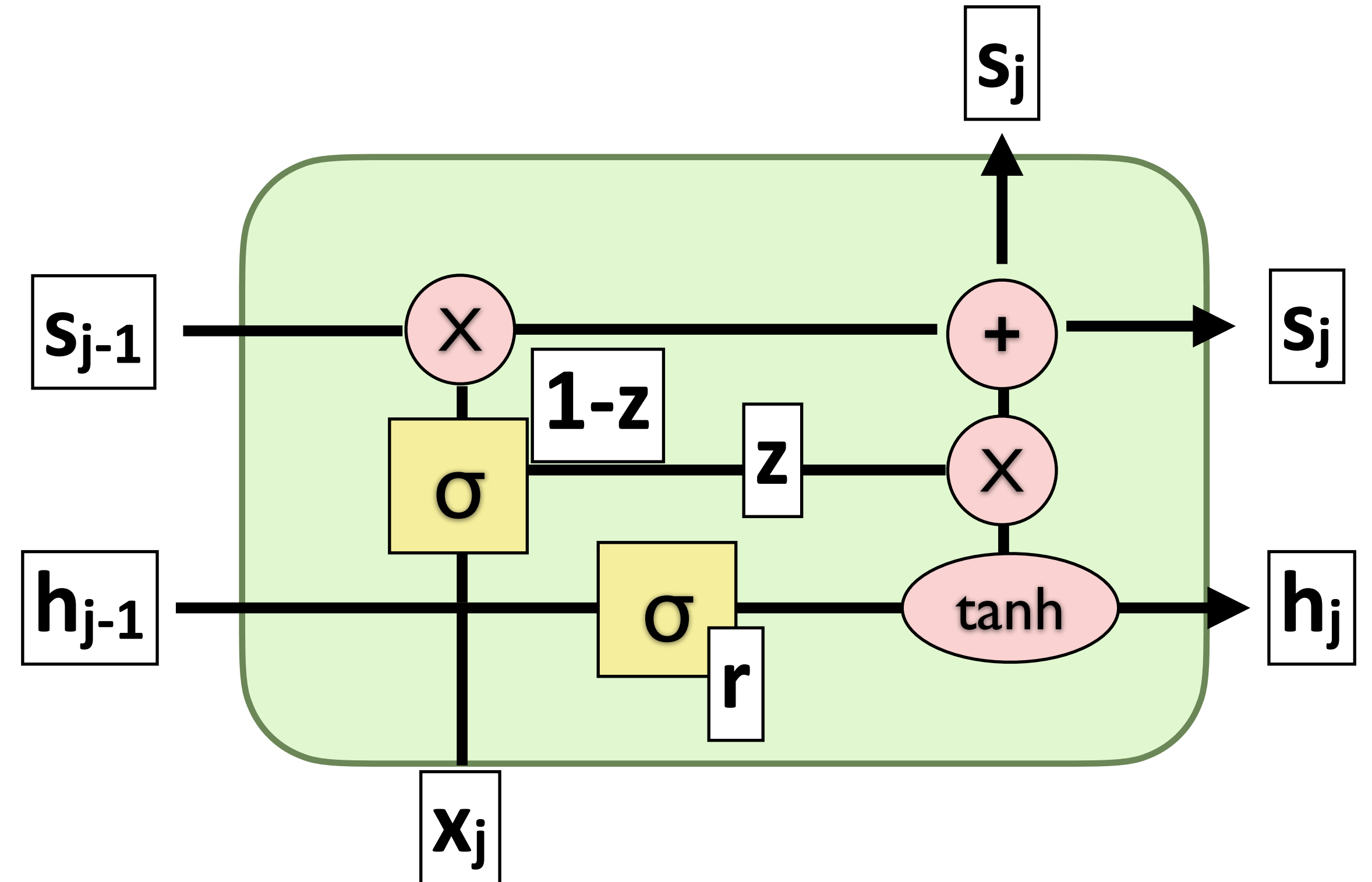


- ▶ GRU: faster, a bit simpler

Gated Recurrent Units (GRUs)



- ▶ LSTM: more complex and slower, may work a bit better



- ▶ GRU: faster, a bit simpler
- ▶ Two gates: z (forget, mixes s and h) and r (mixes h and x)

GRUs

- ▶ Also solves the vanishing gradient problem, simpler than LSTM

GRUs

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$$\mathbf{h}_t = (\mathbf{1} - \mathbf{z}) \odot \mathbf{h}_{t-1} + \mathbf{z} \odot \text{func}(\mathbf{x}_t, \mathbf{h}_{t-1})$$

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- ▶ Also solves the vanishing gradient problem, simpler than LSTM

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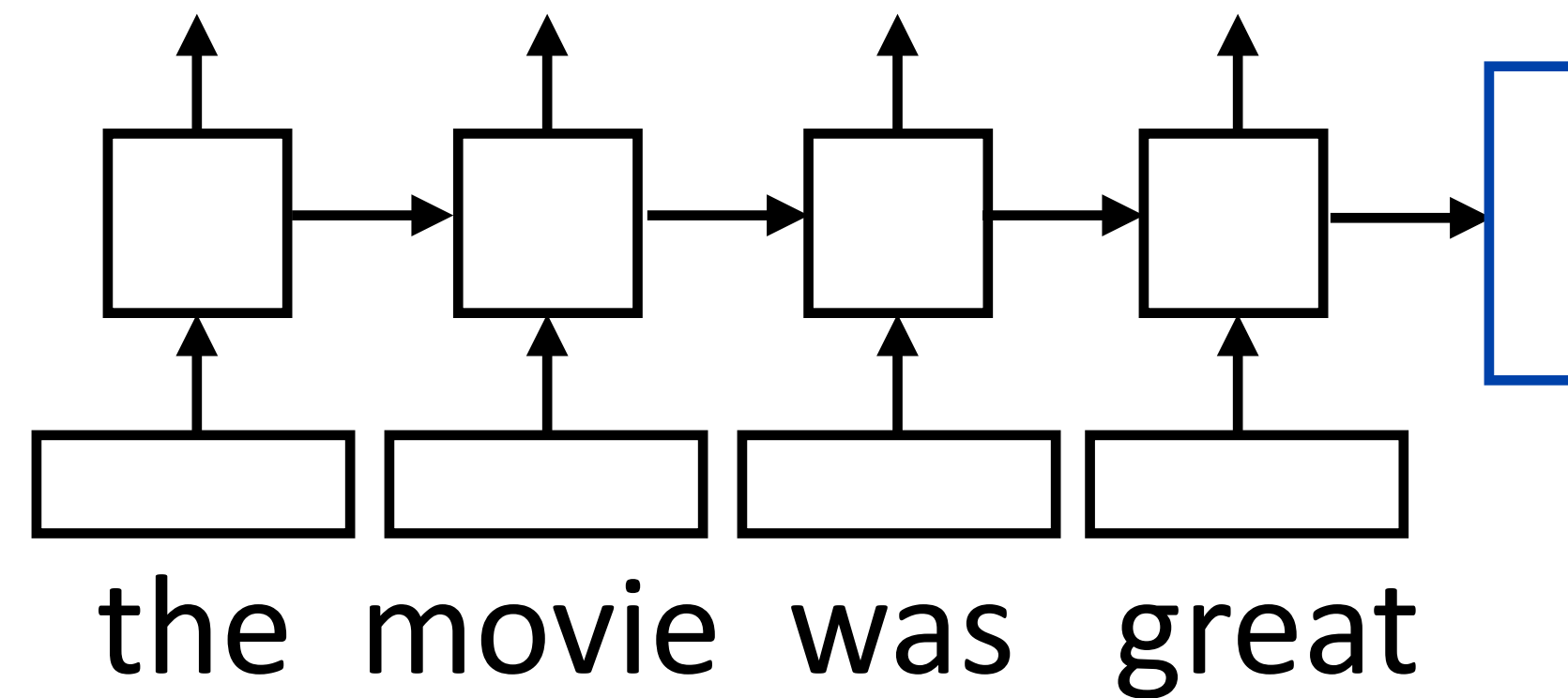
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- ▶ $\mathbf{z}$ controls mixing of hidden state $\mathbf{h}$ with new input $\mathbf{x}$
- ▶ Faster to train and sometimes work better (most times not) than LSTMs
- ▶ Other variants of LSTMs:
 - ▶ multiplicative LSTMs (2017), rotational unit of memory (RUM; 2019), ...

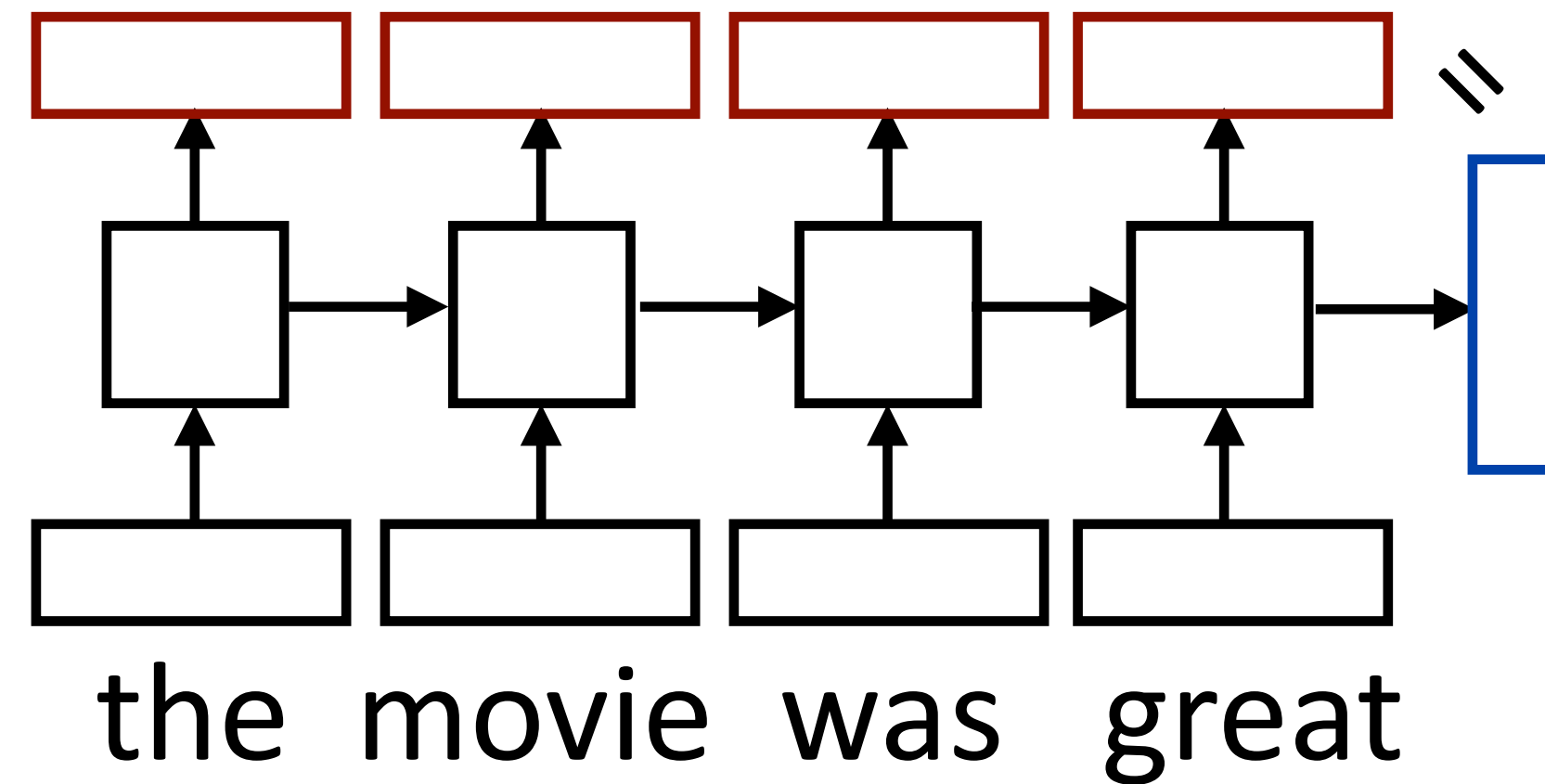
Applications

What do RNNs produce?



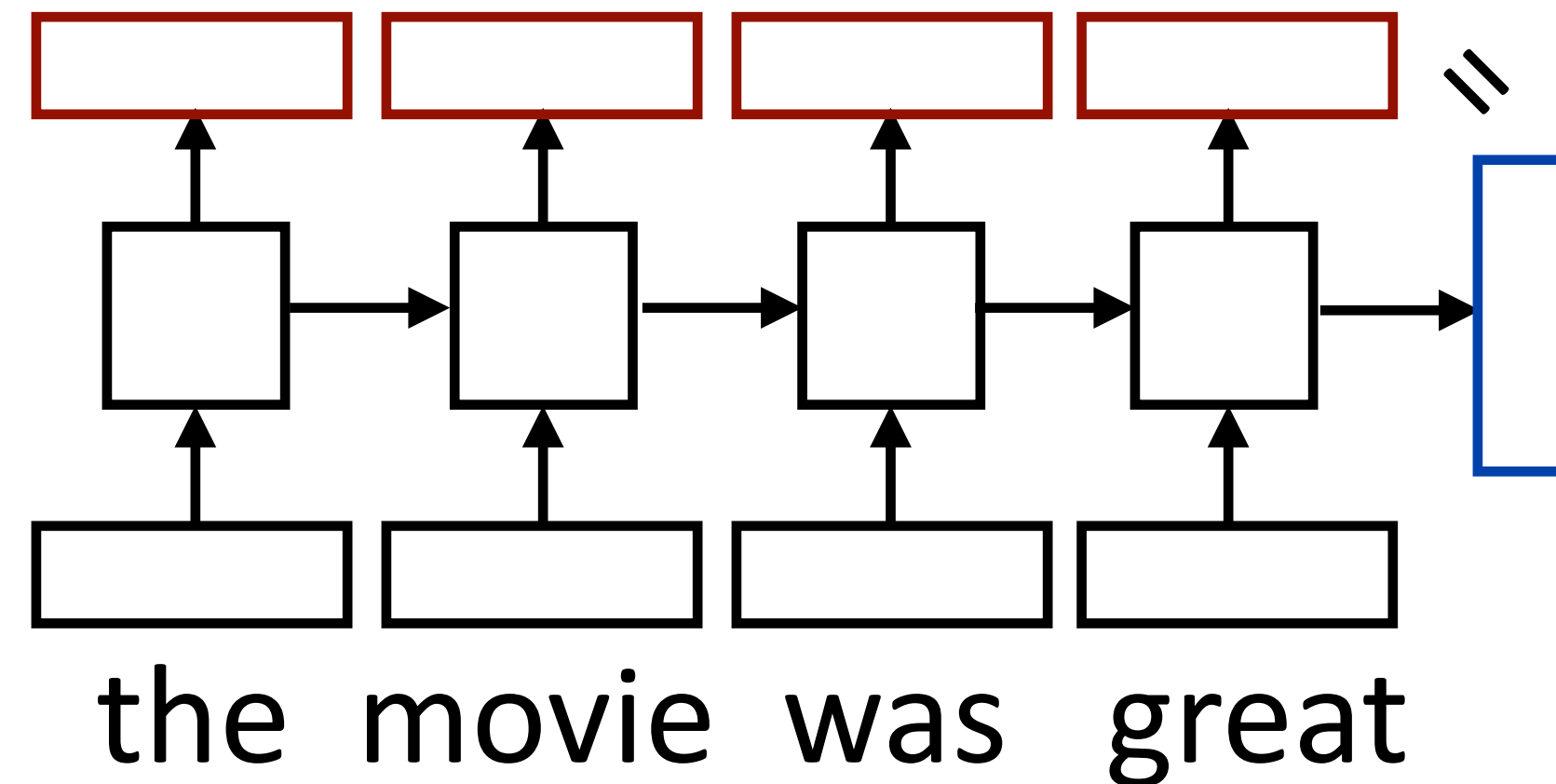
- **Encoding of the sentence** — can pass this a decoder or make a classification decision about the sentence

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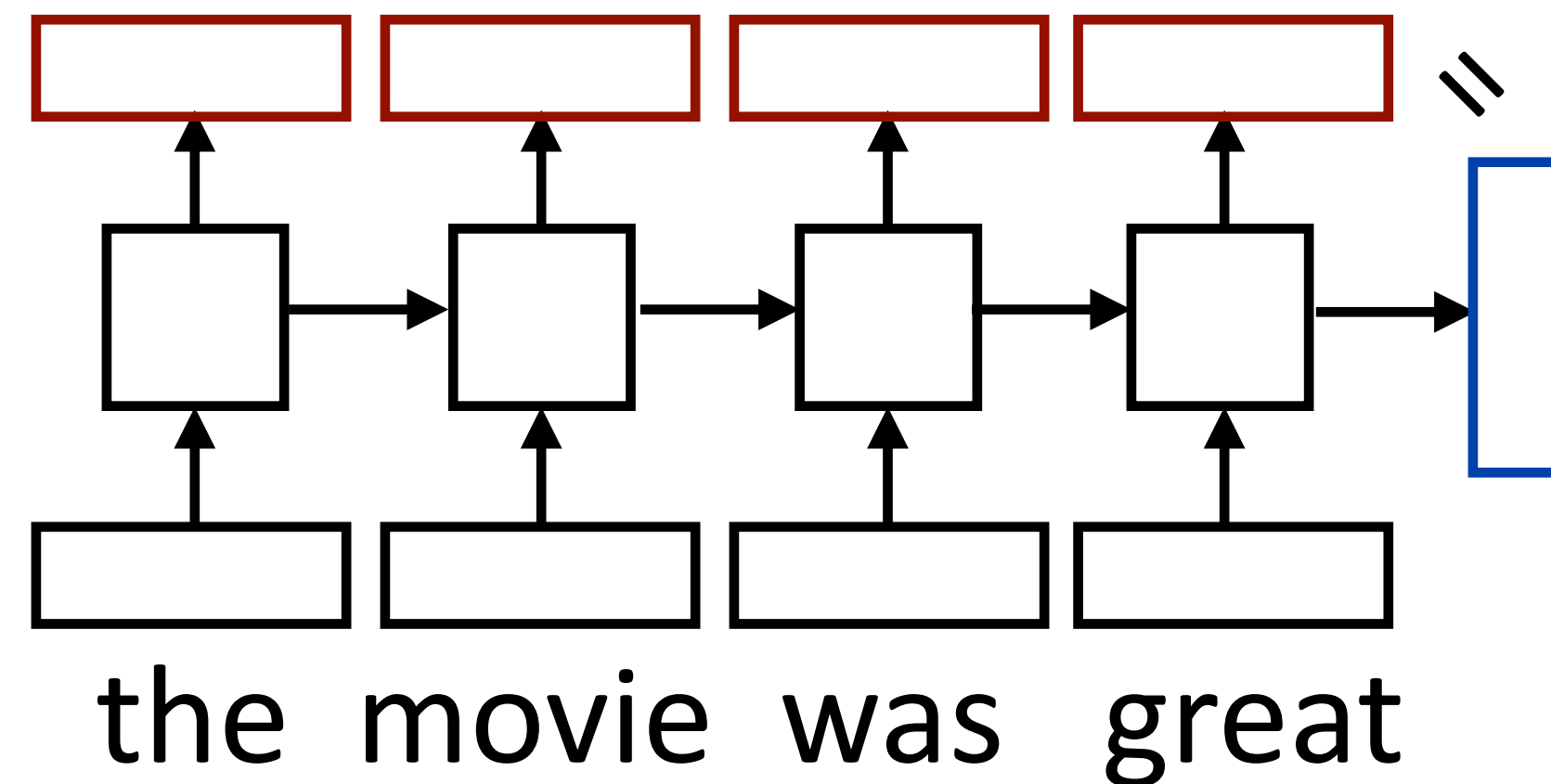
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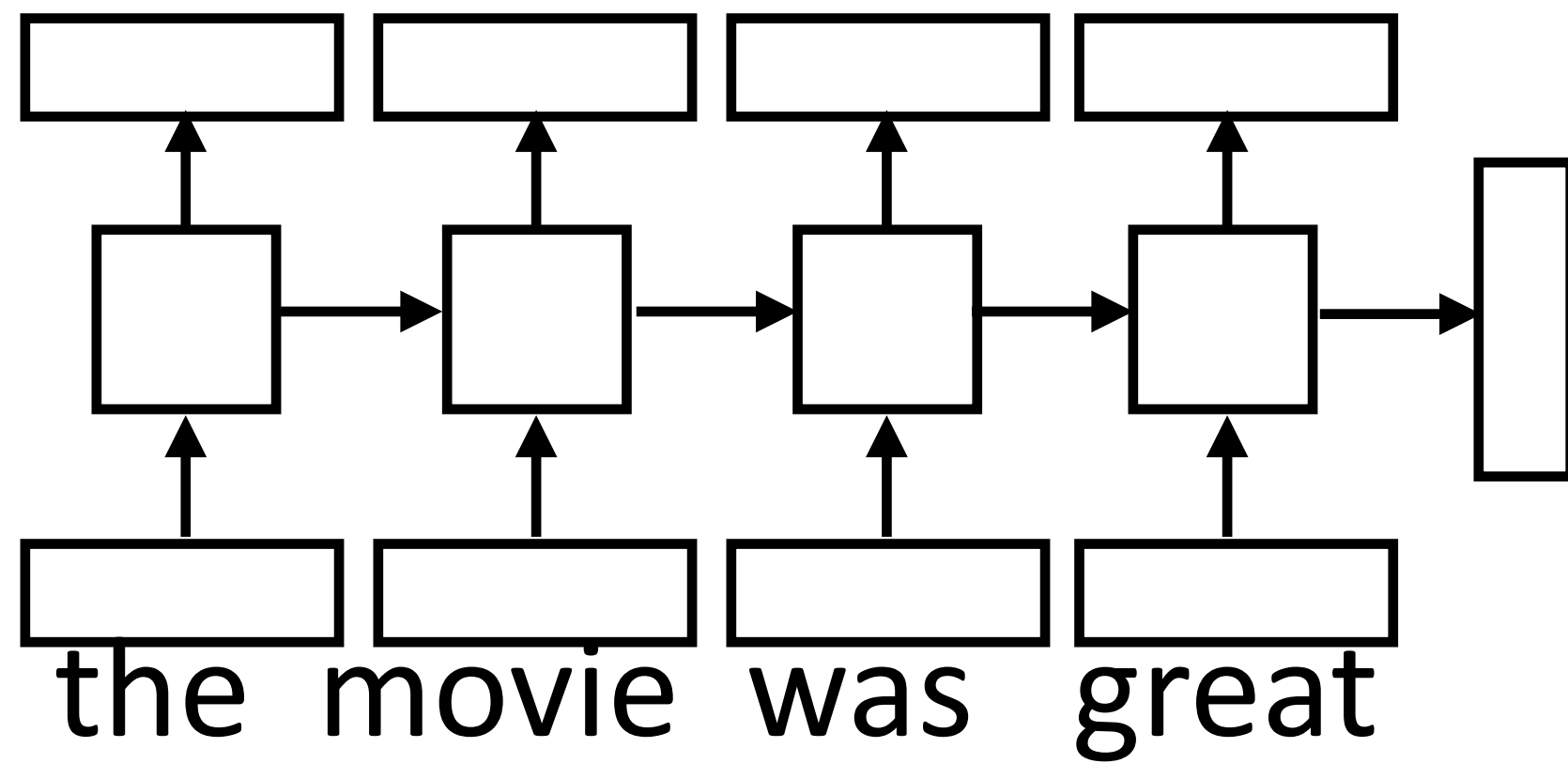
- ▶ **Encoding of the sentence** — can pass this a decoder or make a classification decision about the sentence
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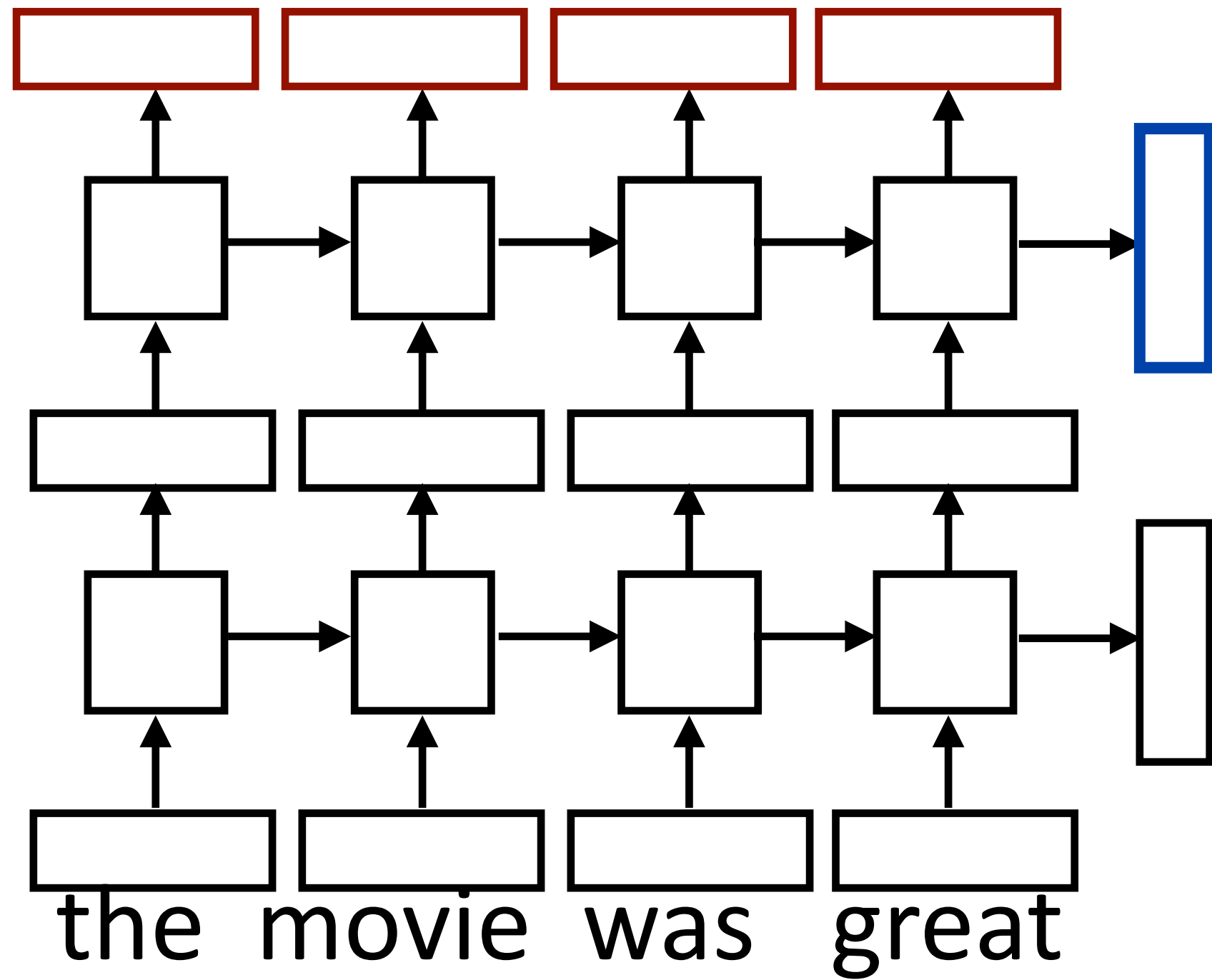


- ▶ **Encoding of the sentence** — can pass this a decoder or make a classification decision about the sentence
- ▶ **Encoding of each word** — can pass this to another layer to make a prediction (can also pool these to get a different sentence encoding)
- ▶ RNN can be viewed as a transformation of a sequence of vectors into a sequence of context-dependent vectors

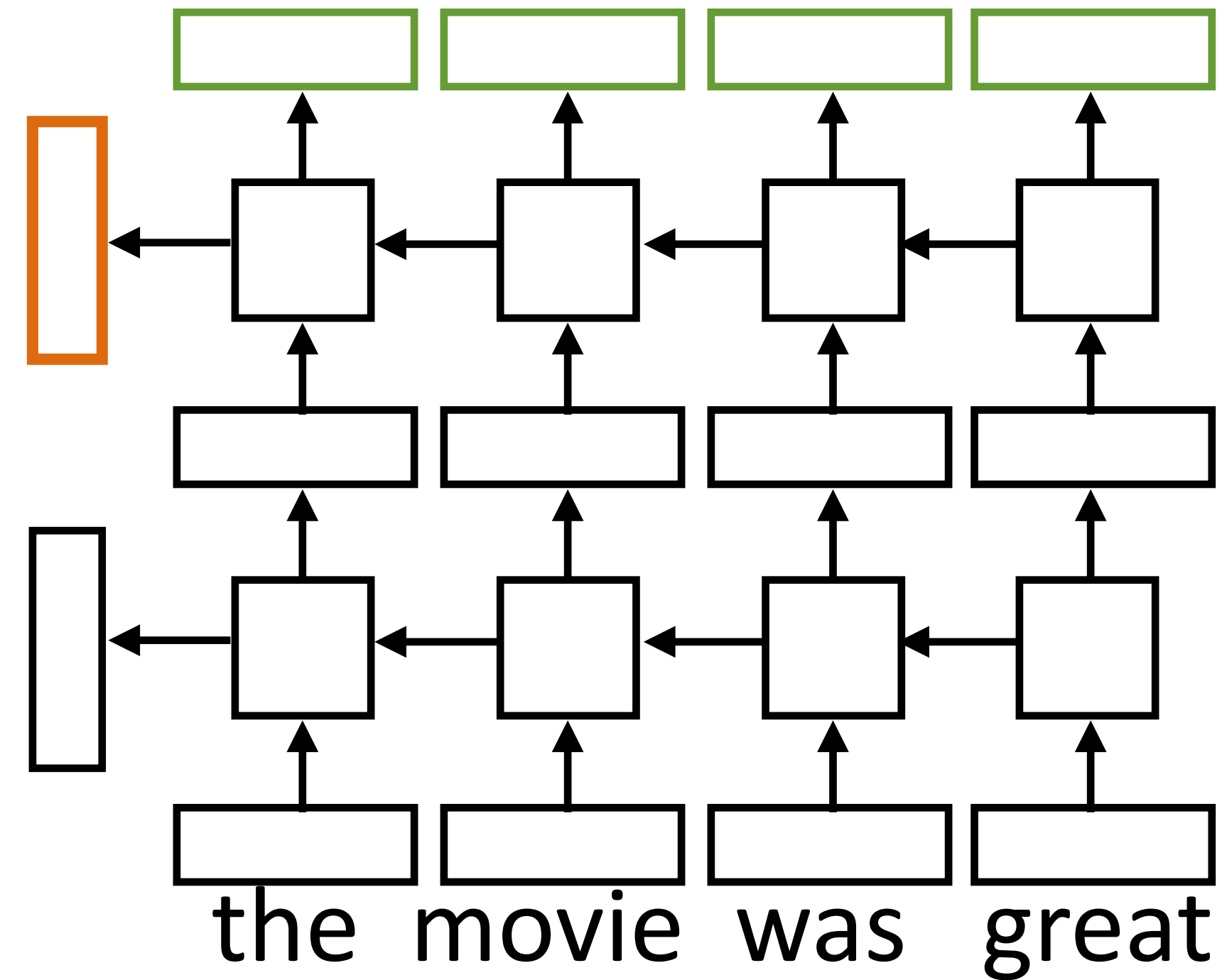
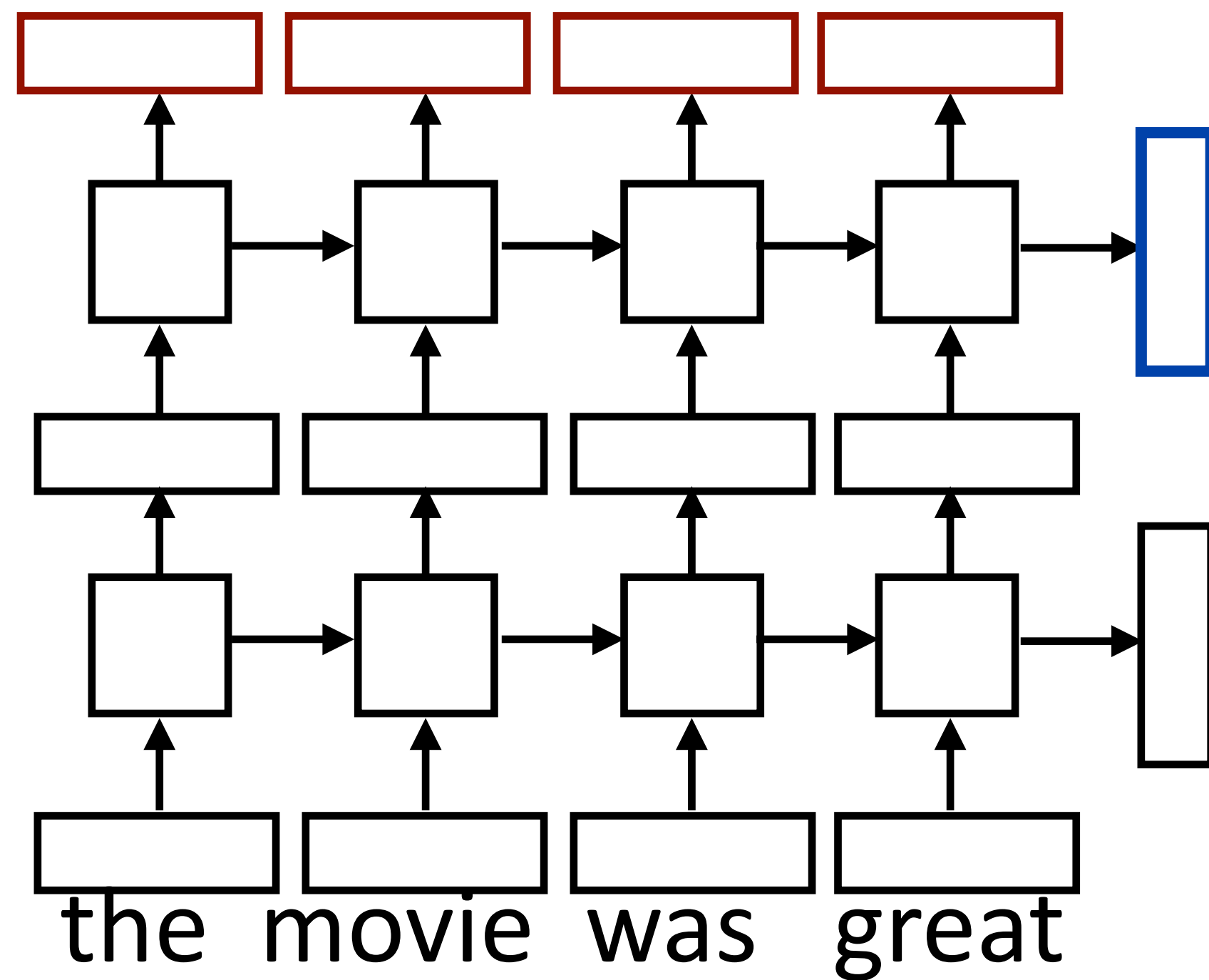
Multilayer Bidirectional RNN



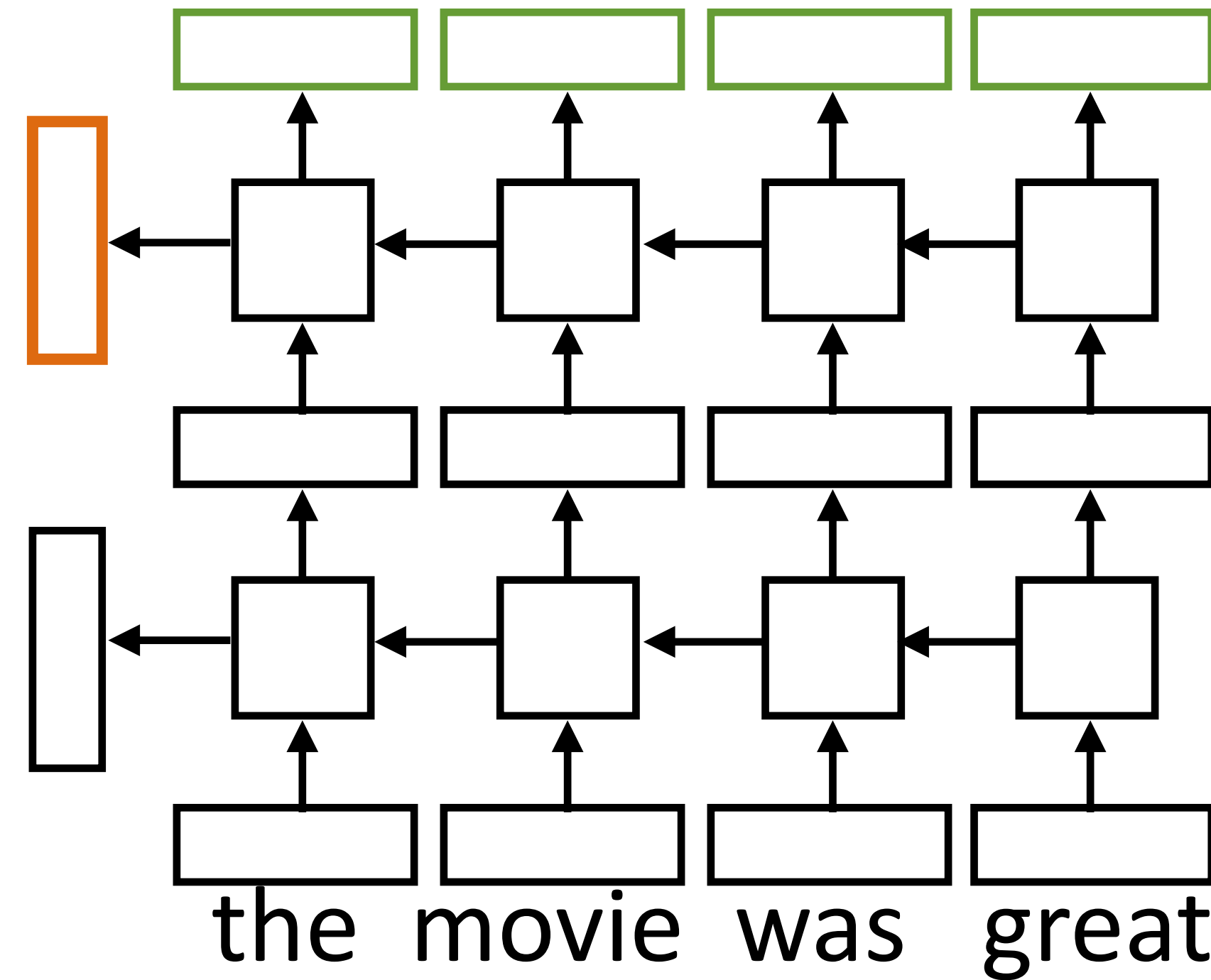
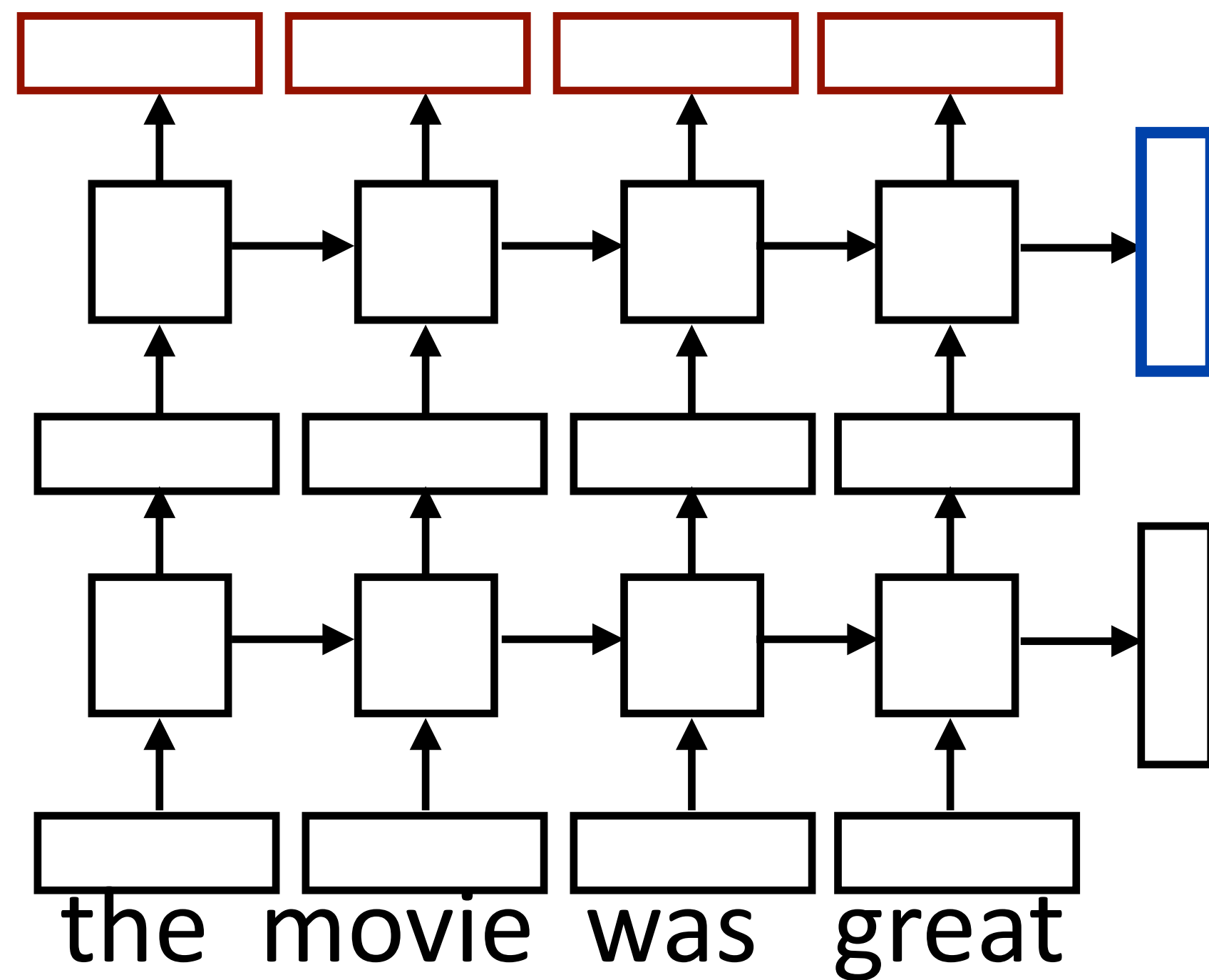
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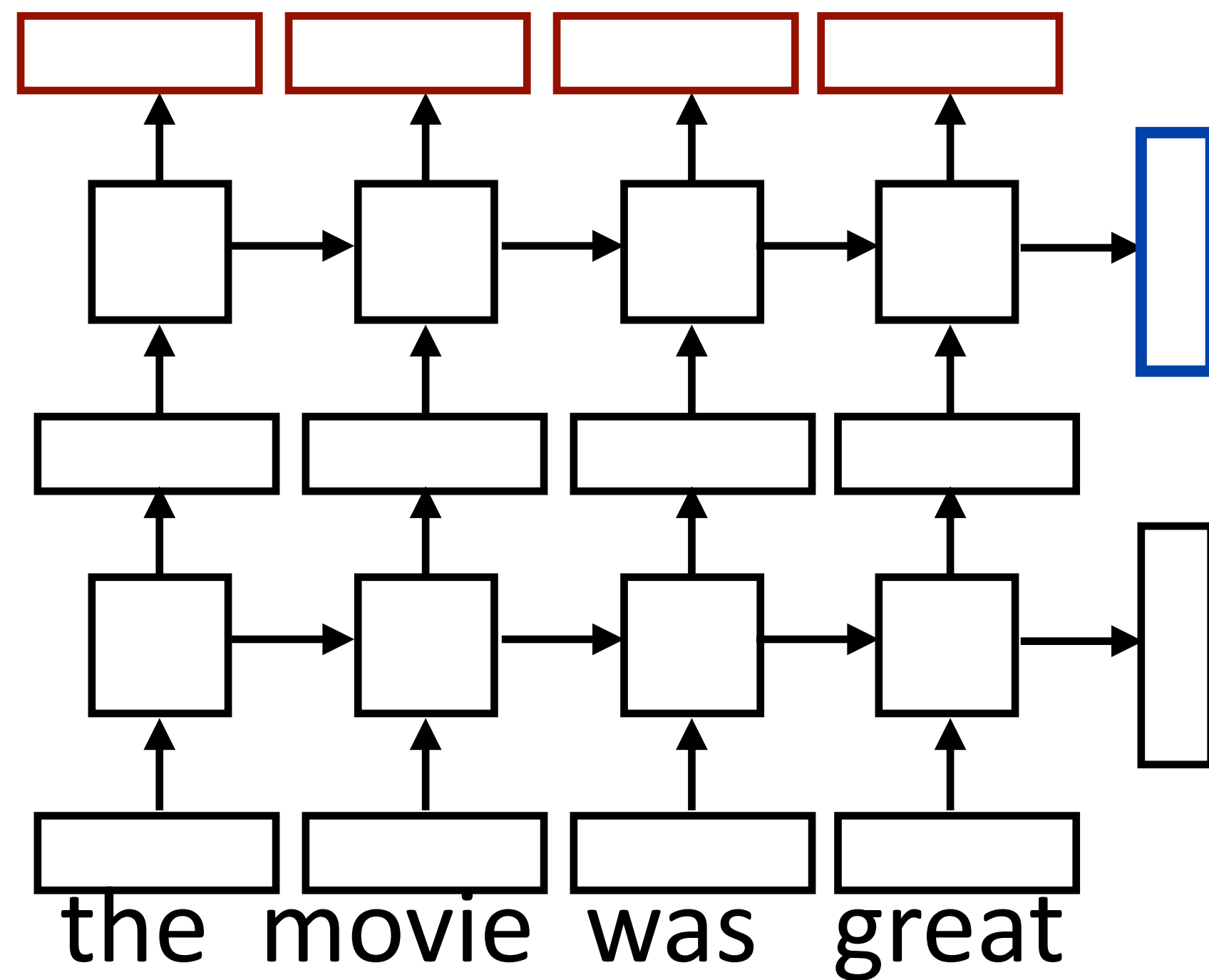
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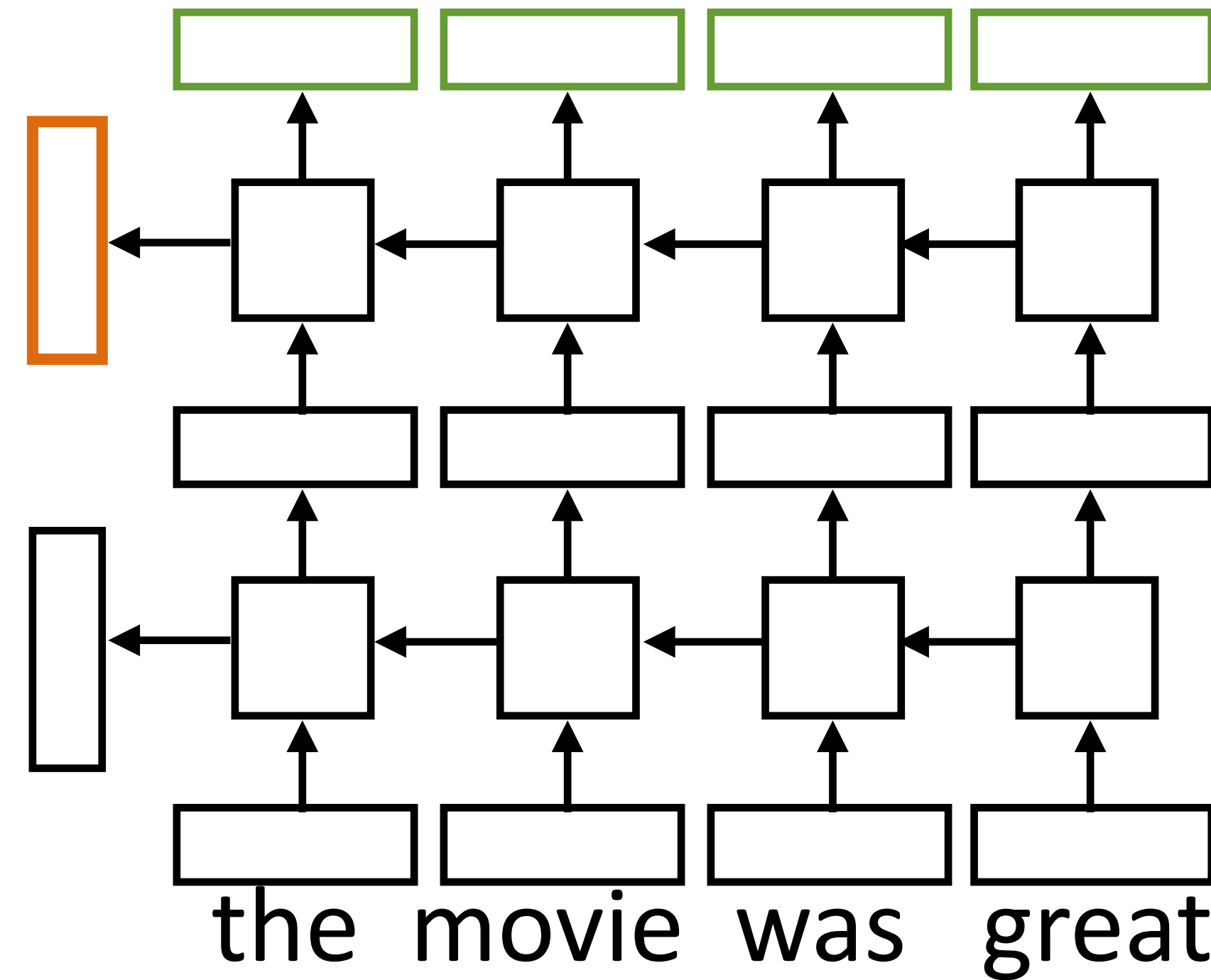
- Sentence classification based on concatenation of both final outputs



Multilayer Bidirectional RNN



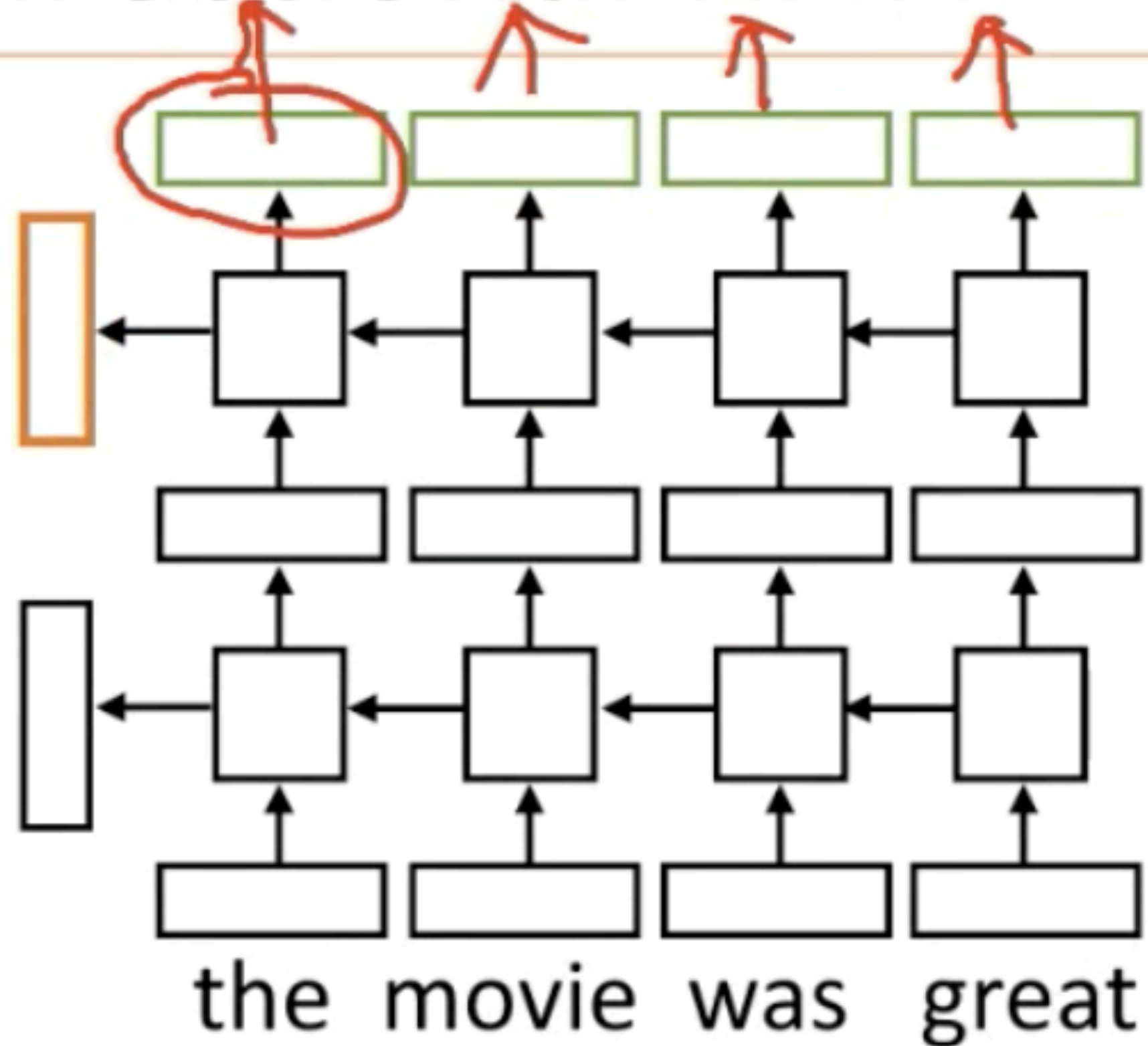
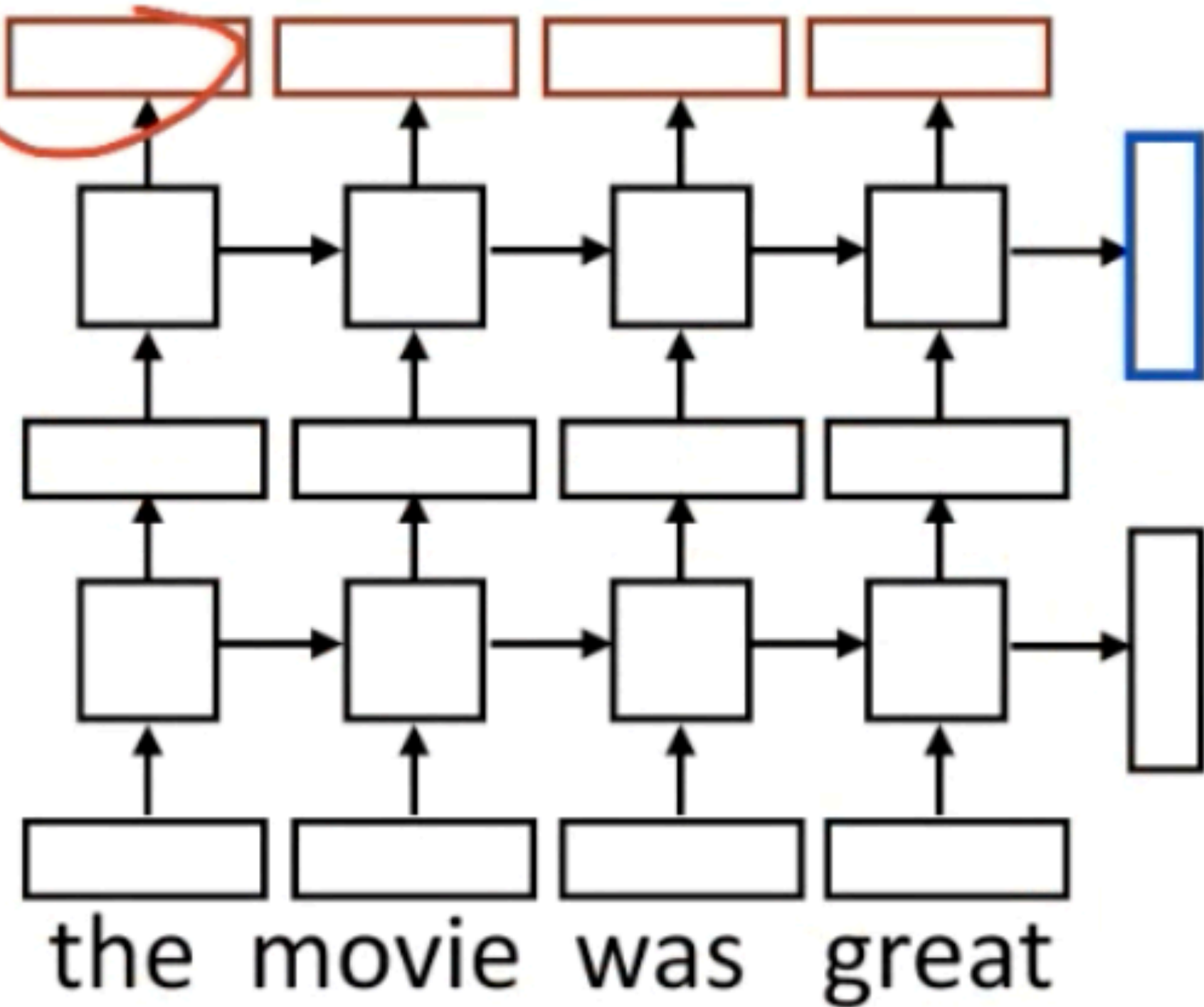
- ▶ Sentence classification based on concatenation of both final outputs



- ▶ Token classification based on concatenation of both directions' token representations



Multilayer Bidirectional RNN



- Sentence classification based on concatenation of both final outputs

$$\text{softmax}(W(h_T, h_1) + b)$$

- Token classification based on concatenation of both directions' token representations

Natural Language Inference

Premise

A boy plays in the snow

Hypothesis

A boy is outside

Natural Language Inference

Premise

A boy plays in the snow

Hypothesis

A boy is outside

entails

Natural Language Inference

Premise

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A boy plays in the snow

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A boy is outside

A man inspects the uniform of a figure

The man is sleeping

Natural Language Inference

Premise

Hypothesis

A boy plays in the snow

entails

A boy is outside

A man inspects the uniform of a figure

contradicts

The man is sleeping

Natural Language Inference

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An older and younger man smiling

Two men are smiling and
laughing at cats playing

Natural Language Inference

Premise

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- ▶ Long history of this task: “Recognizing Textual Entailment” challenge in 2006 (Dagan, Glickman, Magnini)

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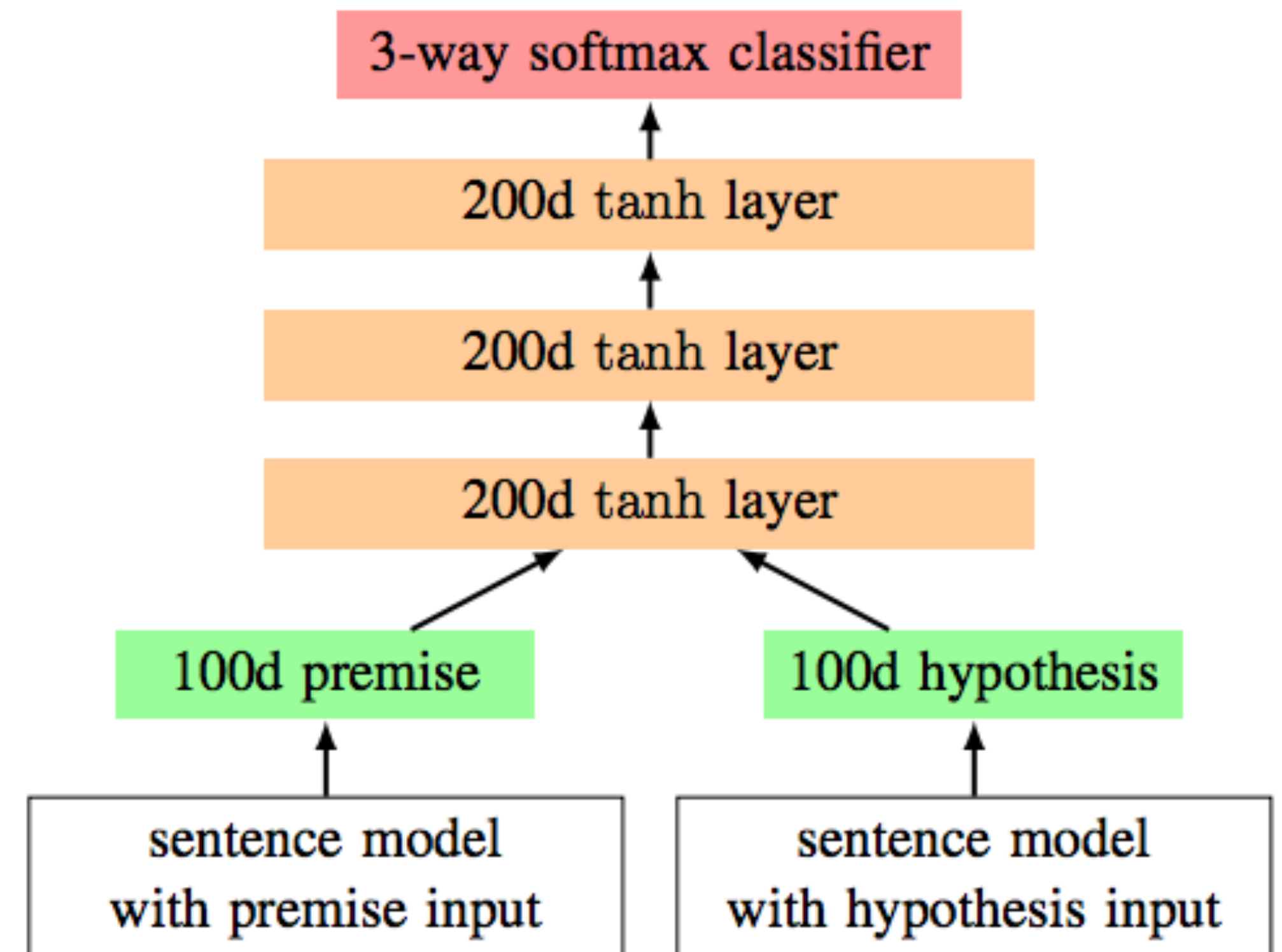
- ▶ Long history of this task: “Recognizing Textual Entailment” challenge in 2006 (Dagan, Glickman, Magnini)
- ▶ Early datasets: small (hundreds of pairs), very ambitious (lots of world knowledge, temporal reasoning, etc.)

SNLI Dataset

- ▶ Show people captions for (unseen) images and solicit entailed / neutral / contradictory statements
- ▶ >500,000 sentence pairs

SNLI Dataset

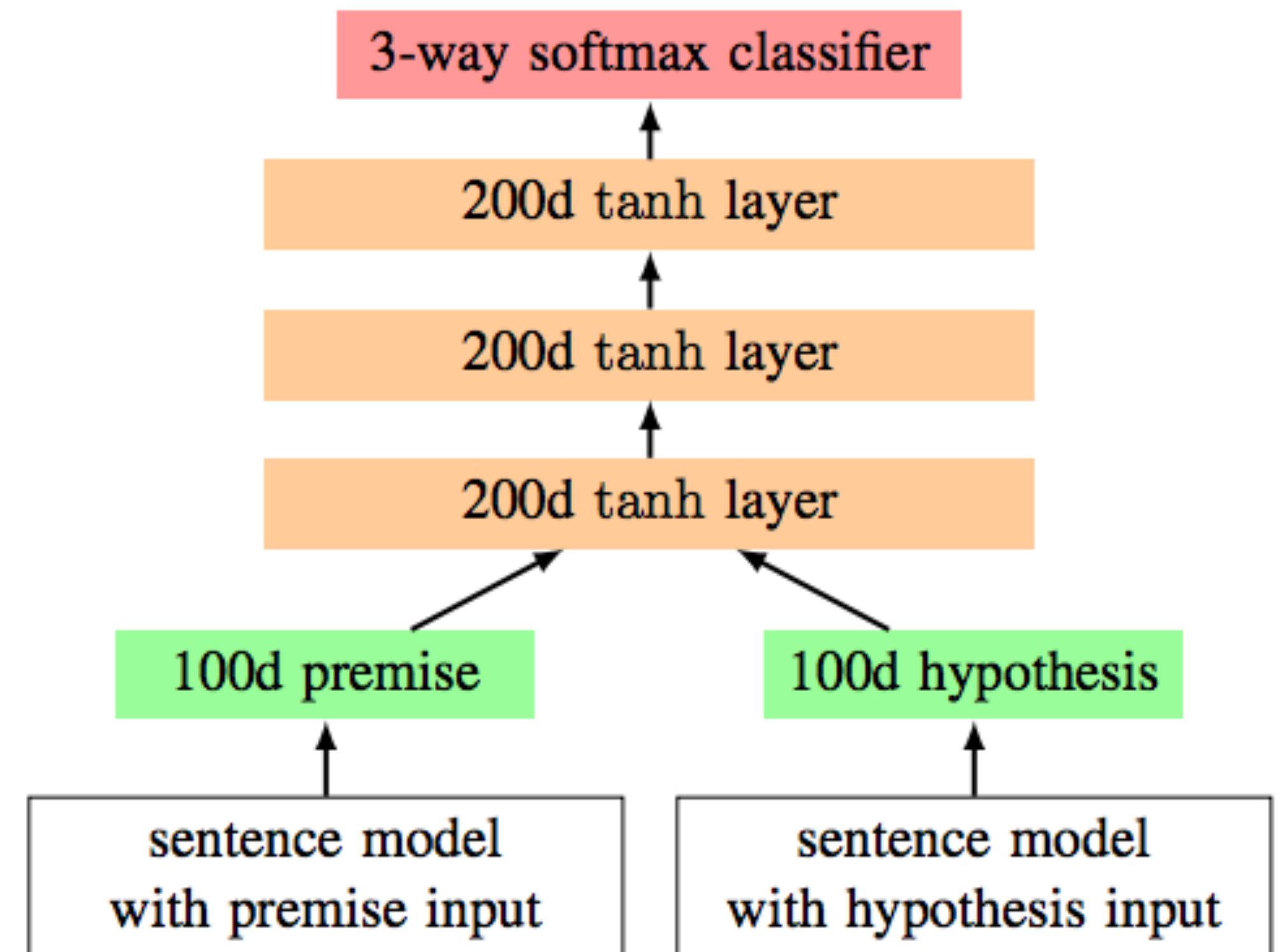
- ▶ Show people captions for (unseen) images and solicit entailed / neural / contradictory statements
- ▶ >500,000 sentence pairs
- ▶ Encode each sentence and process



Bowman et al. (2015)

SNLI Dataset

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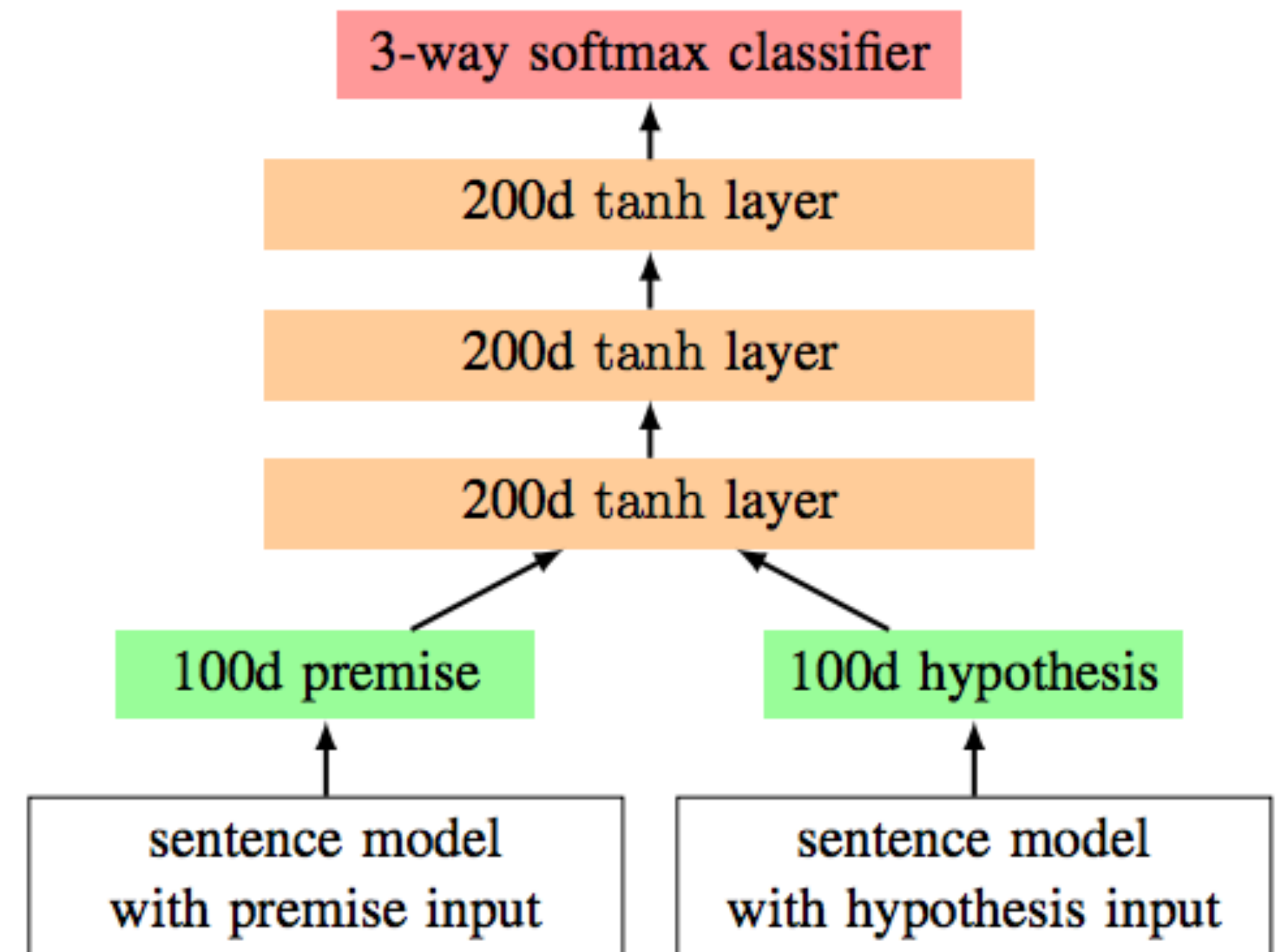
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100D LSTM: 78% accuracy

300D LSTM: 80% accuracy

(Bowman et al., 2016)



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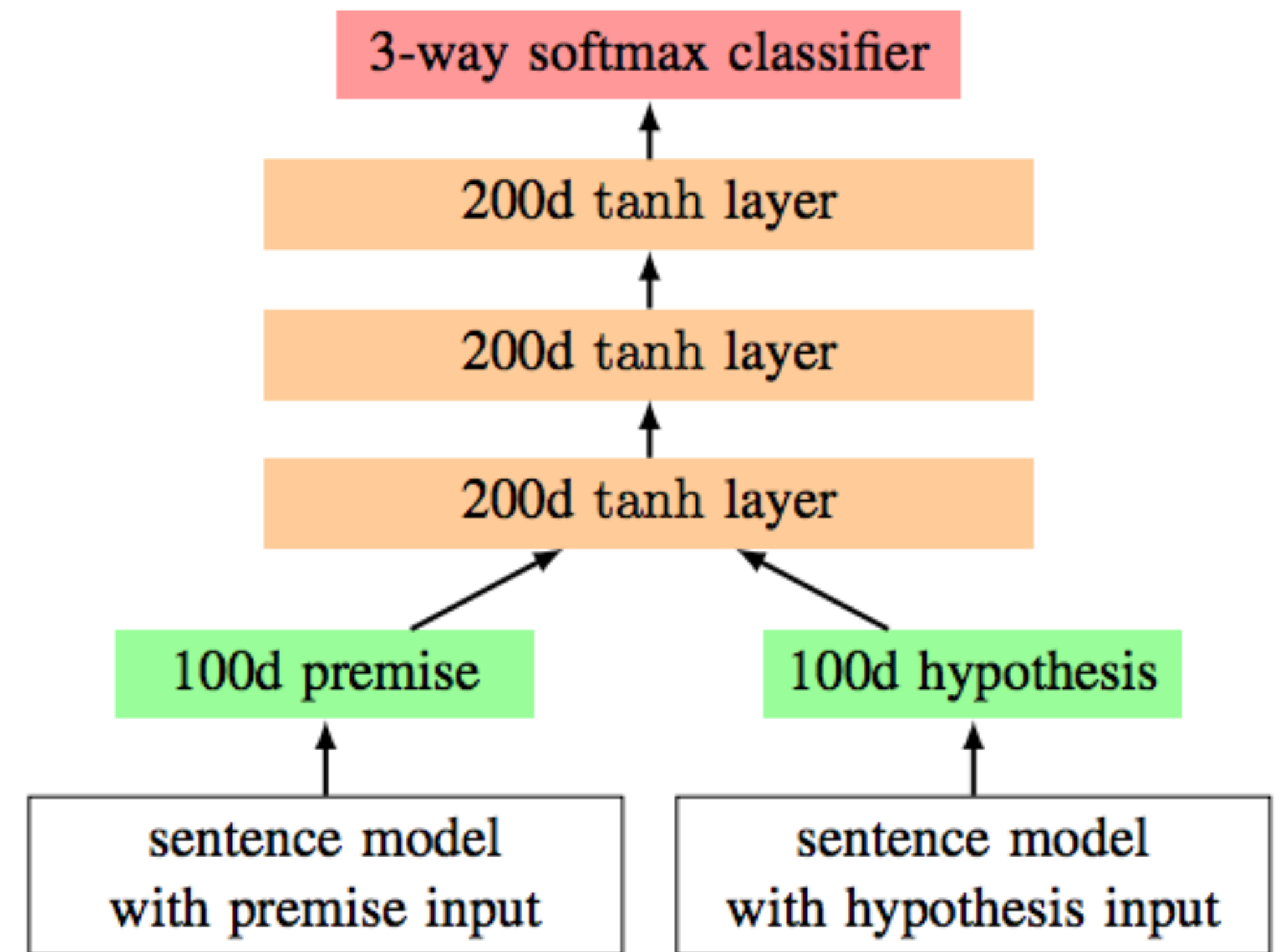
100D LSTM: 78% accuracy

300D LSTM: 80% accuracy

(Bowman et al., 2016)

300D BiLSTM: 83% accuracy

(Liu et al., 2016)



Bowman et al. (2015)

SNLI Dataset

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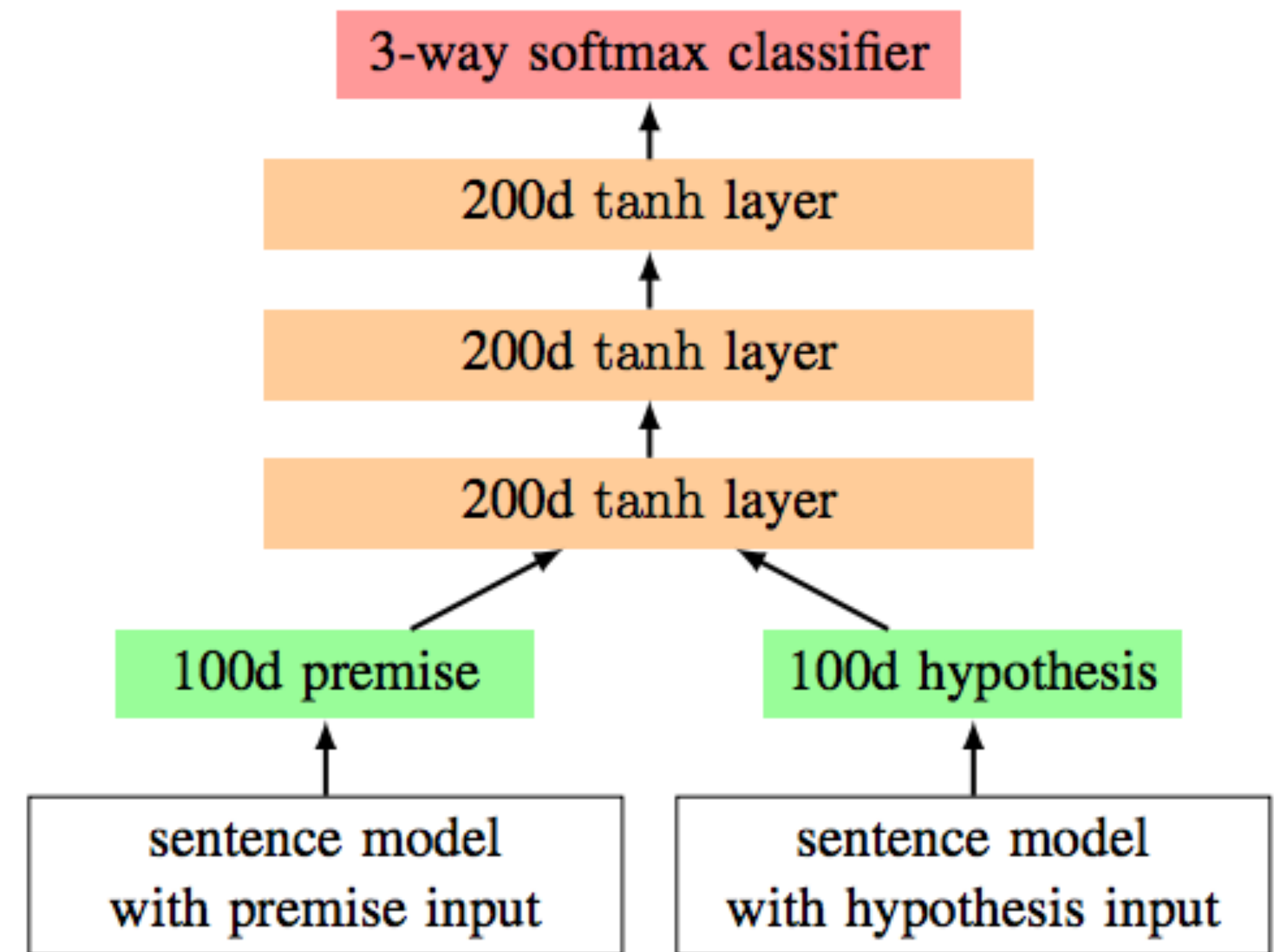
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(Bowman et al., 2016)

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(Liu et al., 2016)

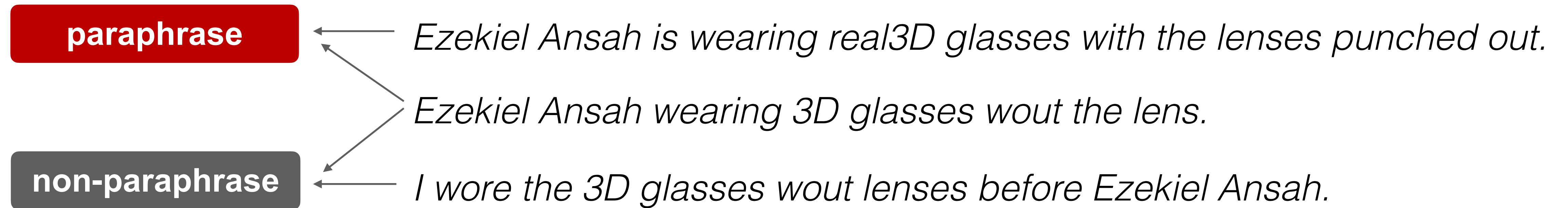
- ▶ Later: better models for this



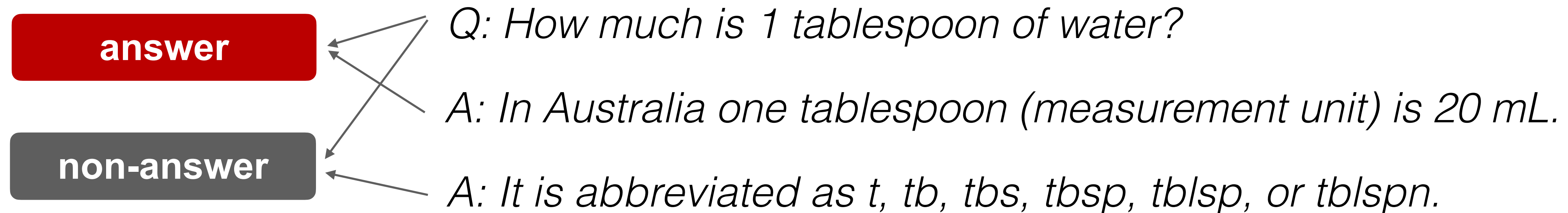
Bowman et al. (2015)

Semantic Similarity

Paraphrase Identification

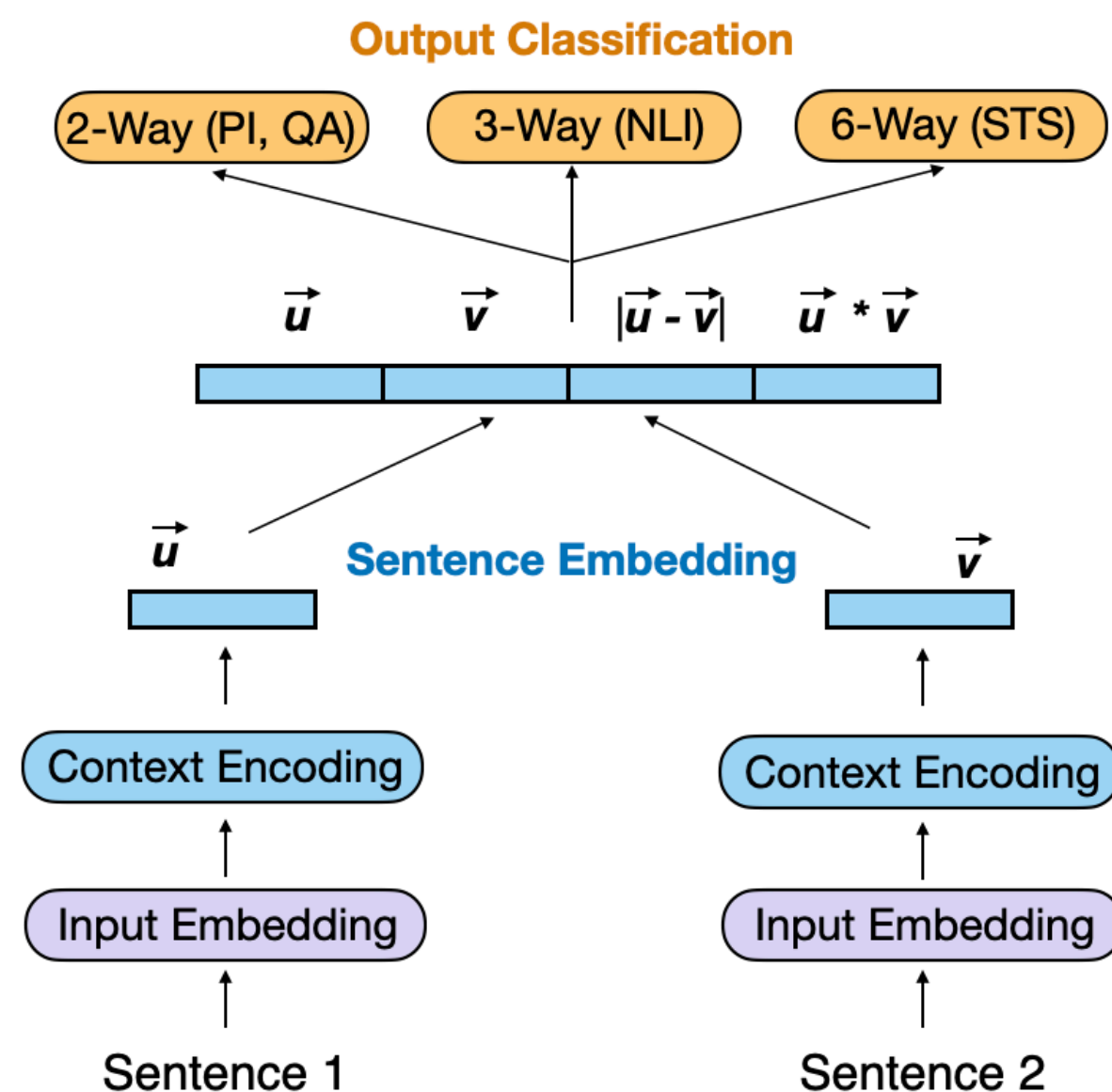


Question Answering

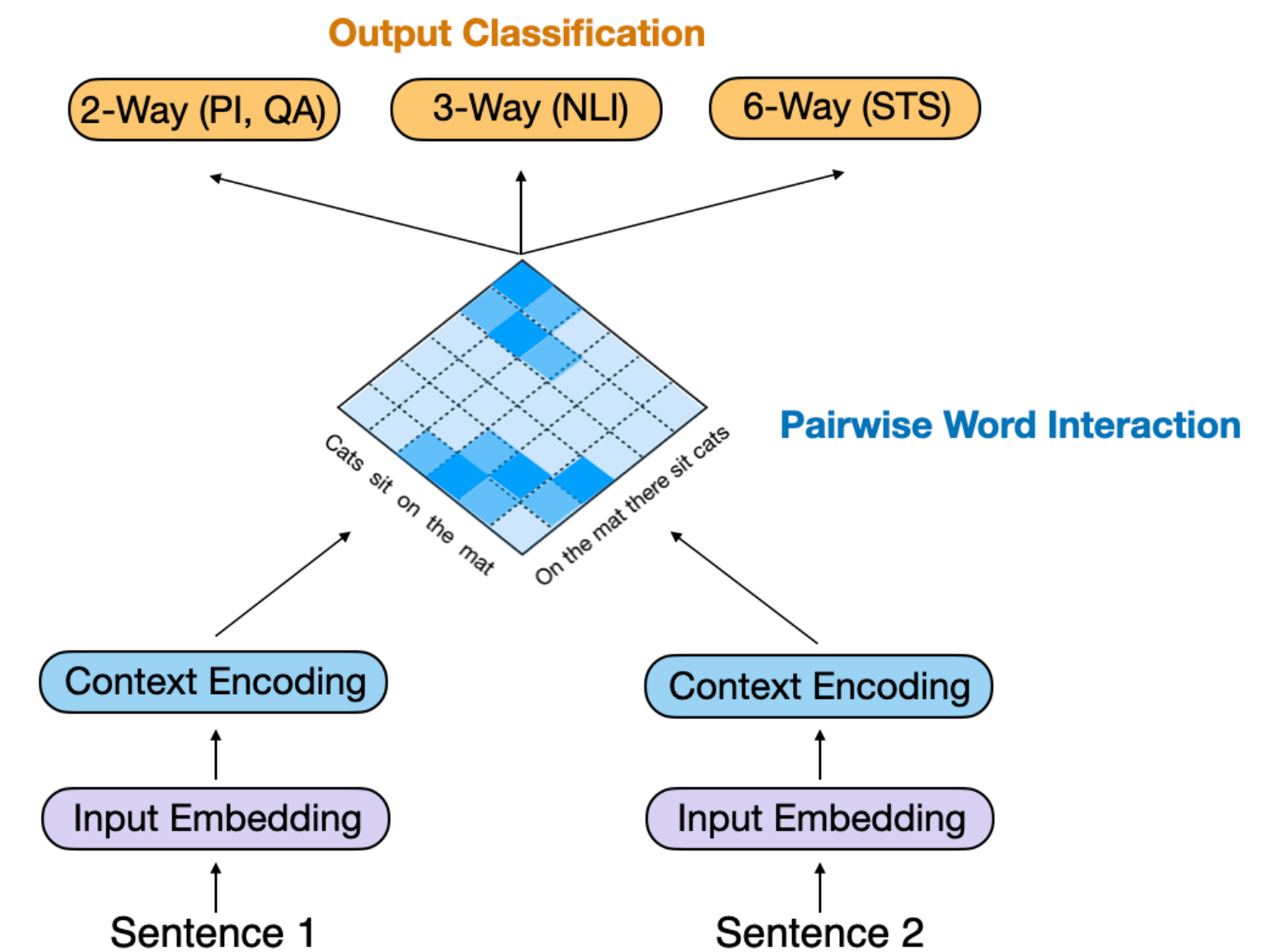


Sentence Pair Classification

Type I: Sentence Encoding-based Models



Type II: Word Interaction-based Models



- semantic relation between two sentences depends largely on aligned words/phrases

What can LSTMs model?

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- ▶ Sequential tagging (e.g., POS/NER), or Language models
 - ▶ Move left-to-right, per-token prediction
- ▶ Translation/Generation
 - ▶ Encode sentence + then decode, use token predictions for attention weights (later in the course)

Neural CRF

NER Revisited

B-PER	I-PER	O	O	O	B-LOC	O	O	O	B-ORG	O	O
<i>Barack Obama will travel to Hangzhou today for the G20 meeting .</i>											
PERSON					LOC		ORG				

NER Revisited



- Features in CRFs: $I[\text{tag}=\text{B-LOC} \ \& \ \text{curr_word}=\text{Hangzhou}]$,
 $I[\text{tag}=\text{B-LOC} \ \& \ \text{prev_word}=\text{to}]$, $I[\text{tag}=\text{B-LOC} \ \& \ \text{curr_prefix}=\text{Han}]$

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- ▶ Linear model over features
- ▶ Downsides:
 - ▶ Lexical features mean that words need to be seen in the training data
 - ▶ Linear model can't capture feature conjunctions as effectively (doesn't work well to look at more than 2 words with a single feature)

LSTMs for NER

B-PER I-PER O O O B-LOC O O O B-ORG O O

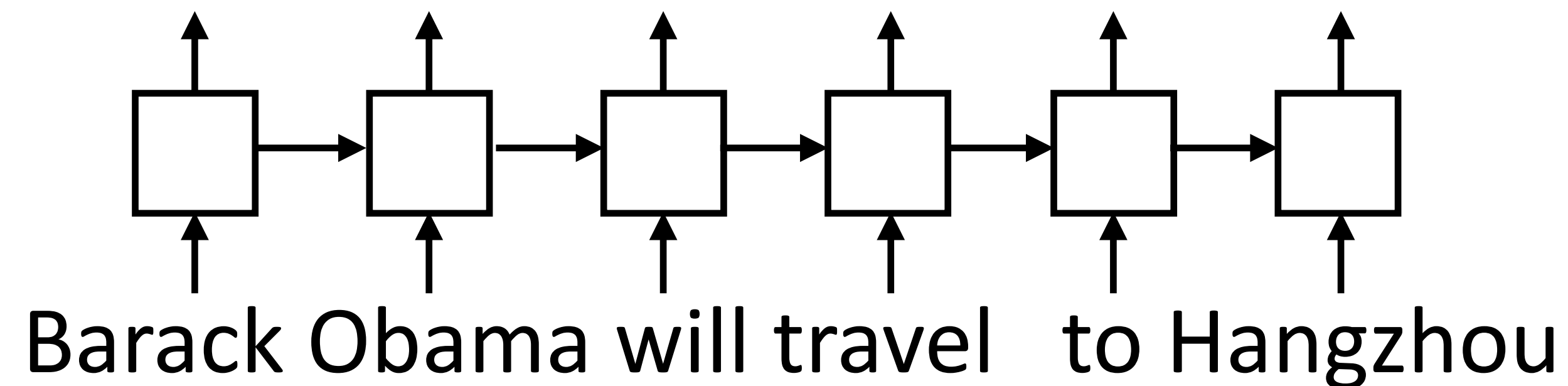
Barack Obama will travel to Hangzhou today for the G20 meeting .

PERSON

LOC

ORG

B-PER I-PER O O O B-LOC



- Transducer (LM-like model)

LSTMs for NER

B-PER I-PER O O O B-LOC O O O B-ORG O O

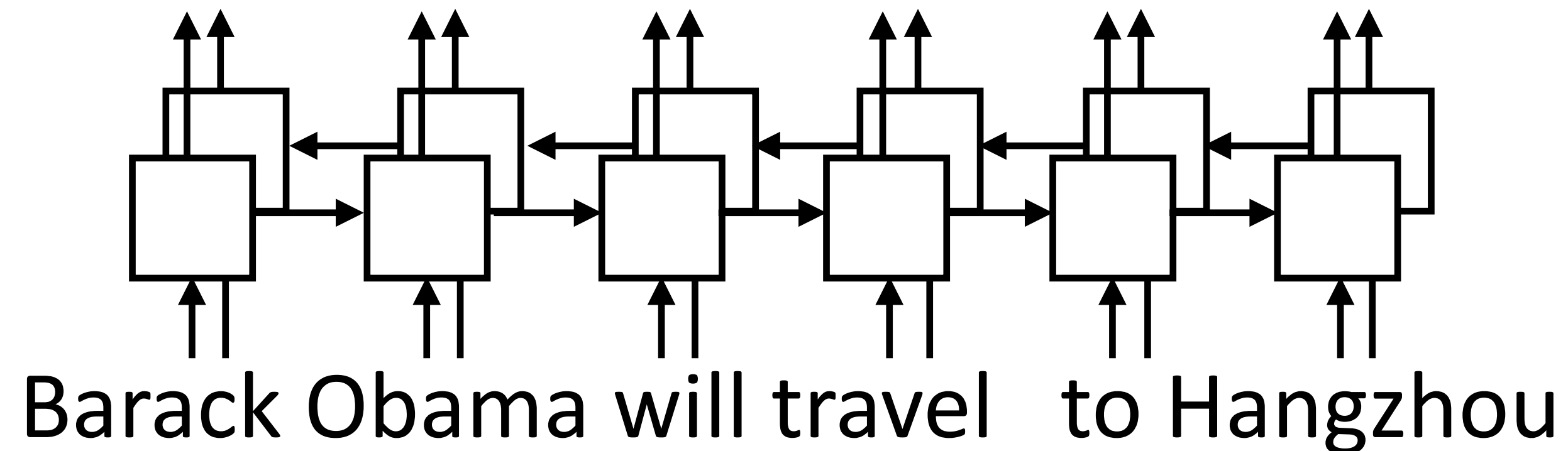
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ORG

B-PER I-PER O O O B-LOC



- ▶ Bidirectional transducer model
- ▶ Q: What are the strengths and weaknesses of this model compared to the linear CRFs?

Neural CRFs

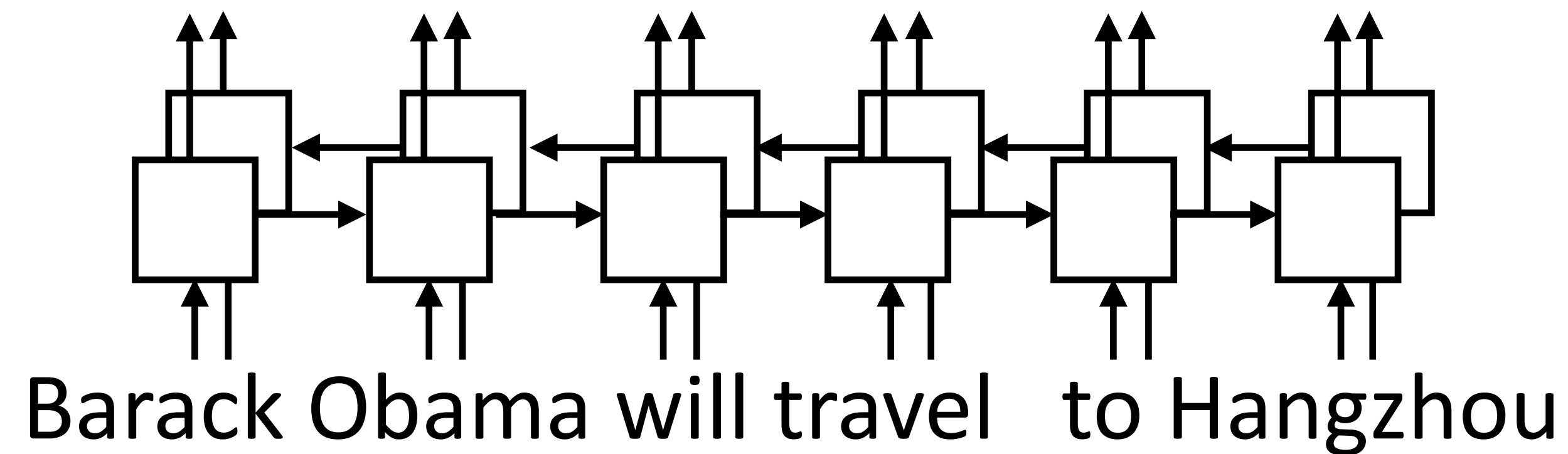
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Neural CRFs

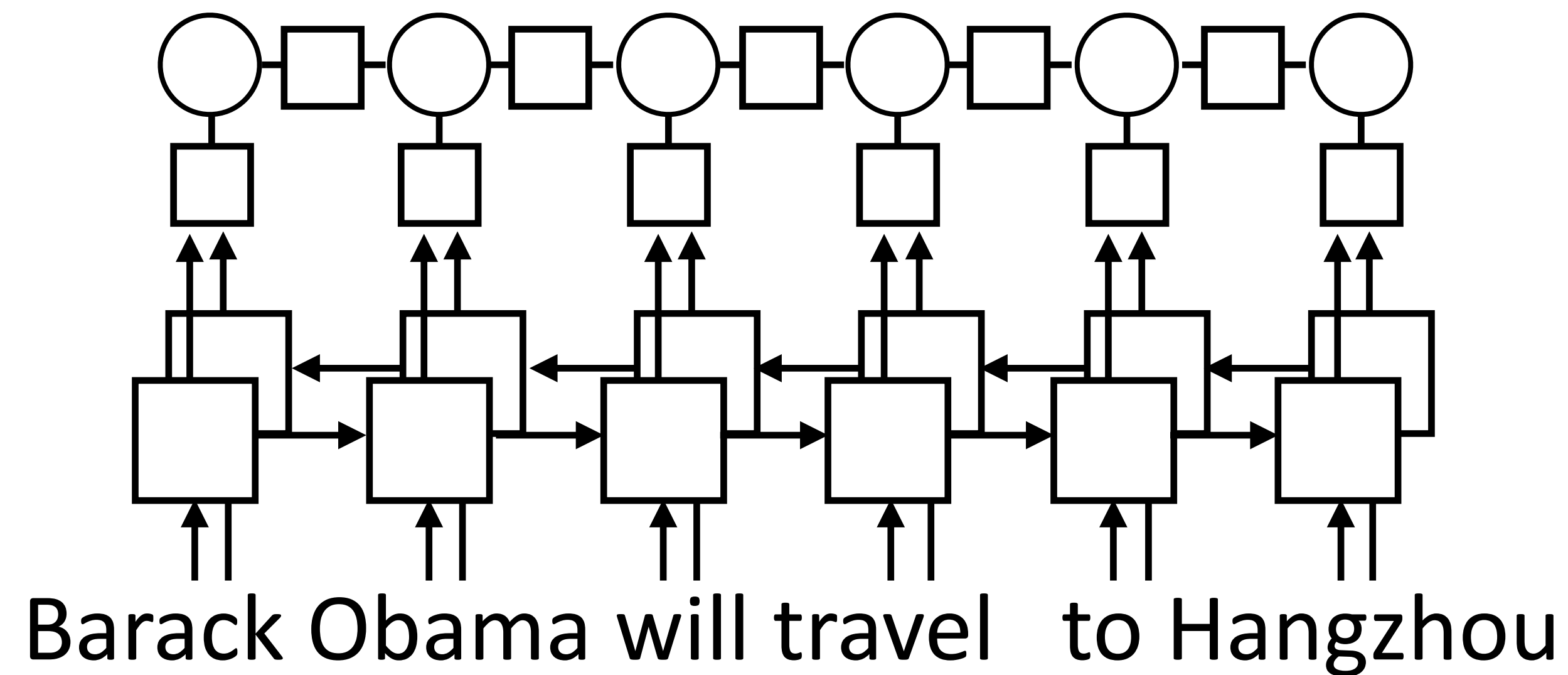
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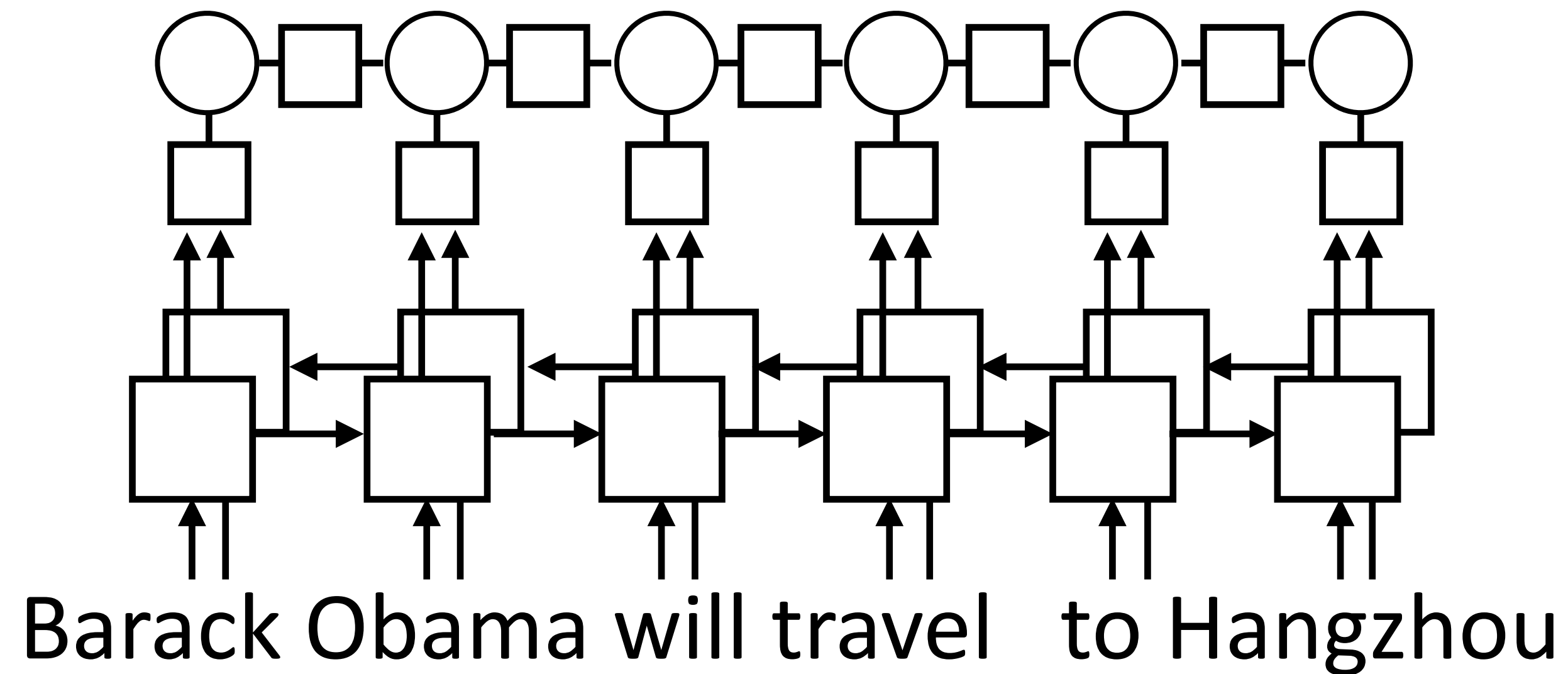
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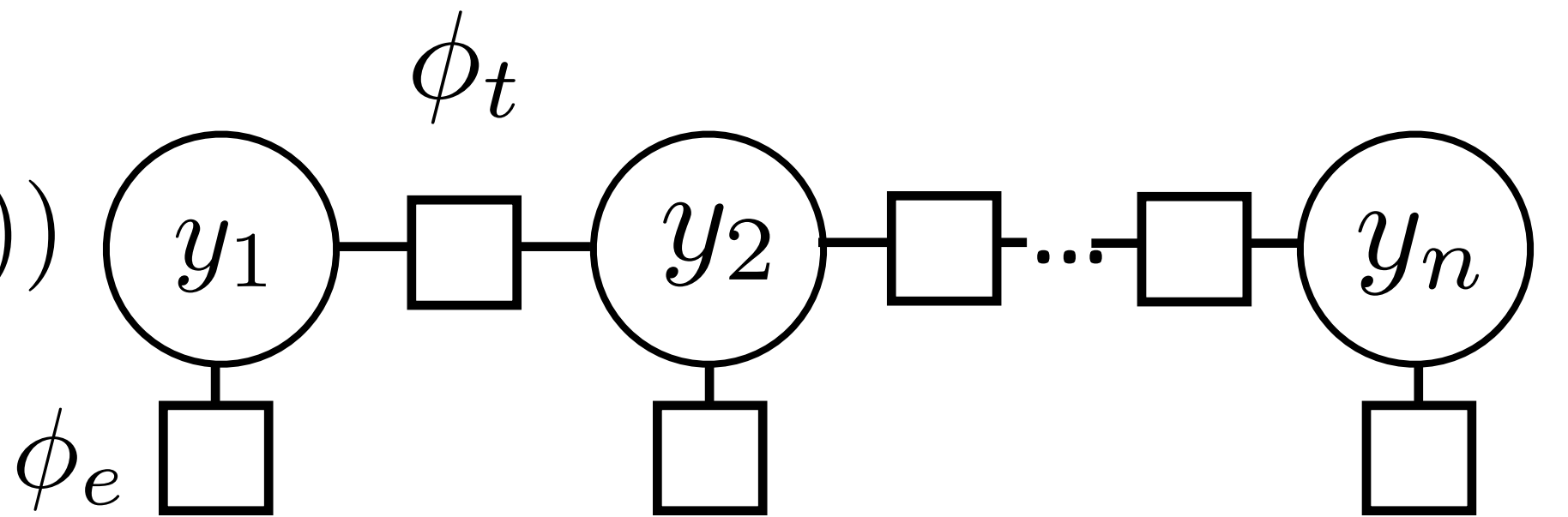
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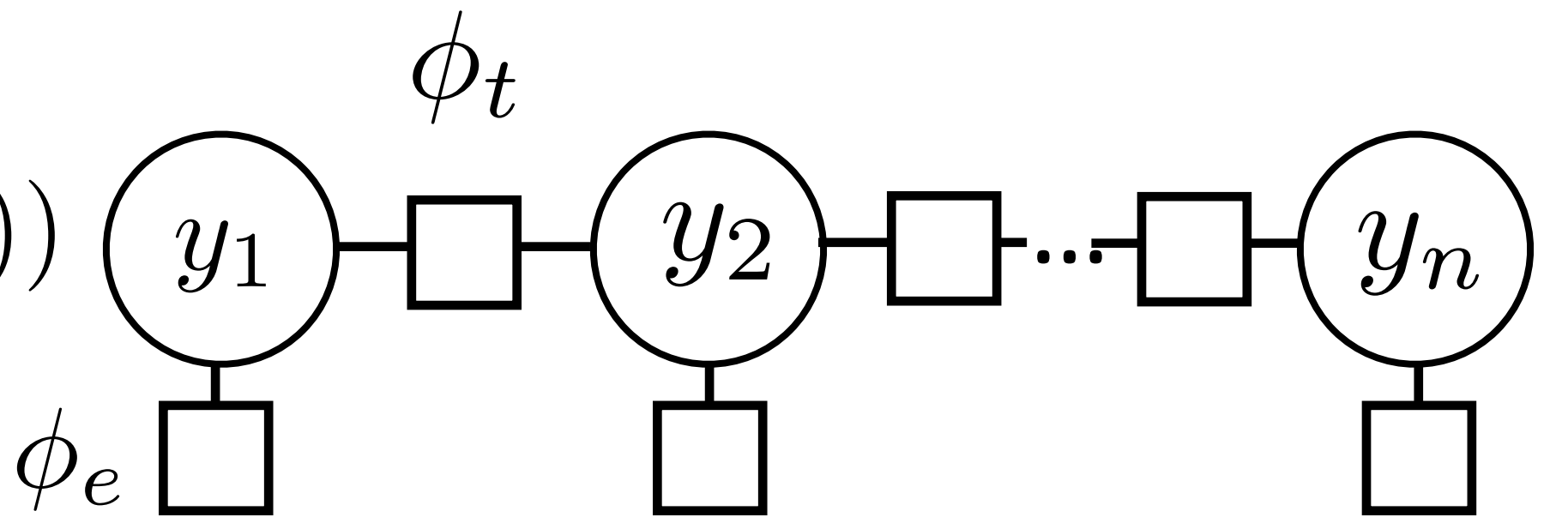
- Neural CRFs: bidirectional LSTMs (or some NN) compute emission potentials, capture structural constraints in transition potentials

Neural CRFs

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


- ▶ Linear model: $\phi_e(y_i, i, \mathbf{x}) = w^\top f_e(y_i, i, \mathbf{x})$

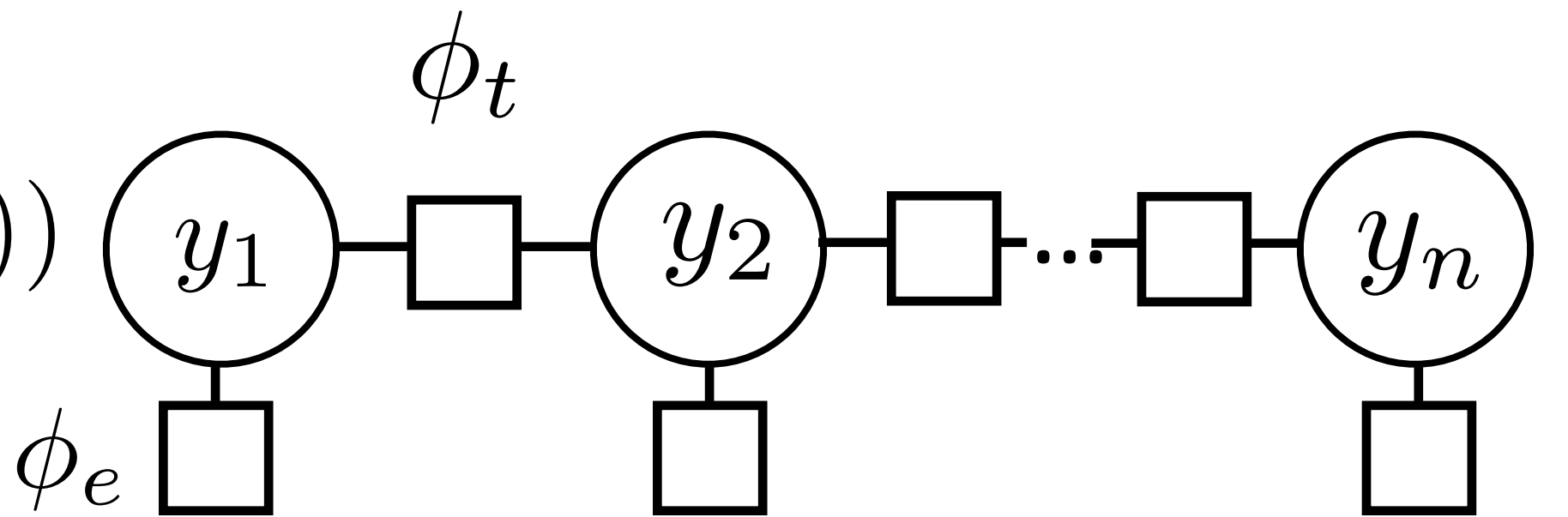
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The diagram illustrates a sequence of nodes $y_1, y_2, \dots, y_n$ represented as circles. Each node y_i is connected to the next node y_{i+1} by a horizontal line, with a square node in between. Additionally, each node y_i is connected to a square node below it. The transition between y_{i-1} and y_i is labeled ϕ_t , and the emission for y_i is labeled ϕ_e .

- ▶ Linear model: $\phi_e(y_i, i, \mathbf{x}) = w^\top f_e(y_i, i, \mathbf{x})$
- ▶ Neural: $\phi_e(y_i, i, \mathbf{x}) = W_{y_i}^\top f(i, \mathbf{x})$ W is a num_tags x len(f) matrix

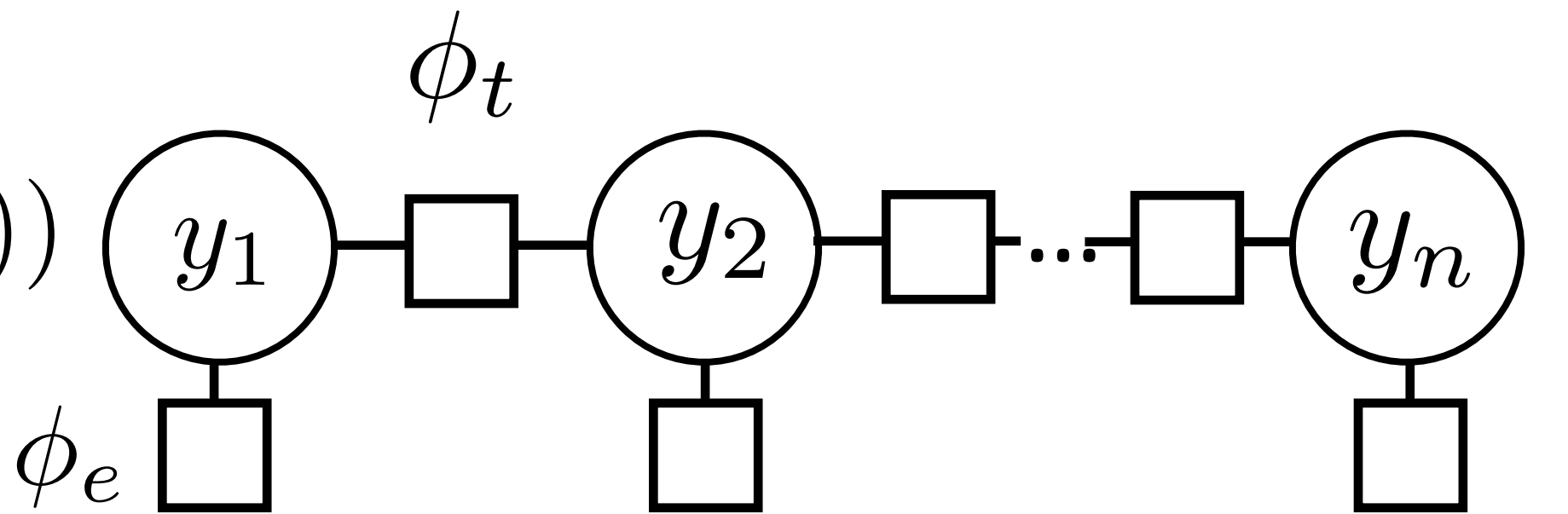
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The diagram illustrates a sequence of nodes $y_1, y_2, \dots, y_n$ represented as circles. These nodes are connected by a chain of square nodes, representing transitions. Below each circle node y_i is a square node representing an emission. The top transition node is labeled ϕ_t , and the bottom emission node for y_1 is labeled ϕ_e .

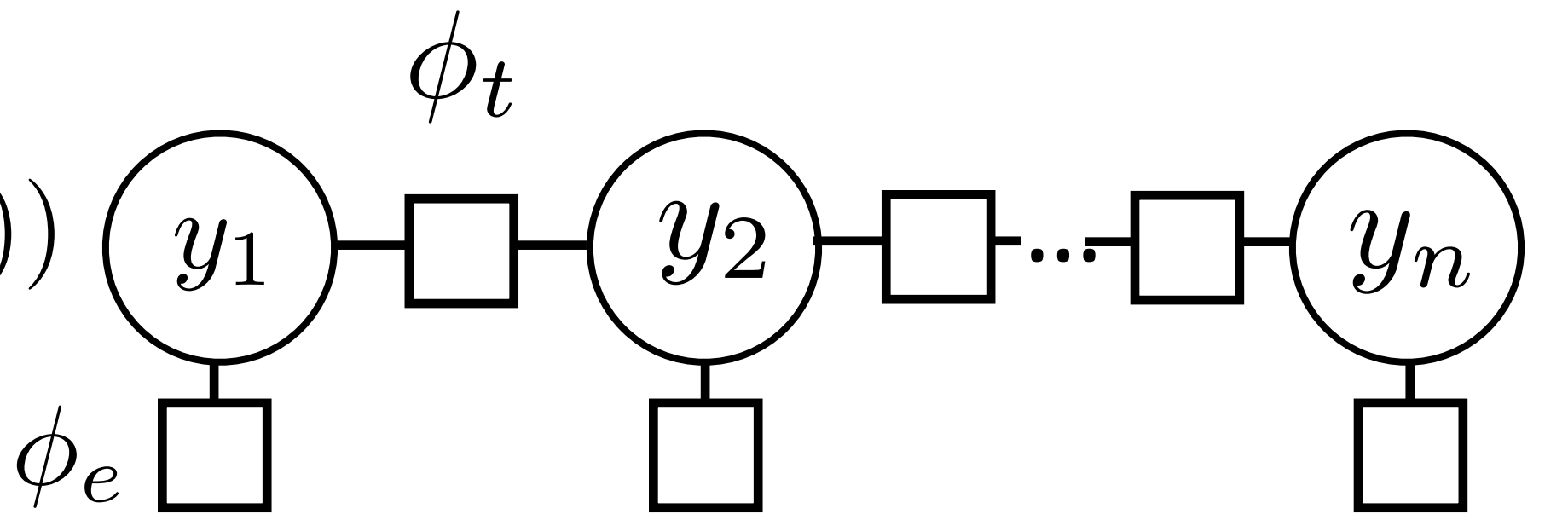
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- ▶ $f(i, \mathbf{x})$ could be the output of a feedforward neural network looking at the words around position i , or the i th output of an LSTM, ...

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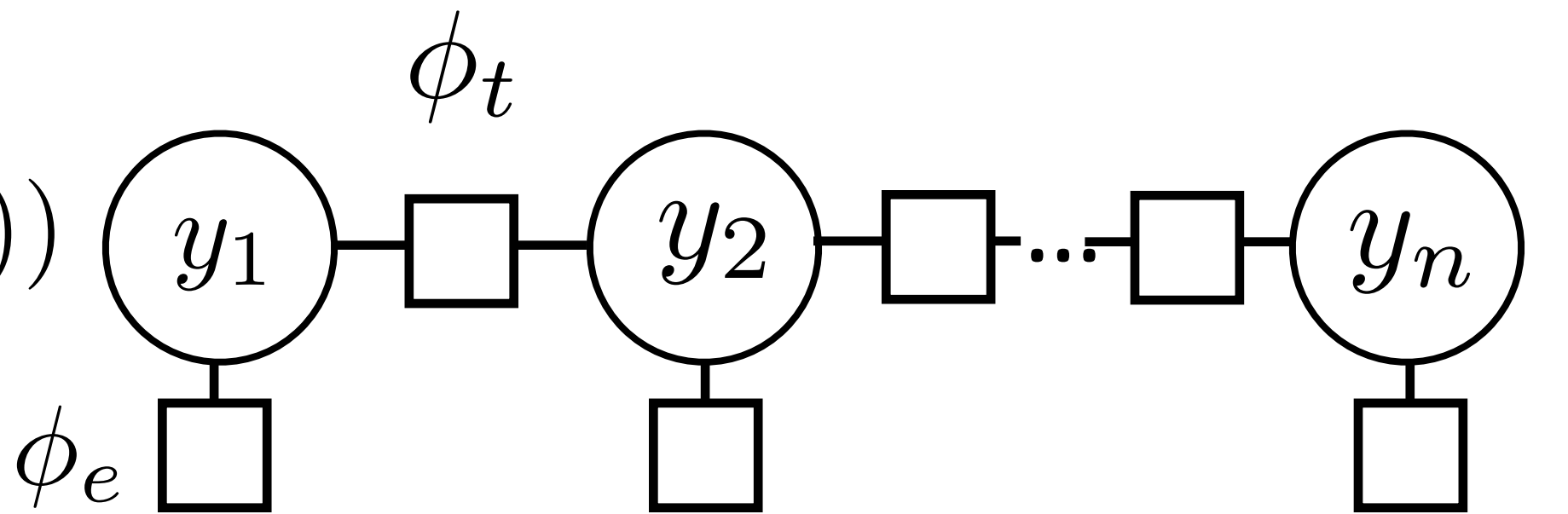
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- ▶ Neural network computes unnormalized potentials that are consumed and “normalized” by a structured model

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- ▶ Neural network computes unnormalized potentials that are consumed and “normalized” by a structured model
- ▶ Inference: compute f , use Viterbi

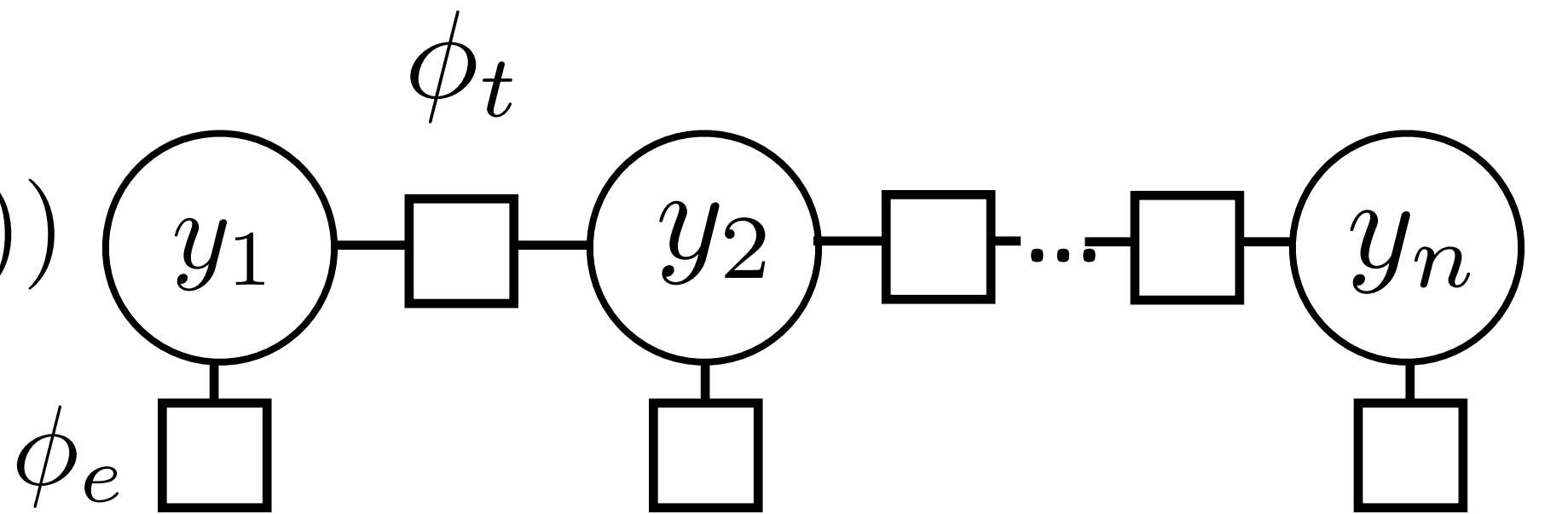
Computing Gradients

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


The diagram illustrates a sequence model. It features a horizontal chain of nodes. The first node is a circle labeled y_1 , with a square node below it labeled ϕ_e . This is followed by a square node, then a circle node labeled y_2 , which also has a square node below it. This pattern continues with an ellipsis and another square node, followed by a final circle node labeled y_n with a square node below it. Above the first transition square is the label ϕ_t .

- ▶ Conventional: $\phi_e(y_i, i, \mathbf{x}) = w^\top f_e(y_i, i, \mathbf{x})$
- ▶ Neural: $\phi_e(y_i, i, \mathbf{x}) = W_{y_i}^\top f(i, \mathbf{x})$

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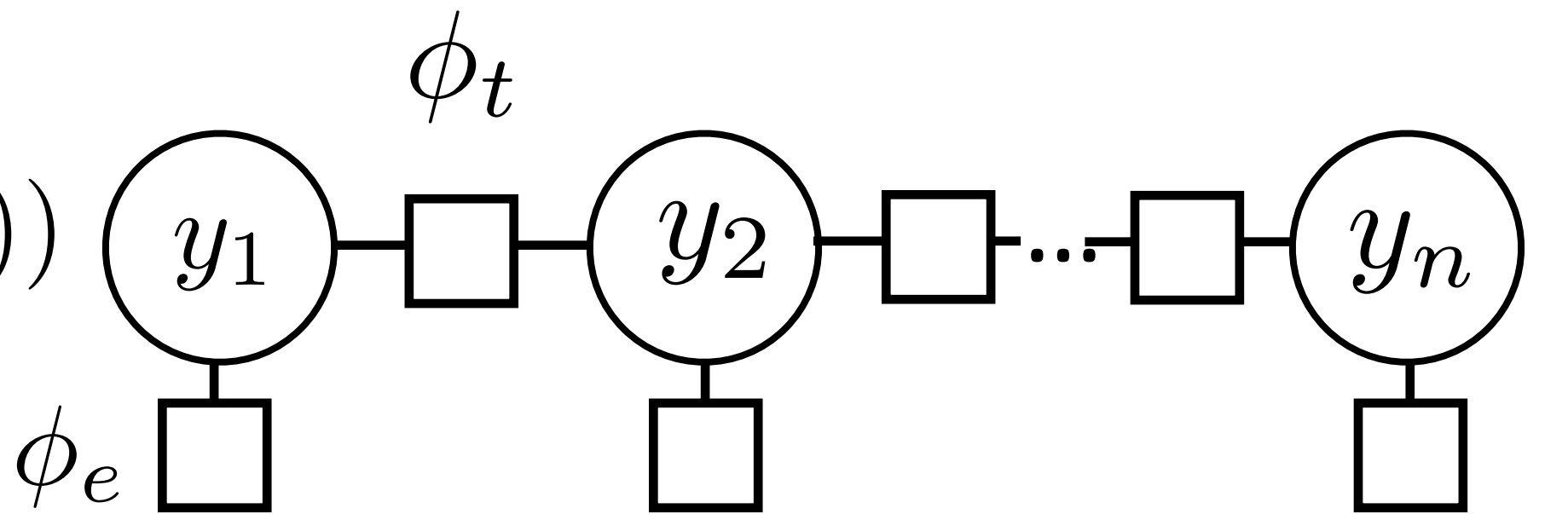
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The diagram illustrates a Markov chain structure. It consists of a sequence of circular nodes labeled $y_1, y_2, \dots, y_n$ connected by horizontal lines. Between each pair of consecutive circular nodes y_i and y_{i+1} , there is a square node. Below each circular node y_i , there is another square node. The transition between y_i and y_{i+1} is labeled ϕ_t . The emission from y_i is labeled ϕ_e .

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$$\frac{\partial \mathcal{L}}{\partial \phi_{e,i}} = -P(y_i = s|\mathbf{x}) + I[s \text{ is gold}] \quad \text{“error signal”, compute with F-B}$$

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$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z} \prod_{i=2}^n \exp(\phi_t(y_{i-1}, y_i)) \prod_{i=1}^n \exp(\phi_e(y_i, i, \mathbf{x}))$$


The diagram illustrates a Markov chain structure. It consists of a sequence of circular nodes labeled $y_1, y_2, \dots, y_n$ connected by square nodes. Below each circular node y_i is another square node. The label ϕ_t is positioned above the first transition square, and the label ϕ_e is positioned to the left of the first bottom square.

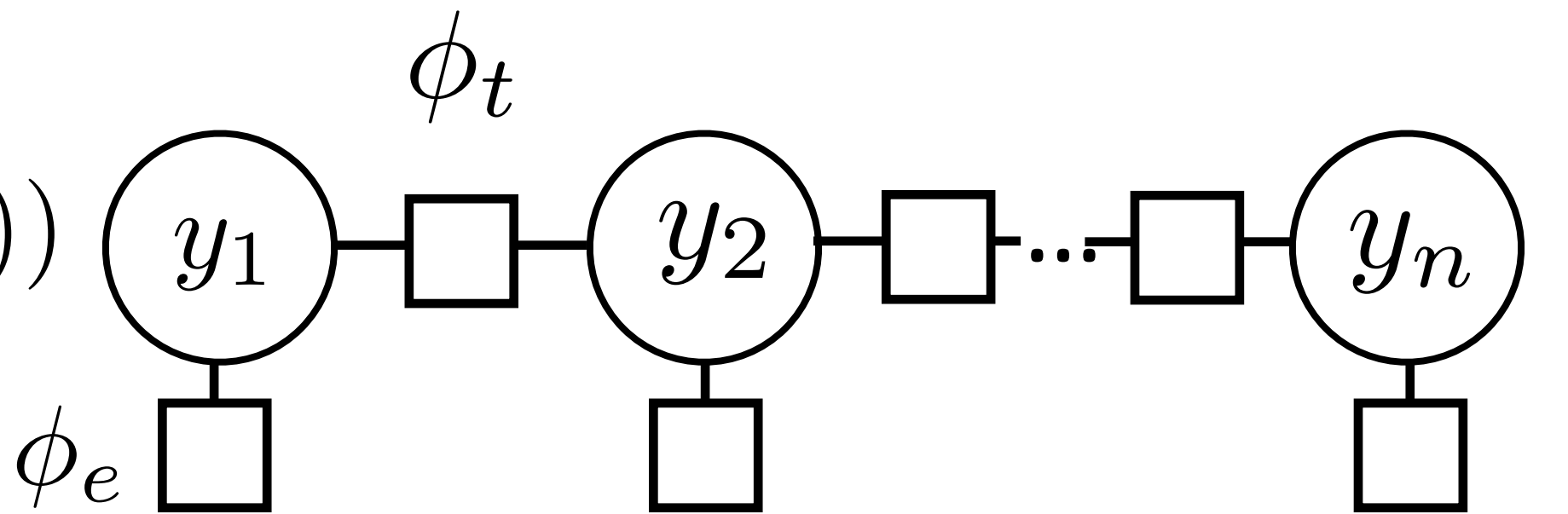
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$$\frac{\partial \mathcal{L}}{\partial \phi_{e,i}} = -P(y_i = s|\mathbf{x}) + I[s \text{ is gold}] \quad \text{“error signal”, compute with F-B}$$

► For linear model: $\frac{\partial \phi_{e,i}}{w_i} = f_{e,i}(y_i, i, \mathbf{x})$

Computing Gradients

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► For linear model: $\frac{\partial \phi_{e,i}}{w_i} = f_{e,i}(y_i, i, \mathbf{x})$

chain rule say to multiply together, gives our update

► For neural model: compute gradient of phi w.r.t. parameters of neural net

LSTM Neural CRFs

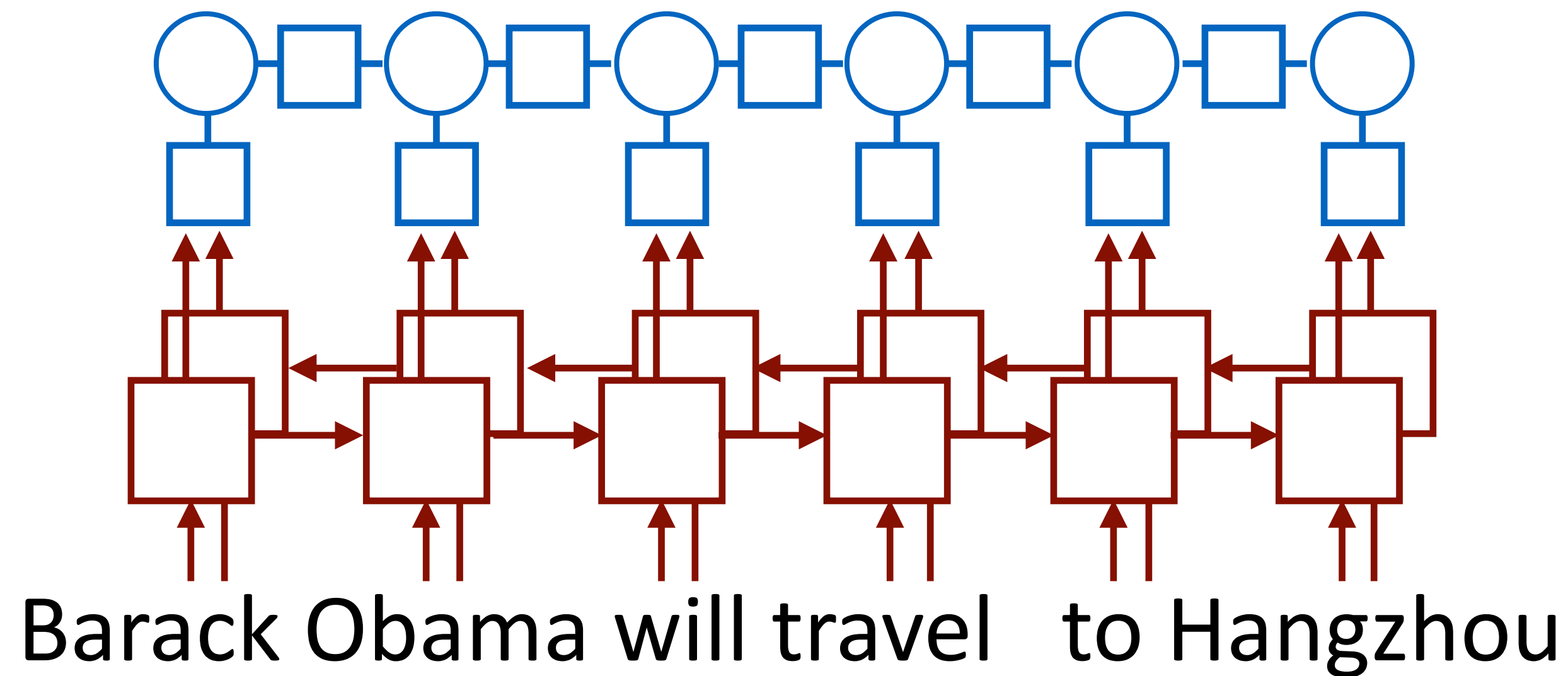
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ORG



LSTM Neural CRFs

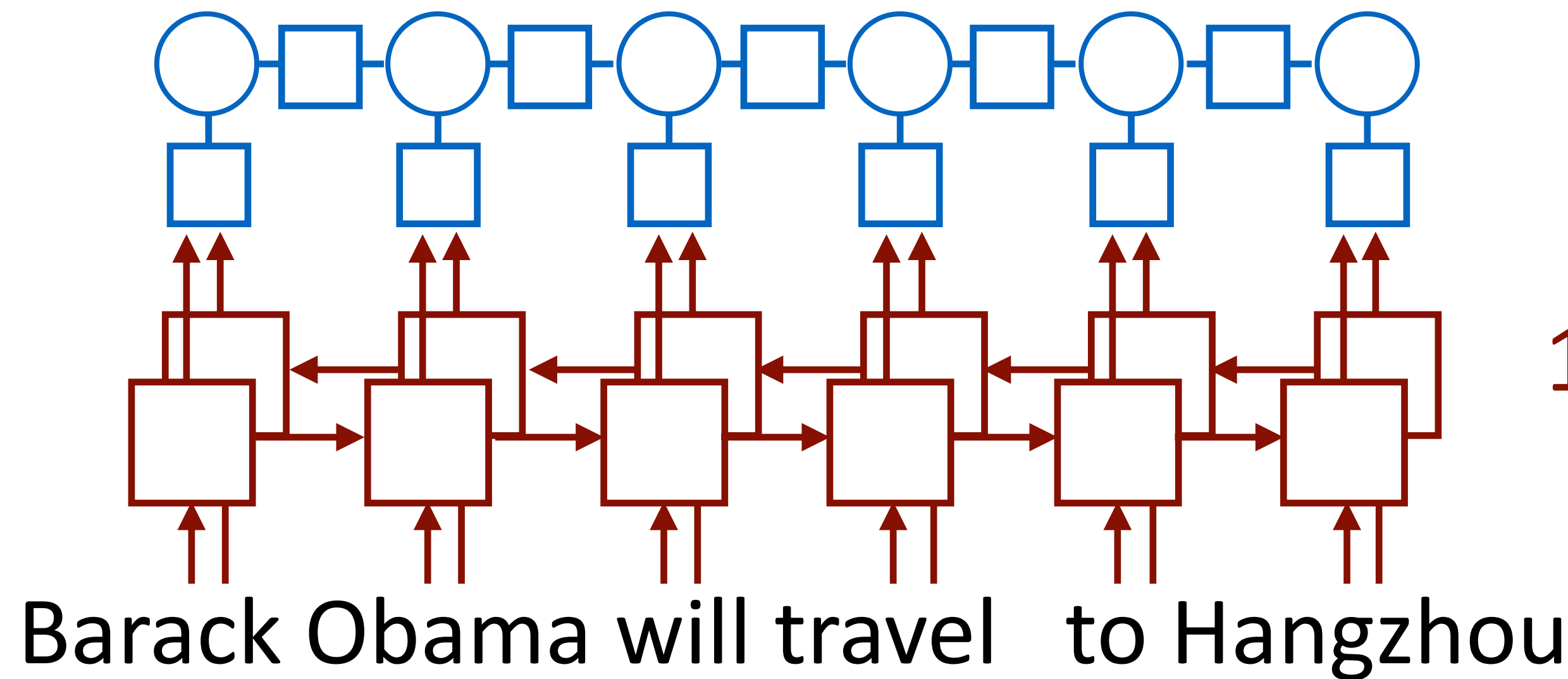
B-PER I-PER O O O B-LOC O O O B-ORG O O

Barack Obama will travel to Hangzhou today for the G20 meeting .

PERSON

LOC

ORG



1) Compute $f(\mathbf{x})$

LSTM Neural CRFs

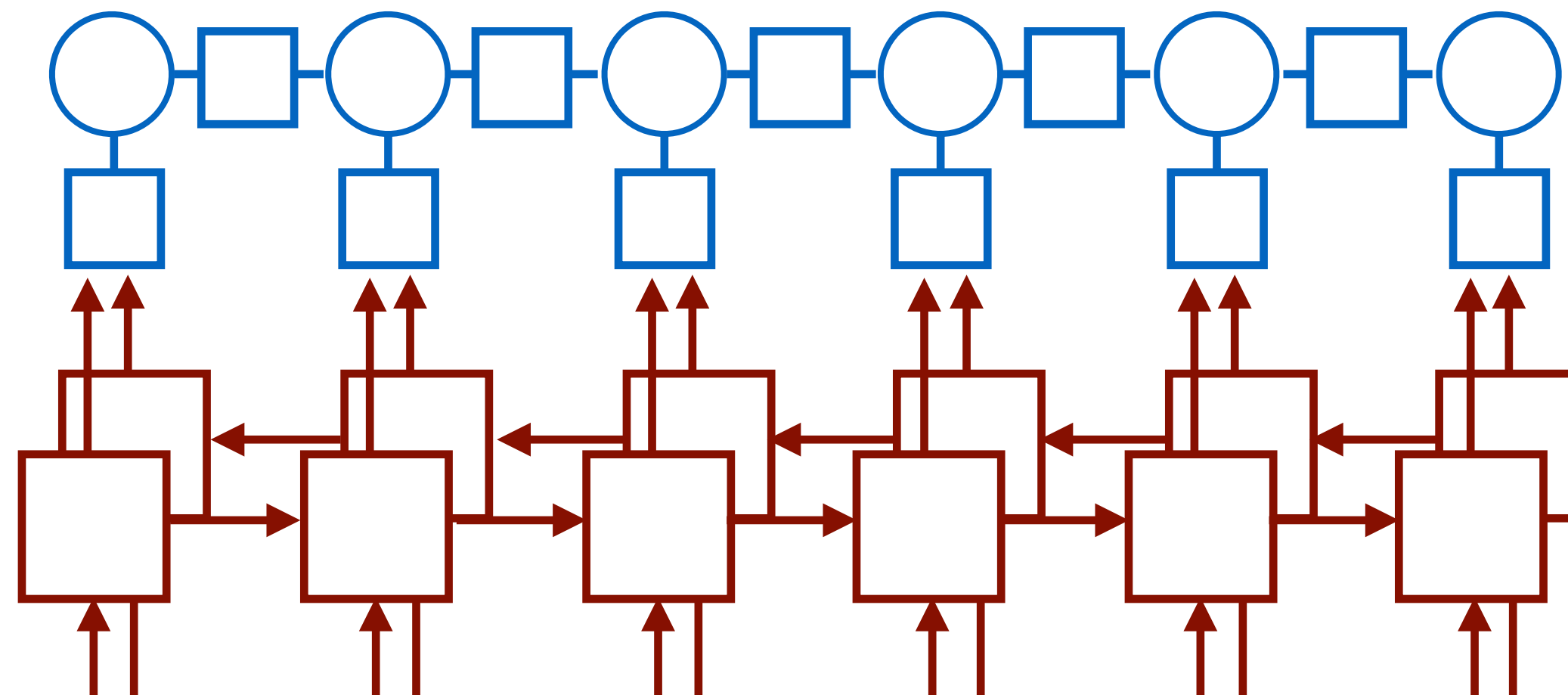
B-PER I-PER O O O B-LOC O O O B-ORG O O

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PERSON

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Barack Obama will travel to Hangzhou

2) Run forward-backward

1) Compute $f(\mathbf{x})$

LSTM Neural CRFs

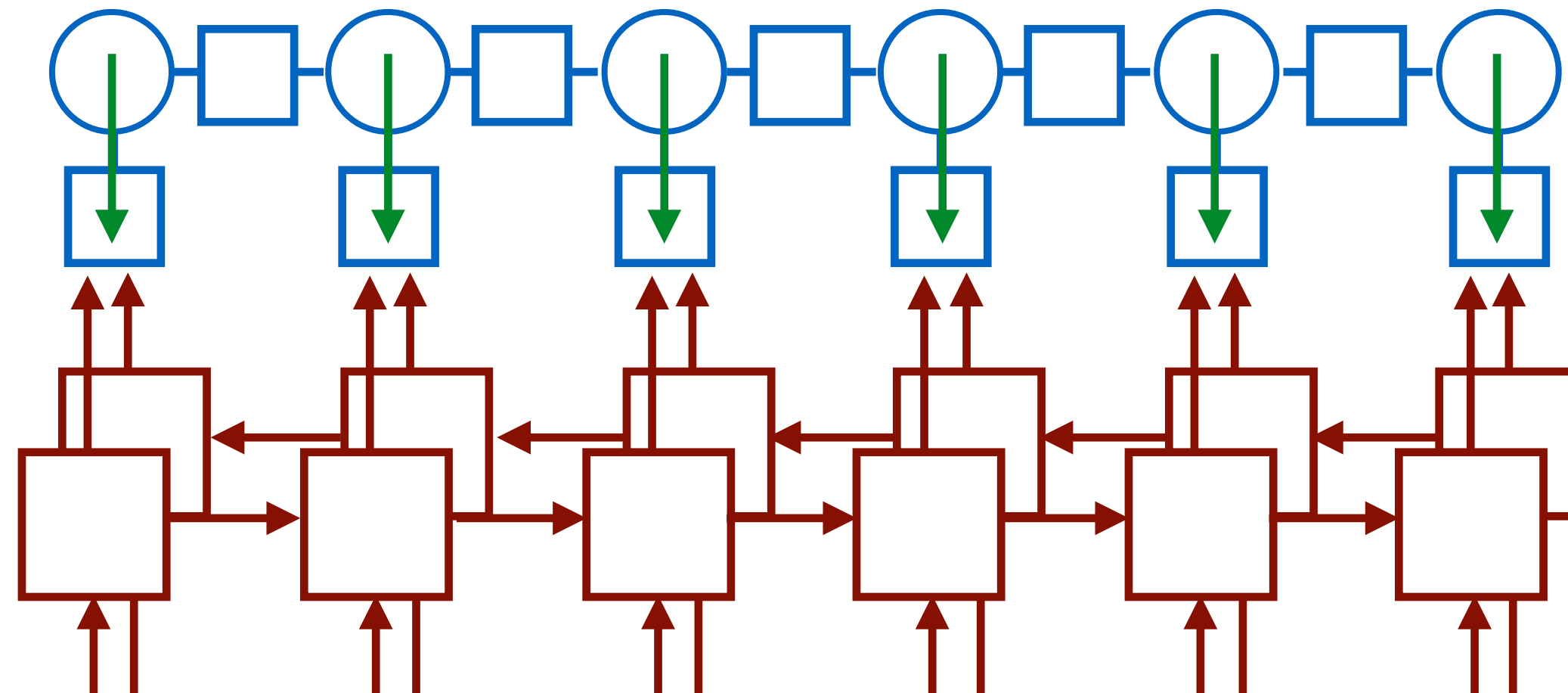
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2) Run forward-backward

3) Compute error signal

1) Compute $f(\mathbf{x})$

LSTM Neural CRFs

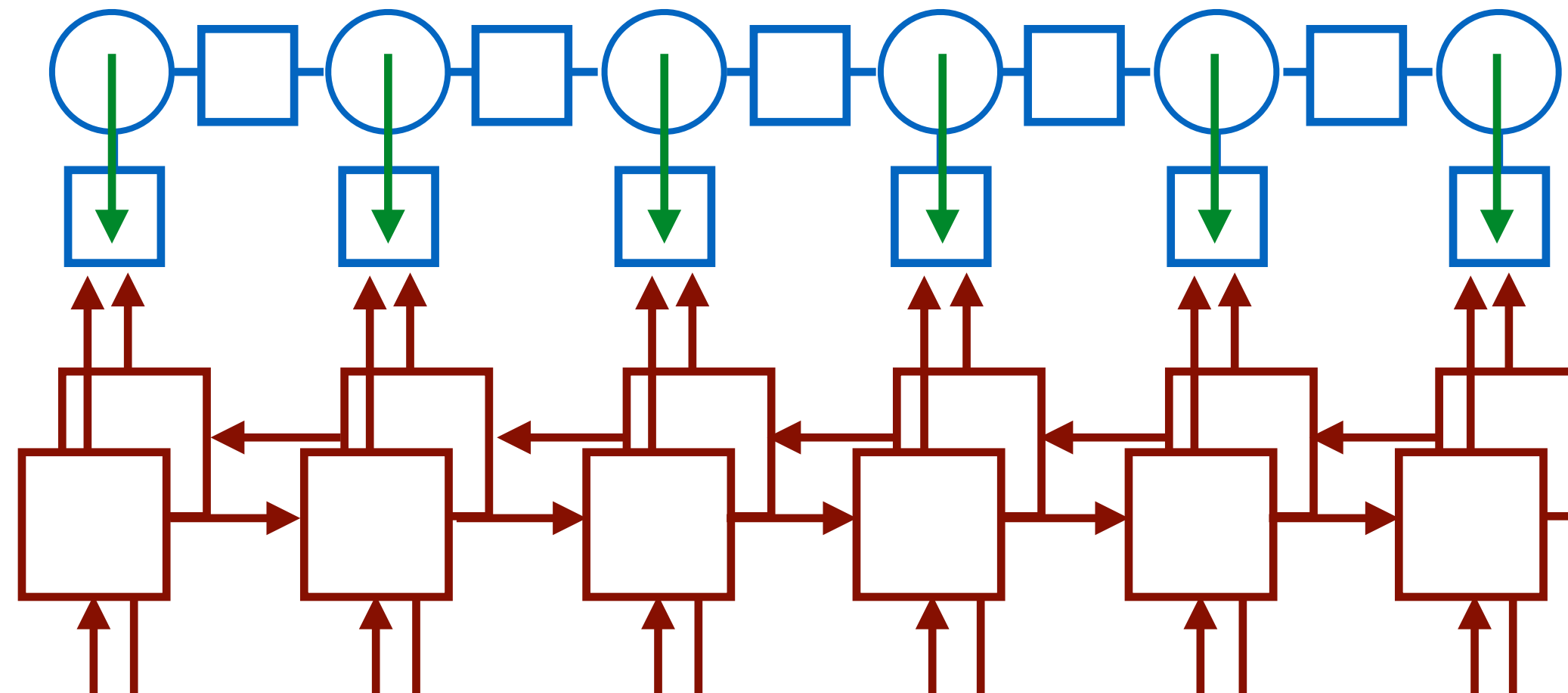
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Barack Obama will travel to Hangzhou

2) Run forward-backward

3) Compute error signal

1) Compute $f(\mathbf{x})$

4) Backprop (no knowledge of sequential structure required)

FFNN Neural CRF for NER

B-PER	I-PER	O	O	O	B-LOC	O	O	O	B-ORG	O	O	
<i>Barack Obama will travel to Hangzhou today for the G20 meeting .</i>												
PERSON					LOC	ORG						

FFNN Neural CRF for NER

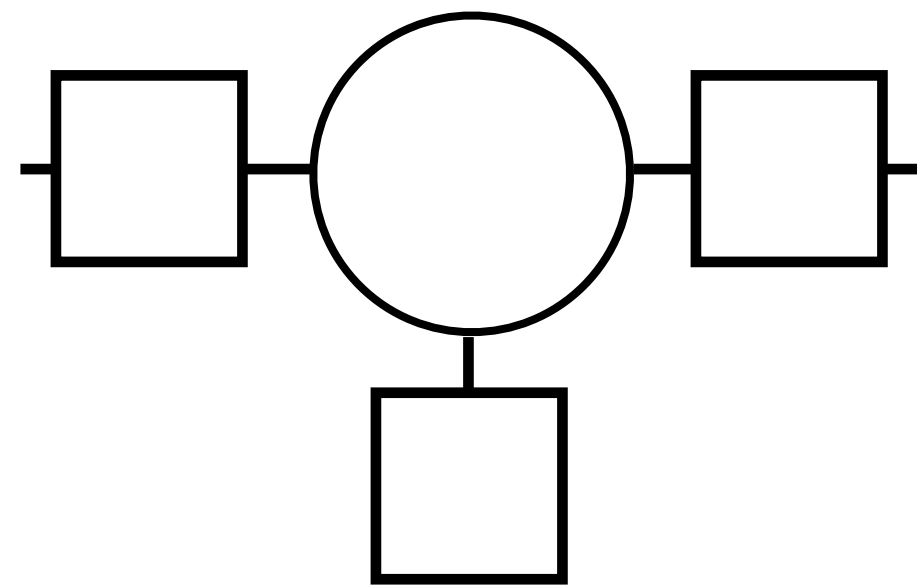
B-PER I-PER O O O B-LOC O O O B-ORG O O

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LOC

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to Hangzhou today

FFNN Neural CRF for NER

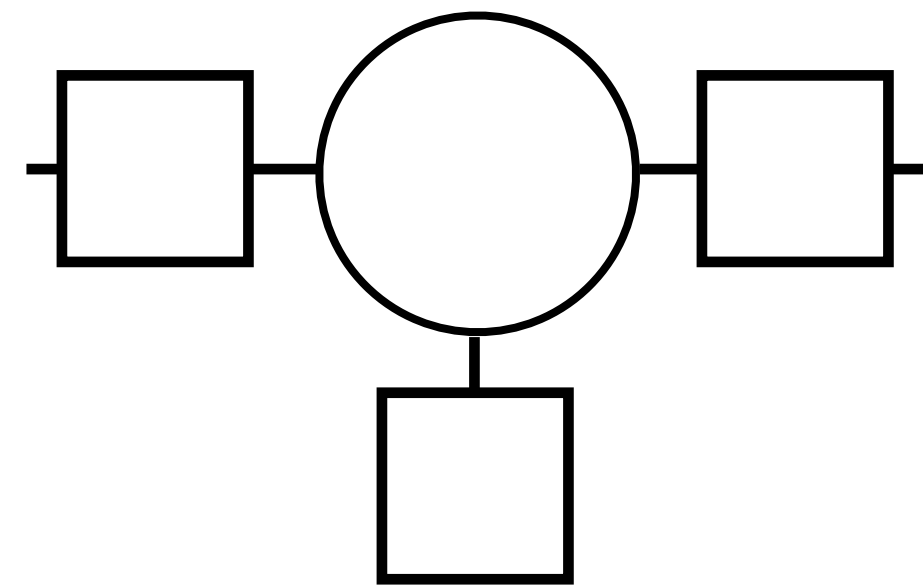
B-PER I-PER O O O B-LOC O O O B-ORG O O

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PERSON

LOC

ORG



$$f(\mathbf{x}, i) = [\text{emb}(\mathbf{x}_{i-1}), \text{emb}(\mathbf{x}_i), \text{emb}(\mathbf{x}_{i+1})]$$



previous word curr word next word

to Hangzhou today

FFNN Neural CRF for NER

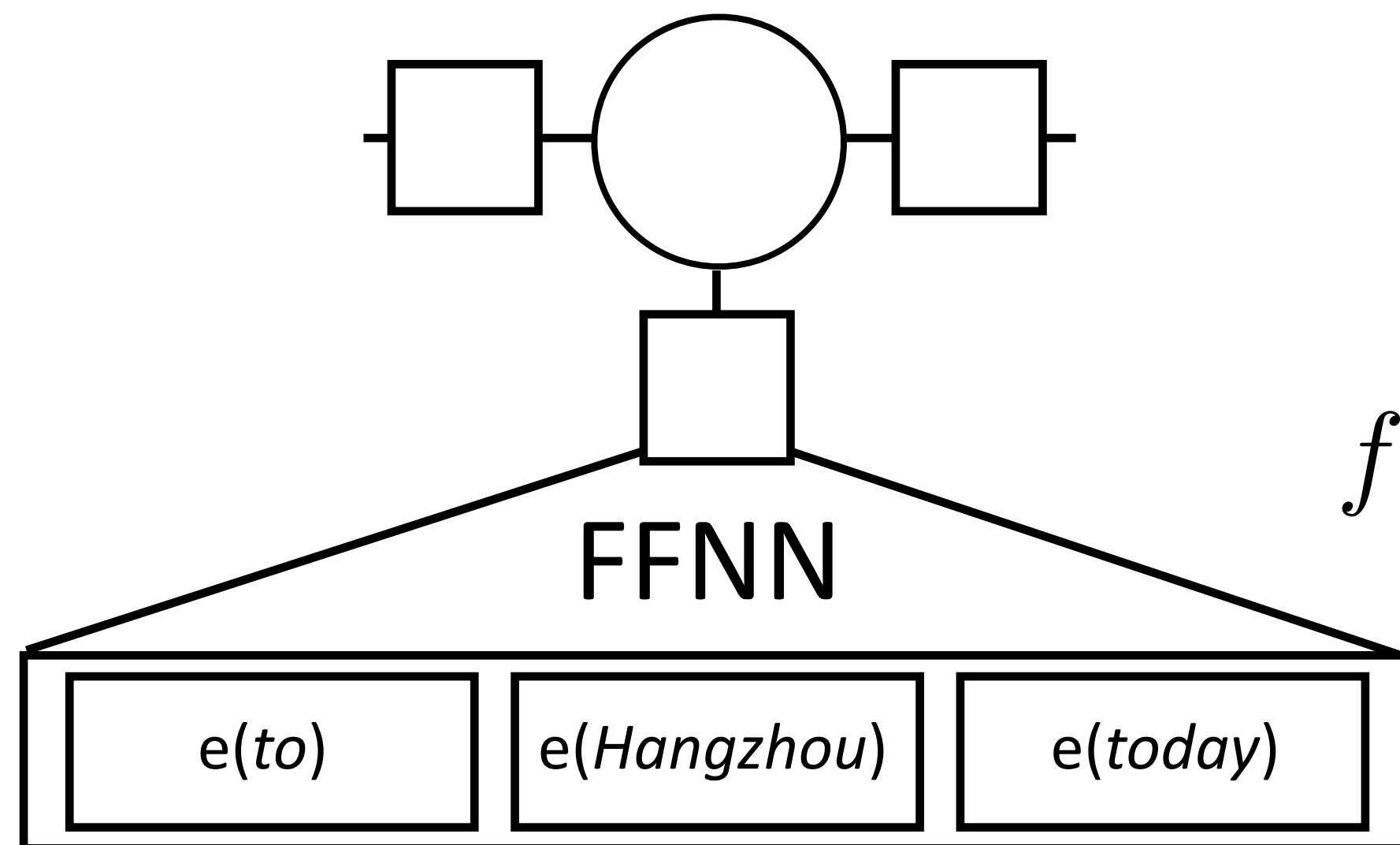
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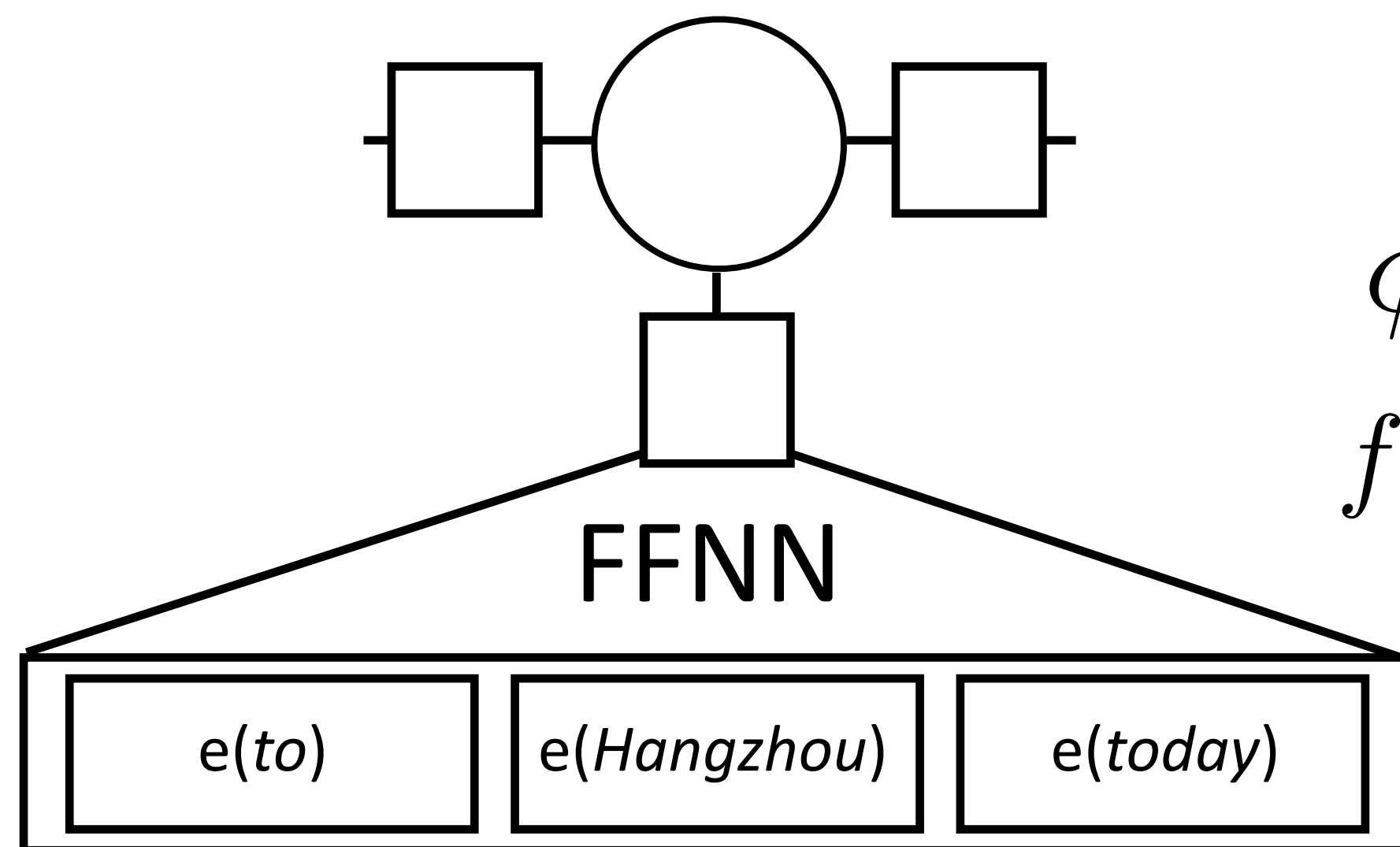
B-PER I-PER O O O B-LOC O O O B-ORG O O

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$$\phi_e = Wg(Vf(\mathbf{x}, i))$$

$$f(\mathbf{x}, i) = [\text{emb}(\mathbf{x}_{i-1}), \text{emb}(\mathbf{x}_i), \text{emb}(\mathbf{x}_{i+1})]$$

previous word curr word next word

to Hangzhou today

RNN Language Modeling

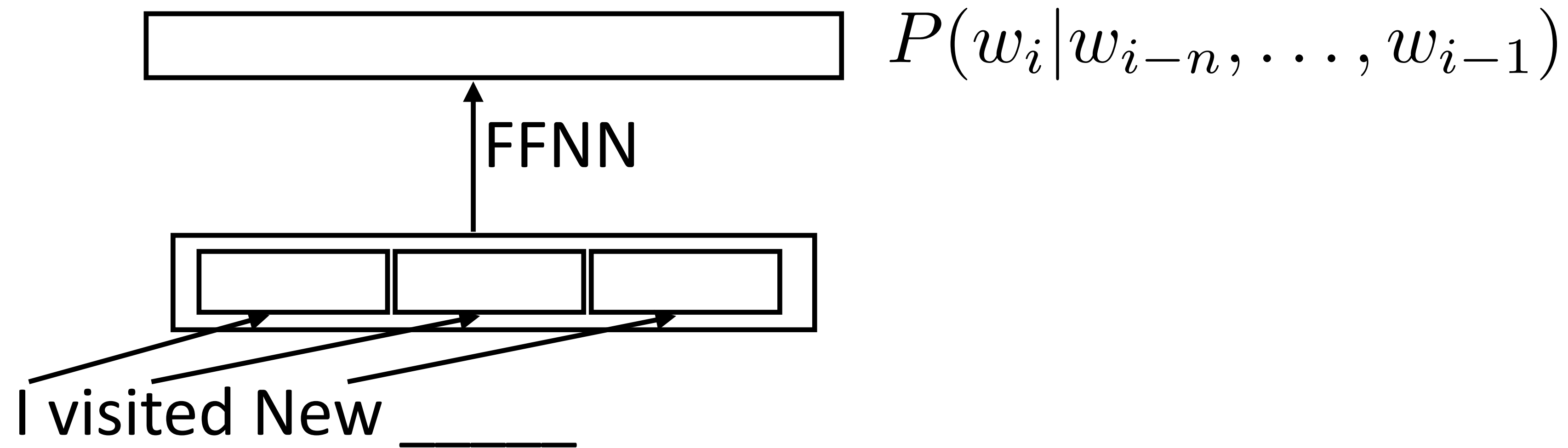
Neural Language Models

- ▶ Early work: feedforward neural networks looking at context

Bengio (2003), Mnih and Hinton (2003)

Neural Language Models

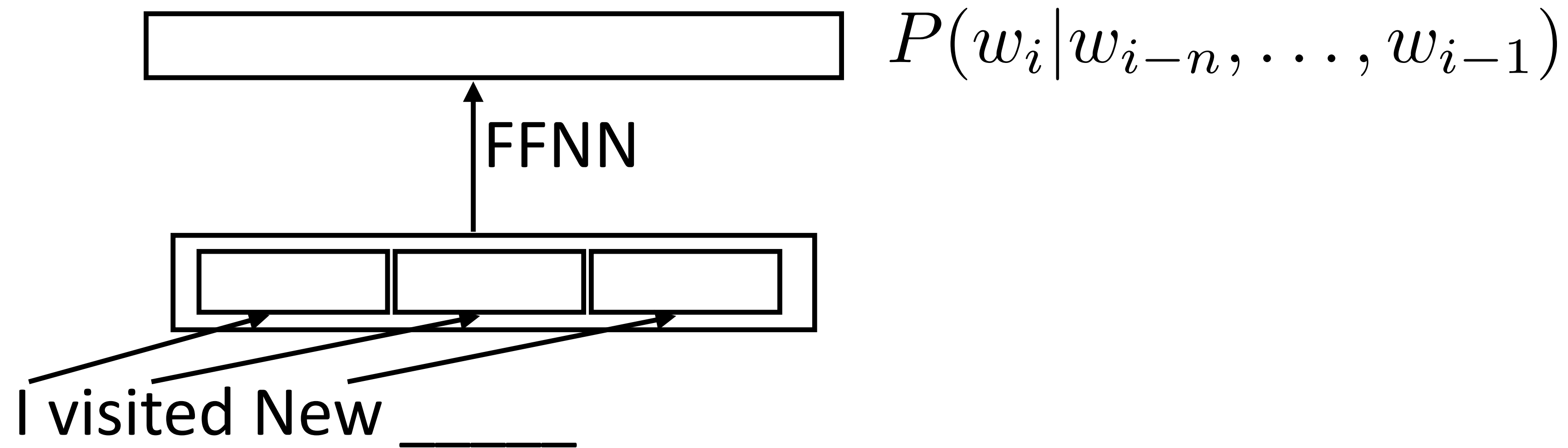
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Bengio (2003), Mnih and Hinton (2003)

Neural Language Models

- ▶ Early work: feedforward neural networks looking at context

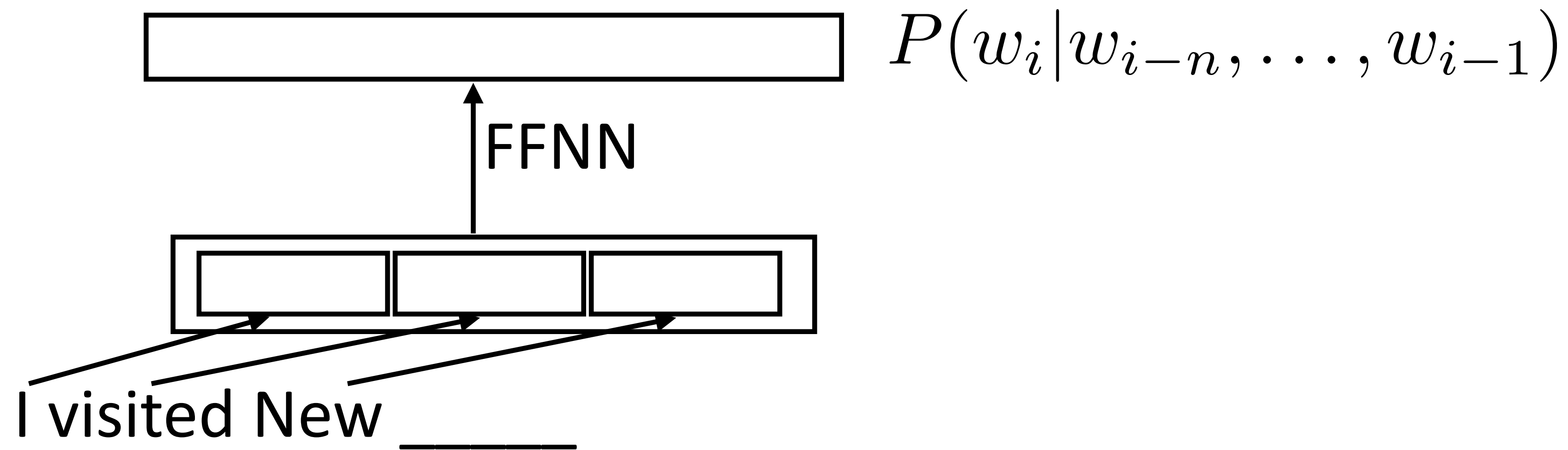


- ▶ Slow to train over lots of data!

Bengio (2003), Mnih and Hinton (2003)

Neural Language Models

- ▶ Early work: feedforward neural networks looking at context



- ▶ Slow to train over lots of data!
- ▶ Still only look at a fixed window of information...can we use more?

Bengio (2003), Mnih and Hinton (2003)

Neural Probabilistic Language Model

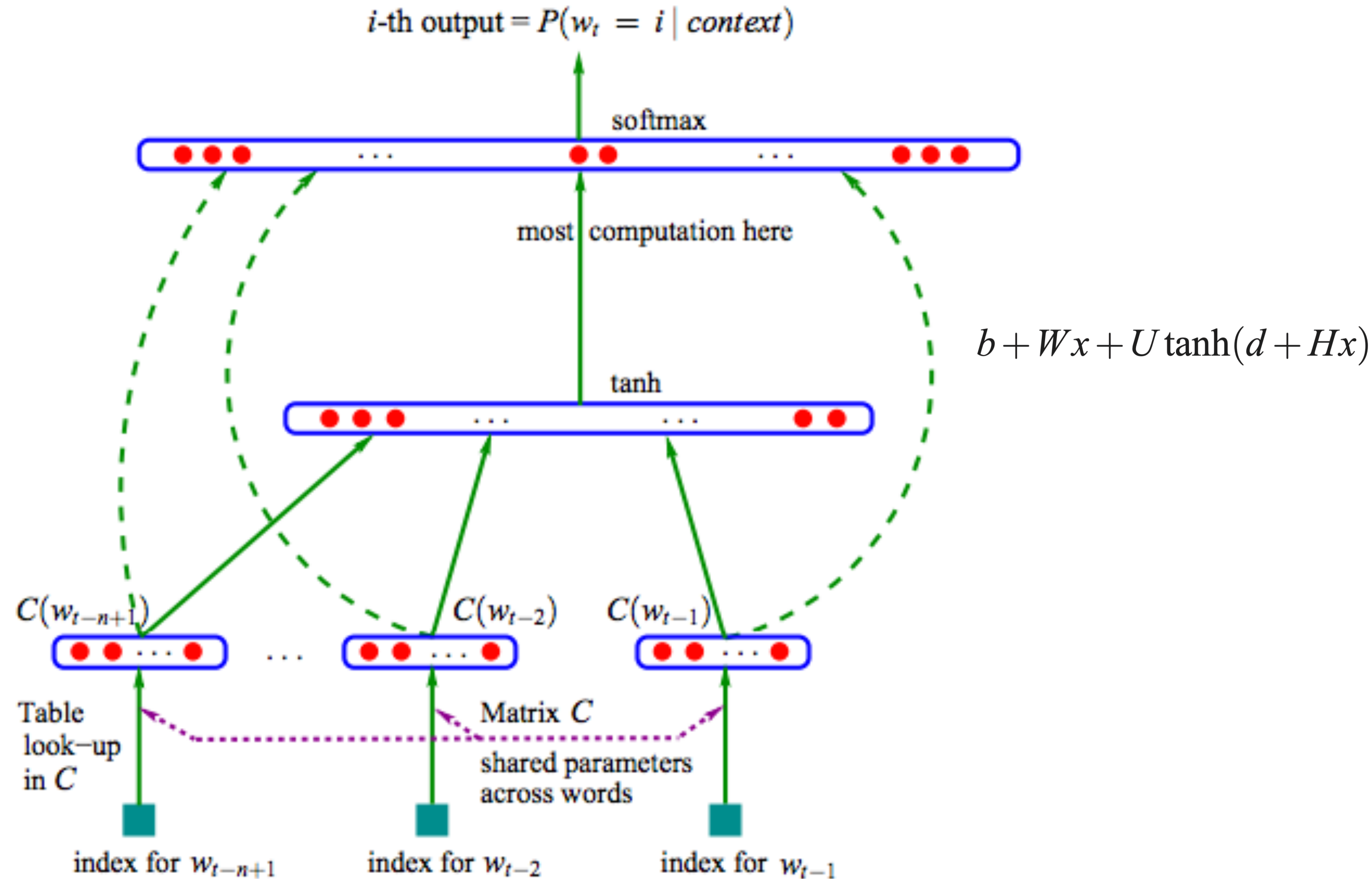
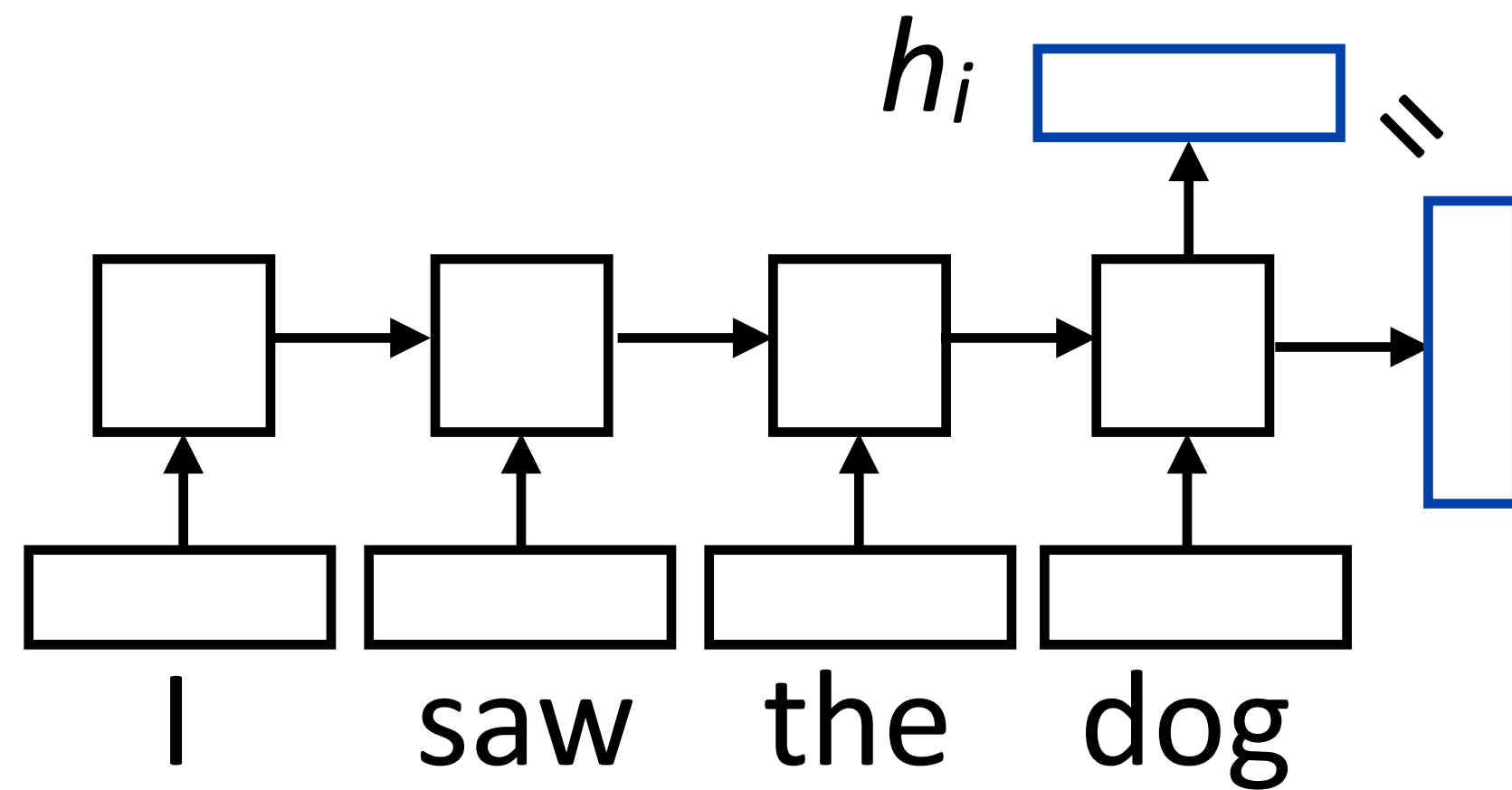


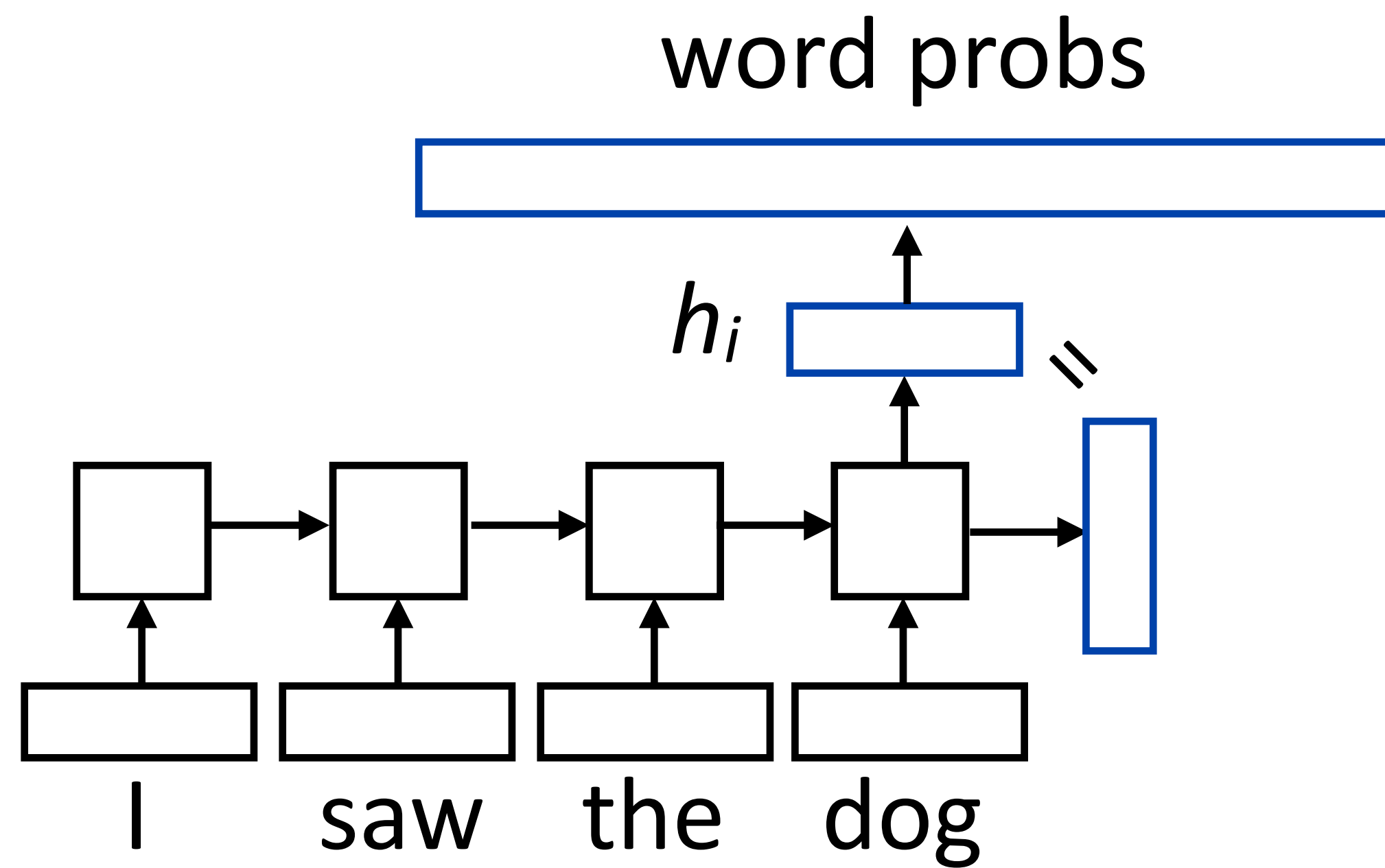
Figure 1: Neural architecture: $f(i, w_{t-1}, \dots, w_{t-n+1}) = g(i, C(w_{t-1}), \dots, C(w_{t-n+1}))$ where g is the neural network and $C(i)$ is the i -th word feature vector.

Bengio et al. (2003)

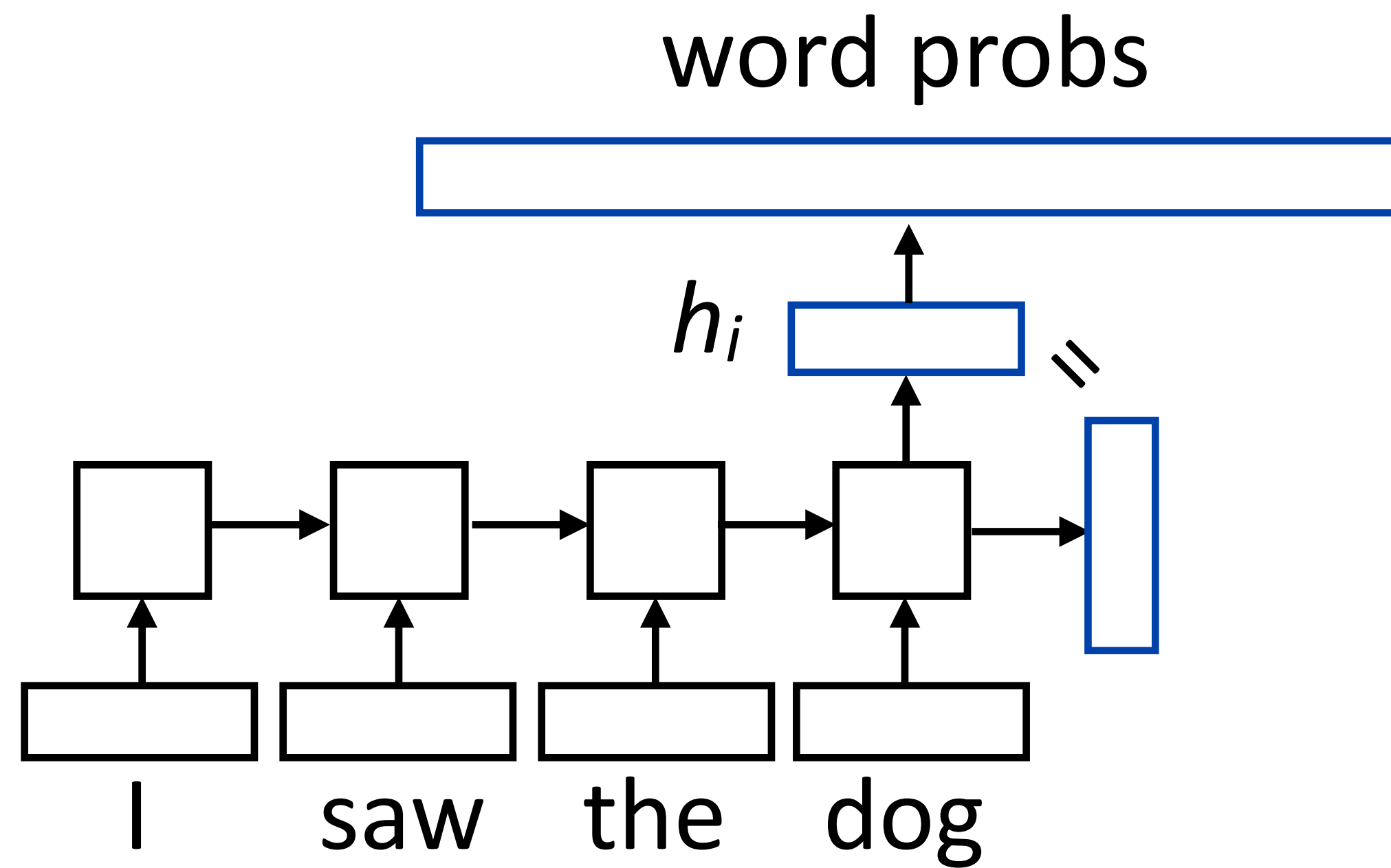
RNN Language Modeling



RNN Language Modeling



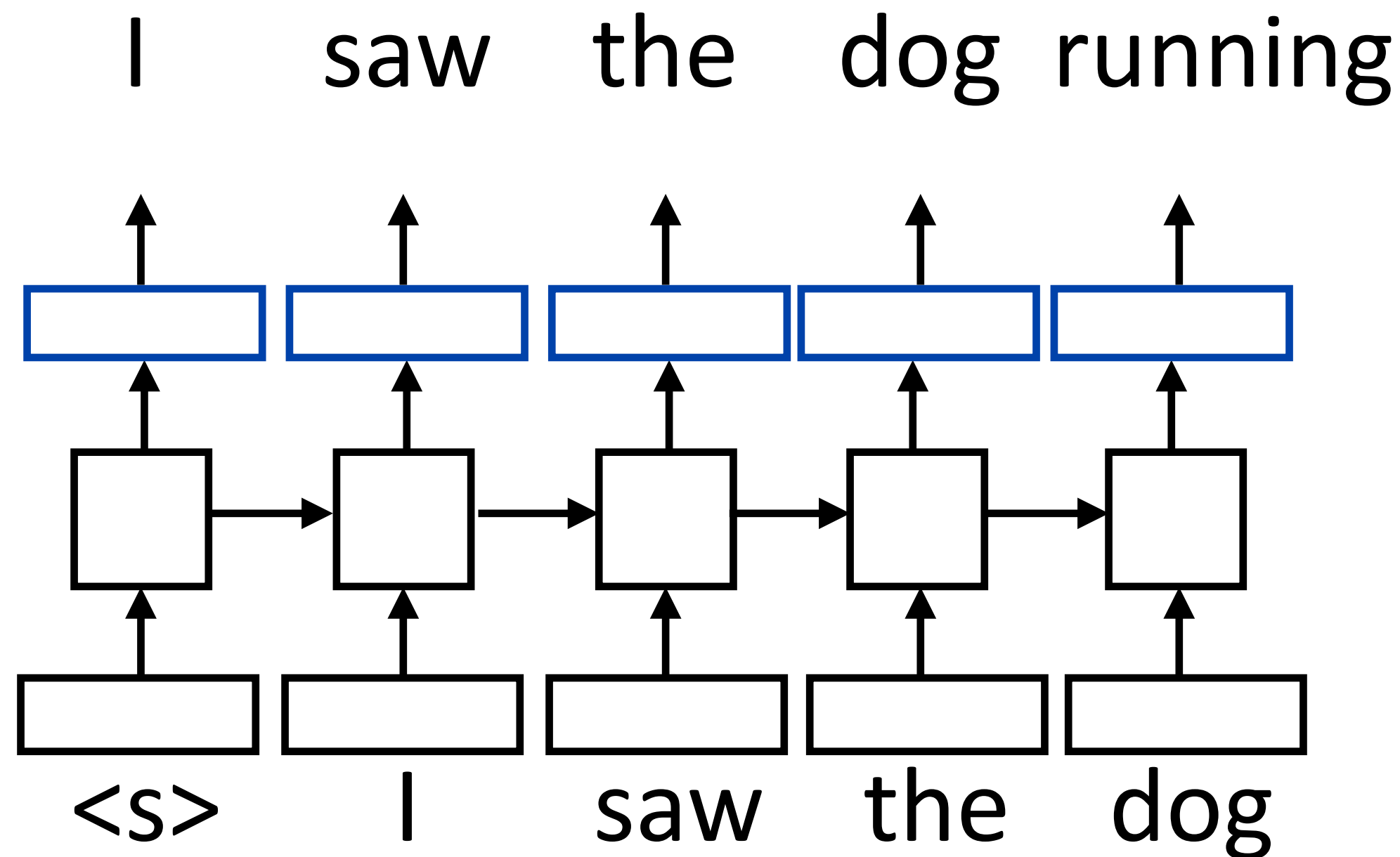
RNN Language Modeling



$$P(w|\text{context}) = \text{softmax}(W\mathbf{h}_i)$$

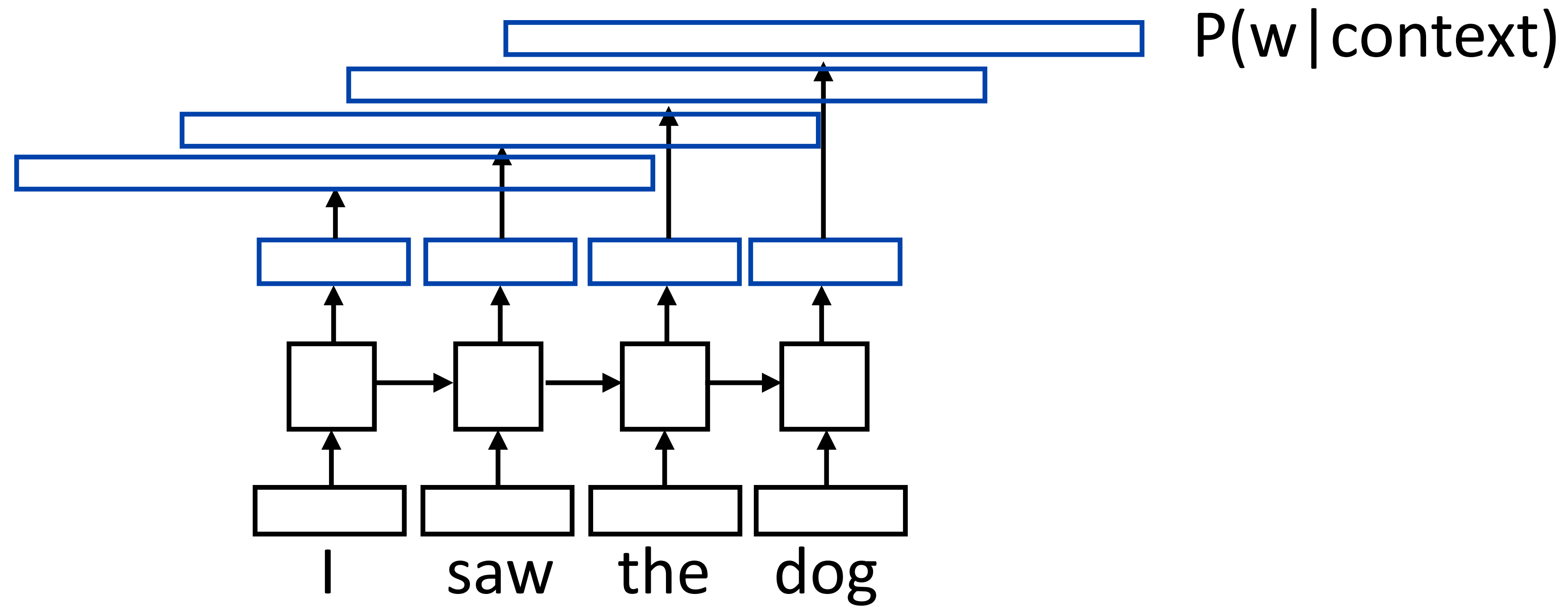
- ▶ W is a (vocab size) x (hidden size) matrix

Training RNNLMs

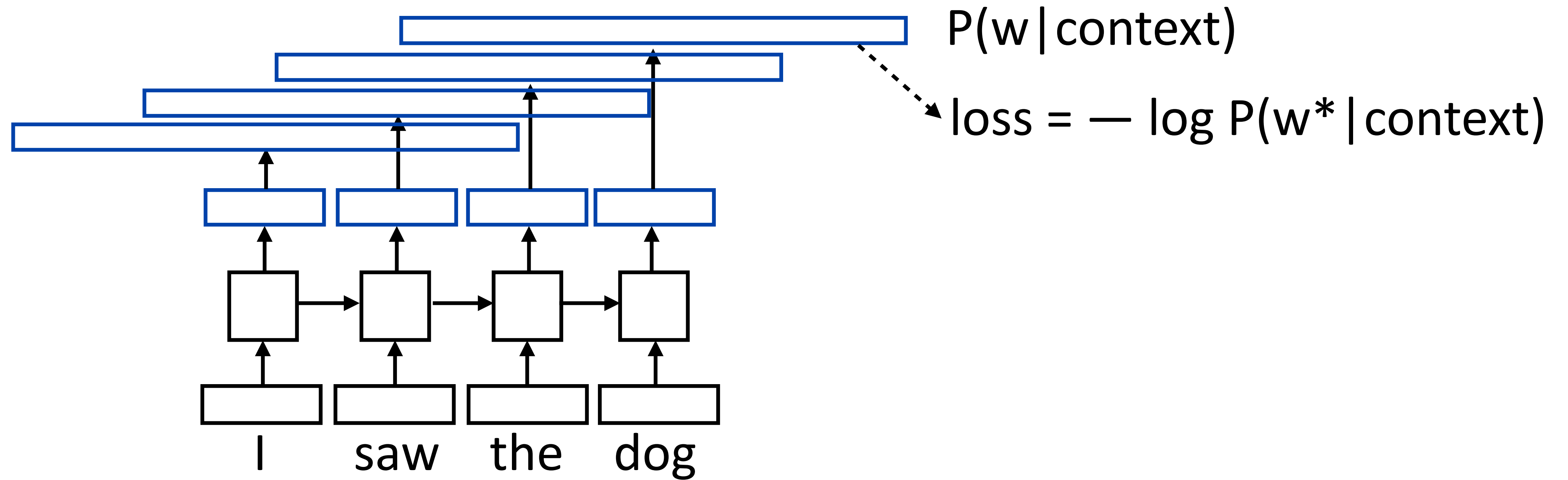


- ▶ Input is a sequence of words, output is those words shifted by one,
- ▶ Allows us to efficiently batch up training across time (one run of the RNN)

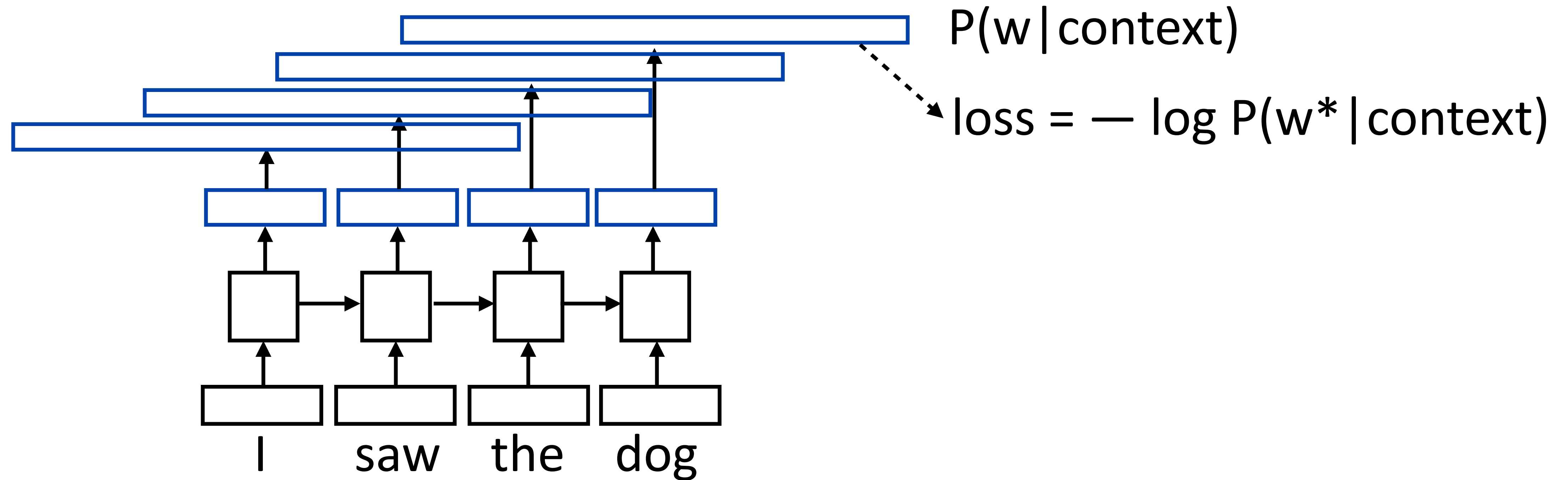
Training RNNLMs



Training RNNLMs

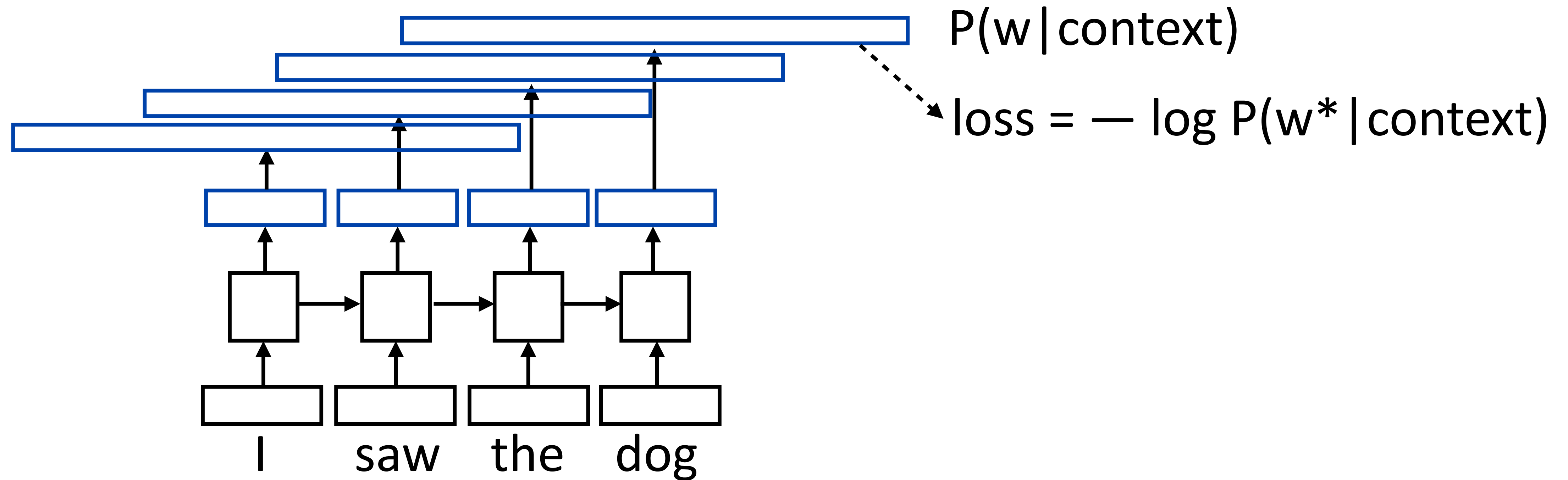


Training RNNLMs



- ▶ Total loss = sum of negative log likelihoods at each position

Training RNNLMs



- ▶ Total loss = sum of negative log likelihoods at each position
- ▶ Backpropagate through the network to simultaneously learn to predict next word given previous words at all positions

Batched LM Training

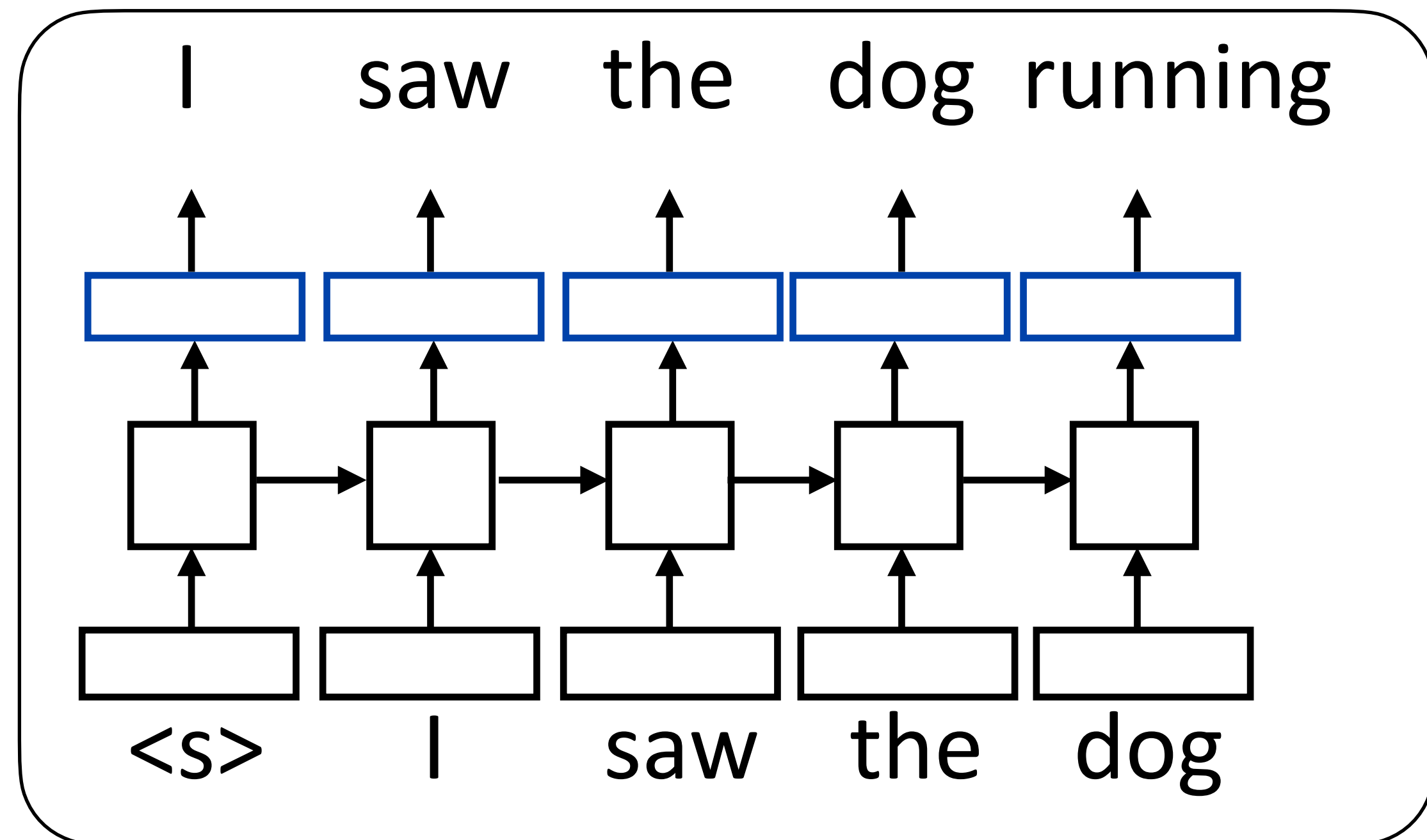
I saw the dog running in the park and it looked very excited to be there

Batched LM Training

I saw the dog running in the park and it looked very excited to be there

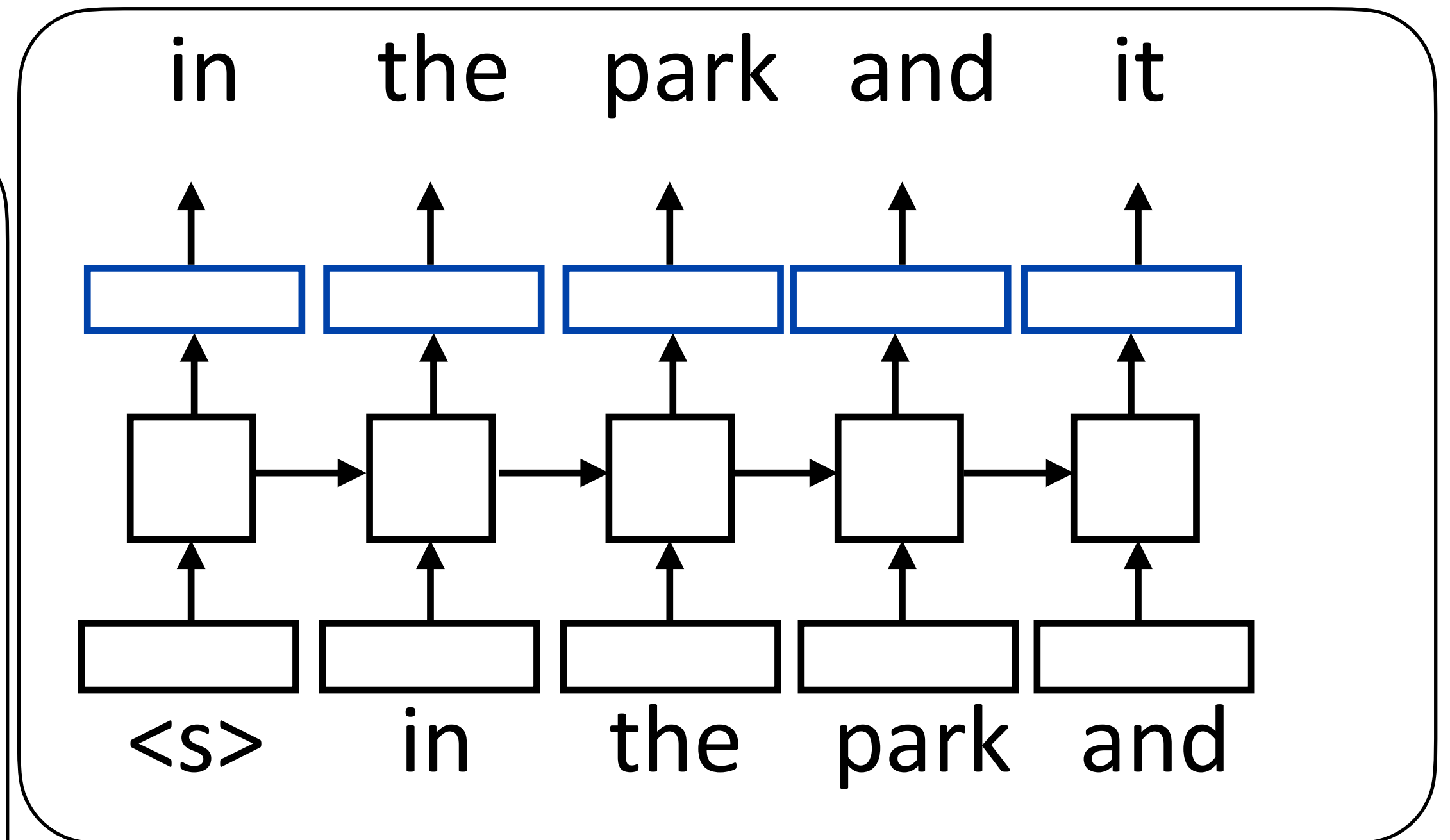
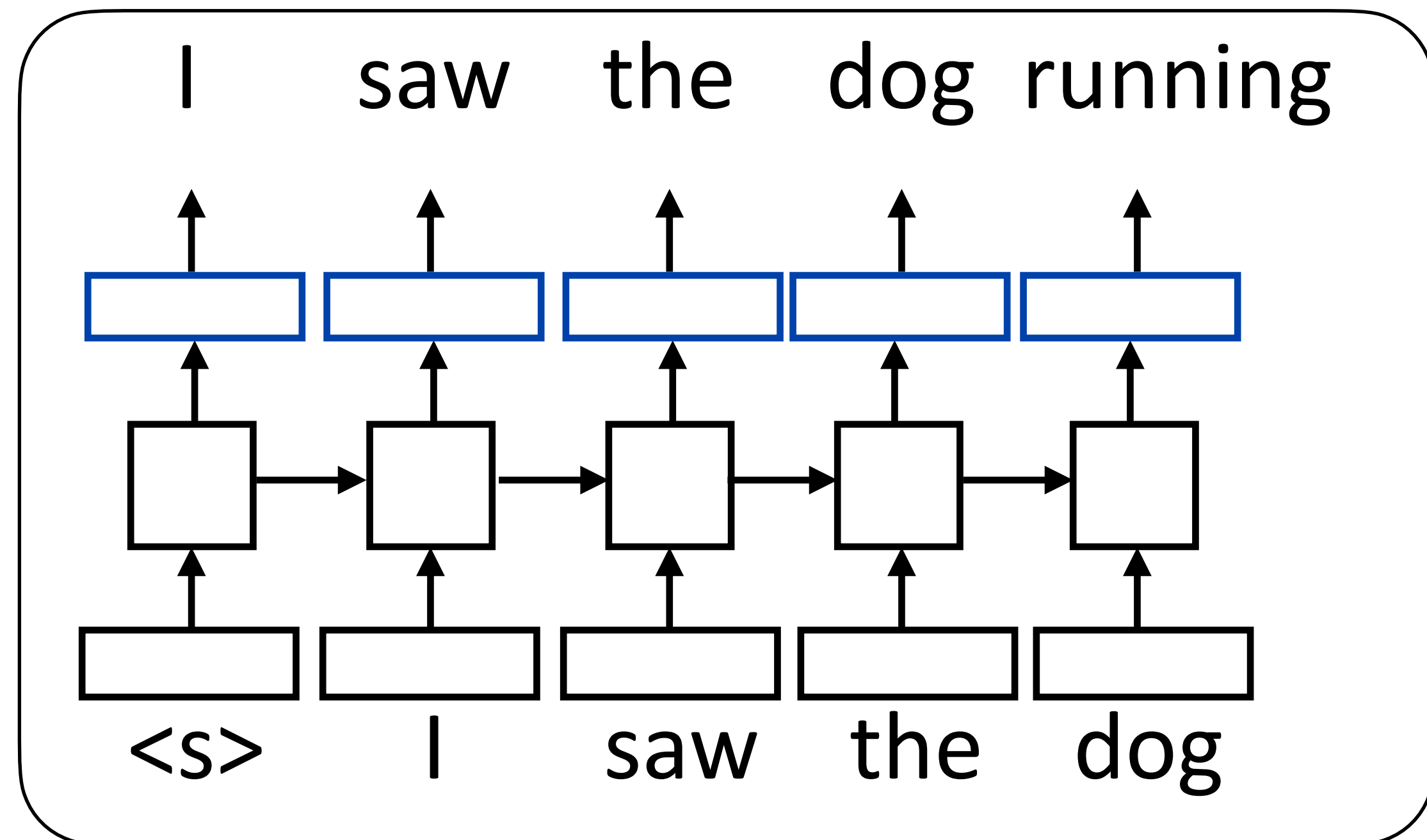
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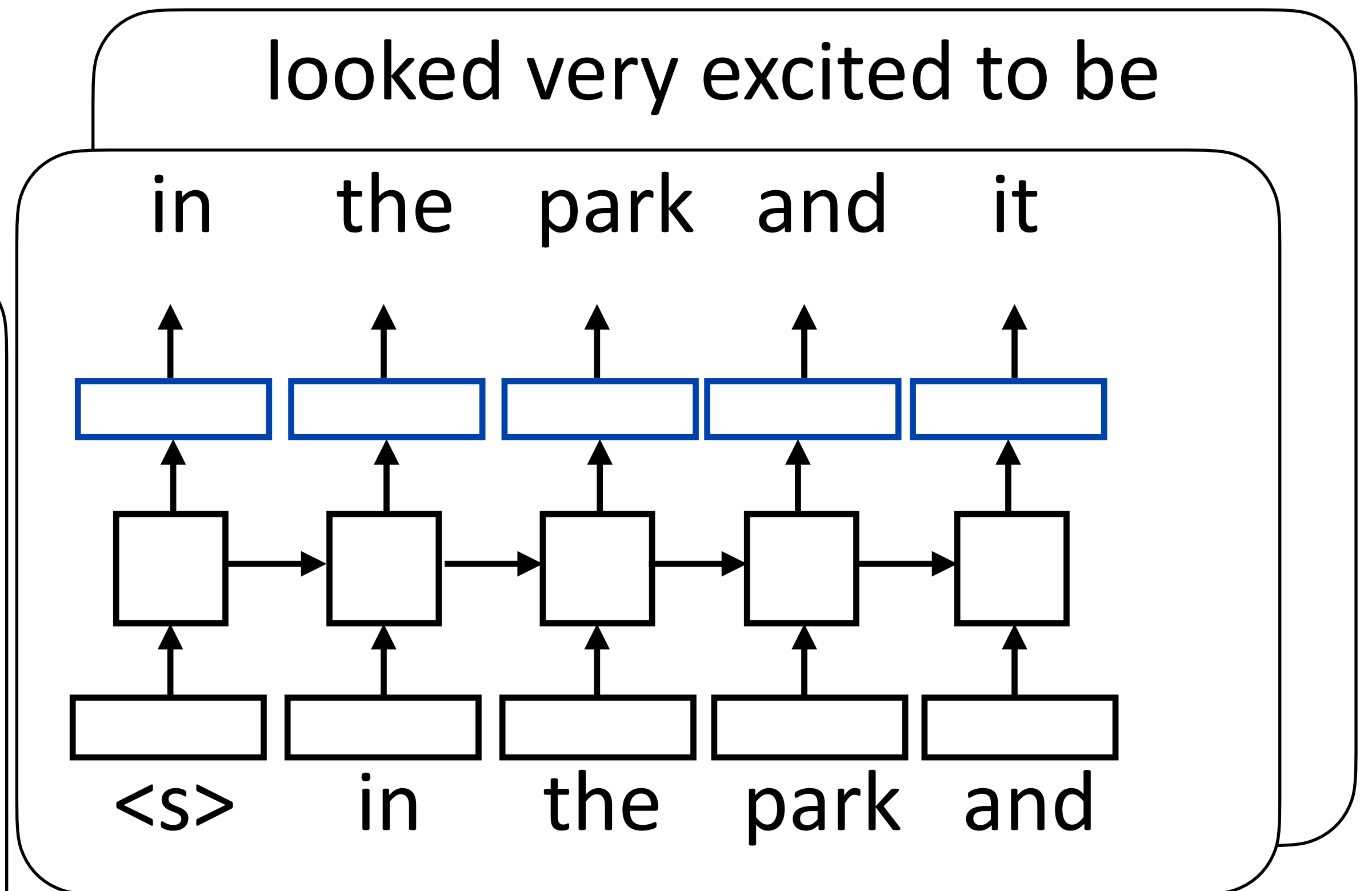
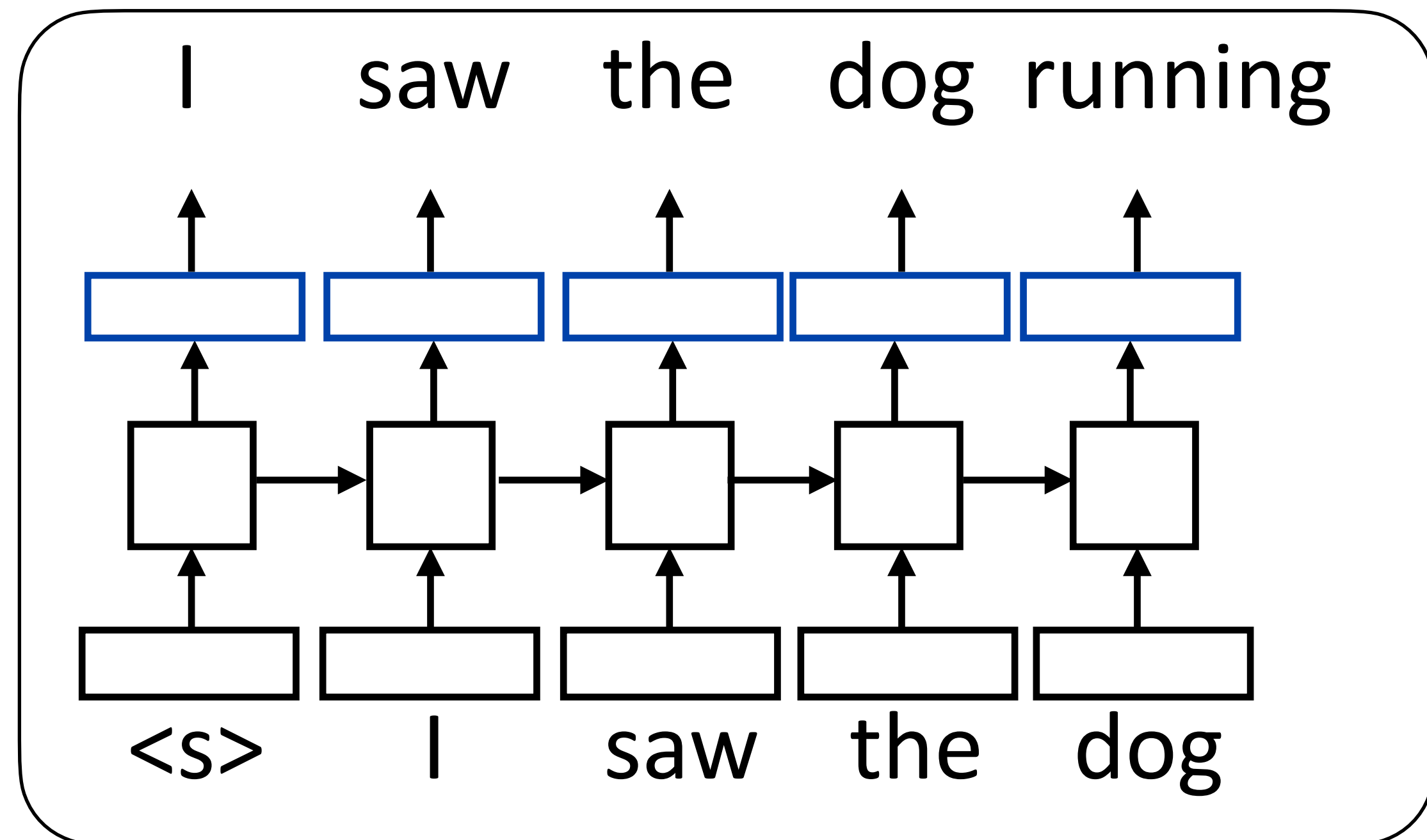
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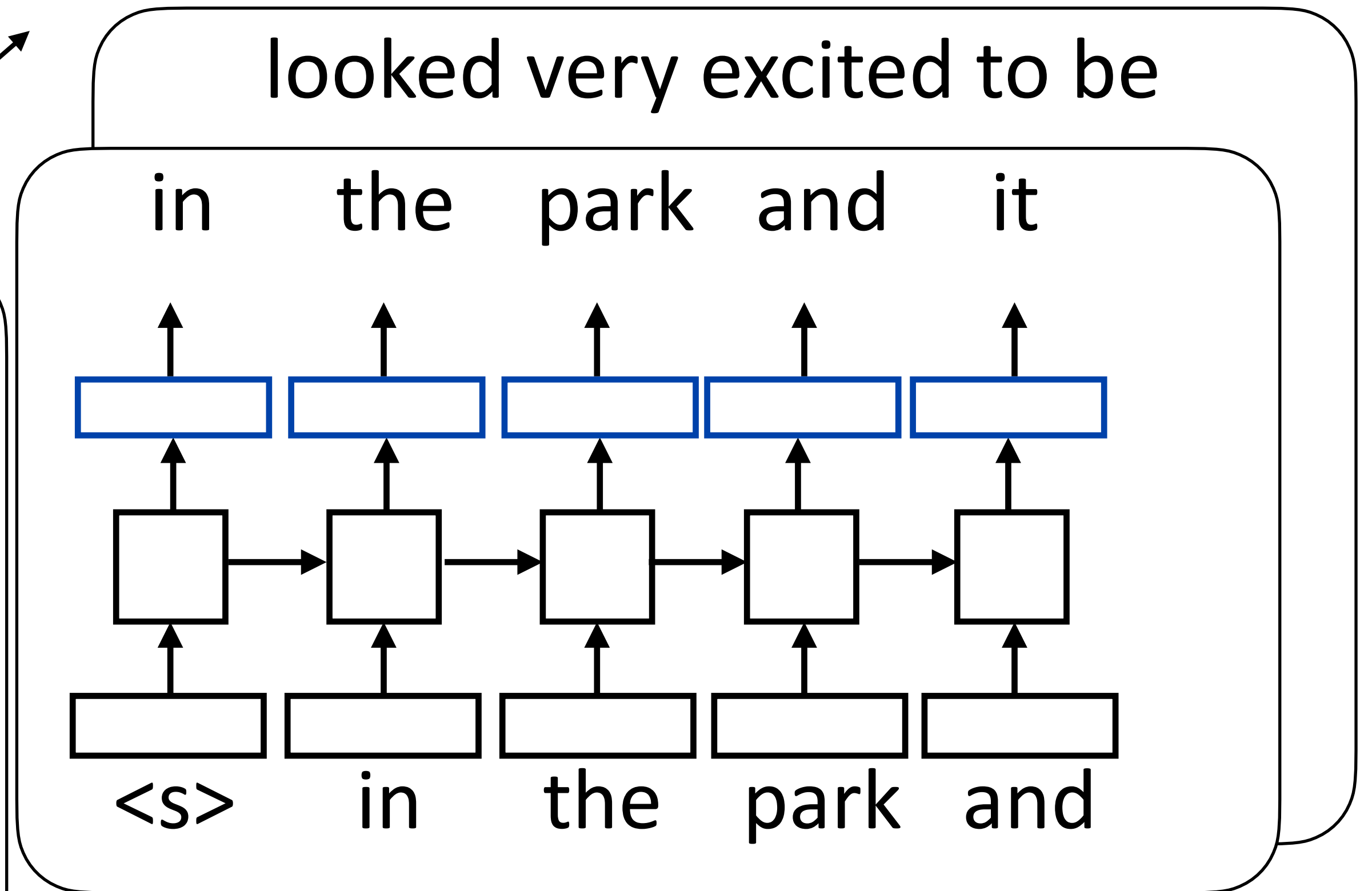
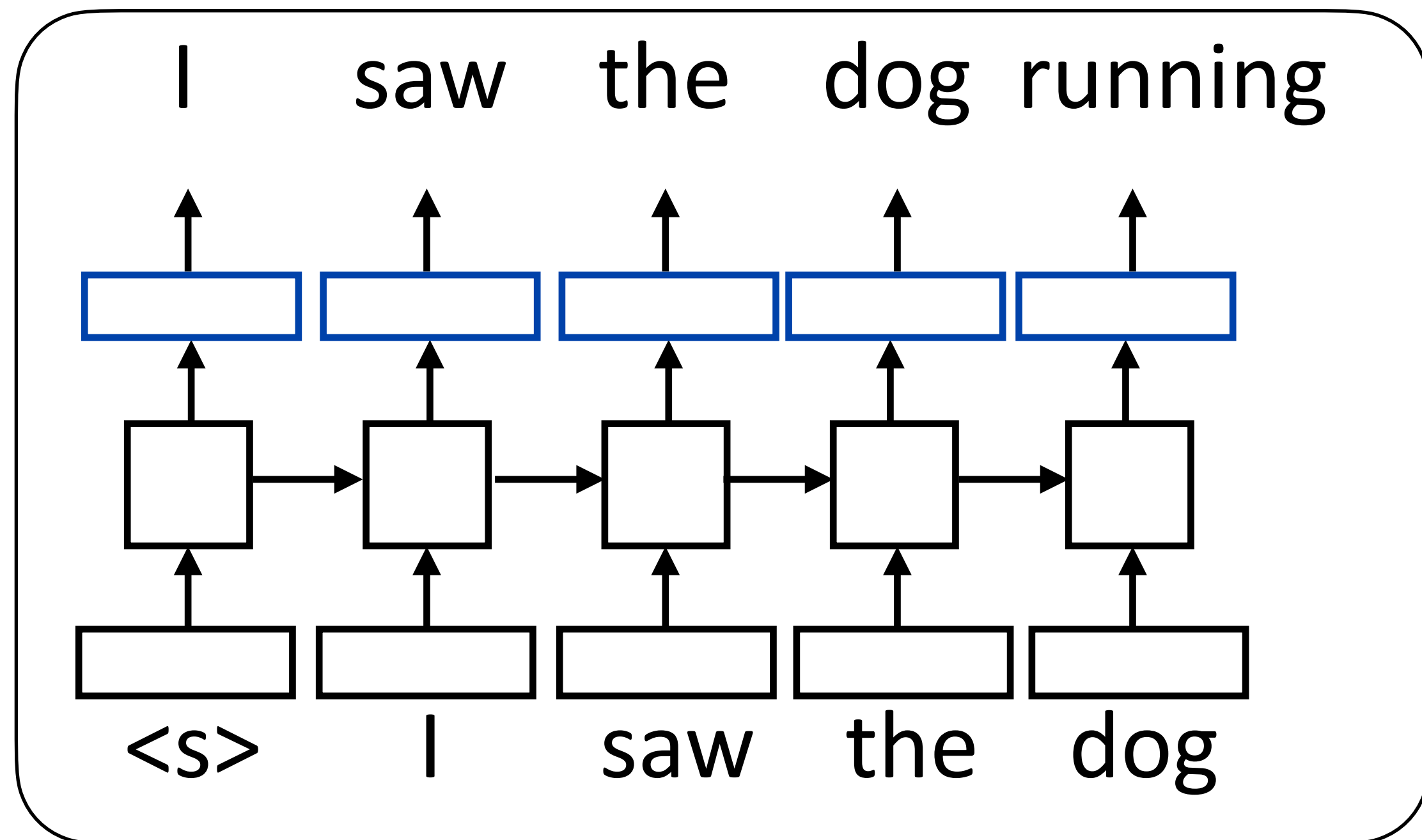
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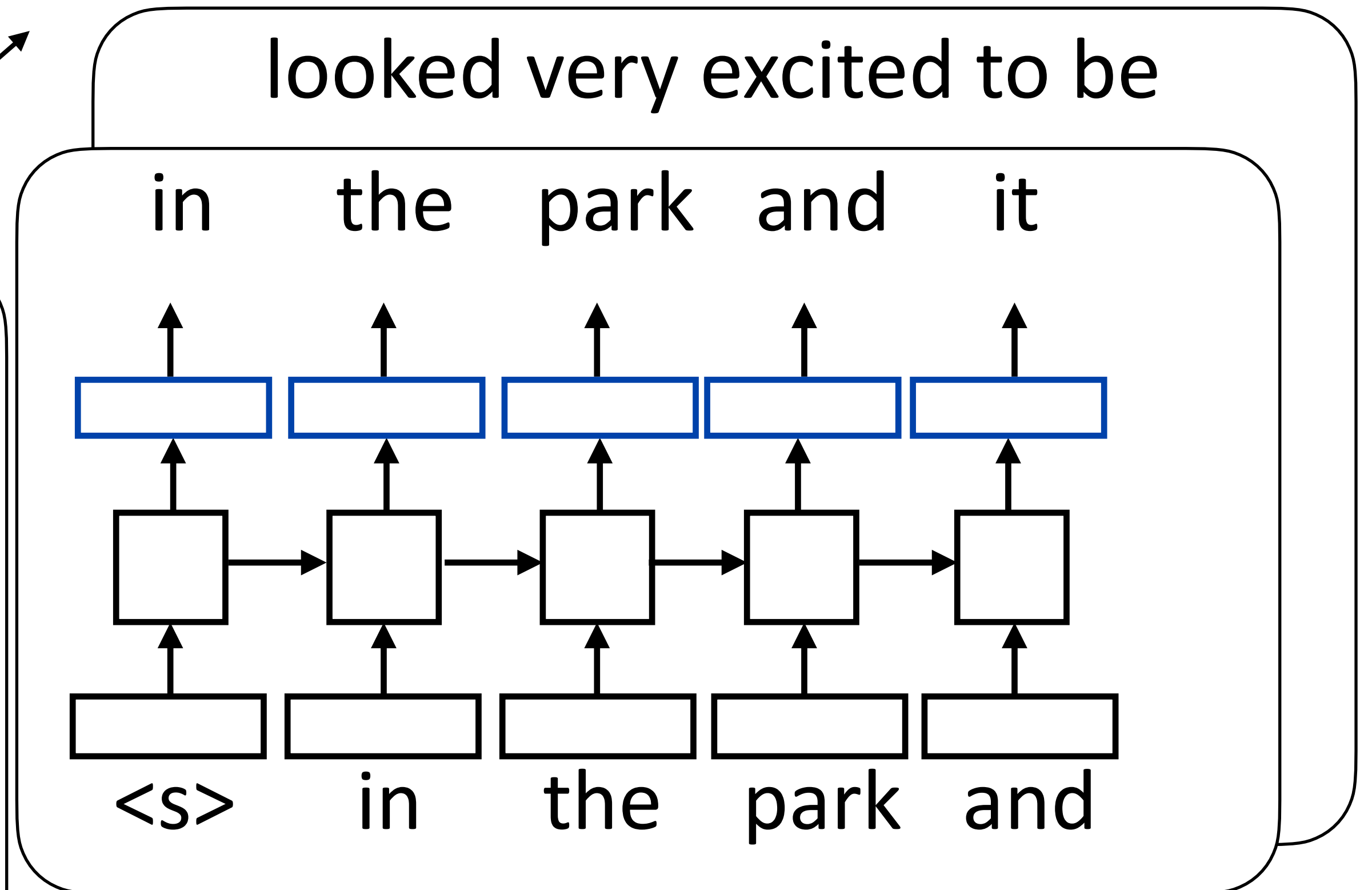
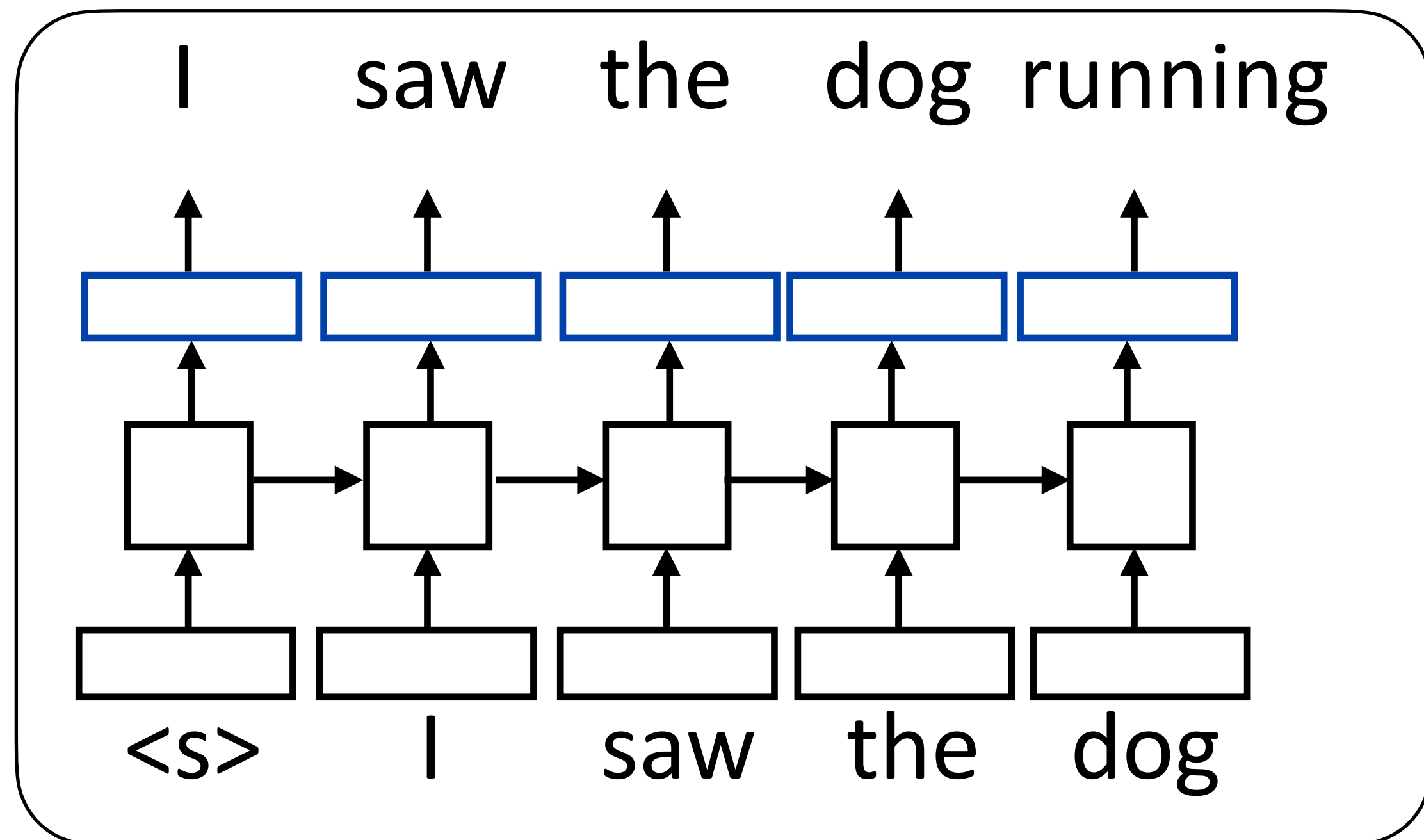
batch dim



Batched LM Training

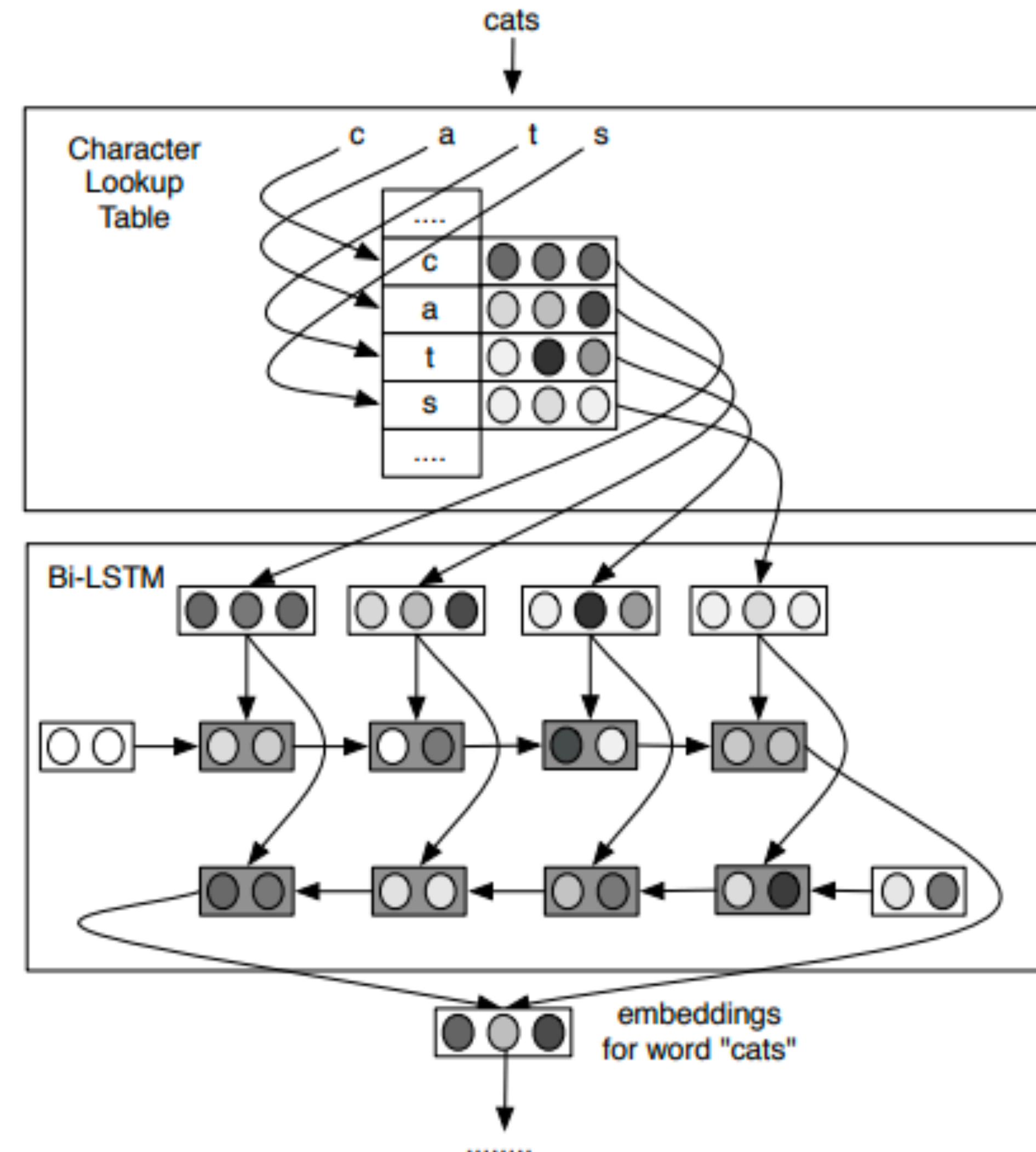
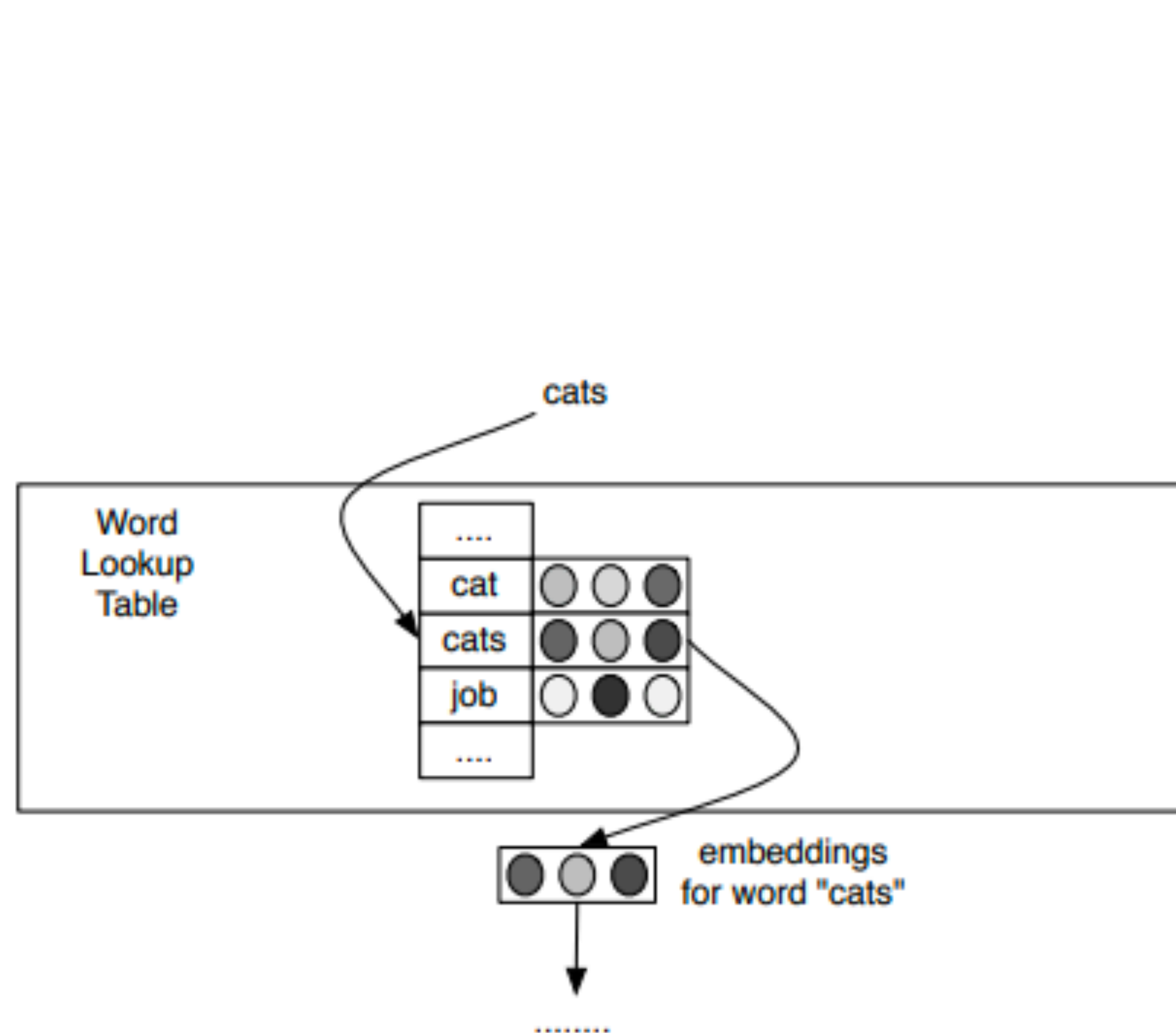
I saw the dog running in the park and it looked very excited to be there

batch dim



- ▶ Multiple sequences **and** multiple timestamps per sequence

Subword Models



Ling et al. (2015)

Padding

- ▶ Prepending or appending zeros
- ▶ To create batches of equal length for faster training time

▶ Padding for character-level LSTM

Adylov [2 7 13 8 10 12]

Sloan [4 8 10 5 9]

Harb [3 5 11 6]

San [4 5 9]



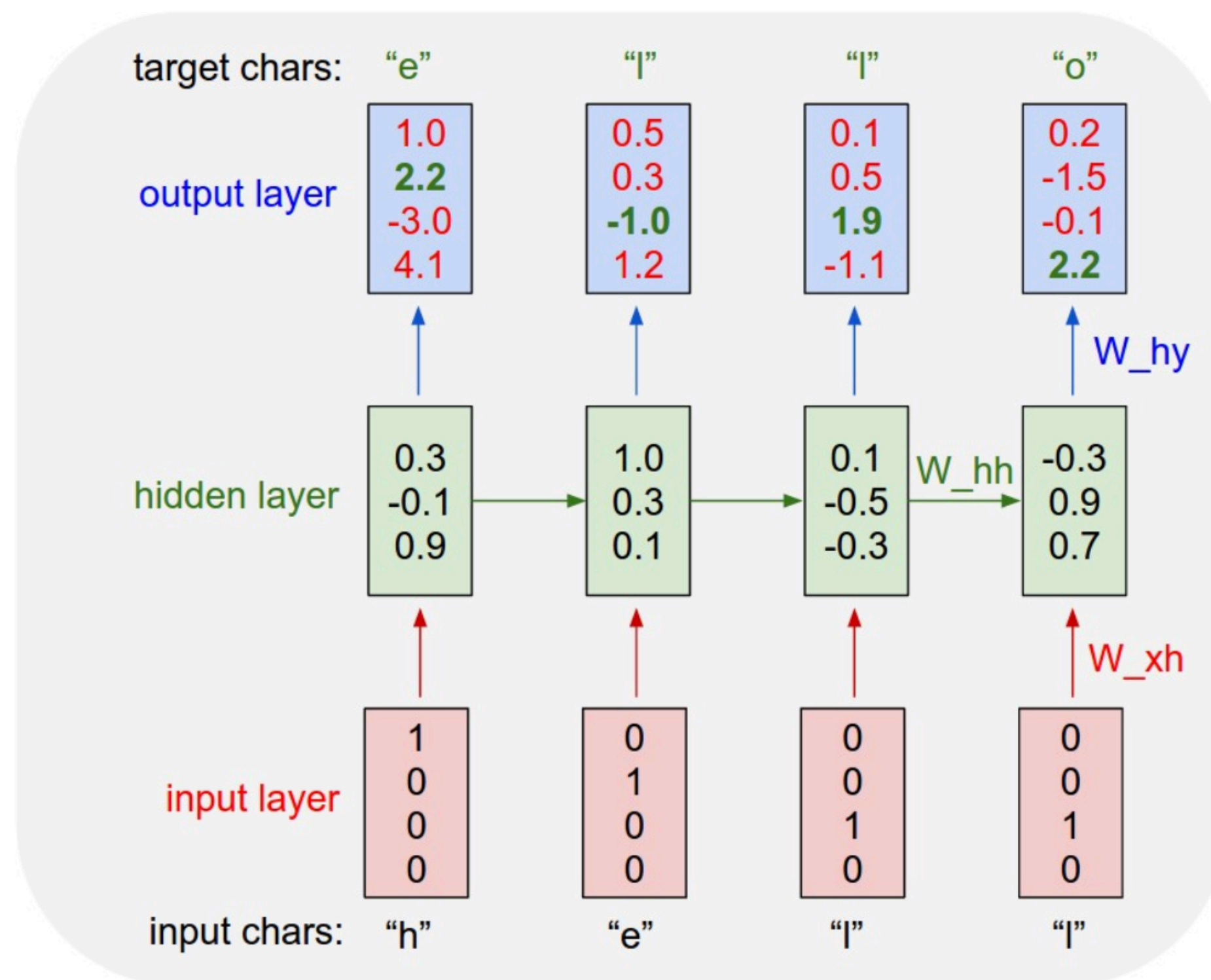
Adylov [2 7 13 8 10 12]

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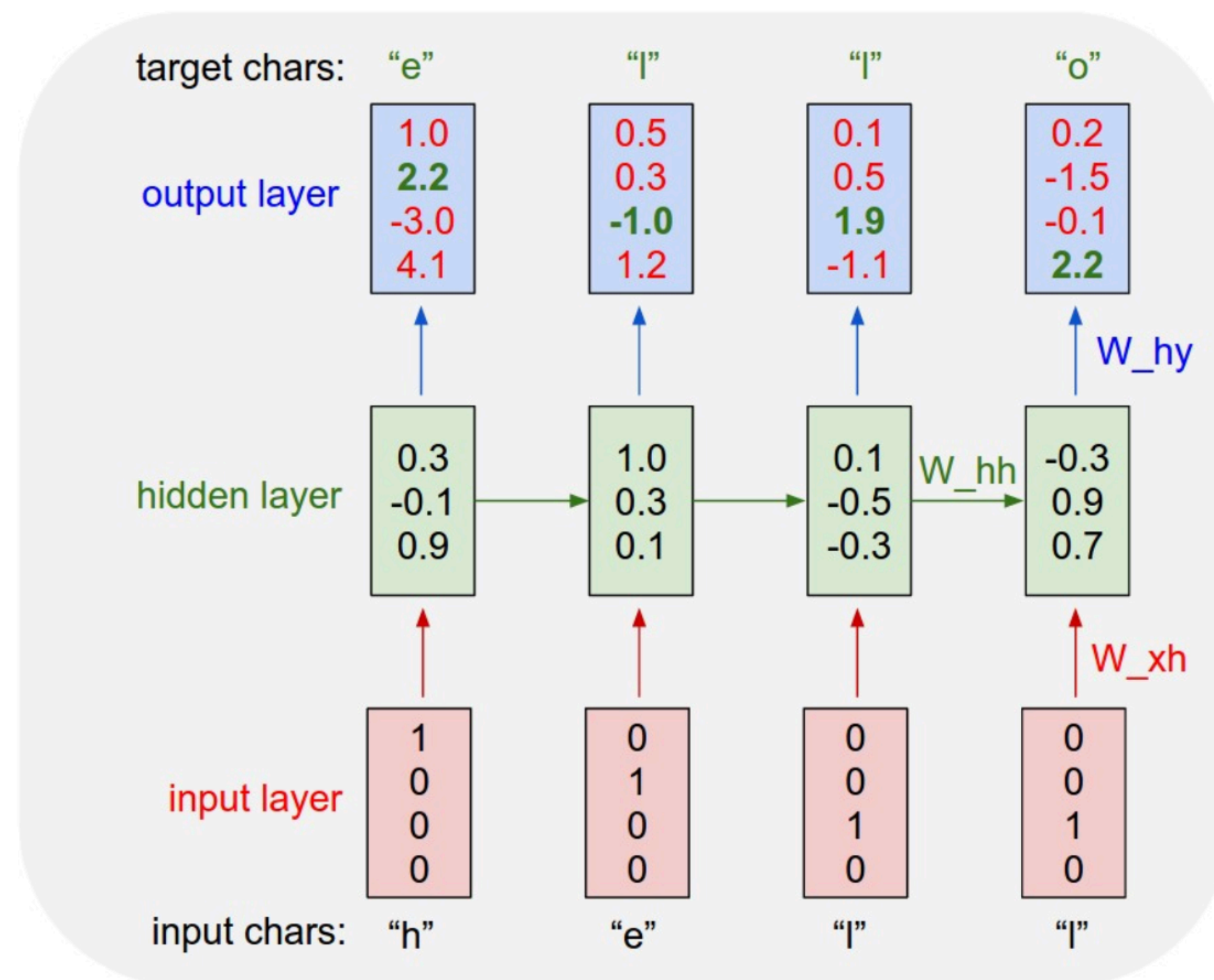
Visualizing LSTMs



An example RNN with 4-dimensional input and output layers, and a hidden layer of 3 units (neurons). This diagram shows the activations in the forward pass when the RNN is fed the characters "hell" as input. The output layer contains confidences the RNN assigns for the next character (vocabulary is "h,e,l,o"); We want the green numbers to be high and red numbers to be low.

Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code



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- ▶ Visualize activations of specific cells (components of **c**) to understand them

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The sole importance of the crossing of the Berezina lies in the fact that it plainly and indubitably proved the fallacy of all the plans for cutting off the enemy's retreat and the soundness of the only possible line of action--the one Kutuzov and the general mass of the army demanded--namely, simply to follow the enemy up. The French crowd fled at a continually increasing speed and all its energy was directed to reaching its goal. It fled like a wounded animal and it was impossible to block its path. This was shown not so much by the arrangements it made for crossing as by what took place at the bridges. When the bridges broke down, unarmed soldiers, people from Moscow and women with children who were with the French transport, all--carried on by vis inertiae--pressed forward into boats and into the ice-covered water and did not, surrender.

Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code
- ▶ Visualize activations of specific cells (components of **c**) to understand them
- ▶ Counter: know when to generate \n

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Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code
- ▶ Visualize activations of specific cells to see what they track

"You mean to imply that I have nothing to eat out of.... On the contrary, I can supply you with everything even if you want to give dinner parties," warmly replied Chichagov, who tried by every word he spoke to prove his own rectitude and therefore imagined Kutuzov to be animated by the same desire.

Kutuzov, shrugging his shoulders, replied with his subtle penetrating smile: "I meant merely to say what I said."

Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code
- ▶ Visualize activations of specific cells to see what they track
- ▶ Binary switch: tells us if we're in a quote or not

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#ifdef CONFIG_AUDITSYSCALL
static inline int audit_match_class_bits(int class, u32 *mask)
{
    int i;
    if (classes[class]) {
        for (i = 0; i < AUDIT_BITMASK_SIZE; i++)
            if (mask[i] & classes[class][i])
                return 0;
    }
    return 1;
}
```

Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code
- ▶ Visualize activations of specific cells to see what they track
- ▶ Stack: activation based on indentation

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```
/* Unpack a filter field's string representation from user-space
 * buffer. */
char *audit_unpack_string(void **bufp, size_t *remain, size_t len)
{
    char *str;
    if (!*bufp || (len == 0) || (len > *remain))
        return ERR_PTR(-EINVAL);
    /* Of the currently implemented string fields, PATH_MAX
     * defines the longest valid length.
     */
}
```

Visualizing LSTMs

- ▶ Train *character* LSTM language model (predict next character based on history) over two datasets: War and Peace and Linux kernel source code
- ▶ Visualize activations of specific cells to see what they track
- ▶ Uninterpretable: probably doing double-duty, or only makes sense in the context of another activation

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Takeaways

- ▶ RNNs can transduce inputs (produce one output for each input) or compress the whole input into a vector
- ▶ Useful for a range of tasks with sequential input: sentiment analysis, natural language inference, machine translation, language modeling